Landau Geometry of Computational Resources

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Abstract

We construct foundations for a differential geometry of computational resources using effective domain theory internal to Hyland's effective topos. Rational-valued functions on inputs give a ring of bounded coefficients and an algebra of eventual germs. Embedded resource modules represent asymptotic budgets; a coherent locale classifies their families, and measured execution supplies the implementations witnessing feasibility. Compactness and supercompactness characterize finite dominating portfolios and optimal implementations. Finite maxima over word-length fibres recover conventional worst-case time and workspace, and constructed clocks identify the polynomial budget with **P**. Implementationwise double-negation closure gives the exact classical reading of universal lower bounds while retaining the internal witness-bearing judgments.

Polynomial event valuations connect execution to differential geometry: their measured Taylor coefficients agree with geometric jets. Embedded budgets have formal deformation classifiers, and finite systems of budget and valuation constraints have intrinsic lifting obstructions. Overhead localization acts on these obstructions through their annihilators.

The primary infinitesimal geometry represents formal principal parts and completed diagonal cochains. An explicit contraction compares the latter with de Rham forms; integrable connections yield stratifications and crystalline jet coalgebras. The comparisons retain resource parameters, coefficients, and uniformly tracked finite outputs on coefficient names. A measured family exhibits polynomial bounds at every finite order without one polynomial budget for the complete coefficient family; its generating function also provides a nonzero logarithmic de Rham class and crystalline transport. Event instrumentations exhibit curvature, while logarithmic resource connections distinguish asymptotic pole orders.

Keywords: resource semantics; asymptotic complexity; effective domains; categorical semantics; synthetic differential geometry; deformation theory.
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Introduction

Computational complexity compares the resources required by programs that solve the same problem. An asymptotic bound records how a cost grows with the input, while allowing constant factors and finitely many exceptional inputs to be ignored. This leaves further questions. How does a cost change when its measurements depend on parameters? Which changes preserve a resource bound? Can the bound itself vary, and can successive finite approximations be controlled by one common bound? We develop a differential geometry in which these are questions about measured computations and their resource constraints.

The connection begins with familiar Landau notation. The relation $f = O(g)$ says that a bounded multiple of g controls f eventually. After identifying functions that eventually agree, the same relation says that f is a bounded multiple of g. Thus an asymptotic bound can be represented by a module of allowed costs. This retains both the costs admitted by the bound and the coefficients used to compare them. This module description is the starting point for the geometry developed here.

A bound must also retain the computation and input convention to which it applies. For example, the register program for $n \mapsto 2n$ in Section 13.4 takes $4n + 1$ updates under the stated instruction convention. Its cost is linear in the numerical input value, but exceeds every polynomial in the input's binary length. The same program therefore meets one polynomial budget and fails the other. Throughout the paper, measured executions supply the witnesses for complexity bounds, and changes of input representation retain their measured overheads.

From bounds to a space of resources. We first construct a classifying space for asymptotic budgets. Its points describe individual budgets, and maps into it describe budgets varying with parameters. The space carries one universal family from which every such family is obtained. We use a locale, so that this classification is formulated through open conditions without requiring the space to be determined by its points.

For the cofinite filter on numerical inputs, every positive (Σ-valued) observation of a growth class is constant and every growth ideal is admitted (Proposition 2.30). Computa-

tional information is retained on names and measured executions, while resource membership remains $\Omega_{\mathcal{E}}$-valued and need not be Σ-classified.

For a fixed problem, each implementation contributes the budgets that admit its cost. Their union is the open condition that the problem can be solved within the budget. Its compactness has a computational meaning: compactness is equivalent to the existence of a finite portfolio that asymptotically dominates all implementations, while supercompactness is equivalent to the existence of one optimal implementation. The corresponding constructions and classification theorems appear in Theorems 3.5, 4.25 and 5.3.

Conventional worst-case complexity. For finite binary inputs, maximum over each word-length fibre is left adjoint to pulling a bound back along length. This operation descends through eventual equality and Landau growth, producing a measured span whose upper predicates are exactly the conventional deterministic time and workspace classes. Explicit polynomial clocks, tape simulations, encodings, and composition identify its polynomial budget with **P** (Section 6 and Theorems 6.7, 6.12 and 6.18). Universal lower bounds retain a separate logical issue: an internal witness transformation and classical implementationwise existence have different meanings. Theorem 6.10 constructs the exact classical predicate by closing the implementationwise relation under double negation before universal quantification. No extra logical axiom is imposed.

From measured variation to differential geometry. An execution produces a finite history. Assigning polynomial weights to its events makes the measured cost depend on parameters. Expanding these weights gives the Taylor coefficients of that cost, and Theorem 7.1 identifies them with geometric jets. This gives differentiation a direct computational interpretation: it measures the response of the same execution to changes in its event weights.

We then allow the budget to vary as well. The first-order equation describes simultaneous changes of the valuation and the allowed costs. At each higher order, an obstruction records whether the chosen variation extends by one more coefficient. Formal classifying objects organize these budget deformations and their compatible extensions. The same construction applies to finite systems of related budgets and to changes in the overheads treated as bounded (Theorems 8.3, 8.16 and 8.23 and Proposition 8.20).

From finite variations to formal geometry and transport. Finite Taylor coefficients do not by themselves describe functions on an entire formal neighbourhood. We construct a geometry of nilpotent thickenings, which retain displacements through a specified finite order. Its function objects recover the usual algebras of principal parts, and compatible truncations give formal completion. An explicit contraction compares functions on successive infinitesimal displacements with the de Rham complex of differential forms (Theorems 10.8 and 11.6).

Connections describe how resource data are transported between nearby parameter values. We prove that integrable connections are equivalent to compatible formal transport, and identify their associated jet coactions. These comparisons connect synthetic differential geometry, differential cohesion, de Rham theory, and crystalline jets within the same resource geometry. Their finite coefficients retain the effective observations of the original computations (Theorems 9.5, 11.11 and 12.11 and Propositions 10.20 and 12.9).

The distinction between finite order and completion already appears in a measured

example. A uniformly constructed family of finite programs has weighted costs

$$F_r(z) = \sum_{j=0}^{r} (n+1)^j z^j.$$

For every fixed order, its coefficients admit a polynomial bound in n. No single polynomial degree bounds the complete coefficient family. The same example later supplies a nonzero logarithmic de Rham class and compatible formal transport. Other examples show how the order of two changes to event measurement produces curvature, and how coupled budgets can obstruct a variation that each budget separately allows (Examples 8.21, 10.14 and 12.12 and Propositions 11.12 and 11.15).

Relation to existing work. The computational foundation is effective domain theory internal to Hyland's effective topos (Smyth 1977; Abramsky and Jung 1994; Hyland 1982; van Oosten 2008). It supplies partial computation, recursion, and numerical names. We construct the completion used here and compare its recursion with finite execution; the distinction between completeness and execution is also studied in (van Oosten and Simpson 2000).

For complexity theory, Blum's machine-independent measures provide the numerical comparison: the measured register execution satisfies the two measure properties, and speedup gives examples without an optimal implementation (Blum 1967). Complexity spaces relate resource comparison to domain-theoretic completion (Schellekens 1995); resource monoids and resource-bounded realizability supply related structures for bounds (Dal Lago and Hofmann 2011). Finite portfolios also have an established role in algorithm selection (Gomes and Selman 2001). Here budgets are embedded modules, and their classifying locale relates the existence of optimal implementations and finite portfolios to topological compactness. For conventional worst-case classes, we use the standard running-time definitions of Barak (2023, Chapter 13) and the constructive tape simulation of Hennie and Stearns (1966). The logical reading of universal bounds uses recursive realizability (Kleene 1945; Hyland 1982).

The connection between cost semantics and execution builds on Niu et al. (2024) and the distinction between cost and behaviour in (Niu et al. 2022; Grodin et al. 2024). Recording histories lets several measurements share the same termination domain and returned values. The additional comparison here identifies polynomially weighted histories with differential coefficients. Cost-recurrence extraction (Danner et al. 2015) and representation-sensitive operator complexity (Kawamura and Cook 2012) motivate the comparisons with size bounds and changes of input names.

The coefficient algebra connects with formal treatments of O-notation (Avigad and Donnelly 2004), filter-based asymptotic verification (Guéneau et al. 2018), and asymptotic gauges and their changes of scale (Giordano and Luperi Baglini 2016; Luperi Baglini and Giordano 2017). Proposition 2.34 gives the gauge comparison. Growth and differentiation also occur in the theory of transseries (Aschenbrenner et al. 2017); the coefficient rings here retain zero divisors.

On the geometric side, we use the synthetic differential geometry of Kock (2006) and the classical theories of principal parts, de Rham forms, and crystalline stratifications (Grothendieck 1966; Berthelot and Ogus 1978; Stacks Project Authors 2026a). The construction of jets from infinitesimal shape follows the approach of Khavkine and Schreiber (2027). Our contribution is to construct the relevant coefficient objects and comparison maps over asymptotic resources, relate them to measured execution, and use them to classify variations of resource constraints. The formal comparisons include the finite matrix contraction, completed coefficients, and the transport associated to finite-projective modules.

Route through the paper. Sections 1–5 construct the computational foundation, resource algebra, budget classifier, and implementation witnesses. Section 6 constructs the conventional worst-case instance and its classical predicates. Sections 7–8 connect measured Taylor coefficients to variations of valuations and budgets. Sections 9–12 construct the infinitesimal geometry, completion, de Rham comparison, and jets. Section 13 returns to numerical programs, input sizes, and asymptotic expansions. The appendices provide the naming and finite-coefficient constructions used in these comparisons.

1 Effective domains and cost data

The later resource constructions need partial and recursive execution, measurements, and positive computational observations. The selected ideal completion supplies the domains used by execution; observations and names specify their computational presentations. Costs will enter the resource algebra of Section 2.

1.1 The effective topos and internal reasoning

We work throughout in Hyland's effective topos $\mathcal{E} = \mathbf{Eff} = \mathrm{RT}(K_1)$, with the usual partial-recursive application (Hyland 1982; van Oosten 2008). Effective domain theory supplies the order-theoretic background (Smyth 1977; Abramsky and Jung 1994). We construct the selected completion below; the supporting naming and coding arguments are given in Section D.

Internal conventions. We write $\Omega_{\mathcal{E}}$ for the subobject classifier and Env for an object of computational environments. The notation Hom_B denotes B-linear maps and $\mathrm{Hom}_{B\text{-alg}}$ denotes B-algebra maps. Internal existence is expressed by a regular image. Equations on local representatives descend through a quotient after the kernel-pair equations have been proved. All constructions retain their parameter context; computational comparisons also retain the environment and input-size maps.

Write $\mathbb{N}$ for the natural numbers object of $\mathcal{E}$. Internal subsets, images, and quotients are those of $\mathcal{E}$.

Definition 1.1 (Internal families and generalized elements)**.** An **internal family over** Z is an object $p : Y \to Z$ of the slice $\mathcal{E}/Z$. A **generalized element of** X **with parameters in** Z is an arrow $x : Z \to X$. Substitution of parameters is pullback in the corresponding slice.

Effective quotients allow a map to descend exactly when it coequalizes the kernel pair of its presentation. A total functional relation determines a unique arrow, because its projection is both monic and regular epic.

The exact/regular construction from assemblies supplies effective quotients, while retaining all realizing witnesses supplies canonical partitioned-assembly covers. The cover, modest-target lifting, and covered-span constructions are proved in Section D. Unrestricted projectivity of arbitrary-carrier partitioned assemblies uses external choice in the ex/lex description (Birkedal 2000, Section 10.6); the dependence is identified in Hofstra (2003, Section 2.4). The constructions below use covers and effective descent rather than that unrestricted lifting assertion.

Definition 1.2 (Partitioned assemblies and trackers)**.** A **partitioned assembly** consists of a carrier and its numerical naming map; a morphism has a single partial-recursive **tracker** on all valid names.

Definition 1.3 (Finite-hit truth values and overtness)**.** The **Rosolini object** $i : \Sigma \hookrightarrow \Omega_{\mathcal{E}}$ is the image of

$$h : 2^{\mathbb{N}} \longrightarrow \Omega_{\mathcal{E}}, \qquad h(\alpha) \iff \exists n : \mathbb{N}\, \alpha(n) = 1.$$

A **Σ-classified partial map** is a span whose domain mono is classified by Σ. An object X is **Σ-overt** when existential quantification restricts uniformly to $\exists_X : \Sigma^X \to \Sigma$.

Proposition 1.4 (Dominance and uniform natural-number search)**.** *In $\mathcal{E}$, Σ is closed under dependent conjunction, and $\mathbb{N}$ is Σ-overt. Thus Σ is a dominance and classified partial maps compose. Finite products, finite sums, regular-epimorphic images, and Σ-classified subobjects of overt objects are overt.*

Proof.

(1) **Construct finite truth operations.** Constant streams and finite interleaving give false, true, finite conjunction, and finite disjunction.

(2) **Prove closure under dependent conjunction.** To check dependent conjunction, pull two successive classified monos back to the canonical partitioned cover P of their ambient object. Since $2^{\mathbb{N}}$ is modest, Lemma D.1 lifts the first classifier through $2^{\mathbb{N}} \twoheadrightarrow \Sigma$, obtaining a total tracked stream α_p. On its hit assembly $H_\alpha = \{(p, n) : \alpha_p(n) = 1\}$, lift the second classifier to a total tracked stream $\beta_{p,n}$. The same modest-target lemma applies to the displayed partitioned hit subassembly. The stream

$$\gamma_p(\langle n, k\rangle) = \begin{cases} \beta_{p,n}(k), & \alpha_p(n) = 1, \\ 0, & \alpha_p(n) = 0 \end{cases}$$

is total: its first branch is evaluated only on a valid hit name. Its hit image is the composite subobject. Its classifier therefore coequalizes the kernel pair of P and descends through $i : \Sigma \hookrightarrow \Omega_{\mathcal{E}}$. This proves dominance in every parameter context.

(3) **Construct uniform natural-number search.** For $\exists_{\mathbb{N}}$, take the canonical partitioned cover of $\Sigma^{\mathbb{N}}$. Its product with $\mathbb{N}$ is partitioned, so Lemma D.1 lifts evaluation to streams $\alpha_{p,n}(k)$. The single total stream $\gamma_p(\langle n, k\rangle) = \alpha_{p,n}(k)$ represents the desired existential. Its classifier descends for the same reason. These constructions use the represented pairing, finite tests, and guarded evaluation, so they have uniform partial-recursive trackers on the supplied numerical names.

(4) **Preserve overtness under finite constructions.** Finite sums use disjunction; products and dependent sums use iterated restricted existential quantification.

(5) **Descend overtness along epimorphisms.** For an epic $q : X \twoheadrightarrow Y$, $\exists_y u(y) \leftrightarrow \exists_x u(qx)$ proves the image assertion.

(6) **Restrict overtness and verify coherence.** For a classified subobject, use dependent conjunction before the existential over its ambient overt object. The corresponding substitution, Frobenius, and Fubini equations hold in $\Omega_{\mathcal{E}}$ and are reflected through the monomorphism i.

□

The subsequent internal constructions apply in the Rosolini-dominant recursion setting thereby realized; they retain arbitrary parameter contexts. In particular, a pointwise external list of semidecisions is not substituted for a map into $\Sigma^{\mathbb{N}}$.

Definition 1.5 (Classified partial elements)**.** A **classified partial element** of Y consists of an extent $p : \Sigma$ and a map $p \to Y$. Its object is (Escardó and Knapp 2017)

$$L_\Sigma Y = \sum_{p:\Sigma} Y^p. \tag{1.1}$$

Its unit has true extent. Multiplication sends a nested partial element with extents p and $q(u)$ to extent $\exists u : p\, q(u)$ and the corresponding value. Strength retains an unused parameter. These formulas represent classified spans and prove the strong, commutative monad laws by reassociation of dependent extents.

1.2 The selected ideal completion

Ordinary ideal completion supplies directed suprema, but the computational construction must also track positive observations of those suprema. We therefore select ideals by the map assigning truth values to their positive upper tests.

Definition 1.6 (Ideals and positive upper tests)**.** For an internal poset P, an **ideal** is an inhabited directed lower subset; $\mathrm{Idl}\, P$ denotes the object of all such ideals. The object of **positive upper tests** and its pairing are

$$U_\Sigma P = \mathrm{Pos}(\mathcal{E})(P, \Sigma), \qquad \chi_I(u) = \exists p : P\, [I(p) \wedge u(p)].$$

Definition 1.7 (Admitted ideal)**.** An **admitted ideal** is an element of $J_\Sigma P$, defined by the pullback

$$J_\Sigma P = \mathrm{Idl}\, P \times_{\Omega_\mathcal{E}^{U_\Sigma P}} \Sigma^{U_\Sigma P}.$$

Thus admission is uniform factorization of the map χ_I of positive upper observations. The underlying ideal membership can remain $\Omega_\mathcal{E}$-valued.

The defining factorization is the cartesian square

$$\begin{array}{ccc} J_\Sigma P & \longrightarrow & \Sigma^{U_\Sigma P} \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Idl}\, P & \xrightarrow[I \mapsto \chi_I]{} & \Omega_\mathcal{E}^{U_\Sigma P}. \end{array}$$

The lower arrow assigns χ_I to each ideal; admission is its uniform factorization through Σ.

Theorem 1.8 (Selected completion and its algebra category)**.** *The formulas*

$$\eta_P(p) = \downarrow p, \qquad (J_\Sigma f)(I) = \downarrow f(I), \qquad \mu_P(\mathcal{I}) = \bigcup_{I \in \mathcal{I}} I$$

define a locally monotone monad on internal posets, with $J_\Sigma \eta_P \leq \eta_{J_\Sigma P}$ and a monad inclusion $J_\Sigma \hookrightarrow \mathrm{Idl}$. It preserves finite products and disjoint finite sums. Its algebra category $\mathcal{A}_\Sigma = \mathrm{Alg}(J_\Sigma)$ is cartesian closed; $[A, B]_\Sigma$ is the poset of maps preserving admitted suprema, with pointwise order and pointwise admitted suprema.

Proof.

(1) **Admit the monad operations.** For principal ideals $\chi_{\downarrow p}(u) = u(p)$. For lower images, $\chi_{\downarrow f(I)}(u) = \chi_I(uf)$. For multiplication define $\widehat{u}(I) = \chi_I(u)$, a monotone map $J_\Sigma P \to \Sigma$. Then

$$\chi_{\mu_P(\mathcal{I})}(u) = \chi_{\mathcal{I}}(\widehat{u}) \in \Sigma.$$

This proves closure of all three operations.

(2) **Verify the monad laws and KZ structure.** Their laws are the lower-image and union identities in ordinary ideal completion, reflected through the monic inclusion. If $K \in (J_\Sigma \eta_P)(I)$, then $K \subseteq\!\downarrow p \subseteq I$ for some $p \in I$; this proves the displayed KZ inequality. Its algebra equations identify an algebra map $a : J_\Sigma A \to A$ with the supremum of each admitted ideal, and its morphisms with maps preserving these suprema.

(3) **Preserve finite products.** An ideal of $P \times Q$ is the product of its two lower-image projections: directedness supplies a common bound for any two coordinate witnesses. Conversely

$$\chi_{I\times J}(u) = \chi_I(p \mapsto \chi_J(q \mapsto u(p,q)))$$

proves admission of that product. The terminal case follows from the unique ideal of 1.

(4) **Preserve disjoint finite sums.** An inhabited directed ideal of $P + Q$ lies in a unique summand; the local summand choice descends by uniqueness. Extending a test by false on the other summand proves admission in both directions. The empty poset has no ideals.

(5) **Construct the exponential algebra.** Let $H \subseteq B^A$ be the object of monotone supremum-preserving maps and $I \in J_\Sigma H$. Define $f_I(x) = \sup_{h\in I} h(x)$. For $J \in J_\Sigma A$, the two iterated suprema over I and J coincide: both are the supremum of the admitted lower image of $I \times J$ under evaluation. Thus f_I preserves the supremum of J, and the pointwise operation makes H an algebra.

(6) **Verify evaluation and currying.** The same calculation proves continuity of evaluation. For $k : C \times A \to B$ continuous, the ideal $\downarrow c \times J$ proves continuity of $k(c,-)$ in an arbitrary context $c : C$. The ideals $I\times \downarrow x$ prove continuity of its transpose $C \to H$. The ordinary evaluation and transposition identities restrict, giving the exponential universal property.

(7) **Retain parameter dependence.** Every construction used internal images, pullbacks, or uniquely specified suprema, so its equations retain their free parameters.

□

Definition 1.9 (Computational algebras)**.** A **computational algebra** is an algebra for the selected ideal-completion monad J_Σ. Their category is $\mathcal{A}_\Sigma = \mathrm{Alg}(J_\Sigma)$; its morphisms preserve admitted suprema. A **pointed computational algebra** has a least element. A morphism of pointed algebras is **strict** when it preserves that element.

The Kock–Zöberlein structure. The inequality $J_\Sigma \eta \leq \eta J_\Sigma$ identifies the selected completion as a Kock–Zöberlein monad on the locally ordered category of internal posets. Its algebra structure is characterized by the adjunction

$$a_A : J_\Sigma A \rightleftarrows A : \eta_A, \qquad a_A \dashv \eta_A, \qquad a_A \eta_A = \mathrm{id}_A\,.$$

Indeed $a_A(I) \leq x$ holds exactly when $I \subseteq\downarrow x$. This characterizes each admitted supremum by its universal property. The algebra and algebra-morphism equations are therefore the following commutative diagrams:

$$\begin{array}{ccc} J_\Sigma J_\Sigma A & \xrightarrow{\mu_A} & J_\Sigma A \\ {\scriptstyle J_\Sigma a_A}\downarrow & & \downarrow{\scriptstyle a_A} \\ J_\Sigma A & \xrightarrow[a_A]{} & A \end{array} \qquad \begin{array}{ccc} J_\Sigma A & \xrightarrow{J_\Sigma f} & J_\Sigma B \\ {\scriptstyle a_A}\downarrow & & \downarrow{\scriptstyle a_B} \\ A & \xrightarrow[f]{} & B. \end{array} \tag{1.2}$$

The free-algebra adjunction gives $\mathcal{A}_\Sigma(J_\Sigma P, A) \cong \mathrm{Pos}(\mathcal{E})(P, U_{\mathcal{A}_\Sigma} A)$. It is the extension principle used below: an operation specified on generators extends uniquely when it respects the admitted suprema. This principle supplies the continuous extension of computational operations from their generators.

A discrete internal object X has only singleton inhabited directed ideals. Their unique elements define its algebra $\mathrm{Disc}\, X$. Every underlying map from it to an algebra preserves those ideals; hence $\mathrm{Disc} \dashv U_{\mathcal{A}_\Sigma}$. A preorder is treated by its effective antisymmetrization before applying J_Σ; monotone maps and the displayed ideal formulas descend.

Proposition 1.10 (Lifting and least solutions)**.** *The partiality monad L_Σ lifts to a strong commutative monad on $\mathcal{A}_\Sigma$. For every pointed algebra A,*

$$\mathrm{lfp}_A : [A, A]_\Sigma \longrightarrow A, \qquad f \longmapsto \bigvee_{n:\mathbb{N}} f^n(\bot)$$

is an algebra morphism selecting the least fixed point. The construction commutes with parameters and strict intertwining maps.

Proof.

(1) **Construct the distributive-law map.** Order partial elements by extension and increase of their values. For $I \in J_\Sigma(L_\Sigma P)$, its extent is $p_I = \chi_I(\mathrm{def}) \in \Sigma$. Under this extent form the lower ideal V_I generated by the defined values of members of I. It is inhabited and directed. For $u : P \to \Sigma$ monotone, the positive extension $\widetilde{u}(z) = \exists d : \mathrm{def}(z)\, u(z(d))$ is Σ-valued by dominance, and $\chi_{V_I}(u) = \chi_I(\widetilde{u})$ under p_I. Consequently (p_I, V_I) defines $\lambda_P : J_\Sigma L_\Sigma P \to L_\Sigma J_\Sigma P$.

(2) **Verify the distributive-law identities.** Principal ideals and total partial elements give the two unit identities. Flattening either nested ideals or nested extents gives the same existential extent and the same ideal of defined values; these are the two multiplication identities. Thus λ is a distributive law, and $L_\Sigma a \lambda_A$ is the lifted algebra structure.

(3) **Lift strength and commutativity.** Strength is continuous because, once one directed member is defined, the defined members are cofinal among both relevant coordinate families. Its equations, and commutativity, follow on extents and values.

(4) **Construct the least fixed point.** An increasing sequence (x_n) generates an admitted lower ideal: its pairing with u is $\exists_n u(x_n)$, using Proposition 1.4. This constructs the displayed least fixed point. Continuity gives the fixed-point equation; induction bounds all its iterates by any pre-fixed point.

(5) **Prove continuity of least fixed points.** For an admitted ideal I of endomorphisms put $g = \sup I$ and $y = \sup_{h \in I} \mathrm{lfp}(h)$. Given $f, h \in I$, a common upper member k gives $f(\mathrm{lfp}(h)) \leq k(\mathrm{lfp}(k)) \leq y$. Continuity and the pointwise supremum give $g(y) \leq y$. Leastness and monotonicity yield $\mathrm{lfp}(g) = y$, proving the algebra-morphism assertion.

(6) **Verify naturality.** Strict intertwining identifies the zeroth iterates and then all successors; preservation of their supremum proves naturality.

□

Lemma 1.11 (Finite returns and least solutions)**.** *For $f : X \to L_\Sigma(Y + X)$, let D_Y, D_X be its exit and continuation domains, with source $c_X : D_X \hookrightarrow X$, continuation $n : D_X \to X$, and exit value $v : D_Y \to Y$. Define $R_0 = D_Y$ and $R_{k+1} = D_X \times_X R_k$, with source $\rho_{k+1} = c_X\pi_1$ and return value $v_{k+1} = v_k\pi_2$. The map $\sum_{k:\mathbb{N}} R_k \to X$ is a Σ-classified mono. Its returned partial map is the least solution of $a(x) = f(x)\,\mathrm{bind}\,[\eta, a]$ (with the usual Kleisli ordering). For algebra morphisms f, it agrees with Proposition 1.10.*

Proof.

(1) **Construct the classified return domain.** Pullback and composition make each ρ_k a classified mono. Exit and continuation are disjoint, and $R_{k+1} \times_X R_{l+1} \cong D_X \times_X (R_k \times_X R_l)$. Induction proves disjointness of distinct lengths. Their single internally indexed dependent sum is therefore monic; its classifier is $\exists k\, \rho_k(x)$ and lies in Σ.

(2) **Identify the least solution.** The jth bottom approximant has domain $\sum_{k<j} R_k$ and value v_k on its kth component. The union of these domains gives the asserted partial map. Each solution agrees with v_k there by induction, so this map is least.

(3) **Compare continuous recursion and parameters.** For continuous f, the lifted supremum has the same extent and stabilized value. All recursions and images commute with parameters.

□

The completion now provides the domains on which the computational operations act. To relate their semantic elements to effective data, we next specify the available observations and the names that present the resulting effective points.

1.3 Observations, effective points, and names

Definition 1.12 (Observation structures)**.** An **observation structure** on a poset is an internal family of upper predicates containing the Σ-constants and closed under finite meets and overt-indexed unions. Its **generated closure** $\mathrm{cl}_\Sigma(G)$ is the intersection, inside the appropriate power object, of all such structures containing G.

The definition is bounded: overt subfamilies are subobjects of the given predicate object, and overtness says that their existential map factors through Σ. A family in the intersection is mapped to an overt family in each member, proving closure of the intersection. Inverse image preserves generating observations and all these operations.

Definition 1.13 (Observed dcpos and effective points)**.** An **observed dcpo** $(D, \mathcal{O}_D)$ has $\mathcal{O}_D \hookrightarrow \mathrm{Sc}(D)$ an observation structure of Scott opens. Its maps are Scott-continuous maps whose inverse images preserve the specified observations; their category is $\mathbf{EDcpo}_\Sigma$. Its object of **effective points** is

$$\mathrm{Pt}_\Sigma\, D = D \times_{\Omega_\mathcal{E}^{\mathcal{O}_D}} \Sigma^{\mathcal{O}_D}. \tag{1.3}$$

It has the uniform evaluation $\mathrm{Obs}_D : \mathrm{Pt}_\Sigma\, D \times \mathcal{O}_D \to \Sigma$.

The effective-point construction and its evaluation fit into

$$\begin{array}{ccc} \mathrm{Pt}_\Sigma\, D & \longrightarrow & \Sigma^{\mathcal{O}_D} \\ \big\downarrow & \lrcorner & \big\downarrow \\ D & \xrightarrow[\mathrm{ev}^\flat]{} & \Omega_{\mathcal{E}}^{\mathcal{O}_D}. \end{array}$$

Transposing the upper map gives the positive evaluation.

Lemma 1.14 (Testing generators in context)**.** *When $\mathcal{O}_D = \mathrm{cl}_\Sigma(G)$, a parameterized $x : Z \to D$ factors through $\mathrm{Pt}_\Sigma\, D$ exactly when its evaluation on $Z \times G$ factors through Σ. The factorization is unique. Effective points are functorial and preserve the limits constructed with componentwise directed suprema and projection-generated observations.*

Proof.

(1) **Extend generator tests to all observations.** In context $z : Z$, the upper predicates whose value at $x(z)$ lies in Σ contain the Σ-constants and are closed under finite meets. An overt union of such predicates has value in Σ by restricted existential quantification. This is an observation structure in that context, hence contains $\mathrm{cl}_\Sigma(G)$ when it contains G.

(2) **Assemble the unique positive evaluation.** Monicity of $\Sigma \hookrightarrow \Omega_{\mathcal{E}}$ and unique choice assemble the joint evaluation map $Z \times \mathcal{O}_D \to \Sigma$; transposition gives (1.3). The reverse implication is restriction.

(3) **Construct the functorial action.** For $F : D \to D'$ use $\mathrm{Obs}_{D'}(Fx, U) = \mathrm{Obs}_D(x, F^*U)$. Uniqueness gives the functor laws.

(4) **Identify effective points of limits.** At a limit, the generator condition says precisely that every projected point is effective, uniformly in the index; the pullback universal property supplies the limit comparison.

□

For a poset P the canonical observed free completion is

$$F_\Sigma P = (\mathrm{Idl}\, P, \{u^\dagger : u \in U_\Sigma P\}), \qquad u^\dagger(I) = \exists p\, [I(p) \wedge u(p)].$$

Restriction to principal ideals recovers u; dagger preserves finite meets and overt unions, and its predicates are Scott open. Thus the displayed family is an observation structure and $\mathrm{Pt}_\Sigma(F_\Sigma P) = J_\Sigma P$ by the two defining pullbacks. For a discrete C, $\mathrm{Idl}\, C \cong C$, whence $\mathrm{Pt}_\Sigma\, F_\Sigma(C_{\mathrm{disc}}) \cong C$; its observations are exactly the maps $C \to \Sigma$.

For completeness the semantic comparison on retracts is explicit. A J_Σ-Kleisli map $r : P \to J_\Sigma Q$ extends to $r^\# : \mathrm{Idl}\, P \to \mathrm{Idl}\, Q$, $I \mapsto \bigcup_{p \in I} r(p)$. The inverse image of $u^\dagger$ is $v^\dagger$, where $v(p) = \chi_{r(p)}(u) \in \Sigma$. Conversely an observed map between these free completions restricts on principal ideals to an admitted ideal and is recovered by directed extension. Hence the free observed category is the Kleisli category of J_Σ, equivalently the category of free J_Σ-algebras. Splitting an idempotent by its fixed subobject on both sides extends this comparison to their retracts. This is the precise comparison used when an algebra map is represented by a semantic observed map; a general observed domain retains its separate observation data.

Definition 1.15 (Name presentations and partial implementations)**.** A **name presentation** of $\mathrm{Pt}_\Sigma\, D$ is a regular epimorphism $q_D : C_D \twoheadrightarrow \mathrm{Pt}_\Sigma\, D$ from a partitioned assembly. Canonical covers and covered-span presentations of arrows are constructed in Proposition D.2; modest target name covers admit tracked lifts by Lemma D.1. A **partial implementation** is an actual tracked partial map on the retained name carriers, with its returned-value equation and equality of its domain with the pullback of the semantic domain. The finite implementation descriptions below construct these maps; an arbitrary semantic arrow is presented by its covered span.

Names, their sizes, and their costs are not identified merely because their denotations agree. For represented operators this distinction is central in the complexity theory of analysis (Kawamura and Cook 2012).

1.4 Parameters and cost data

Proposition 1.16 (Parameter coherence)**.** *The selected completion, its monad structure, and the observation and partial-return constructions commute with substitution between parameter slices of $\mathcal{E}$.*

Proof.

(1) **Identify the slice constructions.** For $\sigma : \Delta \to \Gamma$, pullback preserves finite limits and regular images. The slice power-object and dependent-product Beck–Chevalley comparisons give

$$\sigma^*\mathcal{P}_\Gamma(P) \cong \mathcal{P}_\Delta(\sigma^* P), \qquad \sigma^*(Y^X) \cong (\sigma^* Y)^{\sigma^* X}.$$

(2) **Transport the defining predicates and maps.** These comparisons transport inhabitedness, directedness, lower closure, and the map of upper observations. Hence they identify $\sigma^*(J_\Sigma^\Gamma P)$ with $J_\Sigma^\Delta(\sigma^* P)$, including the principal-ideal unit and union multiplication. They also transport the defining pullback for effective points and the classified extents and values of partial returns.

(3) **Verify substitution coherence.** Identity and composite substitutions give the same comparison by their defining universal properties.

□

Execution uses these effective domains for partiality, observations, and recursion. Its finite measurements take values in internal monoids; numerical measurements use $\mathbb{N}$ or $\mathbb{Q}_{\geq 0}$. The resource algebra below is formed directly from rational-valued functions and their eventual germs. The same construction is used in every parameter context.

Definition 1.17 (Numerical costs and bound functions)**.** A **cost function** assigns a measured nonnegative cost to each execution input in its return domain. A **bound function** specifies a proposed upper bound. General elements of $\mathbb{Q}^X$ are rational-valued functions. A **positive computational test** is a predicate valued in Σ. Semantic incidence takes values in $\Omega_\mathcal{E}$; a positive observation includes its specified factorization through Σ.

2 Landau algebra and observed resource modules

An asymptotic budget should record which cost functions it contains and how those functions combine. We obtain this module description by identifying Landau domination with

multiplication by bounded coefficients. The construction proceeds from rational-valued functions to eventual germs and then to growth classes. Keeping these stages distinct lets us recover both coarse bounds and finer information about coefficients and errors. The resulting embedded modules will be the objects classified in the next section.

2.1 Filters and eventual quotients

Definition 2.1 (Filters and filtered maps)**.** For $X \in \mathcal{E}$, a **filter** is a predicate on subobjects closed upward and under finite intersection and containing X. Write $\mathrm{Filt}(X)$ for their object and $\mathrm{Ev}(P)$ for membership of a predicate P in a displayed filter. A **proper filter** additionally satisfies $\neg\mathrm{Ev}(\varnothing)$. These form a subobject of $\mathrm{Filt}(X)$; the following results are families over that subobject.

A **filtered map** $u : (Y, \mathrm{Ev}_Y) \to (X, \mathrm{Ev}_X)$ is a map satisfying $\mathrm{Ev}_X(P) \Rightarrow \mathrm{Ev}_Y(u^{-1}P)$ for every subobject P.

Substitution proves the identity and composition laws.

Definition 2.2 (Eventual equality and germs)**.** For rational-valued functions on (X, Ev), **eventual equality** is $f =_{\mathrm{Ev}} g \iff \mathrm{Ev}(f = g)$. An **eventual germ**, written $[f]_{\mathrm{Ev}}$, is a class in $K = \mathbb{Q}^X/{=_{\mathrm{Ev}}}$. The **nonnegative cone** $K^+ \subseteq K$ is the image of $(\mathbb{Q}_{\geq 0})^X$.

Proposition 2.3 (Germs and substitution)**.** *Eventual equality is an effective equivalence relation compatible with the ring operations, so K is an internal rational algebra. Moreover $K^+ \cong (\mathbb{Q}_{\geq 0})^X/{=_{\mathrm{Ev}}}$. Filtered maps induce algebra homomorphisms by precomposition and preserve this cone.*

Proof.

(1) **Verify the eventual equivalence relation.** Reflexivity uses $\mathrm{Ev}(X)$, symmetry uses symmetry of equality, and transitivity uses intersection and upward closure.

(2) **Descend the ring operations.** Intersect the extents of two representative equalities to verify each operation. The effective quotient therefore inherits the operations and their finite equations.

(3) **Construct filtered substitution.** The defining implication for a filtered map preserves representative equality and every operation. Identity and composition hold on representatives.

□

Definition 2.4 (Numerical and length filters)**.** For $X = \mathbb{N}$, the **cofinite filter** is

$$\mathrm{Ev}(P) \equiv \exists N\, \forall n \geq N\, P(n).$$

For finite words, the **length filter** is $\mathrm{Ev}(P) \equiv \exists N\, \forall x\, (|x| \geq N \Rightarrow P(x))$.

These filters are proper because a purported eventual empty extent, with cutoff N, is contradicted respectively by $n = N$ and by a word of length N. Thus the displayed formulas construct inhabitants of the proper-filter subobject. For a general size map the proper-filter fibre is retained as part of the context. Arbitrary input maps always act on functions; a constant input map need not act on cofinite germs.

2.2 Functions, germs, and growth classes

The successive comparisons forget different information. The inclusion $(\mathbb{Q}_{\geq 0})^X \hookrightarrow \mathbb{Q}^X$ retains pointwise values. Eventual equality then identifies functions that agree on one eventual extent, and mutual domination makes the further growth quotient. For example, $n+1$, $2(n+1)$, and $n+2$ have the same growth class but distinct germs. Leading coefficients and small relative errors must therefore be compared before passing to growth classes.

2.3 Growth order and admitted resources

Definition 2.5 (Landau domination and growth classes)**.** For rational germs, **Landau domination** is defined by

$$f \preceq g \quad \Longleftrightarrow \quad \exists a \in \mathbb{N}_{\geq 1}\, \mathrm{Ev}(|f| \leq a|g|). \tag{2.1}$$

A **growth class** $[g]$ is a class of nonnegative germs under mutual domination. Write L_{Ev} for this quotient and order it by domination. Its ideals are **growth ideals**.

Reflexivity uses $a = 1$; transitivity intersects the two eventual extents and multiplies their integer bounds. Thus this is a preorder, independent of representatives. Its join is $[f + g]$, its meet is $[\min(f, g)]$, and its bottom is $[0]$. Common upper bounds on f, g bound $f + g$ after adding their constants; common lower bounds bound their minimum after taking the larger constant. The distributive identities follow from those for minimum and maximum, with $f + g$ growth-equivalent to $\max(f, g)$. Multiplication is monotone and distributes over joins.

Definition 2.6 (Admitted resources)**.** The object of **admitted growth ideals** is $\mathsf{Res}_{\mathrm{Ev}} = J_\Sigma \mathsf{L}_{\mathrm{Ev}}$. Its members are the growth ideals selected by Definition 1.7. The full semantic ideal object is $\mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}})$.

Admission requires the map χ_I of positive upper observations to factor uniformly through Σ. General semantic growth ideals need not satisfy this condition. In Section 2.6 these ideals will be identified with marked B-submodules of K, our budgets.

2.4 Landau calculus

2.4.1 Definitions and their quantifiers

In (2.1), positive rational constants give the same predicate by rational rounding. The notation $f = O(g)$ is shorthand for membership $f \in O(g)$, and similarly for the other class notations.

Definition 2.7. The **Landau classes** $O, \Omega, \Theta, o, \omega$ and **asymptotic equality** $\sim$ are defined by

$$\begin{aligned} f \in O(g) &\iff f \preceq g, & f \in \Omega(g) &\iff g \preceq f, \\ f \in \Theta(g) &\iff f \preceq g \ \wedge\ g \preceq f, & f \in o(g) &\iff \forall k \geq 1\, \mathrm{Ev}(k|f| \leq |g|), \\ f \in \omega(g) &\iff g \in o(f), & f \sim g &\iff f - g \in o(g). \end{aligned}$$

The symbol $\ll$ has its approximation-theoretic meaning; Landau domination uses $\preceq$. For nonnegative costs Ω is the eventual lower-bound convention of (Knuth 1976); $\sim$ is asymptotic equality in the sense of relative error (NIST 2026).

These predicates apply to germs with arbitrary support. For instance $o(0) = O(0) = \{0\}$ as germs. The object $K_{\mathrm{nz}} = \{g : \mathrm{Ev}(g \neq 0)\}$ is constructed by its displayed predicate. On this object rational division, defined arbitrarily as zero off the support, gives the usual statements: $f \in o(g)$ exactly when $|f/g|$ tends to zero, and $f \sim g$ exactly when f/g tends to one. The support condition makes these formulas independent of the off-support extension.

2.4.2 Arithmetic and replacement laws

Proposition 2.8 (Landau arithmetic)**.** *Domination is a preorder and direct smallness is transitive. Each of $O, \Omega, \Theta, o, \omega$ is unchanged by replacing either argument by a Θ-equivalent germ. For germs f, g, h, k one has*

$$\begin{aligned} O(g) + O(h) &\subseteq O(|g| + |h|), \quad O(g)O(h) \subseteq O(gh), \\ o(g) + o(h) &\subseteq o(|g| + |h|), \quad o(g)O(h) \subseteq o(gh), \\ O(g)o(h) &\subseteq o(gh). \end{aligned}$$

Furthermore $f \sim g$ is an equivalence relation, implies $f \in \Theta(g)$, and is preserved by multiplication of two asymptotically equivalent pairs. For nonnegative germs, $f \sim g$ and $h \sim k$ imply $f + h \sim g + k$.

Proof.

(1) **Prove transitivity and replacement.** Multiply domination constants and intersect their extents. To compose smallness with domination by a, use the smallness inequality at ka. To compose two smallness relations, use smallness of one and the $k = 1$ bound of the other. These arguments also prove replacement under two-sided domination.

(2) **Establish the sum and product estimates.** For sums, use the two smallness inequalities at $2k$; for products, multiply the inequalities after intersecting their extents. Absolute values satisfy the needed triangle and product identities pointwise.

(3) **Deduce tightness from asymptotic equality.** If $f - g \in o(g)$, the inequality at $k = 2$ gives eventually $|g|/2 \le |f| \le 3|g|/2$. Hence $f \in \Theta(g)$.

(4) **Prove equivalence of asymptotic equality.** Replacement and $|f - g| = |g - f|$ give symmetry. For transitivity write $f - h = (f - g) + (g - h)$, replace the reference germ g by h, and use the sum law. Reflexivity is zero error.

(5) **Preserve asymptotic equality under products.** Write $f = g + r$ and $h = k + s$ with $r \in o(g)$ and $s \in o(k)$. Then $fh - gk = rk + gs + rs \in o(gk)$ by the product and sum laws.

(6) **Preserve asymptotic equality under nonnegative sums.** For nonnegative sums the error is $r + s$ and the reference germ is $g + k$, giving the assertion. For signed sums the preceding estimate uses $|g| + |k|$; cancellation is retained in the reference germ.

□

The definition of $\sim$ uses the signed difference, not merely the two absolute values. For example $-g$ and g are Θ-equivalent, while their relative error is -2 on the eventually nonzero locus.

2.4.3 Logical refinements and uniformity

Definition 2.9 (Double-negation smallness)**.** The **double-negation smallness relation** is

$$f \in o_{\neg\neg}(g) \iff f \preceq g \ \wedge \ \forall k \ge 1 \, \neg\neg \mathrm{Ev}(k|f| \le |g|).$$

Throughout the paper, o denotes the direct smallness relation of Definition 2.7.

We have $o(g) \subseteq o_{\neg\neg}(g) \subseteq O(g)$. Finite conjunction is stable under double negation, so the arithmetic and replacement arguments above give the analogous closure statements for $o_{\neg\neg}$. Quantifiers are left in their displayed order.

Definition 2.10 (Smallness moduli)**.** For cofinite input, a **smallness modulus** is an element of

$$\mathrm{Mod}_o(f,g) = \{m : \mathbb{N}_{\geq 1} \to \mathbb{N} : \forall k\, \forall n \geq m(k),\ k|f(n)| \leq |g(n)|\}.$$

It maps to the direct smallness predicate. A **represented computable modulus** carries its implementation as well. The modulus object and its represented points record these two levels of data.

Definition 2.11 (Uniform asymptotic comparisons)**.** For $f, g : A \times \mathbb{N} \to \mathbb{Q}$, **uniform domination** is the predicate

$$\exists c, N\ \forall a \in A\ \forall n \geq N, \quad |f(a,n)| \leq c|g(a,n)|.$$

Pointwise-in-A domination has $\forall a$ before $\exists c, N$. **Uniform smallness** has $\forall k\, \exists N\, \forall a\, \forall n \geq N$.

All the arithmetic proofs preserve uniformity because only finitely many bounds are combined at each tolerance. The internal pointwise-in-A predicate still retains effective witness dependence on a parameter name; it is distinct from choosing one constant for all parameters. The exact classical reading for implementation families is constructed in Section 6.4.

2.4.4 Oscillation and infinitely-often bounds

Definition 2.12 (Infinitely-often lower bounds)**.** On cofinite numerical input, an **infinitely-often lower bound** is defined by

$$f \in \Omega_{\mathrm{io}}(g) \iff \exists c \in \mathbb{Q}_{>0}\, \forall N\, \exists n \geq N, \quad |f(n)| \geq c|g(n)|.$$

For signed one-sided statements replace the last inequality by $f(n) \geq c|g(n)|$ or $f(n) \leq -c|g(n)|$, giving $\Omega_{\mathrm{io},+}$ and $\Omega_{\mathrm{io},-}$. These definitions use witnesses of arbitrarily late inputs. They are not defined by negating smallness.

Let $g(n) = n + 1$ and let $f(n) = n + 1$ on even inputs and 1 on odd inputs. Then $f \in O(g) \cap \Omega_{\mathrm{io}}(g)$, while $f \notin o(g)$ and $f \notin \Theta(g)$. Each failure has explicit arbitrarily large parity witnesses. If instead $f(n) = (n+1)^2$ on even inputs and 1 on odd inputs, then $f \notin O(g)$ and $f \notin \omega(g)$. Thus lower and strict lower bounds are predicates, not complements of upper bounds. The eventual/infinitely-often distinction is also the distinction emphasized in (Knuth 1976).

Filtered substitution preserves every direct eventual comparison, and the arithmetic proofs commute with it. An isomorphism of filtered inputs reflects the comparisons by its inverse. Infinitely-often comparisons use their own displayed witness transport and are not silently inferred from preservation of eventual filters.

The comparison laws can now be expressed inside the germ algebra. Bounded coefficients will describe O-bounds, vanishing coefficients will describe o-bounds, and their units will describe the corresponding equivalence relations. This converts the calculus of inequalities into module membership and group actions.

2.5 Bounded coefficients, vanishing errors, and units

2.5.1 Bounded division and support

Definition 2.13 (Bounded coefficients and support)**.** The **bounded coefficient ring** is $B = O(1) \subseteq K$. For a function g, its **support idempotent** e_g is the indicator germ of $g \neq 0$; put $z_g = 1 - e_g$.

Bounds add and multiply, and negation preserves them, so B is a unital rational subalgebra of K. Rational equality is decidable, so the indicator is a bounded rational function and $e_g^2 = e_g$. These formulas descend through eventual equality.

Lemma 2.14 (Bounded division)**.** *For $f, g \in K$,*

$$f = O(g) \quad \iff \quad f = bg \text{ for some } b \in B.$$

Multiplication by g identifies Be_g with Bg.

Proof.

(1) **Construct the bounded quotient.** Define $b(x) = f(x)/g(x)$ where $g(x) \neq 0$, and $b(x) = 0$ otherwise. An eventual bound $|f| \leq a|g|$ bounds $|b|$ by a and ensures $f = bg$ on the same extent.

(2) **Recover domination from a multiplier.** Conversely any bound on $|b|$ gives domination.

(3) **Identify the supported principal module.** On Bg, this division is a B-linear inverse to $Be_g \to Bg$, since the quotient of cg by g, extended by zero, is ce_g. The formulas respect eventual equality and thus define maps on germs.

□

Proposition 2.15 (Bounded coefficients and the total quotient ring)**.** *The ring B is reduced, $\mathrm{Ann}_B(g) = Bz_g$ for every $g \in K$, and K is the total quotient algebra of B.*

Proof.

(1) **Prove reducedness.** If $b^r = 0$ for a positive integer r, rational arithmetic gives $b = 0$ on the same eventual extent.

(2) **Calculate annihilators.** If $bg = 0$, then $b = bz_g$ as germs; the converse follows from $z_g g = 0$.

(3) **Invert regular bounded coefficients.** Hence a regular element $b \in B$ has $z_b = 0$ and is eventually nonzero. Its rational reciprocal, set to zero off its support, is an inverse in K.

(4) **Prove the total-quotient universal property.** For arbitrary $f \in K$, the germ $s = (1 + |f|)^{-1}$ is regular and bounded, and $sf \in B$. Thus every element of K is a fraction of bounded germs with regular denominator. Regularity reflects equality of these fractions, proving the localization universal property.

□

2.5.2 Units and the vanishing ideal

Lemma 2.16 (Bounded inverse)**.** *For $u \in B$, the following are equivalent:*

$$u \in B^\times; \qquad \exists a \in \mathbb{N}_{\geq 1}\, \mathrm{Ev}(1 \leq a|u|).$$

Thus the units are exactly the bounded germs bounded away from zero. In particular $B^\times = \Theta(1)$.

Proof.

(1) **Extract a lower bound from an inverse.** If $uv = 1$ and $|v| \le a$ eventually, then $1 \le a|u|$ on the intersection.

(2) **Construct and descend the reciprocal.** Conversely define $v = 1/u$ where $u \neq 0$ and $v = 0$ elsewhere. The displayed bound gives $|v| \le a$ and $uv = 1$ eventually. These equations descend to the germ ring. The last assertion includes the already specified boundedness of u.

□

Definition 2.17 (Vanishing coefficients)**.** The **vanishing coefficient ideal** is $\mathfrak{m} = o(1)$. Its **double-negation variant** is $\mathfrak{m}_{\neg\neg} = o_{\neg\neg}(1)$.

The sum inequalities at $2k$ and the product inequality at ka, for a bound a on a multiplier, prove that both are ideals of B. No infinite intersection of eventual extents is used.

Definition 2.18 (Jacobson radical)**.** The **Jacobson radical** is defined by the unit condition

$$\mathrm{Jac}(B) = \{a \in B : \forall b \in B,\ 1 - ab \in B^{\times}\}.$$

Theorem 2.19 (Vanishing errors and the Jacobson radical)**.**

$$\mathrm{Jac}(B) = \mathfrak{m}.$$

In particular $1 + \mathfrak{m}$ *is a subgroup of* $B^{\times}$*.*

Proof.

(1) **Show that vanishing germs lie in the radical.** Let $a \in \mathfrak{m}$ and let $|b| \le c$ eventually. The smallness bound at $2c$ gives $|ab| \le 1/2$ eventually, so $|1 - ab| \ge 1/2$. The bounded-inverse lemma makes $1 - ab$ a unit.

(2) **Construct a cutoff multiplier.** Conversely, fix $a \in \mathrm{Jac}(B)$ and $k \ge 1$. On a representative define

$$b_k(x) = \begin{cases} a(x)^{-1}, & k|a(x)| > 1, \\ 0, & k|a(x)| \le 1. \end{cases}$$

This is bounded by k and descends to a bounded germ.

(3) **Deduce vanishing from radical membership.** The element $q_k = 1 - ab_k$ is the indicator of $k|a| \le 1$, hence idempotent. It is invertible by the definition of $\mathrm{Jac}(B)$. Multiplying $q_k^2 = q_k$ by its inverse gives $q_k = 1$, which is precisely $\mathrm{Ev}(k|a| \le 1)$. This holds for each k, proving $a \in \mathfrak{m}$.

(4) **Verify the subgroup of units.** For $a, b \in \mathfrak{m}$, one has $(1 + a)(1 + b) = 1 + (a + b + ab)$ with $a + b + ab \in \mathfrak{m}$. Also $(1 + a)^{-1} - 1 = -a(1 + a)^{-1} \in \mathfrak{m}$. These formulas prove the subgroup assertion.

□

The theorem uses a definition of the radical by units and finite rational tests. The quotient $B/\mathfrak{m}$ retains nontrivial idempotents. For example, on cofinite numerical input the parity indicator and its complement are neither zero nor vanishing: the tolerance $1/2$ and arbitrarily late inputs of each parity prove this directly. Its class is therefore an idempotent distinct from zero and one. The residue algebra is not treated as a field.

Theorem 2.20 (Multiplier and orbit formulas)**.** *For every signed germ g,*

$$O(g) = Bg, \qquad o(g) = \mathfrak{m}g, \qquad o_{\neg\neg}(g) = \mathfrak{m}_{\neg\neg}g,$$
$$\Theta(g) = B^{\times}g, \quad \{f : f \sim g\} = g + \mathfrak{m}g = (1+\mathfrak{m})g.$$

Consequently

$$f \in \Omega(g) \iff Bg \subseteq Bf, \qquad f \in \omega(g) \iff g \in \mathfrak{m}f.$$

Proof.

(1) **Identify the bounded orbit.** Bounded division proves the first formula.

(2) **Identify the vanishing orbit.** For $f \in o(g)$ use the same ratio $a = f/g$ on the support of g, zero elsewhere. Every smallness inequality implies $k|a| \leq 1$ on its eventual extent, while the $k = 1$ inequality ensures $f = ag$ as germs. Thus $a \in \mathfrak{m}$. Conversely multiply the inequalities for a by $|g|$.

(3) **Identify the double-negated orbit.** For $o_{\neg\neg}$ the explicit domination supplies the germ equation and boundedness; applying double negation to each implication gives the remaining bounds.

(4) **Identify the supported unit orbit.** If $f \in \Theta(g)$, the two domination inequalities imply that f and g have the same support eventually. Define $u = f/g$ where both are nonzero and $u = 1$ elsewhere. On the intersection of these eventual extents it is bounded, its reciprocal is bounded, and $f = ug$. The bounded-inverse lemma gives $u \in B^{\times}$. Conversely a unit and its bounded inverse give both domination bounds. This argument includes $g = 0$.

(5) **Identify the asymptotic unit orbit.** Finally $f \sim g$ means $f - g \in \mathfrak{m}g$, proving the coset and orbit descriptions.

(6) **Recover the lower-bound relations.** The two reversed comparisons follow immediately from the definitions and bounded division.

□

For nonnegative germs the unit orbit may be restricted to eventually positive units. The construction of u in the proof has this property. Signed absolute-value comparison also records sign-changing units, while the subgroup $1 + \mathfrak{m}$ retains the relative sign in asymptotic equivalence.

2.6 Resource modules and admitted growth ideals

To collect several or indefinitely many asymptotic bounds into one budget, we use ideals of growth classes. The following correspondence identifies these ideals with embedded B-submodules of K, so the marking records exactly which costs a budget contains.

Definition 2.21 (Resource budgets)**.** A **resource budget** is a marked B-submodule $M \hookrightarrow K$. A **principal budget** has the form Bg. For a growth ideal $I \in \mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}})$, its **resource module** is

$$M_I = \{f \in K : [|f|] \in I\}.$$

Here an ideal is inhabited, lower, and directed; equivalently it contains zero and is closed under finite joins.

Theorem 2.22 (Resource representation)**.** *The assignments*

$$I \longmapsto M_I, \qquad M \longmapsto \{[g] : g \geq 0,\ g \in M\}$$

are inverse order isomorphisms $\mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}}) \cong \mathrm{Sub}_B(K)$*. Ideal joins correspond to module sums, meets to intersections, and ideal products to products inside* K*. For every* g*,*

$$\mathrm{Ann}_B(g) = Bz_g, \qquad Bg \cong B/Bz_g \cong Be_g,$$

and

$$Bg + Bh = B(|g| + |h|), \qquad Bg \cap Bh = B\min(|g|, |h|).$$

Proof.

(1) **Pass between resource modules and growth classes.** For every $f \in K$, the sign of a rational representative defines a bounded germ $\mathrm{sgn}(f) \in B$. The identities

$$|f| = \mathrm{sgn}(f)\, f, \qquad f = \mathrm{sgn}(f)\, |f|$$

show that $f \in M$ if and only if $|f| \in M$ for every B-submodule $M \subseteq K$. Its nonnegative growth classes are lower by bounded division and directed by addition. Conversely the directed union of principal budgets is a submodule. These constructions are inverse on elements, giving the order isomorphism.

(2) **Identify principal supports.** The annihilator was calculated in Proposition 2.15; the idempotent splitting gives the principal isomorphism.

(3) **Establish the lattice formulas.** Both g, h lie in $B(|g| + |h|)$, and their absolute values belong to $Bg + Bh$, proving the sum formula. Membership in both Bg and Bh gives two eventual bounds; intersect their extents and take the larger constant to obtain a bound by $\min(|g|, |h|)$. Bounded division proves the intersection formula and its converse. Finite sums and products describe the general lattice operations.

□

Definition 2.23 (Displayed finite-projective modules)**.** Throughout, an internally **finite-projective module** means a **displayed finite-free retract**: specified finite maps $M \xrightarrow{s} B^n \xrightarrow{r} M$ with $rs = \mathrm{id}_M$, or its idempotent matrix sr. A **finite-projective resource cell** is a marked resource module with such a presentation.

Such a presentation constructs a finite dual basis, dualizability, and flatness. Indeed $\mathrm{Hom}_B(M, N) \cong M^\vee \otimes_B N$, where $M^\vee = \mathrm{Hom}_B(M, B)$; this functor is exact because $M^\vee$ is a finite-free retract. This is the finite-projective property used below. In an arbitrary topos it is not identified with unrestricted lifting against every regular epimorphism of internal modules. Standard finite-projective and flat-colimit calculations are cited in (Stacks Project Authors 2026a), with the internal finite-witness and displayed-retract arguments supplied where needed.

Theorem 2.24 (Resource tensors and flatness)**.** *For marked resources* $M, N \subseteq K$*, multiplication induces an isomorphism*

$$M \otimes_B N \xrightarrow{\cong} MN \subseteq K.$$

Every resource is a directed union of displayed finite-projective principal modules and is flat over B*.*

Proof.

(1) **Identify principal resource tensors.** For principal resources, bounded division gives $Bg \cong Be_g$ and $Bh \cong Be_h$. Multiplication identifies

$$Be_g \otimes_B Be_h \cong Be_g e_h.$$

Its inverse sends $be_g e_h$ to $be_g \otimes e_h$: this is independent of the choice of b, since balancing moves e_h to the first factor. The two composites are identities by balancing and idempotence. Since $e_{gh} = e_g e_h$, the support isomorphisms identify this map with $Bg \otimes_B Bh \to Bgh$, $f \otimes k \mapsto fk$. In particular it commutes with inclusions of principal resources.

(2) **Construct the directed principal presentation.** For any finite family $f_1, \dots, f_n \in M$, put $g = \sum_i |f_i|$. The sign multipliers lie in B, so $g \in M$, and bounded division puts every f_i in Bg. Thus the principal submodules form an inhabited directed system with union M.

(3) **Deduce flatness from finite witnesses.** Each $Bg \cong Be_g$ is a displayed retract of B and is flat. Filtered colimits are exact internally: a finite relation in a colimit is witnessed at a common stage, where the corresponding finite relation can be checked. This is the finite-witness argument for flat filtered colimits (Stacks Project Authors 2026a, Tag 05UT). Hence M is flat.

(4) **Assemble the marked tensor isomorphism.** Tensor preserves colimits in each variable by its balanced-map universal property. Taking the directed colimit of the principal multiplication isomorphisms gives $M \otimes_B N \cong \bigcup_{g \in M, h \in N} Bgh$. This union equals MN: any finite sum of products lies in a common principal product stage, by choosing common stages in both factors. The resulting map is multiplication into K, proving the assertion with its marking.

□

Definition 2.25 (The category of marked resource modules)**.** The **category of marked resource modules** $\mathsf{Emb}_B(K)$ has objects $i_M : M \hookrightarrow K$ and B-linear maps $f : M \to N$ satisfying $i_N f = i_M$.

Corollary 2.26 (Monoidal resource representation)**.** *The ideal-to-module correspondence is a strong symmetric monoidal equivalence*

$$(\mathrm{Idl}\, L, \star, \downarrow [1]) \simeq (\mathsf{Emb}_B(K), \otimes_B, B).$$

The tensor marking is multiplication into K; it is monic by the resource tensor theorem. Joins correspond to embedded module sums, meets to intersections, and directed joins to directed unions.

Proof.

(1) **Identify the monoidal equivalence.** The equivalence of Theorem 2.22 has tensor comparison $M_I \otimes_B M_J \cong M_{I \star J}$ given by multiplication, by Theorem 2.24, and unit comparison $M_{\downarrow[1]} = B$.

(2) **Verify coherence.** Both associativity composites send $(f \otimes g) \otimes h$ to fgh in K. The unit and symmetry equations are $1f = f$ and $fg = gf$. The markings are monic, so these equalities prove coherence.

(3) **Preserve the lattice operations.** Sums and intersections correspond by Theorem 2.22; the membership formula for M_I sends directed unions of ideals to directed unions of modules.

□

Restricting to $J_\Sigma \mathsf{L}_{\mathrm{Ev}}$ selects the resources whose positive upper-observation maps factor through Σ. Natural-valued costs embed conservatively: their domination formula is unchanged by rational inclusion.

Theorem 2.27 (Constructive finite-generator splitting)**.** *Every finitely generated submodule $N \subseteq B^n$ has a displayed finite-free retract, hence a finite idempotent-matrix presentation. More precisely, if $w_1, \dots, w_m \in B^n$ generate N and $p : E \to N$ is a B-linear map with displayed lifts $v_i \in E$ satisfying $p(v_i) = w_i$, then these finite data construct a B-linear section $s : N \to E$ of p. All operations commute with substitution of coefficient and generator parameters.*

Proof.

(1) **Initialize the induction.** Induct on n. For $n = 0$ the zero module has its unique section.

(2) **Split a module with one coordinate.** For $n = 1$ put $h = \sum_i |w_i|$, so $N = Bh \cong Be_h$ by bounded division. Write $q(c, h) = c/h$ on the support of h, zero elsewhere, and define

$$s(c) = q(c, h) \sum_i \mathrm{sgn}(w_i) v_i \qquad (c \in Bh).$$

The quotient is a well-defined B-linear function of $c \in Bh$: if $c = bh$, then $q(c, h) = be_h$. Since $\sum_i \mathrm{sgn}(w_i) w_i = h$, we have $ps(c) = q(c, h)h = c$.

(3) **Split the final coordinate.** For $n > 1$ let $\pi : N \to B$ be the final-coordinate map, $a_i = \pi(w_i)$, and $h = \sum_i |a_i|$. The rank-one formula, using the same displayed lifts v_i, constructs $j : Bh \to E$ with $(\pi p)j = \mathrm{id}_{Bh}$. Put

$$q_0 = \mathrm{id}_E - j\pi p, \qquad q_0^2 = q_0, \qquad \pi p q_0 = 0.$$

(4) **Generate the kernel of the final coordinate.** The elements $w_i' = pq_0(v_i)$ have final coordinate zero and generate $N' = \ker(\pi : N \to Bh)$. Indeed, for $w \in N'$, generation of N gives locally $w = \sum_i b_i w_i$. Then

$$\sum_i b_i w_i' = \sum_i b_i (w_i - pj\pi w_i) = w - pj\pi w = w.$$

(5) **Apply the induction hypothesis.** The corestriction $pq_0 : E \to N'$ has displayed lifts v_i of these generators. Induction gives $s' : N' \to E$ with $pq_0 s' = \mathrm{incl}_{N'}$. Put $\bar{s}' = q_0 s'$, so $p\bar{s}' = \mathrm{incl}_{N'}$.

(6) **Reconstruct a section of the original surjection.** For $w \in N$ set

$$s(w) = j(\pi w) + \bar{s}'\big(w - pj(\pi w)\big).$$

The second argument lies in N', and direct substitution gives $ps(w) = pj(\pi w) + w - pj(\pi w) = w$.

(7) **Obtain finite projectivity and parameter stability.** Taking $E = B^m$ and p to be the canonical generating surjection gives $ps = \mathrm{id}_N$ and the idempotent sp on B^m, whose image is N. Every displayed operation is finite support division, finite arithmetic, or induction, and the equations hold uniformly in parameters.

□

Corollary 2.28 (The algebra needed by the later geometry)**.** *Every embedded resource is flat over B, and multiplication identifies $M \otimes_B N$ with $MN \subseteq K$. The ring B is reduced and semihereditary, and K is its total quotient algebra.*

Proof.

(1) **Collect flatness and the coefficient identities.** Flatness and the tensor formula are Theorem 2.24; the coefficient-ring assertions are Proposition 2.15.

(2) **Prove semihereditariness.** Every finitely generated ideal of B is principal by the sum formula in Theorem 2.22, and its support displays it as a retract of B. This proves semihereditariness.

□

In particular $f \in \Theta(g)$ says that f and g generate the same marked principal module. Asymptotic equivalence is equality with g in the quotient $Bg/\mathfrak{m}Bg$. The lower-bound incidences $\Omega(g)$ and $\omega(g)$ exclude zero when $g \neq 0$.

The lattice of all resources is a frame, since it is the ideal completion of the distributive growth lattice. The admitted subobject is selected by J_Σ. For numerical costs the passage to rational resources is the following explicitly typed comparison.

Proposition 2.29 (Conservative rational extension of growth ideals)**.** *Let $\mathsf{L}_{\mathrm{Ev}}^{\mathbb{N}}$ be the growth poset of nonnegative integer-valued functions and let $\ell : \mathsf{L}_{\mathrm{Ev}}^{\mathbb{N}} \to \mathsf{L}_{\mathrm{Ev}}$ be induced by $\mathbb{N} \hookrightarrow \mathbb{Q}_{\geq 0}$. It is order-reflecting. For an ideal $I \in \mathrm{Idl}(\mathsf{L}_{\mathrm{Ev}}^{\mathbb{N}})$, its rational lower image is*

$$\ell_! I = \{q \in \mathsf{L}_{\mathrm{Ev}} : \exists n \in I,\ q \leq \ell(n)\}.$$

Then, for a natural-valued function f,

$$[f]_{\mathbb{N}} \in I \quad \Longleftrightarrow \quad f \in M_{\ell_! I}.$$

The map $J_\Sigma \ell$ carries admitted ideals to admitted ideals and reflects their inclusion order; its underlying ideal map is $\ell_!$.

Proof.

(1) **Compare rational and integer bounds.** A rational eventual domination coefficient is bounded above by an integer coefficient; hence rational domination between integer functions reflects integer domination. Therefore

$$\ell([f]_{\mathbb{N}}) \in \ell_! I \iff \exists n \in I\ ([f]_{\mathbb{N}} \leq n) \iff [f]_{\mathbb{N}} \in I.$$

By definition of $M_{\ell_! I}$ this is the displayed comparison.

(2) **Reflect admitted resources and their operations.** Functoriality of J_Σ gives admission of the lower image. Moreover, for every f, $[f]_{\mathbb{N}} \in I \iff \ell([f]_{\mathbb{N}}) \in (J_\Sigma \ell)(I)$; these tests recover inclusion of ideals and prove order reflection. All statements are internal and uniform in the filter and input.

□

2.6.1 The observation domain of computational resources

The module classification describes resource membership semantically. We next specify which observations of a resource are positive computational tests. Put $L = \mathsf{L}_{\mathrm{Ev}}$. Its free observed domain from Section 1.3 has the type and observation family

$$F_\Sigma L = (\mathrm{Idl}\, L, \mathcal{O}_{F_\Sigma L}), \qquad \mathcal{O}_{F_\Sigma L} = \{u^\dagger : u \in \mathrm{Mon}(L, \Sigma)\},$$

where

$$u^\dagger(I) = \exists a : L\, [I(a) \wedge u(a)].$$

Its effective-point object is $\mathrm{Pt}_\Sigma(F_\Sigma L) = J_\Sigma L$. Thus the uniform positive pairing is a map

$$\chi : J_\Sigma L \times \mathrm{Mon}(L, \Sigma) \to \Sigma. \tag{2.2}$$

By contrast, the marked-resource membership and measured incidences have types

$$\mu : \mathrm{Idl}\, L \times K \to \Omega_{\mathcal{E}}, \qquad \mu(I, f) \iff [|f|] \in I, \qquad (I, e) \mapsto \mu(I, \Phi(e)) : J_\Sigma L \times P \to \Omega_{\mathcal{E}}.$$

Proposition 2.30 (Positive observations in the cofinite numerical fibre)**.** *For the cofinite filter on numerical input, evaluation at zero gives canonical isomorphisms*

$$\Sigma^B \cong \Sigma, \qquad \Sigma^M \cong \Sigma \quad (M \in \mathrm{Mod}_B), \qquad \Sigma^L \cong \Sigma.$$

They hold uniformly in parameter contexts. Consequently

$$\mathrm{Mon}(L, \Sigma) \cong \Sigma, \qquad J_\Sigma L = \mathrm{Idl}(L), \qquad \chi_I(u) = u(0).$$

The isomorphisms concern positive observations of germs and growth classes. Resource membership and its marked geometry retain their $\Omega_{\mathcal{E}}$-valued predicates.

Proof.

(1) **Obtain one uniform semidecision on binary names.** Let $v : Z \times B \to \Sigma$. Compose it with the map sending a binary sequence to its bounded germ, and pull the parameter object back to its canonical partitioned cover. The numerical support of $2^{\mathbb{N}}$ is partitioned. By Lemma D.1, the resulting classifier lifts to binary hit streams. Thus one partial-recursive procedure $w(z, e)$ semidecides $v(z, [\varphi_e])$ on total binary sequence names e, uniformly in the parameter name.

(2) **Identify the observations of zero and one.** The recursion theorem with parameters (Terwijn 2024, Section 2) constructs a name e for the total binary sequence

$$h_e(n) = \begin{cases} 1, & w(z, e) \text{ has not accepted within } n \text{ steps}, \\ 0, & w(z, e) \text{ has accepted within } n \text{ steps}. \end{cases}$$

The bounded test makes h_e total independently of whether acceptance occurs. If $v(z, 1)$ holds and $w(z, e)$ never accepts, then h_e is constantly one, contradicting correctness of w. Thus it accepts; h_e is thereafter zero, so its germ is zero and $v(z, 0)$ holds. Reversing the binary values proves the reverse implication. A witness transformation searches the target hit stream; the argument just given proves that this search terminates on every valid source witness. These programs are uniform in the parameter names. Equality of their hit supports therefore descends through the cover, giving $v(z, 0) = v(z, 1)$ in Σ. This termination verification is in the usual external classical metatheory of the fixed effective-topos model; it does not posit an internal Markov principle or a choice principle.

(3) **Use scalar multiplication to contract module tests.** For $u : Z \times M \to \Sigma$, apply the preceding equality to $(z, m, b) \mapsto u(z, bm)$. Its values at $b = 1$ and $b = 0$ give $u(z, m) = u(z, 0)$. Hence the constant-map operation and evaluation at zero are inverse, proving $\Sigma^M \cong \Sigma$ in every parameter context. Taking $M = B$ gives the first isomorphism.

(4) **Contract growth tests and compute admission.** Bounded scalar multiplication gives a well-defined action $B \times L \to L$, $(b, a) \mapsto [|b|]a$: eventual equality of b and mutual domination of representatives of a preserve the resulting growth class. Its values at zero and one are 0 and a. Applying the binary equality to $b \mapsto u(z, [|b|]a)$ proves $u(z, a) = u(z, 0)$. Every ideal contains 0, so for every positive upper test

$$\chi_I(u) = \exists a \in I \, u(a) = u(0) \in \Sigma$$

uniformly in I, u. This is exactly the admission pullback, proving $J_\Sigma L = \mathrm{Idl}(L)$.

□

In this cofinite fibre, admission therefore imposes no further restriction on growth ideals. The general construction of J_Σ remains the one used for computational domains and other input filters. Finite tests of a sequence representative or a computation are retained on its name carrier; they descend to the germ or growth quotient exactly when their kernel-pair equation holds. The proposition shows that every descended positive test is constant.

Proposition 2.31 (Effective transport of resource observations)**.** *A monotone map $r : L \to L'$ induces an effective Scott map $F_\Sigma r : F_\Sigma L \to F_\Sigma L'$ and its admitted-point map $J_\Sigma r : J_\Sigma L \to J_\Sigma L'$. Explicitly,*

$$(F_\Sigma r)(I) = {\downarrow} r(I), \qquad \chi_{J_\Sigma r(I)}(u) = \chi_I(u \circ r).$$

Proof.

(1) **Transport ideals and observations.** The lower-image construction of Theorem 1.8 commutes with directed unions. For a monotone $u : L' \to \Sigma$,

$$\chi_{{\downarrow} r(I)}(u) = \chi_I(u \circ r),$$

since $b \le r(a)$ and $u(b)$ imply $u(r(a))$, while the converse existential uses $b = r(a)$. Thus $(F_\Sigma r)^*(u^\dagger) = (u \circ r)^\dagger$.

(2) **Restrict to admitted points.** For admitted I this pairing factors uniformly through Σ, giving $J_\Sigma r$. Identity and composition are substitution of tests.

□

Proposition 2.32 (Admitted membership is not generally positive)**.** *In the effective numerical fibre with the cofinite input filter, the predicate that a uniformly given rational-valued function lies in the principal budget B, represented by ${\downarrow}[1]$, need not be Σ-classified.*

Proof.

(1) **Construct the clocked family of germs.** The bounded numerical interpreter of Propositions D.9 and 13.3 gives a Boolean test $H(e, e, n)$ for halting within n updates. Define the primitive-recursive family

$$r_e(n) = \begin{cases} 1, & \neg H(e, e, n), \\ n + 1, & H(e, e, n). \end{cases}$$

(2) **Characterize boundedness of the family.** If the diagonal computation never halts, r_e is constantly one. If it halts, r_e is eventually $n+1$ and cannot be eventually bounded.

(3) **Deduce failure of semidecidability.** Hence $r_e = O(1)$ is exactly the complement of diagonal halting. A Σ factorization of this membership predicate would semidecide that complement, contradicting Lemma D.10.

(4) **Verify admission of the principal resource.** The principal ideal $\downarrow [1]$ is admitted by the unit of J_Σ.

□

The pairing (2.2) supplies the stipulated positive resource observations. In the cofinite numerical fibre they are constant by Proposition 2.30. Matrix equations over germ coefficients and resource membership are interpreted in $\Omega_{\mathcal{E}}$; finite computational tests retain their sequence or execution names.

2.7 Gauge localization, soft notation, and coefficient descent

2.7.1 Comparison with asymptotic gauges

The asymptotic gauges of (Giordano and Luperi Baglini 2016) define moderate rings by domination by a member of a scale family. The following comparison uses rational-valued functions and the same eventual filter as our germ algebra. It concerns moderate elements before a negligible-element quotient.

Definition 2.33 (Rational asymptotic gauge)**.** A **rational asymptotic gauge** is a family $\mathcal{G}$ of rational germs with a member tending to $+\infty$, such that for $g, h \in \mathcal{G}$ there are $k, l \in \mathcal{G}$ with $gh = O(k)$ and $|g| + |h| = O(l)$. Rational rescaling is absorbed by the O-relation. Its generated growth ideal is $I_{\mathcal{G}} = \downarrow \{[|g|] : g \in \mathcal{G}\}$.

Proposition 2.34 (Gauge-moderate germs)**.** *The family $I_{\mathcal{G}}$ is an inhabited directed growth ideal, and its marked resource module is*

$$M_{I_{\mathcal{G}}} = \{f \in K : \exists g \in \mathcal{G},\ f = O(g)\} = \bigcup_{g \in \mathcal{G}} B|g|.$$

It is a unital B-subalgebra of K. Thus moderate rational-valued functions modulo eventual equality give exactly this marked algebra. If $\mathcal{G}$ and $\mathcal{H}$ dominate one another memberwise, their ideals and marked algebras coincide.

Proof.

(1) **Verify directedness of the moderate resource.** The divergent member makes $\mathcal{G}$ inhabited and dominates 1. The sum condition provides a common upper growth bound for any two generators, so their lower closure is directed.

(2) **Identify its resource module.** Membership in M_I means precisely that the growth of the absolute value belongs to I; bounded division identifies this with membership in $B|g|$ for some g. This proves both equalities.

(3) **Construct the moderate subalgebra.** The sum and product conditions respectively close the displayed union under addition and multiplication; bounded multiplication and negation preserve domination. The divergent member puts 1, hence all of B, in this union.

(4) **Descend multiplication to eventual classes.** Eventual equality preserves every domination condition, proving the quotient statement. Memberwise domination gives inclusions of the generated lower ideals in both directions.

□

A general resource module need not have the gauge's multiplicative closure: on cofinite numerical input Bn contains n but not n^2. Also B itself contains no divergent germ. These facts distinguish the full resource classifier from its gauge-generated subclass of resource algebras. Admission of $I_{\mathcal{G}}$ is still the uniform positive-pairing property of Definition 1.7; it is not inferred solely from the gauge conditions. Colombeau constructions additionally quotient moderate elements by negligible ones, whereas K uses eventual equality. The comparison above therefore does not identify those quotient algebras with K or with its Weil thickenings. Functorial changes of asymptotic gauges are studied in (Luperi Baglini and Giordano 2017); the input and coefficient transports below retain their specified filter and module maps.

2.7.2 Overhead gauges

A chosen family of overheads changes which multipliers count as bounded. We encode that family before changing the coefficient ring.

Definition 2.35 (Overhead gauges)**.** An **overhead gauge** is a multiplicative family $\mathcal{P}$ of functions $p \geq 1$ containing 1. These families form a subobject of $\mathcal{P}((\mathbb{Q}_{\geq 0})^X)$. For example, powers of the numerical bit length function below form such a family by NNO recursion. Its downward growth closure is closed under finite joins, since $p + q \leq 2pq$. Define

$$f \in O_{\mathcal{P}}(g) \iff \exists p \in \mathcal{P}, \quad f \in O(pg),$$

and define $\Omega_{\mathcal{P}}$ by reversal and $\Theta_{\mathcal{P}}$ by mutual comparison. The corresponding coefficient ring inside K is

$$B_{\mathcal{P}} = \bigcup_{p \in \mathcal{P}} Bp.$$

The displayed directed union is a ring: products multiply gauges, and sums use the gauge pq.

Lemma 2.36 (Admitted multiplication of growth ideals)**.** *Let $L = \mathsf{L}_{\mathrm{Ev}}$ with its monotone multiplication $m : L \times L \to L$. For $P, I \in J_\Sigma L$ put $P \star I = J_\Sigma m(P \times I)$. Then $P \star I \in J_\Sigma L$; these products are associative and commutative with principal unit $\downarrow [1]$. For every positive upper predicate $u : L \to \Sigma$,*

$$\chi_{P \star I}(u) = \chi_P(p \mapsto \chi_I(x \mapsto u(px))).$$

Consequently an admitted multiplicatively closed growth ideal P containing $[1]$ induces an inflationary idempotent operation $I \mapsto P \star I$ on admitted ideals.

Proof.

(1) **Construct the admitted product and its pairing.** Product preservation and functoriality of J_Σ admit $J_\Sigma m(P \times I)$. The product-pairing and lower-image formulas of Theorem 1.8 give the displayed iterated pairing.

(2) **Verify the product laws.** Lower images of the multiplication on L inherit associativity, commutativity, and the principal unit. If $[1] \in P$ and P is multiplicatively closed, then $P \star P = P$: closure gives one inclusion and multiplication by $[1]$ gives the other. Hence $I \leq P \star I$ and $P \star (P \star I) = P \star I$.

□

Theorem 2.37 (Gauge localization)**.** *Let $D = \{p^{-1} : p \in \mathcal{P}\} \subseteq B$. Then*

$$B_{\mathcal{P}} \cong D^{-1}B, \qquad O_{\mathcal{P}}(g) = B_{\mathcal{P}}g, \qquad B_{\mathcal{P}} \otimes_B M_I \cong M_{\mathcal{P}I}. \tag{2.3}$$

The induced resource saturation is an inflationary, idempotent, finite-meet-preserving operation.

Proof.

(1) **Identify the gauge localization.** Each p^{-1} is bounded, regular, and invertible in K. A fraction b/p^{-1} is bp, and every element of the displayed union has this form. This proves the localization isomorphism and its universal property.

(2) **Compute localized resource modules and tensors.** Bounded division gives the second formula. The tensor-product theorem for marked resources, or its balanced-map proof on principal cells, gives the third, with marking $a \otimes m \mapsto am$.

(3) **Prove saturation and finite-meet preservation.** Containment of 1 makes saturation inflationary, and multiplicative closure makes it idempotent. Localization is flat: a finite relation of fractions is cleared by one product of denominators and is then a relation before localization. Thus it preserves the kernel representing an intersection of two submodules. This proves finite-meet preservation.

(4) **Restrict saturation to admitted resources.** For an admitted multiplicative growth ideal, the induced operation on admitted resources is precisely the J_Σ-product of Lemma 2.36. Its outputs are admitted by that lemma; finite-meet preservation on full resource modules is the localization calculation above.

□

Equivalently an admitted multiplicative ideal P containing $[1]$ has coefficient algebra M_P. Its denominators can be taken as $(1+g)^{-1}$ for $[g] \in P$, $g \geq 0$:

$$f = \frac{f/(1+g)}{(1+g)^{-1}} \quad \text{whenever } f \in O(g).$$

The product of denominators for g, h is the denominator for $g + h + gh$. This proves the same localization directly from the ideal presentation.

2.7.3 Soft Landau notation

Definition 2.38 (Soft Landau notation)**.** On numerical input put $\lambda(n) = 1+\text{bitlength}(n+1)$. This is a computable integer-valued function, comparable to $1 + \log_2(n+1)$ by fixed positive constants. Its finite powers form a gauge. The **soft Landau classes** are

$$\widetilde{O}(g) = \bigcup_{k \geq 0} O(\lambda^k g), \quad f \in \widetilde{\Omega}(g) \iff g \in \widetilde{O}(f), \quad \widetilde{\Theta} = \widetilde{O} \cap \widetilde{\Omega}.$$

Then $\widetilde{O}(g) = B[\lambda]g$, where $B[\lambda] = B[(\lambda^{-1})^{-1}]$ inside K. In particular $f \in \widetilde{\Theta}(g)$ means equality of their principal modules after this localization. The function is obtained by finite bit length computation.

Proposition 2.39 (The admitted polylogarithmic gauge)**.** *In $\mathcal{E}$, the growth ideal*

$$P_\lambda = \downarrow \{[\lambda^k] : k \in \mathbb{N}\}$$

is J_Σ-admitted, contains $[1]$, and is closed under growth multiplication and finite joins. Consequently $I \mapsto P_\lambda \star I$ is an inflationary, idempotent soft-resource operation on $J_\Sigma \mathsf{L}_{\mathrm{Ev}}$.

Proof.

(1) **Prove admission of the power ideal.** The powers are increasing because $\lambda \geq 1$, and their lower closure is an ideal. For a positive upper predicate $u : \mathsf{L}_{\mathrm{Ev}} \to \Sigma$,

$$\chi_{P_\lambda}(u) = \exists k \in \mathbb{N}\, u([\lambda^k]) \in \Sigma$$

by overtness of the NNO, uniformly in u.

(2) **Verify multiplicative closure and induced saturation.** The unit is the zeroth power; multiplication adds exponents. Also $[\lambda^j] \vee [\lambda^k] \leq [\lambda^{j+k}]$. The admitted-growth-product lemma now constructs the stated operation and proves its laws.

□

Proposition 2.40 (Vanishing under overhead inversion)**.** *If $s \in \mathfrak{m}$ is inverted by a gauge localization, then $\mathfrak{m}B_{\mathcal{P}} = B_{\mathcal{P}}$. In particular this occurs when $s = t$, $t(n) = 1/(n+1)$, is inverted, and when $s = \lambda^{-1}$ is inverted.*

Proof.

(1) **Show that inversion makes the ideal contain the unit.** The element $1 = s^{-1}s$ belongs to the extended ideal.

(2) **Verify the stated vanishing functions.** The functions t and λ^{-1} satisfy each vanishing bound by their explicit unbounded denominators.

□

Thus gauge-relative domination is calculated in the localized ring, while original small-error incidences remain recorded in the original module over B. The extended ideal $\mathfrak{m}B_{\mathcal{P}}$ is not redefined to be the vanishing ideal of the original numerical filter. This formula specifies exactly how its marking changes.

The resource algebra now supplies embedded modules, their tensor and lattice operations, and the coefficient localizations associated with overhead gauges. We next classify these resources in families, so that one construction can express a budget varying over a parameter space.

3 The universal structured resource space

The preceding section describes individual budgets as embedded modules. To vary a budget over parameters, we need a classifier carrying a universal family. The growth order already determines its topology: a basic open expresses availability of one growth class.

Definition 3.1 (The resource locale)**.** For $L = \mathsf{L}_{\mathrm{Ev}}$, form the frame of upper predicates

$$\mathrm{Up}(L) = \{U : L \to \Omega_{\mathcal{E}} : \forall a \leq b,\ U(a) \Rightarrow U(b)\}.$$

The **resource locale** is the internal locale $\mathfrak{C}_{\mathrm{Ev}}$ with $\mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}) = \mathrm{Up}(L)$.

We use the standard internal locale and localic-sheaf constructions (Vickers 1989; Mac Lane and Moerdijk 1992). The following results identify its generalized points with marked resource families and construct their universal module. Thus the geometry is determined by growth comparison itself.

3.1 Ideal classification and the Scott frame

Definition 3.2 (Compact and supercompact opens)**.** In an internal frame, an open W is **supercompact** if $W \leq \bigvee_{i:J} W_i$ implies $\exists i : J\ (W \leq W_i)$, and **compact** if it implies $\exists n \in \mathbb{N}\, \exists i_1, \ldots, i_n : J\ W \leq \bigvee_{k=1}^{n} W_{i_k}$. The empty family is included in these definitions.

Theorem 3.3 (Canonical classifying frame)**.** *For every internal frame H, frame maps $\mathrm{Up}(L) \to H$ are naturally equivalent to maps $I : L \to H$ satisfying*

$$I(0) = \top, \qquad a \leq b \Rightarrow I(b) \leq I(a), \qquad I(a) \wedge I(b) = I(a \vee b). \tag{3.1}$$

Consequently

$$\mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Idl}(L), \qquad \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sc}(\mathrm{Idl}(L)).$$

The principal opens $U_a = \{b : a \leq b\}$ are supercompact, are closed under finite intersections, and generate the frame. Thus $\mathfrak{C}_{\mathrm{Ev}}$ is a coherent internal locale.

Proof.

(1) **Extract an ideal from a frame map.** Every upper predicate has the canonical expansion $U = \bigvee_{a \in U} U_a$. Because L has bottom and finite joins,

$$U_0 = \top, \qquad U_a \wedge U_b = U_{a \vee b}.$$

Given a frame map $\theta : \mathrm{Up}(L) \to H$, put $I_\theta(a) = \theta(U_a)$. The displayed identities and $U_b \subseteq U_a$ for $a \leq b$ prove (3.1).

(2) **Construct a frame map from an ideal.** Conversely, given I with those equations, define

$$\theta_I(U) = \bigvee_{a \in U} I(a).$$

Arbitrary-join preservation is immediate. Frame distributivity gives

$$\theta_I(U) \wedge \theta_I(V) = \bigvee_{a \in U, b \in V} I(a \vee b) = \theta_I(U \cap V).$$

For the last equality, $a \vee b \in U \cap V$ gives one inequality, while each $c \in U \cap V$ is obtained with $a = b = c$ for the other. Preservation of top follows from $I(0) = \top$.

(3) **Verify the classification and its points.** The inverse equations are

$$\theta_{I_\theta}(U) = \bigvee_{a \in U} \theta(U_a) = \theta(U), \qquad I_{\theta_I}(a) = \bigvee_{b \geq a} I(b) = I(a).$$

For $H = \Omega_{\mathcal{E}}$, conditions (3.1) say exactly that I is an inhabited directed lower ideal: bottom belongs to it, and the join of two members is a member. This gives the asserted internal point comparison.

(4) **Identify the Scott frame.** For the Scott-frame comparison, map an upper predicate U to

$$\widehat{U}(I) \iff \exists a \in I\ U(a).$$

It is upper and preserves inhabited directed joins of ideals, which are unions. Conversely send a Scott-open W to $u_W(a) = W(\downarrow a)$. Each ideal is the directed union $I = \bigvee_{a \in I} \downarrow a$, so Scott openness gives $W(I) \iff \exists a \in I\, W(\downarrow a)$. Also $\widehat{U}(\downarrow a) \iff U(a)$ by upperness. These are the two inverse equations.

(5) **Prove supercompactness and coherence.** Finally, if $U_a \leq \bigvee_j V_j$, evaluating at a gives locally a witness j with $a \in V_j$; upperness gives $U_a \leq V_j$. This proves supercompactness internally. The principal opens form a finite-meet-stable generating family, including top, so the locale is coherent.

□

3.2 Embedded modules and the universal resource family

The ideal classification now extends to families of embedded modules.

Definition 3.4 (Budget families)**.** For an internal locale Z, write $p_Z : \mathrm{Sh}_{\mathcal{E}}(Z) \to \mathcal{E}$ for its localic geometric morphism and $\Delta_Z = p_Z^*$ for inverse image. A **budget family over** Z is a $\Delta_Z B$-submodule of $\Delta_Z K$ in $\mathrm{Sh}_{\mathcal{E}}(Z)$; this allows its membership conditions to vary locally with the parameter.

The family-valued moduli problem is the contravariant indexed poset

$$\mathcal{B} : \mathrm{Loc}_{\mathcal{E}}^{\mathrm{op}} \longrightarrow \mathbf{Pos}, \qquad \mathcal{B}(Z) = \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K), \qquad \mathcal{B}(f) = f^*.$$

Each object retains its embedding into the ambient module.

The next theorem represents this indexed poset by $\mathfrak{C}_{\mathrm{Ev}}$. Its naturality says that reindexing a budget family is composition with its classifying map, and hence determines the universal family.

Theorem 3.5 (Localic resource-module classification)**.** *There is a natural order isomorphism*

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \;\cong\; \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K), \tag{3.2}$$

where the right-hand side is the poset of $\Delta_Z B$-submodules of the fixed ambient module $\Delta_Z K$. Under it, the identity of $\mathfrak{C}_{\mathrm{Ev}}$ classifies a canonical marked module

$$\mathscr{M} \hookrightarrow \Delta_{\mathfrak{C}_{\mathrm{Ev}}} K \qquad \textit{in } \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}). \tag{3.3}$$

For a generalized ideal point I, its pullback is $M_I \hookrightarrow K$.

Proof.

(1) **Express subobjects by open-valued membership.** The localic subobject-classifier formula $p_{Z*}\Omega_{\mathrm{Sh}_{\mathcal{E}}(Z)} \cong \mathcal{O}(Z)$ and the adjunction $\Delta_Z \dashv p_{Z*}$ give

$$\mathrm{Sub}_{\mathrm{Sh}_{\mathcal{E}}(Z)}(\Delta_Z K) \cong \mathcal{E}(K, \mathcal{O}(Z)).$$

Thus a subobject has an open-valued membership function $\lambda : K \to \mathcal{O}(Z)$. It is a module precisely when the equalities and implications expressing membership of zero, addition, negatives, and multiplication by B hold in this frame-valued logic.

(2) **Descend membership to growth classes.** The sign multiplier $\mathrm{sgn}(f) \in B$ satisfies $|f| = \mathrm{sgn}(f) f$ and $f = \mathrm{sgn}(f)|f|$; hence $\lambda(f) = \lambda(|f|)$. If $f, g \geq 0$ have equal growth classes, the bounded-division theorem supplies multipliers in both directions, giving $\lambda(f) = \lambda(g)$. Therefore membership restricted to K^+ descends across the effective growth quotient $K^+ \twoheadrightarrow L$ to a map $I_\lambda : L \to \mathcal{O}(Z)$.

(3) **Verify the ideal equations.** It satisfies $I_\lambda(0) = \top$. Bounded division gives lowerness. On nonnegative elements, module closure gives $\lambda(f) \wedge \lambda(g) \leq \lambda(f+g)$, and both f, g are bounded multiples of $f + g$, giving

$$\lambda(f) \wedge \lambda(g) = \lambda(f+g).$$

Since $[f+g] = [f] \vee [g]$, these are precisely (3.1).

(4) **Recover the module from its ideal.** Conversely an $\mathcal{O}(Z)$-valued ideal I defines $\lambda_I(f) = I([|f|])$. The triangle inequality, $|bf| \preceq |f|$ for $b \in B$, and (3.1) prove closure under all module operations. The two constructions are inverse on their membership predicates.

(5) **Classify modules by locale maps.** By Theorem 3.3, the resulting $\mathcal{O}(Z)$-valued ideals are precisely frame maps $\mathrm{Up}(L) \to \mathcal{O}(Z)$, equivalently locale maps $Z \to \mathfrak{C}_{\mathrm{Ev}}$. Every step is natural in Z by pullback of membership predicates.

(6) **Identify the universal module and its fibres.** At $Z = \mathfrak{C}_{\mathrm{Ev}}$, the identity frame map defines (3.3), with membership of $f \in K$ given by the open

$$[\![f \in \mathscr{M}]\!] = U_{[|f|]}. \tag{3.4}$$

Evaluating this at an ideal point I gives $[|f|] \in I$, exactly the membership predicate of M_I from Section 2.6. This proves the fibre assertion.

□

For $b : Z \to \mathfrak{C}_{\mathrm{Ev}}$, the classified family is

$$\mathscr{M}_b = b^*\mathscr{M} \hookrightarrow \Delta_Z K, \qquad \mathscr{M}_{b\circ f} = f^*\mathscr{M}_b. \tag{3.5}$$

Thus the category of marked budget families and cartesian maps is $\mathrm{Loc}_{\mathcal{E}}/\mathfrak{C}_{\mathrm{Ev}}$, with universal family corresponding to $\mathrm{id}_{\mathfrak{C}_{\mathrm{Ev}}}$.

3.3 Generalized points as families of budgets

The classification can be used directly to evaluate a resource condition on a varying family. Retain the indexed budget functor $\mathcal{B}$ and the coordinate identities

$$U_a \wedge U_b = U_{a\vee b}, \qquad U_0 = \top, \qquad U_a(I) \iff a \in I, \qquad M_{\downarrow[g]} = Bg = O(g).$$

Write $I \models U$ for evaluation of an open U at an ideal point I. This notation expresses an internal truth value.

The budget functor of points is the classification

$$h_{\mathfrak{C}_{\mathrm{Ev}}}(Z) = \mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K) \tag{3.6}$$

of Theorem 3.5. The following example evaluates a resource condition on a nonconstant family.

A nonconstant family of budgets. To exhibit a nonconstant budget family, take the Sierpiński locale $Z = \mathbb{S}$ itself, and write $u \in \mathcal{O}(Z)$ for its distinguished proper open. Let $a_0 = [1]$ and $a_1 = [n+1]$; thus $a_0 < a_1$ in the numerical cofinite growth order. Construct the following $\mathcal{O}(Z)$-valued ideal on L:

$$I_Z(c) = [\![c \le a_0]\!]_Z \ \vee \ (u \wedge [\![c \le a_1]\!]_Z), \qquad c \in L. \tag{3.7}$$

Here $[\![P]\!]_Z$ denotes the constant open of Z classified by the internal truth value P; it is not assumed that arbitrary order relations on L are decidable. Since $a_0 \le a_1$, the expression in (3.7) sends 0 to top, is lower in c, and satisfies $I_Z(c) \wedge I_Z(d) = I_Z(c \vee d)$. Indeed the component with no factor u requires both $c, d \le a_0$, whereas all remaining terms require u and both $c, d \le a_1$. By Theorems 3.3 and 3.5 it defines a generalized point $b : Z \to \mathfrak{C}_{\mathrm{Ev}}$.

Its budget is constant-bounded at the closed specialization and linear-bounded on the distinguished open u.

$$b^*U_{a_1} = [\![a_1 \le a_0]\!]_Z \vee (u \wedge [\![a_1 \le a_1]\!]_Z) = u. \tag{3.8}$$

Thus a resource condition can hold on a proper open of the parameter locale. The family-valued classification records this varying locus, whereas an individual ideal point records only its specialization. In Section 13.4 the same family will be probed by a measured numerical implementation.

Theorem 3.6 (Flat universal resource module)**.** *The universal module $\mathscr{M}$ is an internal directed colimit of finite projective principal modules and is flat over $\Delta_{\mathfrak{C}_{\mathrm{Ev}}}B$.*

Proof.

(1) **Construct the principal stages.** Work internally in $\mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$. Pullback transports the bounded-division maps and support idempotents of Lemma 2.14 and Proposition 2.15. Thus each $f \in \mathscr{M}$ generates $Bf \cong Be_f$, a displayed retract of B.

(2) **Take their directed union.** The stage $B(|f| + |g|)$ contains Bf and Bg, and $f \in Bf$. These principal stages therefore have directed union $\mathscr{M}$.

(3) **Deduce flatness.** The internal finite-relation argument of Theorem 2.24 applies to this directed union of flat principal stages and proves that $\mathscr{M}$ is flat.

□

3.4 The ringed resource topos and its diagram presentation

The universal family now equips the budget classifier with coefficients.

Definition 3.7 (The structured resource space)**.** Put $\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$, $\bar{B} = \Delta_{\mathfrak{C}_{\mathrm{Ev}}}B$ and $\bar{K} = \Delta_{\mathfrak{C}_{\mathrm{Ev}}}K$. The **structured resource space** is $(\mathscr{S}, \bar{B})$ with its marked module $\mathscr{M} \hookrightarrow \bar{K}$. A **cartesian map of budget families** $(Z, N) \to (Z', N')$ is a locale map $f : Z \to Z'$ with $N = f^*N'$ as marked submodules.

By Theorem 3.5, these families form a category equivalent to $\mathrm{Loc}_{\mathcal{E}}/\mathfrak{C}_{\mathrm{Ev}}$. Allowing $N \subseteq f^*N'$ instead gives the indexed poset of resources.

The marking fixes the image in the ambient module. In particular, $i\alpha = i$ forces a marked automorphism α to be the identity. Choices of a presentation can still have change-of-frame maps. The following diagram description makes both the coefficients and their budget transitions explicit.

Theorem 3.8 (Resource sheaves as budget diagrams)**.** *There is an equivalence of topoi over $\mathcal{E}$*

$$\mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \simeq [L, \mathcal{E}], \qquad F \longmapsto (a \longmapsto F(U_a)). \tag{3.9}$$

The right side is the category of internal covariant diagrams on the internal poset L. Under this equivalence $\bar{B}, \bar{K}$ are constant diagrams and

$$\mathscr{M}(a) = M_a := M_{\downarrow a}, \qquad (a \le b) \longmapsto (M_a \hookrightarrow M_b).$$

At a generalized ideal point I, inverse image is $I^*F \cong \varinjlim_{a \in I} F(a)$.

Proof.

(1) **Construct a sheaf from a budget diagram.** The principal opens form a basis, and $a \leq b$ gives the restriction $F(U_a) \to F(U_b)$. Supercompactness says that a covering sieve on this basis contains the identity internally; thus no additional matching condition is imposed on a diagram. Explicitly, extend a diagram D by

$$\widetilde{D}(U) = \{(x_a)_{a \in U} : x_a \in D(a),\ D(a \leq b)x_a = x_b \text{ for } a \leq b \text{ in } U\}.$$

(2) **Verify gluing and the inverse equivalence.** These compatible families glue uniquely across unions of opens. On U_a, the element a is initial in its indexing poset, so $\widetilde{D}(U_a) \cong D(a)$. Principal decomposition of an open proves the other inverse equation. All constructions take place in the internal logic, including in parameter slices.

(3) **Identify constant and universal module diagrams.** Constant diagrams represent $\Delta_{\mathfrak{C}_{\mathrm{Ev}}}$, and (3.4) gives the stated module diagram.

(4) **Compute the fibre at an ideal.** The inverse image at I is the colimit over its principal neighbourhoods; I is inhabited and directed. Equivalently this colimit preserves finite limits by taking a common stage for a finite tuple and its equations, and sends U_b to the truth value $b \in I$. It is therefore the inverse image classified by I.

□

Evaluation at a generalized ideal point is compatible with the diagram presentation:

$$\begin{array}{ccc} \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) & \xrightarrow{\ \simeq\ } & [L, \mathcal{E}] \\ {\scriptstyle I^*}\downarrow & \cong & \downarrow{\scriptstyle \varinjlim_{a \in I}} \\ \mathcal{E} & = & \mathcal{E}. \end{array}$$

The commutativity is the canonical natural isomorphism of inverse-image functors, including their action on marked modules and witness sheaves.

This equivalence preserves the observation data specified in Section 2.6.1: the effective points are $J_\Sigma L$ with their positive-test pairing. Membership in a principal open U_a remains the corresponding $\Omega_{\mathcal{E}}$-valued resource condition.

The individual resource operations now lift to universal families. Tensor multiplication, sums, and intersections must preserve the markings as well as the underlying modules. The next results construct these operations on the classifier and compare them with admission.

3.5 Multiplication, lattice structure, and admission

Theorem 3.9 (Localic resource multiplication)**.** *There is a commutative monoid structure*

$$\mu : \mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}} \to \mathfrak{C}_{\mathrm{Ev}}, \qquad e : 1 \to \mathfrak{C}_{\mathrm{Ev}},$$

whose generalized-point formulas are $\mu(I, J) = I \star J = \downarrow \{ab : a \in I, b \in J\}$ *and* $e = \downarrow [1]$*. On principal opens,*

$$\mu^* U_g = \bigvee_{g \leq ab} (\pi_1^* U_a \wedge \pi_2^* U_b). \tag{3.10}$$

For the universal module there is a canonical coherent isomorphism of marked modules over $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$*:*

$$\mu^* \mathscr{M} \ \cong\ \pi_1^* \mathscr{M} \otimes_{\Delta B} \pi_2^* \mathscr{M}. \tag{3.11}$$

Finite lattice joins and meets of resources likewise define locale maps, represented by sums and intersections of their universal modules.

Proof.

(1) **Classify the product ideal.** For the universal frame-valued ideals I_1, I_2 over $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$, set

$$(I_1 \star I_2)(g) = \bigvee_{g \le ab} (I_1(a) \wedge I_2(b)).$$

It contains zero and is lower. Two witnesses $a_1 b_1, a_2 b_2$ have common upper witness $(a_1 \vee a_2)(b_1 \vee b_2)$. Thus Theorem 3.3 constructs μ with inverse image (3.10).

(2) **Verify the monoid laws.** Associativity, commutativity, and the unit follow from multiplication in L and uniqueness of the classifying map.

(3) **Identify the universal tensor.** Apply the principal-stage multiplication and directed-colimit argument of Theorem 2.24 internally over $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$, using Theorem 3.6. It identifies the product-ideal module with $\pi_1^* \mathscr{M} \otimes_{\Delta B} \pi_2^* \mathscr{M}$ by $f \otimes g \mapsto fg$, proving (3.11). Associativity, unit, and symmetry are the marked multiplication equalities of Corollary 2.26.

(4) **Classify sums and intersections.** Their frame-valued ideals have principal-open formulas

$$U_g \longmapsto \pi_1^* U_g \wedge \pi_2^* U_g, \qquad U_g \longmapsto \bigvee_{g \le a \vee b} (\pi_1^* U_a \wedge \pi_2^* U_b),$$

respectively for intersections and sums. The ideal equations classify locale maps, and Theorem 2.22 identifies their marked modules.

□

The universal tensor isomorphism preserves the markings:

$$\begin{array}{ccc} \pi_1^* \mathscr{M} \otimes_{\Delta B} \pi_2^* \mathscr{M} & \xrightarrow{\cong} & \mu^* \mathscr{M} \\ \downarrow & & \downarrow \\ \Delta K \otimes_{\Delta B} \Delta K & \xrightarrow[\text{mult}]{} & \Delta K. \end{array}$$

All objects in this square live over $\mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}_{\mathrm{Ev}}$. Associativity of this square is the family-valued version of the monoidal coherence in Corollary 2.26.

The admission doctrine identifies the effective points of this locale. In the cofinite numerical fibre all ideal points are admitted by Proposition 2.30; this does not make the principal-open membership predicates positive tests.

Proposition 3.10 (Admitted points and positive observation)**.** *The comparison* $\mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Idl}(L)$ *identifies* $J_\Sigma L$ *with the subobject of points* I *for which*

$$\chi_I(u) = \exists a : L\, [I(a) \wedge u(a)] \in \Sigma \quad \textit{uniformly in monotone } u : L \to \Sigma.$$

Localic multiplication restricts on this subobject to the admitted product $J_\Sigma m$ *of Lemma 2.36. The canonical map from the discrete locale on* $J_\Sigma L$ *to* $\mathfrak{C}_{\mathrm{Ev}}$ *pulls the universal module back to* $I \mapsto M_I$*. For cofinite numerical input this subobject is all of* $\mathrm{Idl}(L)$*, and its positive upper tests are constant.*

Proof.

(1) **Identify admitted points and their product.** Under the point classification, the displayed condition is the defining pullback of $J_\Sigma L$ in Definition 1.7. The product ideal of Theorem 3.9 is the lower image under m, so Lemma 2.36 proves that it restricts to the admitted product $J_\Sigma m$.

(2) **Pull back the universal family.** The inclusion $J_\Sigma L \hookrightarrow \mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$ transposes to the canonical map from its discrete locale. Pullback of (3.4) gives membership $[|f|] \in I$, hence the family $I \mapsto M_I$.

(3) **Identify the cofinite specialization.** Apply Proposition 2.30 to obtain $J_\Sigma L = \mathrm{Idl}(L)$ and $\chi_I(u) = u(0)$. The universal membership predicate remains the displayed $\Omega_{\mathcal{E}}$-valued predicate.

□

Proposition 3.11 (Admitted points are locale-dense)**.** *Let $a : \mathrm{Disc}(J_\Sigma L) \to \mathfrak{C}_{\mathrm{Ev}}$ be the canonical locale map from admitted points. Its inverse-image map is*

$$a^* : \mathrm{Up}(L) \longrightarrow \Omega_{\mathcal{E}}^{J_\Sigma L}, \qquad (a^*U)(I) = \exists x \in I\, U(x).$$

It preserves arbitrary joins and finite meets, and is order-reflecting. Consequently the localic image nucleus $a_* a^*$ is the identity on $\mathrm{Up}(L)$: the localic image is all of $\mathfrak{C}_{\mathrm{Ev}}$.

Proof.

(1) **Verify preservation of frame operations.** Joins commute with existential quantification. For finite meets, two witnesses $x, y \in I$ have their join $x \vee y \in I$, and upperness of U, V gives the common witness; the reverse inclusion follows by projecting a witness. This proves that a^* is a frame homomorphism.

(2) **Prove order reflection using principal ideals.** Every principal ideal $\downarrow x$ is J_Σ-admitted: its pairing with a positive upper test u is $u(x)$, as in Section 1. Evaluation on it gives

$$(a^*U)(\downarrow x) \iff U(x).$$

Hence $a^*U \leq a^*V$ implies $U \leq V$, so a^* is order-reflecting.

(3) **Establish density by the frame adjunction.** Since a^* preserves arbitrary joins, it has a right adjoint a_*. The adjunction unit gives $U \leq a_* a^* U$, while the counit gives $a^* a_* a^* U \leq a^* U$; order reflection yields $a_* a^* U \leq U$. Thus $a_* a^* = \mathrm{id}$. This proves density of the admitted points.

□

Finite diagrams of the classifier parameterize compatible budgets for compiler maps and resource deformations.

3.6 Classifying finite diagrams of budgets

Theorem 3.12 (Finite resource-diagram classifiers)**.** *The equalizer*

$$\mathrm{Flag}_2 = \mathrm{Eq}(\vee, \pi_2 : \mathfrak{C}_{\mathrm{Ev}}^2 \rightrightarrows \mathfrak{C}_{\mathrm{Ev}})$$

classifies pairs of marked modules $M_0 \subseteq M_1$ over every localic parameter. More generally, a finite diagram of inclusions between resource modules has a classifier obtained by finite limits from products of $\mathfrak{C}_{\mathrm{Ev}}$. Finite equations and inclusions between resource expressions formed using sums, intersections, and products have classifiers by the same construction.

Proof.

(1) **Classify a single inclusion.** A map to $\mathfrak{C}_{\mathrm{Ev}}^2$ classifies two marked modules. Factoring through the equalizer says $M_0 + M_1 = M_1$, equivalently $M_0 \subseteq M_1$, by Theorem 3.9.

(2) **Classify finite diagrams and resource equations.** For a finite graph of inclusions, impose this equation for each edge and take their common equalizer. Commutativity is automatic for inclusions into the fixed ambient module. Resource expressions are interpreted by the already constructed locale operations; their equations are equalizers, and $R \subseteq S$ is the equation $R + S = S$. These constructions commute with parameter pullback because their stated universal properties do.

□

For a monotone growth transformation $r : L \to L'$, the inverse-image frame map r^{-1} constructs $s_r : \mathfrak{C}_{\mathrm{Ev}} \to \mathfrak{C}'_{\mathrm{Ev}}$. The same construction gives a locale

$$\mathfrak{R}_r = \mathrm{Eq}(\vee \circ (s_r \pi_1, \pi_2), \pi_2 : \mathfrak{C}_{\mathrm{Ev}} \times \mathfrak{C}'_{\mathrm{Ev}} \rightrightarrows \mathfrak{C}'_{\mathrm{Ev}}), \tag{3.12}$$

classifying $\downarrow r(I) \subseteq J$. An implemented compiler will act over this budget-compatibility locale in Section 4.10, with its code supplied separately.

3.7 Gauge saturation as a classifying sublocale

Coefficient localization closes a resource under chosen overheads. We now identify the families fixed by this operation and their classifying sublocale.

Definition 3.13 (Gauge saturation)**.** Let $P \in \mathrm{Idl}(L)$ be a multiplicatively closed ideal containing [1], regarded as a point of the object of such ideals. Its coefficient ring is the localization $B_P = M_P \subseteq K$ of Section 2.7. Its **gauge saturation** is

$$S_P : \mathrm{Idl}(L) \longrightarrow \mathrm{Idl}(L), \qquad I \longmapsto P \star I.$$

Theorem 3.14 (Gauge saturation and the localized moduli problem)**.** *The map S_P is Scott-continuous, inflationary and idempotent. Consequently its inverse-image operation*

$$j_P = S_P^{-1} : \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}) \longrightarrow \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}})$$

is a nucleus. It is determined on the principal opens by

$$j_P(U_g) = \bigvee_{\substack{p \in P,\ a \in L \\ g \leq pa}} U_a. \tag{3.13}$$

The nucleus constructs a sublocale $\iota_P : \mathfrak{C}_{\mathrm{Ev}}^{(P)} \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$ satisfying, naturally in every internal locale Z,

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}^{(P)}) \cong \mathrm{Sub}_{\Delta_Z B_P}(\Delta_Z K). \tag{3.14}$$

Its points are precisely the P-saturated ideals $I = P \star I$, and its universal marked module is $\iota_P^* \mathscr{M}$ with its canonical $\Delta_{\mathfrak{C}_{\mathrm{Ev}}^{(P)}} B_P$-action. Resource multiplication restricts to a commutative monoid on $\mathfrak{C}_{\mathrm{Ev}}^{(P)}$ with unit P, and its universal marked tensor comparison is balanced over B_P.

Proof.

(1) **Construct the gauge nucleus.** Products of ideals distribute over directed unions, so S_P preserves directed joins. The inequalities $I \leq P \star I$ and $P \star (P \star I) = I \star (P \star P) = P \star I$ follow from the unit and product closure of P. Under the Scott-frame identification in Theorem 3.3, inverse image along S_P preserves arbitrary joins and finite meets. Since S_P is inflationary and idempotent, its inverse image is also inflationary and idempotent; hence j_P is a nucleus.

(2) **Compute the nucleus formula.** Expanding $g \in P \star I$ gives

$$g \in P \star I \iff \exists p \in P \, \exists a \in I \; (g \leq pa),$$

which proves (3.13).

(3) **Characterize factorization through the sublocale.** A classifying map $f : Z \to \mathfrak{C}_{\mathrm{Ev}}$ corresponds to an open-valued ideal I_f and its marked module $M_{I_f} \subseteq \Delta_Z K$. It factors through the nucleus sublocale precisely when $f^* j_P(U) = f^* U$ for every open U. Since the U_g generate, this is equivalent to

$$(P \star I_f)(g) = I_f(g) \quad \text{for all } g \in L.$$

(4) **Identify localized modules and their universal family.** The module representation and the localization theorem identify this equation with $B_P M_{I_f} = M_{I_f}$. Since $1 \in B_P$, this is exactly closure under the B_P-action in $\Delta_Z K$. The two classifications therefore restrict to the bijection (3.14); pullback proves its naturality and supplies the stated universal module.

(5) **Restrict multiplication and its unit.** If I, J are saturated, associativity gives $P \star (I \star J) = (P \star I) \star J = I \star J$, so multiplication restricts. Its unit is $S_P(\downarrow [1]) = P$.

(6) **Restrict the coherent tensor comparison.** As $B_P \otimes_B B_P \cong B_P$ by localization, for two B_P-modules the balanced tensors over B and B_P agree. The universal tensor comparison of Theorem 3.9 therefore restricts to one balanced over B_P. Its associativity, unit and symmetry follow from multiplication of marked elements.

□

Proposition 3.15 (Reflection into overhead-stable resources)**.** *Extension of marked modules and restriction of their coefficient action give an adjunction with fully faithful right adjoint*

$$B_P(-) : \mathsf{Emb}_B(K) \rightleftarrows \mathsf{Emb}_{B_P}(K) : \mathrm{incl}.$$

Its unit is the marked inclusion $M \hookrightarrow B_P M$, and its counit is the identity on a B_P-submodule. Under Corollary 2.26, the induced idempotent monad is $S_P(I) = P \star I$. This reflection preserves resource products with their localized unit, and finite intersections.

Proof.

(1) **Prove the reflection adjunction.** For an embedded B_P-module N, closure under B_P gives

$$B_P M \subseteq N \quad \iff \quad M \subseteq N.$$

This is the adjunction, including uniqueness of its marked maps.

(2) **Verify idempotence and the comparison formulas.** The multiplication isomorphism $B_P \otimes_B B_P \cong B_P$ gives idempotence. The tensor and intersection comparisons are the resource product and flat localization formulas of Theorem 2.37.

(3) **Identify the commuting classification square.** In particular the square of functors

$$\begin{array}{ccc} \operatorname{Idl} L & \xrightarrow{\simeq} & \mathsf{Emb}_B(K) \\ {\scriptstyle S_P}\downarrow & \cong & \downarrow{\scriptstyle B_P(-)} \\ (\operatorname{Idl} L)^{S_P} & \xrightarrow[\simeq]{} & \mathsf{Emb}_{B_P}(K) \end{array}$$

commutes up to the canonical marked isomorphism. Here the lower left object denotes the fixed ideals, with their inherited maps.

□

The adjunction identifies the same operation in three forms: extension to B_P-coefficients, reflection into overhead-stable budgets, and the classifying nucleus $j_P = S_P^{-1}$. Its unit embeds each budget into its saturation, compatibly with the marking:

$$\begin{array}{ccc} M & \hookrightarrow & B_P M \\ & \searrow & \downarrow \\ & & K \end{array} \qquad \begin{array}{ccc} B_P \otimes_B M & \xrightarrow{\cong} & B_P M \\ \downarrow & & \downarrow \\ B_P \otimes_B K & \xrightarrow[\cong]{} & K. \end{array}$$

Corollary 3.16 (Functorial overhead geometries)**.** *For multiplicative ideals $P \subseteq Q$ containing* [1]*, there is a canonical inclusion $\mathfrak{C}_{\mathrm{Ev}}^{(Q)} \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}^{(P)} \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$, compatible with multiplication and the localized universal marked modules. For numerical cofinite input, the explicitly admitted bit length ideal P_λ of Proposition 2.39 constructs $\mathfrak{C}_{\mathrm{Ev}}^{(P_\lambda)}$, which classifies budgets closed under polylogarithmic overhead.*

Proof.

(1) **Derive the inclusion of fixed-resource locales.** The equations $Q \star P = Q$ and $Q \star Q = Q$ yield $S_Q S_P = S_Q$. Every Q-fixed ideal is therefore P-fixed, which gives the sublocale inclusion by the defining nuclei.

(2) **Verify coefficient and tensor compatibility.** Compatibility of coefficient actions and tensors follows from $B_P \subseteq B_Q \subseteq K$.

(3) **Specialize to the polylogarithmic gauge.** For P_λ use the explicitly constructed admitted multiplicative ideal and $B_{P_\lambda} = B[\lambda]$ of Section 2.7.

□

Definition 3.17 (Localized growth order)**.** For the multiplicative resource ideal P containing [1], the **localized growth preorder** is

$$a \preceq_P b \quad \Longleftrightarrow \quad \exists p \in P\ (a \le pb). \tag{3.15}$$

Its separated quotient L/P is the **localized growth order**, with quotient map $q_P : L \to L/P$.

Theorem 3.18 (The localized growth order)**.** *The relation $\preceq_P$ is a preorder. Finite joins descend to L/P, and there are natural isomorphisms*

$$\operatorname{Idl}(L/P) \cong \{I \in \operatorname{Idl}(L) : P \star I = I\}, \qquad \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}^{(P)}) \cong \operatorname{Up}(L/P). \tag{3.16}$$

In particular the localized principal opens $U_{q_P(a)}^{(P)}$ are supercompact and form a finite-meet-stable basis of $\mathfrak{C}_{\mathrm{Ev}}^{(P)}$.

Proof.

(1) **Verify the localized preorder.** Reflexivity uses $[1] \in P$. If $a \le pb$ and $b \le qc$, then $a \le pqc$ with $pq \in P$, proving transitivity.

(2) **Descend finite joins to the quotient.** To descend joins, replace p, q by $p \vee [1], q \vee [1] \in P$: their product dominates both, so $a \preceq_P b$ and $c \preceq_P d$ imply $a \vee c \preceq_P b \vee d$. The quotient is therefore a finite-join semilattice with bottom $q_P(0)$.

(3) **Identify saturated ideals.** A lower ideal I of L is saturated precisely when it is lower also for $\preceq_P$: if $b \in I$ and $a \le pb$, saturation puts pb and hence a in I; the converse follows by setting $a = pb$. Since q_P preserves finite joins, inverse image along q_P identifies ideals of L/P with exactly these saturated ideals.

(4) **Compute the nucleus on upper predicates.** For an upper predicate U on L, evaluate its nucleus image at a using the principal ideal $\downarrow a$:

$$(j_P U)(a) \iff \exists p \in P\, U(pa).$$

(5) **Identify fixed opens and prove coherence.** The equation $j_P U = U$ is equivalent to U being upward closed also under $\preceq_P$. Indeed, $pa \preceq_P a$ and hence upward closure for $\preceq_P$ makes $U(pa) \Rightarrow U(a)$; conversely, if $a \le pb$ and $U(a)$, original upperness gives $U(pb)$, and the fixed-point equation gives $U(b)$. Such predicates are precisely the pullbacks of upper predicates on the separated quotient. This proves the second isomorphism in (3.16). Applying Theorem 3.3 to the finite-join semilattice L/P proves the principal-open and coherence assertions.

□

3.8 The multiplicative quantale and its support spectrum

The resource product also defines a spectrum. Calculating it will distinguish the support information carried by multiplication from the growth information carried by $\mathfrak{C}_{\mathrm{Ev}}$. Write $\mathfrak{Q} = \mathrm{Idl}(L)$, ordered by inclusion, with resource product $\star$ and unit $\downarrow [1]$. This product is distinct from the finite meet of the Scott frame $\mathcal{O}(\mathfrak{C}_{\mathrm{Ev}})$. Let $\mathrm{Idem}(B) = \{e \in B : e^2 = e\}$, with Boolean operations $e \wedge f = ef$, $e \vee f = e + f - ef$, and $\neg e = 1 - e$.

Definition 3.19 (Multiplicative frame-valued characters)**.** A **unital multiplicative frame-valued character** of $\mathfrak{Q}$ is a map $\chi : \mathfrak{Q} \to H$ to an internal frame H satisfying

$$\chi\Big(\bigvee_i I_i\Big) = \bigvee_i \chi(I_i), \quad \chi(I \star J) = \chi(I) \wedge \chi(J), \quad \chi(\downarrow [1]) = \top.$$

Theorem 3.20 (Quantale and constructive multiplicative spectrum)**.** *The complete lattice $\mathfrak{Q}$ is a commutative unital quantale: multiplication preserves arbitrary joins in each variable. Its unital multiplicative frame-valued characters are classified by the Stone locale of* $\mathrm{Idem}(B)$*:*

$$\mathcal{O}(\mathrm{Sp}_{\otimes}(\mathfrak{Q})) \cong \mathrm{Idl}(\mathrm{Idem}(B)). \tag{3.17}$$

More precisely every such character satisfies

$$\chi(\downarrow [g]) = \chi(\downarrow [e_g]), \qquad \chi(I) = \bigvee_{[g] \in I} \chi(\downarrow [e_g]). \tag{3.18}$$

Thus the unital multiplicative spectrum sees support idempotents, whereas the original classifying locale retains growth comparisons.

Proof.

(1) **Verify the quantale operations.** The join of resource ideals is generated by finite joins of their members. Multiplying such finite joins distributes over their products in L, giving

$$I \star \bigvee_j J_j = \bigvee_j (I \star J_j).$$

The zero ideal is absorbing and $\downarrow [1]$ is a unit; associativity and commutativity are those of function multiplication.

(2) **Extract the principal character equations.** A join-preserving χ is determined by its values $d_g = \chi(\downarrow [g])$, because every ideal is the join of its principal ideals. Finite joins and resource products imply

$$d_0 = \bot, \qquad d_1 = \top, \qquad d_{a \vee b} = d_a \vee d_b, \qquad d_{ab} = d_a \wedge d_b.$$

(3) **Reduce characters to support idempotents.** For a nonnegative germ g let $g^\dagger$ be its rational reciprocal on its nonzero support and zero elsewhere. It is a germ in K^+ with $gg^\dagger = e_g$ and $ge_g = g$. The multiplicative equations give

$$d_g = d_g \wedge d_{e_g} \leq d_{e_g}, \qquad d_{e_g} = d_g \wedge d_{g^\dagger} \leq d_g.$$

Hence $d_g = d_{e_g}$. These idempotents are invariant under growth equivalence, and $e_{g+h} = e_g \vee e_h$ and $e_{gh} = e_g e_h$. The d_e therefore form a Boolean-algebra homomorphism $\mathrm{Idem}(B) \to H$ (the images of e and $1 - e$ are complements).

(4) **Recover a character from Boolean support data.** Conversely any such map φ defines a quantale character by $\chi(I) = \bigvee_{[g] \in I} \varphi(e_g)$: finite joins handle ideal generation, and frame distributivity handles products. These constructions are inverse.

(5) **Classify the support spectrum.** Finally, the ideal completion of a Boolean algebra is the free frame on its bounded-distributive-lattice operations: a map $\varphi : E \to H$ preserving finite meets, joins and bounds extends uniquely to $\widehat{\varphi}(J) = \bigvee_{e \in J} \varphi(e)$. Its inverse is restriction to principal ideals. Taking $E = \mathrm{Idem}(B)$ proves the frame classification (3.17) by its universal property. This is the point-free support/Pierce spectrum; the general quantalic spectrum perspective is discussed in (Manuell 2022).

□

For cofinite numerical input, the parity indicator yields a nontrivial complemented open in the support spectrum. The support spectrum records these idempotent decompositions; $\mathfrak{C}_{\mathrm{Ev}}$ records growth budgets.

3.9 Filtered-input substitution and coherence

For a filtered map $u : (Y, \mathrm{Ev}_Y) \to (X, \mathrm{Ev}_X)$, substitution gives $B_X \to B_Y$ and $K_X \to K_Y$. On a principal resource,

$$B_Y \otimes_{B_X} B_X g \cong B_Y e_{g \circ u} \cong B_Y (g \circ u).$$

Taking directed colimits proves

$$B_Y \otimes_{B_X} M_I \cong M_{\downarrow u^* I}. \tag{3.19}$$

The map sends $b \otimes m$ to $b(m \circ u)$.

Both composite comparisons for successive substitutions or localizations have this same formula on elementary tensors. Their equality proves unit and associativity coherence. Functions still reindex along every input map, as in Section 2.2.

Proposition 3.21 (Natural base change of classifying resources)**.** *For a filtered map $u : (Y, \mathrm{Ev}_Y) \to (X, \mathrm{Ev}_X)$, write $\varphi_u : L_X \to L_Y$ for the map on growth classes induced by precomposition. The frame map $\varphi_u^{-1} : \mathrm{Up}(L_Y) \to \mathrm{Up}(L_X)$ defines $s_u : \mathfrak{C}_X \to \mathfrak{C}_Y$, acting on ideal points by $I \mapsto{\downarrow}\, \varphi_u(I)$. There is a natural isomorphism of marked modules over $\mathfrak{C}_X$,*

$$\Delta_{\mathfrak{C}_X} B_Y \otimes_{\Delta_{\mathfrak{C}_X} B_X} \mathscr{M}_X \;\cong\; s_u^* \mathscr{M}_Y, \tag{3.20}$$

where the marking on the left is multiplication after substitution into $\Delta_{\mathfrak{C}_X} K_Y$. These isomorphisms satisfy the identity and composition laws.

Proof.

(1) **Compute the classifying map.** Preimage along φ_u preserves unions and finite intersections, and

$$s_u^* U_h = \bigvee_{\substack{g \in L_X \\ h \le \varphi_u(g)}} U_g.$$

Evaluation at an ideal gives its lower image under φ_u.

(2) **Extend the principal comparison.** Internally in $\mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_X)$, use the principal stages of Theorem 3.6 and the support comparison

$$B_Y \otimes_{B_X} B_X g \cong B_Y e_{g \circ u} \cong B_Y(g \circ u) \hookrightarrow K_Y.$$

Tensor preserves their directed colimit. Its resulting marking is monic: a finite tensor expression mapping to zero lies locally in the common stage generated by the sum of its input absolute values, where the displayed marking is monic.

(3) **Identify the image.** The principal-open formula gives

$$[\![h \in s_u^* \mathscr{M}_Y]\!] = \bigvee_{\substack{g \in L_X \\ [|h|] \le \varphi_u(g)}} U_g.$$

Thus membership locally supplies a generator $g \in \mathscr{M}_X$ with $h \in B_Y(g \circ u)$. Bounded division writes $h = b(g \circ u)$, the image of $b \otimes g$. This proves (3.20) with its marking.

(4) **Verify coherence.** Each comparison sends $b \otimes m$ to $b(m \circ u)$. Identity and composite substitutions agree on these generators, and hence on the tensors.

□

Filtered substitution intertwines resource representation and scalar extension by the natural comparison

$$\begin{array}{ccc} \mathrm{Idl}(L_X) & \xrightarrow{\simeq} & \mathsf{Emb}_{B_X}(K_X) \\ {\scriptstyle \downarrow \varphi_u(-)} \big\downarrow & \cong & \big\downarrow {\scriptstyle B_Y \otimes_{B_X} -} \\ \mathrm{Idl}(L_Y) & \xrightarrow[\simeq]{} & \mathsf{Emb}_{B_Y}(K_Y). \end{array}$$

The right-hand marking is induced by $K_X \to K_Y$. The theorem proves its required monicity on these resources.

Proposition 3.22 (Extension and restriction of input budgets)**.** *For the filtered substitution $u : Y \to X$, the growth map $\varphi_u : L_X \to L_Y$ preserves 0 and finite joins. There is a locale map $t_u : \mathfrak{C}_Y \to \mathfrak{C}_X$ with*

$$t_u^* U_a^X = U_{\varphi_u(a)}^Y, \qquad t_u(J) = \varphi_u^{-1}(J), \qquad s_u(I) \subseteq J \iff I \subseteq t_u(J).$$

In the diagram presentations,

$$s_u^* G = G \circ \varphi_u, \qquad (t_u^* F)(b) = \varinjlim_{\varphi_u(a) \le b} F(a). \tag{3.21}$$

The coefficient map $B_X \to B_Y$ equips t_u with a ringed-topos map. Its module pullback has a canonical monomorphism into $\mathscr{M}_Y$; at J its marked image is $M_{s_u t_u(J)} \subseteq M_J$.

Proof.

(1) **Construct the reverse locale map and ideal adjunction.** Precomposition preserves zero and sums of functions, hence finite joins of growth classes. The assignment on principal opens preserves their finite-meet relations, so Theorem 3.3 constructs t_u. Its point formula and the displayed ideal adjunction follow by membership.

(2) **Identify the diagram inverse images.** Precomposition of diagrams sends the subterminal U_b to $\varphi_u^{-1}(U_b)$, proving the first formula in (3.21). For the second, the comma poset $\{a : \varphi_u(a) \le b\}$ contains 0 and is directed by joins. Its colimit preserves finite limits by the finite-witness argument in Theorem 3.8. The resulting left Kan extension preserves colimits and sends U_a to $U_{\varphi_u(a)}$, so it is exactly t_u^*.

(3) **Construct the ringed pullback.** It sends the constant coefficient ring B_X to the constant ring B_X; the substitution map to B_Y defines the ringed structure. Module pullback is $\bar{B}_Y \otimes_{t_u^* \bar{B}_X} t_u^*(-)$.

(4) **Compute the resource comparison and its fibres.** On a principal stage a, the marked scalar-extension comparison of Proposition 3.21 identifies its value with $M_{\varphi_u(a)}$. At budget b, these form a directed union inside M_b, proving monicity. At an ideal J this union has growth ideal $\downarrow \varphi_u(\varphi_u^{-1} J)$.

(5) **Verify coherence and observation transport.** Identity and composition follow on elementary tensors and principal opens. The forward map s_u retains the effective observation transport of Proposition 2.31; no assertion that t_u preserves admission is needed or made.

□

3.10 Finite principal-open descent

To use the resource coefficients on affine charts, we need to glue modules and their maps across localizations. For a commutative coefficient algebra A, finite lists of generators and equations construct the category of finitely presented A-algebras. A finite principal cover is displayed by $s_1, \dots, s_r$ and a unit-ideal witness $1 = \sum_i a_i s_i$. The next result supplies the gluing calculation used later for differential coefficients.

Proposition 3.23 (Module gluing)**.** *For every A-module M such a cover gives an equalizer*

$$M \longrightarrow \prod_i M_{s_i} \rightrightarrows \prod_{i,j} M_{s_i s_j}.$$

These equalizers commute with restriction and refinement.

Proof.

(1) **Construct unit combinations of denominator powers.** This is the finite standard-open Čech equalizer (see (Stacks Project Authors 2026a, Tag 01X9)). Its denominator calculation is finite, and hence applies internally. First, for every $d \ge 1$, $1 = \sum_i c_i s_i^d$ follows by expanding $(\sum_i a_i s_i)^{r(d-1)+1}$ and grouping each monomial by an index whose exponent is at least d.

(2) **Prove injectivity of the equalizer map.** If m maps to zero in every M_{s_i}, choose a common d with $s_i^d m = 0$. The unit combination yields $m = 0$, proving injectivity.

(3) **Choose compatible denominator representatives.** For compatible local elements choose a common denominator exponent n and representatives $x_i = m_i/s_i^n$ with $m_i \in M$. Compatibility on every overlap gives a common q such that

$$(s_i s_j)^q (s_j^n m_i - s_i^n m_j) = 0 \qquad (i, j).$$

(4) **Construct the global glued element.** Put $d = n + q$ and $\widetilde{m}_i = s_i^q m_i$. Then $s_j^d \widetilde{m}_i = s_i^d \widetilde{m}_j$. Choose $1 = \sum_i c_i s_i^d$ by the unit-combination calculation above and set $m = \sum_i c_i \widetilde{m}_i$. For each j,

$$s_j^d m = \sum_i c_i s_i^d \widetilde{m}_j = \widetilde{m}_j,$$

so $m/1 = x_j$ in M_{s_j}.

(5) **Verify uniqueness and refinement coherence.** This constructs the gluing map; injectivity proves uniqueness, restriction, and refinement coherence. All common exponents arise from finite maxima.

□

The gluing condition is the equalizer diagram

$$M \hookrightarrow \prod_i M_{s_i} \rightrightarrows \prod_{i,j} M_{s_i s_j}.$$

The parallel arrows restrict to the two factors of each overlap; refinement compares the equalizers through localization.

We now have a classifier of budget families, its marked coefficient module, and compatible resource operations and transports. The remaining computational question is which of these budgets contain the cost of an implementation of a given problem. Answering it requires the measured execution construction that follows.

4 Effective implementations and universal incidence

The budget classifier tells us what a resource constraint is. To ask whether a problem satisfies it, we must supply implementations and measure their execution. We first construct returning histories and their resource valuations, then use finite descriptions and input names to fix the measured implementation family. Its denotation and cost maps form the span from which universal feasibility is obtained.

The partiality and cost construction is related to cost-sensitive synthetic domain semantics (Niu et al. 2024). Here the finite-return argument establishes the required comparison in the selected effective-domain setting. The subsequent sheaf construction carries the resulting descriptions and observation maps over the budget classifier.

4.1 Finite words in the computational category

Histories must record finitely many events in their execution order. We construct their word objects in $\mathcal{A}_\Sigma$, the computational category of Section 1. The classified domains of partial execution remain subobjects in $\mathcal{E}$.

Lemma 4.1 (Internally indexed disjoint sums)**.** *Every object of $\mathcal{E}$ with its discrete order has a canonical J_Σ-algebra. For an internally given family $(P_n)_{n:\mathbb{N}}$ of posets, with the disjoint order on its dependent sum, there is an isomorphism*

$$J_\Sigma\Big(\sum_{n:\mathbb{N}} P_n\Big) \cong \sum_{n:\mathbb{N}} J_\Sigma P_n$$

compatible with principal ideals and ideal unions. Thus an internal family of computational algebras has a componentwise disjoint-sum algebra.

Proof.

(1) **Identify admitted ideals in a discrete poset.** An inhabited directed ideal in a discrete poset has exactly one element: two members with a common upper bound are equal. Unique existence gives its element and identifies $J_\Sigma A$ with A.

(2) **Identify the unique summand and its admitted ideal.** In a disjoint sum, members with a common upper bound have the same index. Inhabitedness and directedness therefore specify a unique n. Restrict the ideal to P_n. To prove its admission, extend each monotone test $P_n \to \Sigma$ by false on the other summands, using decidable equality on $\mathbb{N}$. Conversely, every monotone positive test on the sum restricts to a test on the chosen fibre. These constructions identify the pairing predicates and are inverse.

(3) **Verify monad and algebra compatibility.** Principal ideals and unions retain the unique index, proving the monad compatibility and hence the algebra assertion. The sum is one internally constructed family, not an external countable coproduct.

□

Proposition 4.2 (Computational finite words)**.** *For $A \in \mathcal{A}_\Sigma$, put*

$$A^* = \sum_{n:\mathbb{N}} A^{[n]}, \qquad [n] = \{j : \mathbb{N} : j < n\},$$

ordered componentwise at each fixed length. This is a computational algebra, and empty word, singleton, and concatenation are algebra morphisms

$$[] : 1 \to A^*, \qquad [-] : A \to A^*, \qquad +\!+ : A^* \times A^* \to A^*.$$

It is the free monoid on A in $\mathcal{A}_\Sigma$. In particular, for a monoid (C, e, μ) in $\mathcal{A}_\Sigma$ and an algebra morphism $w : A \to C$, its unique monoid extension $\overline{w} : A^* \to C$ is constructed by

$$\overline{w}([]) = e, \qquad \overline{w}(a :: s) = \mu(w(a), \overline{w}(s)).$$

Proof.

(1) **Construct the word algebra.** Finite powers carry the product algebra of Theorem 1.8, uniformly by recursion on the length. Apply Lemma 4.1.

(2) **Prove continuity of concatenation and the unit.** An admitted ideal of a pair of words has fixed lengths in both components, so preservation of its supremum by concatenation reduces to the finite-product algebra formula. The empty word and singleton satisfy the same equation.

(3) **Construct the continuous fold and prove uniqueness.** At each length the displayed fold is a finite composite of algebra morphisms. The disjoint-sum algebra therefore makes it an algebra morphism on all words. Induction proves preservation of concatenation, and proves uniqueness from the empty and singleton clauses. These constructions retain all free parameters.

□

4.2 Resource-valued partiality

A returning computation now carries both a value and a resource. Sequential execution combines resources using a monoid; commutativity is unnecessary because the execution order is retained.

Definition 4.3 (Resource-valued partiality)**.** Let (C, e, μ) be an internal monoid in $\mathcal{A}_\Sigma$, with algebra morphisms $e : 1 \to C$ and $\mu : C \times C \to C$, and write $cd = \mu(c, d)$. The **resource-valued partiality functor** and its unit are

$$L_{\Sigma,C}A = L_\Sigma(C \times A), \qquad L_{\Sigma,C}u = L_\Sigma(\mathrm{id}_C \times u), \qquad \eta^C_A(a) = \eta_{C\times A}(e, a).$$

For $u \in L_{\Sigma,C}A$ and $g : A \to L_{\Sigma,C}B$, define **resource-valued bind** by

$$u \,\mathrm{bind}_C\, g = u \,\mathrm{bind}_\Sigma\, ((c, a) \mapsto g(a) \,\mathrm{bind}_\Sigma\, ((d, b) \mapsto \eta_{C\times B}(cd, b))). \tag{4.1}$$

Here bind_Σ is the partiality bind of (1.1). The strength has type

$$X \times L_{\Sigma,C}A \longrightarrow L_{\Sigma,C}(X \times A), \qquad (x, u) \longmapsto u \,\mathrm{bind}_\Sigma\, ((c, a) \mapsto \eta(c, (x, a))).$$

Theorem 4.4 (Resource-valued partiality)**.** *These maps make $L_{\Sigma,C}$ a strong monad on $\mathcal{A}_\Sigma$. Cost erasure*

$$\epsilon^C_A = L_\Sigma \pi_A : L_{\Sigma,C}A \to L_\Sigma A$$

is a strong monad morphism. A monoid morphism $h : C \to C'$ induces the strong monad morphism $L_\Sigma(h \times \mathrm{id}_A) : L_{\Sigma,C}A \to L_{\Sigma,C'}A$. For commutative C, the monad is commutative.

Proof.

(1) **Verify continuity of the resource operations.** All operations are composites of the lifted partiality operations, strength, products, and e, μ, so they are algebra morphisms.

(2) **Compute composite domains and costs.** Write a partial value as $(p, (c, a))$ in context $p : \Sigma$, and the next partial value as $(q, (d, b))$ in that context. Its composite has extent $\exists t : p\ q(t)$ and value (cd, b). Dominance classifies this extent.

(3) **Verify associativity, units, and strength.** Three successive computations have the same composite domain under both associations; their values differ only by $(cd)r = c(dr)$. The unit equations are $ec = c = ce$. Strength retains the same unused parameter throughout these domains and values, so the strength equations of Proposition 1.10 prove its laws.

(4) **Verify erasure and change-of-cost morphisms.** Erasure forgets precisely the accumulated monoid value. Applying h to a product gives $h(cd) = h(c)h(d)$ and $h(e) = e$, proving the two monad-morphism claims.

(5) **Prove commutativity for independent computations.** For independent calls the two orders have the same domain by commutativity of L_Σ and costs cd and dc; commutativity of C proves the final assertion.

□

The notation $L_{\Sigma,C}$ distinguishes resource-valued partiality from the symmetric-algebra monad T and the infinitesimal-shape modality $\mathfrak{I}_W$. The monoids $(\mathbb{N}, 0, +)$, $(\mathbb{N}, 0, \max)$, and their product are constructed using the discrete algebra of Lemma 4.1. They accumulate declared event costs, recorded peak occupancy, and joint time–space values, respectively.

The resource-valued bind describes finite sequential composition. Recursive execution requires the same account of costs on every returning branch. We obtain it by attaching resource values to the finite return objects of the selected-domain recursion construction.

4.3 Weighted returns and iteration

Definition 4.5 (Resource-valued iteration)**.** For an equation morphism $f : X \to L_{\Sigma,C}(Y+X)$ in $\mathcal{A}_\Sigma$, the map

$$\Psi : [X, L_{\Sigma,C}(Y+X)]_\Sigma \times [X, L_{\Sigma,C}Y]_\Sigma \longrightarrow [X, L_{\Sigma,C}Y]_\Sigma, \quad \Psi(f,a)(x) = f(x)\ \mathrm{bind}_C\ [\eta^C_Y, a]$$

is an algebra morphism. The second function object is pointed by the everywhere undefined map. The parameterized least-fixed-point operation of Proposition 1.10 therefore constructs the **resource-valued iteration**

$$f^{\ddagger_C} = \mathrm{lfp}(\Psi(f,-)) : X \to L_{\Sigma,C}Y.$$

The operation $f \mapsto f^{\ddagger_C}$ is itself an algebra morphism between the displayed function objects.

To calculate it, represent f by the partial-map span

$$X \xleftarrow{m} D \xrightarrow{o} C \times (Y+X), \qquad D = D_Y + D_X.$$

Its branches have maps in $\mathcal{E}$

$$\begin{array}{lll} c_Y : D_Y \hookrightarrow X, & w_Y : D_Y \to C, & v : D_Y \to Y, \\ c_X : D_X \hookrightarrow X, & w_X : D_X \to C, & n : D_X \to X. \end{array}$$

Both source monos are Σ-classified. Retain exactly the return objects of Lemma 1.11, and attach the indicated valuations:

$$\begin{array}{ll} R_0 = D_Y, & R_{k+1} = D_X \times_{n,X,\rho_k} R_k, \\ \rho_0 = c_Y, & \rho_{k+1} = c_X\pi_1, \\ v_0 = v, & v_{k+1} = v_k\pi_2, \\ a_0 = w_Y, & a_{k+1} = \mu(w_X\pi_1, a_k\pi_2). \end{array}$$

Theorem 4.6 (Resource-valued finite return)**.** *The domain of $f^{\ddagger_C}$ is the Σ-classified mono*

$$D_f = \sum_{k:\mathbb{N}} R_k \hookrightarrow X,$$

and its value on R_k is (a_k, v_k). Consequently

$$\epsilon^C_Y f^{\ddagger_C} = (\epsilon^C_{Y+X} f)^\ddagger. \tag{4.2}$$

These constructions and equations commute with substitution of parameters.

Proof.

(1) **Identify the classified return domain.** Apply Lemma 1.11 to the erased equation. It gives the disjoint return components R_k and their Σ-classified union D_f.

(2) **Compute the weighted approximants.** For $b_0 = \bot$ and $b_{j+1} = \Psi(f, b_j)$, induction gives domain $\sum_{k<j} R_k$ and value (a_k, v_k) on R_k. The exit branch contributes $w_Y e = w_Y$; the continuation branch prefixes its cost to the recursive cost. Later approximants retain each defined value.

(3) **Take the least solution and erase costs.** The admitted supremum is the stated partial map. Erasure retains the return components and their values, proving (4.2). Parameter coherence follows from Proposition 1.16 and the finite cost recursion.

□

Equivalently, define

$$\widetilde{f} : C \times X \to L_\Sigma((C \times Y) + (C \times X))$$

by executing $f(x)$ and sending $(d, \mathsf{inl}\, y)$ to $\mathsf{inl}(cd, y)$ and $(d, \mathsf{inr}\, x')$ to $\mathsf{inr}(cd, x')$. The iteration satisfies

$$f^{\ddagger_C} = \widetilde{f}^{\ddagger} \langle e!, \mathrm{id}_X \rangle.$$

Indeed a return in R_k started with accumulator c ends with ca_k, by induction on k. This identifies the two constructions. The return length is the projection $D_f \to \mathbb{N}$; the predicates R_k need not be Boolean. Unit cost on every call gives $k+1$ on R_k. Unit cost on continuation and zero on return gives k; we specify the convention in each concrete measurement.

Theorem 4.7 (Resource-valued Elgot structure)**.** *Write $g \diamond_C f$ for Kleisli composition and $g \oplus_C h = [L_{\Sigma,C}(\mathsf{inl})g, L_{\Sigma,C}(\mathsf{inr})h]$. For $f : X \to L_{\Sigma,C}(Y + X)$, $g : Y \to L_{\Sigma,C}Z$, and $h : X \to L_{\Sigma,C}((Y + X) + X)$, the iteration operation satisfies*

$$f^{\ddagger_C} = [\eta^C_Y, f^{\ddagger_C}] \diamond_C f, \tag{4.3}$$

$$g \diamond_C f^{\ddagger_C} = ((g \oplus_C \eta^C_X) \diamond_C f)^{\ddagger_C}, \tag{4.4}$$

$$(h^{\ddagger_C})^{\ddagger_C} = (L_{\Sigma,C}[\mathrm{id}_{Y+X}, \mathsf{inr}]h)^{\ddagger_C}. \tag{4.5}$$

For $u : Z \to X$ and $k : Z \to L_{\Sigma,C}(Y + Z)$,

$$fu = L_{\Sigma,C}(\mathrm{id}_Y + u)k \quad \Longrightarrow \quad f^{\ddagger_C} u = k^{\ddagger_C}.$$

Thus $L_{\Sigma,C}$ is a complete Elgot monad. Monoid morphisms commute with this iteration. Coproduct feedback is the algebra morphism

$$\mathrm{Tr}^U_{+,C}(f) = [\eta^C_Y, (f\iota_U)^{\ddagger_C}] \diamond_C f\iota_X \quad (f : X + U \to L_{\Sigma,C}(Y + U)). \tag{4.6}$$

It satisfies the symmetric coproduct-trace equations.

Proof.

(1) **Verify the fixed-point equation.** Exposing the first transition proves the fixed-point equation.

(2) **Verify naturality by matching histories.** For naturality, both sides execute the same returning f-history and then g. Their partial domains agree, including failure of g, and their ordered products append the same g-cost.

(3) **Verify codiagonal by cutting and flattening histories.** For codiagonal, tag each continuation by its inner or outer occurrence of X, and cut a finite history at the outer tags. This gives exactly the successive inner returns of the left side. Flattening the blocks is inverse. Each original h-call occurs once in its original order, so associativity and the unit law identify the costs.

(4) **Transport return domains and costs.** For uniformity, the commuting equation transports transitions along u with identical tags, costs, and outputs. Induction identifies each return domain with the corresponding pullback and identifies its cost. The classified union proves the equation.

(5) **Verify change-of-cost compatibility and continuity.** A monoid morphism preserves the displayed cost recurrence, which proves its iteration compatibility. The operation is an algebra morphism by the construction using Ψ.

(6) **Verify the trace equations.** Formula (4.6) first enters the equation and then iterates the interface branch. For each trace equation, matching the finite-return components retains every primitive call in order: naturality adjoins the same entry and exit calls, dinaturality moves the interface cut, and vanishing cuts or flattens the same blocks. The cost products are consequently equal. Superposing leaves the unused component unchanged. The symmetry in yanking is a pure map followed by η^C, so its cost is the monoid unit. These arguments prove all trace equations, including equality of classified domains, in every parameter context.

□

Functorial change of resource monoid. Changing a resource valuation should commute with recursive execution. The monoid-morphism clause of the theorem gives exactly this compatibility: the construction is a functor to strong complete Elgot monads and iteration-preserving strong monad morphisms,

$$C \longmapsto L_{\Sigma,C}, \qquad h : C \to C' \longmapsto \alpha_h = L_\Sigma(h \times -).$$

Functoriality follows from functoriality of L_Σ; preservation of iteration is the monoid-morphism assertion just proved. In particular the following square commutes:

$$\begin{array}{ccc} [X, L_{\Sigma,C}(Y+X)]_\Sigma & \xrightarrow{\ddagger_C} & [X, L_{\Sigma,C}Y]_\Sigma \\ \downarrow{\scriptstyle (\alpha_h)_*} & & \downarrow{\scriptstyle (\alpha_h)_*} \\ [X, L_{\Sigma,C'}(Y+X)]_\Sigma & \xrightarrow[\ddagger_{C'}]{} & [X, L_{\Sigma,C'}Y]_\Sigma. \end{array} \tag{4.7}$$

Here $(-)_*$ means postcomposition, and all four objects are the constructed computational function objects. Erasure is the case of the unique monoid morphism $C \to 1$.

4.4 Histories and their resource valuations

We can measure several resources from the same execution by recording its events before applying a valuation.

Definition 4.8 (Returning histories)**.** For $A \in \mathcal{A}_\Sigma$ and $f : X \to L_\Sigma(A \times (Y + X))$, replace each event by its singleton word and execute in the free monoid:

$$f_* = L_\Sigma([-] \times \mathrm{id}_{Y+X})f, \qquad H_f = f_*^{\ddagger_{A^*}} : X \to L_\Sigma(A^* \times Y).$$

This is the **returning-history map** with the original domain.

The event weight $w : A \to C$ extends through the free-monoid unit:

$$\begin{array}{ccc} A & \xrightarrow{[-]} & A^* \\ & \searrow^{w} & \downarrow{\scriptstyle \overline{w}} \\ & & C. \end{array} \tag{4.8}$$

The dashed arrow is the unique monoid morphism extending the event weight. It induces $\alpha_{\overline{w}} : L_{\Sigma,A^*} \to L_{\Sigma,C}$. Thus Equation (4.7) says that folding the returning history gives the same result as accumulating its weights during execution. The following theorem records this comparison.

Theorem 4.9 (Finite-history valuation)**.** *For an algebra morphism $w : A \to C$ and its monoid extension $\overline{w} : A^* \to C$,*

$$L_\Sigma(\overline{w} \times \mathrm{id}_Y)H_f = \big(L_\Sigma(w \times \mathrm{id}_{Y+X})f\big)^{\dagger_C}. \tag{4.9}$$

Proof.

(1) **Identify the word and cost recurrences.** The word on R_0 is the singleton return label. On R_{k+1} it is the continuation label followed by the word on R_k. The fold recurrence is precisely the recurrence for a_k in Theorem 4.6.

(2) **Compare values on every return summand.** Induction identifies the values on every summand of the common domain.

□

State events are recorded by a typed history map. Given $b : X \to L_\Sigma(Y + X)$, strength constructs

$$X \to L_\Sigma((X \times (Y + X)) \times (Y + X)), \qquad x \mapsto b(x)\ \mathrm{bind}_\Sigma\ (z \mapsto \eta((x, z), z)).$$

Its words record state events. A finite word may contain domain values or supplied names, so validity of an arbitrary proposed history is not thereby decidable.

Definition 4.10 (History valuations and measured execution)**.** A **history valuation** is a total algebra morphism $\nu : A^* \to C$. Its **measured execution** is $L_\Sigma(\nu \times \mathrm{id})H_f$, even when ν is not a monoid morphism.

Only a monoid fold inherits the sequential product law of (4.9). In particular, peak occupancy uses the maximum of recorded occupancy values, including the initial and final states when relevant; allocation increments alone do not give that measurement.

Proposition 4.11 (Universal classification of event weights)**.** *For an event algebra $A \in \mathcal{A}_\Sigma$ and an internal monoid (C, e, μ) in $\mathcal{A}_\Sigma$, the computational exponential $\mathcal{W}_C = [A, C]_\Sigma$ represents all continuous event weights $w : A \to C$. There is a canonical, parameter-uniform valuation map*

$$\mathrm{fold}_C : \mathcal{W}_C \times A^* \to C$$

whose value on $(w, [a_1, \ldots, a_n])$ is the ordered finite product $w(a_1) \cdots w(a_n)$, with empty product e. This represents monoid morphisms from A^* to C in every parameter context; its pullback at w is the history valuation of Theorem 4.9.

Proof.

(1) **Construct the parametrized fold.** Cartesian closure gives evaluation $\mathrm{ev} : \mathcal{W}_C \times A \to C$. Apply the finite-word fold construction of Proposition 4.2, retaining $w : \mathcal{W}_C$ as parameter. On each fixed length this is a finite composite of evaluation and μ, so Lemma 4.1 makes the resulting map an algebra morphism.

(2) **Apply the free-monoid property.** Restriction to singleton words and extension by this fold are inverse, by Proposition 4.2 in the same parameter context. They give the claimed classification and monoid equations. Evaluation at w recovers the valuation of Theorem 4.9.

□

The execution operations are now available. To form a complexity judgment, we must specify which descriptions and primitive operations are being measured. Fixing that presentation also fixes what is charged by an event valuation and what is supplied by the environment.

4.5 Implementation descriptions

Definition 4.12 (Computational environments)**.** For a typed parameter context, a **computational environment** is a tuple of its supplied name and operation parameters. Its carrier is denoted by Env. An implementation quantifier fixes this tuple; the supplied parameters are part of the implementation context.

Construction 4.13 (Fixed computational environments)**.**

(1) **Form the environment carrier.** The finite description construction of Construction D.5 has typed contexts of supplied name and operation parameters. For such a context, form its environment object Env by the indicated finite products and dependent sums of parameter carriers. Varying a finite context uses the corresponding internally constructed sum of these objects. A description in an environment refers to its supplied parameters by their typed slots. Closed constructor data and implemented programs are still constructed by their finite syntax. Every supplied parameter is recorded in the environment; it is not an extra choice made while taking an implementation fibre.

(2) **Interpret descriptions over the environment.** Pairing a well-typed description with its environment constructs $E_{X,Y} \to \mathsf{Env}$. Interpretation is a morphism over Env, obtained by the finite recursion of Lemma D.6. The primitive signature and its selected execution descriptions are retained in this context. A general history valuation is a point of the computational exponential $[A^*, C]_\Sigma$. By Proposition 4.11, the monoid-fold valuations are precisely those obtained by selecting an event-weight point of $[A, C]_\Sigma$ and taking its unique monoid extension. We work internally at a generic $\theta : \mathsf{Env}$, suppressing θ in the formulas. Thus $E_{X,Y}$ below means $E_{X,Y,\theta}$, and an implementation quantifier fixes θ. Products, function objects, images, and quantifiers in this construction are those of $\mathcal{E}/\mathsf{Env}$.

(3) **Retain the work performed by substitution.** An implemented substitution of environments precomposes these interpretations. A substitution which itself performs charged work is represented by its execution description and contributes its history; ordinary reindexing alone supplies no assertion that this work is free. The numerical instance has the fixed register signature. Program-generated function values are obtained by evaluating their generating descriptions, with that evaluation recorded.

Use the finite typed descriptions with supplied parameters constructed in Construction D.5 and Lemma D.6. Fix a typed fragment of its implemented signature. Its primitive operations have types $b_j : U_j \to L_\Sigma V_j$, and the fragment includes the products, sums, composition, and execution operations it uses. The well-typed subobject of $\mathsf{Tree}_\mathcal{S}$ carries these descriptions, their interpretation, and their name presentation. Parameter carriers remain part of the specified environment; a numerical enumeration of them is not part of the construction.

Definition 4.14 (Logged primitives and descriptions)**.** For the finitely many primitive operation tags form the computational **event object** $A = \sum_j (U_j \times V_j)$. Define the **logged primitive** by

$$b_j^H(u) = b_j(u)\ \mathrm{bind}_\Sigma\ (v \mapsto \eta([\iota_j(u, v)], v)) : U_j \to L_\Sigma(A^* \times V_j).$$

Pure structural maps are followed by η^{A^*}, so have empty word. Interpret composition by $\diamond_{A^*}$, coproduct branching by copairing, and recursive execution by $\ddagger_{A^*}$ or (4.6). A specified sequential pair of computations $f : U \to L_{\Sigma,A^*}V$, $g : U \to L_{\Sigma,A^*}W$ has interpretation

$$u \mapsto f(u)\ \mathrm{bind}_{A^*}\ (v \mapsto g(u)\ \mathrm{bind}_{A^*}\ (w \mapsto \eta^{A^*}(v, w))).$$

Thus its evaluation order is retained in the word. These clauses are operations on the same finite descriptions; no quotient by unweighted semantic equality is performed.

The substitution here is substitution in the description's typed operation parameters. For a context of arrow parameters

$$G = \prod_i [U_i, L_{\Sigma,A^*} V_i]_\Sigma,$$

interpretation of a description $t : X \to Y$ has type $[\![t]\!]_H : G \to [X, L_{\Sigma,A^*} Y]_\Sigma$. For substitutions σ_i from a parameter context G', put $\sigma_H = \langle [\![\sigma_i]\!]_H \rangle_i : G' \to G$. The substitution equation is

$$[\![t[\sigma]]\!]_H = [\![t]\!]_H \sigma_H.$$

Ordinary data parameters are retained in the same constructor maps; their evaluation uses the stated sequencing convention.

Proposition 4.15 (Interpretation of implementation descriptions)**.** *For each input and output pair in the fragment, structural recursion constructs maps in* $\mathcal{E}$

$$[\![-]\!]_H : E_{X,Y} \to [X, L_\Sigma(A^* \times Y)]_\Sigma, \qquad \mathrm{ev}_H : E_{X,Y} \times X \to L_\Sigma(A^* \times Y).$$

They commute with the supplied finite-tree substitutions. Erasing words recovers the original interpretation of every description, including its classified partial domain. A fixed algebra morphism $\nu : A^* \to \mathbb{N}$ *constructs its measured evaluator*

$$\mathrm{ev}_\nu = L_\Sigma(\nu \times \mathrm{id}_Y)\, \mathrm{ev}_H : E_{X,Y} \times X \to L_\Sigma(\mathbb{N} \times Y). \tag{4.10}$$

Proof.

(1) **Interpret the typed constructors.** Each constructor clause above is a morphism on the appropriate computational function objects: composition and pairing use cartesian closure and strength; iteration and feedback use Theorem 4.7. Interpret a well-typed finite tree by its constructors, retaining its parameters. Induction on its typing derivation gives the displayed function-object type. This is the finite structural recursion on the tree carrier of Construction D.5.

(2) **Verify substitution by structural induction.** Substitution at a variable evaluates the substituted tree, substitution at a parameter leaf retains its interpretation, and a compound constructor applies the same operation to the substituted children. Induction proves the substitution equation.

(3) **Verify erasure and the evaluation maps.** Erasure commutes with the primitive clauses, unit, bind, strength, and iteration by Theorems 4.4 and 4.6; induction therefore gives the original domain and value. Evaluation and postcomposition give the last two maps.

□

The signature, valuation, and environment θ remain fixed when we quantify over $E_{X,Y,\theta}$. This specifies what each event measures. A supplied function parameter has an invocation-cost measurement; measuring its internal work requires its execution description and the events of that description. The numerical instance constructs the configuration update in Construction 13.2. Projective lifting supplies a map on names and its domain comparison; the history of an execution supplies its cost.

Fixing the environment is essential to the complexity interpretation. If arbitrary functions could be varied as parameters while application had unit cost, the description $x \mapsto f(x)$ would give every such denotation a one-call bound. Here the supplied operation f is fixed by θ. To measure its internal execution we substitute its supplied description and charge the resulting events. The language and the environment together determine the implementations under comparison.

The word fold is stable under the cut and concatenation operations in the weighted trace proof. Counting outer loop iterations alone is not: regrouping two primitive calls into one body changes that count. Nor does equality after the unweighted denotation quotient imply equal cost. Resource-preserving transformations preserve the recorded word, or carry a proved comparison of its selected valuation.

4.6 Measured name presentations

A represented input may have several names, and execution on those names may have different costs. We therefore carry the name maps through the measurement. In the generic fixed environment, retain the input and output presentations

$$q_X : C_X \twoheadrightarrow X_0 = \mathrm{Pt}_\Sigma\, D_X, \qquad q_Y : C_Y \twoheadrightarrow Y_0 = \mathrm{Pt}_\Sigma\, D_Y.$$

Use the canonical name covers of Proposition D.2 and the actual primitive implementations in the environment constructed in Construction D.5. Each primitive retains its tracker, partial name map, returned-value equation, and equality of its domain with the pullback of the semantic domain. The structural maps are the explicit tuple, tag, substitution, and finite-return algorithms. Finite products and sums of these named sorts provide the typed execution fragment. The discrete admitted algebra on a name carrier allows the finite-word and weighted-execution constructions to be applied to these maps. Let $\widehat{A}$ be their event object: an invocation event records the primitive tag and its input and output names; an unfolded primitive instead records its displayed internal execution events.

Theorem 4.16 (Exact measured execution on names)**.** *Finite recursion on the specified implementation descriptions constructs*

$$\widehat{\mathrm{ev}}_H : E \times C_X \longrightarrow L_\Sigma(\widehat{A}^* \times C_Y). \tag{4.11}$$

Write $\mathrm{ev}^{\mathrm{sem}} : E \times X_0 \rightharpoonup Y_0$ *for the interpretation of Lemma D.6 on effective points. Then, as partial maps,*

$$q_Y \ \mathrm{erase}\, \widehat{\mathrm{ev}}_H = \mathrm{ev}^{\mathrm{sem}}(\mathrm{id}_E \times q_X), \qquad \widehat{D} = (\mathrm{id}_E \times q_X)^* D_{\mathrm{sem}}. \tag{4.12}$$

For the fixed implemented total valuation $\widehat{\nu} : \widehat{A}^* \to \mathbb{N}$*, let* $P \hookrightarrow E$ *classify total named executions. The returned values and measured histories construct*

$$p_0 : P \to Y_0^{X_0}, \qquad \widehat{\Phi}_0 : P \to \mathbb{N}^{C_X}, \qquad P \xrightarrow{p \twoheadrightarrow} V \hookrightarrow Y_0^{X_0}. \tag{4.13}$$

The cost descends to $\mathbb{N}^{X_0}$ *exactly on the subobject of* P *where*

$$q_X c = q_X c' \quad \Longrightarrow \quad \widehat{\Phi}_0(e)(c) = \widehat{\Phi}_0(e)(c') \qquad (c, c' : C_X). \tag{4.14}$$

All constructions commute with substitution of the specified environment and finite-description parameters.

Proof.

(1) **Interpret the measured primitives and finite constructors.** Log the specified name primitives and apply Proposition 4.15. Their domain-and-value squares are those of the constructed implemented environment in Construction D.5. Bind pulls back the next domain along the returned name, so these squares compose. The branching and sequential-pairing clauses preserve them by finite case analysis.

(2) **Verify iteration and substitution.** Induction on return length identifies each named return component, including its value, with the pullback of the semantic component. Theorem 4.6 and pullback stability of the natural-number existential identify their classified unions. Structural induction proves (4.12) and the finite-description substitution equation.

(3) **Construct total executions and descend their values.** For the return classifier $\widehat{d} : E \times C_X \to \Sigma$, put $P(e) \iff \forall c : C_X\ \widehat{d}(e, c)$. Since q_X is regular epic, domain adequacy identifies this with semantic totality. On $P \times C_X$, the returned name followed by q_Y coequalizes the kernel pair of $\mathrm{id}_P \times q_X$, by value adequacy. It therefore descends uniquely to $P \times X_0 \to Y_0$. Transposition and image factorization give p_0, p, V; folding the recorded history gives $\widehat{\Phi}_0 : P \to \mathbb{N}^{C_X}$.

(4) **Characterize cost descent and parameter coherence.** On the subobject defined by (4.14), the cost coequalizes that same kernel pair, so descends uniquely to X_0. Conversely every descended cost coequalizes it. Pullback stability of the quotients, dependent-product Beck–Chevalley, and the interpretation equations give parameter coherence. The selected event valuation supplies the measurement throughout this construction.

□

The return denotation descends by its extensionality equation. Cost descends precisely when it is constant on names of the same input. For a fixed total implementation e, the coequalizer criterion for q_X expresses that condition:

$$\begin{array}{ccccc} C_X \times_{X_0} C_X & \underset{\pi_2}{\overset{\pi_1}{\rightrightarrows}} & C_X & \xrightarrow{\ q_X\ } & X_0 \\ & & \Big\downarrow{\scriptstyle \widehat{\Phi}_0(e)} & \swarrow{\scriptstyle \Phi_0^{\mathrm{ext}}(e)} & \\ & & \mathbb{N} & & \end{array} \tag{4.15}$$

The dashed map exists uniquely when the two cost composites from the kernel pair agree. The value leg has its own descent equation, Equation (4.12). Until the cost equation also holds, the name carrier remains the domain of the measured function.

Example 4.17 (Value descent without cost descent)**.** Take $q_X : \mathbb{N} \times \mathbb{N} \twoheadrightarrow \mathbb{N}$, $q_X(n, k) = n$. In the retained signature a description can perform k implemented idle invocations and return 0. For their counting valuation its cost is k. Its erased name map $(n, k) \mapsto 0$ has true domain and commutes with q_X and the identity presentation of the output. It therefore implements the constant semantic primitive; unfolding this particular implementation records its name-dependent waiting history. The returned value descends to $\mathbb{N}$, but the cost fails (4.14). This is a name-level measurement, not an extensional function of n.

4.7 Total implementations and the measured image

We now collect the total implementations of the fixed measured evaluator and separate their denotations from their costs.

Definition 4.18 (Total implementations and their measurements)**.** First take execution input and return carriers X, Y. Write $E = E_{X,Y}$, or use the numerical family of Construction 13.2 or the Turing-machine family of Construction 6.4, with the corresponding evaluator of the form (4.10). If $d : E \times X \to \Sigma$ classifies its domain, the **total implementations** form the subobject

$$P \hookrightarrow E, \qquad P(e) \iff \forall x : X \; d(e, x).$$

Totality is an $\Omega_{\mathcal{E}}$-valued predicate. On a returning execution write

$$H_e(x) \in A^*, \qquad T_{\nu,e}(x) = \nu(H_e(x)). \tag{4.16}$$

Both expressions have exactly the return domain of the represented execution.

Proposition 4.19 (Measured implementation image)**.** *The construction gives canonical arrows*

$$p_0 : P \to Y^X, \qquad \Phi_0 : P \to \mathbb{N}^X, \qquad P \xrightarrow{p \twoheadrightarrow} V \hookrightarrow Y^X, \quad V = \mathrm{im}(p_0).$$

Their uncurried values are the returned value and the selected finite resource measurement. The cost map has domain P.

Proof.

(1) **Restrict the evaluator to its totality locus.** Restriction of the evaluator to $P \times X$ has true extent, so its partial-map span is a unique total map $P \times X \to \mathbb{N} \times Y$.

(2) **Construct the implementation image and cost map.** Transposing its two projections gives Φ_0, p_0. Regular image factorization in $\mathcal{E}$ gives p and V.

□

For represented inputs, use Theorem 4.16 instead: the denotation lies in $Y_0^{X_0}$ and the measured cost function in $\mathbb{N}^{C_X}$. From now on X in a cost function denotes the execution input carrier, so $X = C_X$ in this case. The notation $p : P \twoheadrightarrow V$, $\Phi_0 : P \to \mathbb{N}^X$ covers both constructions. All these objects remain over the fixed environment θ of Construction 4.13.

The order of this construction matters: the language, environment, and event valuation are fixed before the denotation image is formed. Otherwise arbitrary clocks or unit-cost semantic primitives would trivialize the measurement. For example, the clock constantly equal to true with return value $g(x)$ has first hit zero for every total $g : X \to Y$. The concrete numerical clocks of Proposition 13.3 instead come from the specified transition system. The general return theorem Theorem 4.6 does not require those clocks.

For Landau comparison, rational inclusion and the eventual quotient of Proposition 2.3 give the cost-germ map

$$\Phi : P \xrightarrow{\Phi_0} \mathbb{N}^X \longrightarrow {\mathbb{Q}_{\geq 0}}^X \xrightarrow{q_+} K^+.$$

Each measurement is fixed before taking the existential over its implementation fibre. Joint resources use a product valuation on the same histories and retain the same implementation witness.

The proper-filter context used in K is a context on this input carrier X. Bounds specified by an extensional input or by an input-size map are pulled back to X. See Proposition 13.7.

The measured implementation span.

Definition 4.20 (The measured implementation span)**.** For a fixed computational environment, a total implementation determines both a denotation and a measured cost. These assignments have the same source and form the **measured implementation span** in $\mathcal{E}/\mathsf{Env}$

$$V \xleftarrow{\ p\ } P \xrightarrow{\ \Phi\ } K. \tag{4.17}$$

Its apex retains the total implementations. To impose a budget, we will pull its marked module back along Φ, then compose with p to obtain the family of implementations meeting that budget. Taking the image will give the feasible denotations.

Before taking germs, the cost endpoint is $\mathbb{N}^X$. Rational inclusion and the germ quotient change it to K while preserving the apex. The pair of legs also factors through its regular image:

$$P \xrightarrow{\ (p,\Phi)\ } \mathscr{R}_{p,\Phi} \hookrightarrow V \times K.$$

The relation records which denotation–cost pairs occur. The span carries the witnesses of those pairs, including distinct implementations with the same denotation and cost. The following slice construction preserves this information until the feasibility image is taken.

Proposition 4.21 (Witness transport and its feasibility image)**.** *The measured span induces the slice functor*

$$\mathsf{Wit}_{p,\Phi} = p_! \Phi^* : \mathcal{E}/K \longrightarrow \mathcal{E}/V, \qquad \mathsf{Wit}_{p,\Phi}(M \to K) = (P \times_K M \to V).$$

Here $p_!$ is dependent sum in the slice. On marked subobjects, taking the image gives the monotone map

$$\mathsf{Feas}_{p,\Phi} = \exists_p \Phi^* : \mathrm{Sub}(K) \longrightarrow \mathrm{Sub}(V).$$

Both constructions commute with substitution of environment parameters. A commuting regular-epimorphic refinement of the apex induces a regular-epimorphic map of witness families and preserves their feasibility images.

Proof.

(1) **Construct witness transport and its feasibility image.** The pullback and subsequent projection are

$$\begin{array}{ccc} P \times_K M & \longrightarrow & M \\ \downarrow & \lrcorner & \downarrow \\ P & \xrightarrow[\Phi]{} & K \\ {\scriptstyle p}\downarrow & & \\ V & & \end{array} \tag{4.18}$$

and define $p_! \Phi^*$. Their regular image is $\exists_p \Phi^* M$ by the definition of existential image.

(2) **Verify parameter substitution.** Beck–Chevalley for pullbacks and stability of regular images give parameter substitution.

(3) **Prove invariance under a covering presentation.** If $r : P' \twoheadrightarrow P$ satisfies $p' = pr$ and $\Phi' = \Phi r$, the square

$$\begin{array}{ccc} P' \times_K M & \twoheadrightarrow & P \times_K M \\ \downarrow & \lrcorner & \downarrow \\ P' & \xrightarrow[r]{} & P \end{array}$$

is a pullback. Its upper arrow is therefore regular epic. The composites to V have the same image.

□

For simultaneous measurements $\Phi_T : P \to K_T$ and $\Phi_S : P \to K_S$, the right leg is their pairing and the joint witness family is the pullback

$$\begin{array}{ccc} Z_{M_T,M_S} & \longrightarrow & M_T \times M_S \\ \downarrow & \lrcorner & \downarrow \\ P & \xrightarrow[(\Phi_T,\Phi_S)]{} & K_T \times K_S. \end{array}$$

The pullback imposes both constraints on the same implementation. Its image in V is therefore simultaneous feasibility. This fixes the order of the operations: combine the resource conditions first, then quantify over their shared witness.

4.8 Universal implementation incidence and its fibres

The measured span can now be applied to the universal budget family. Write its fixed-environment maps as $p : P \twoheadrightarrow V$, $\Phi : P \to K^+ \hookrightarrow K$, and $\kappa : P \to L$; with the parameter displayed they are $p_\theta : P_\theta \twoheadrightarrow V_\theta$ and Φ_θ. For represented inputs, these are the maps of Theorem 4.16, with germs formed on the name carrier.

Definition 4.22 (Universal incidence and feasibility)**.** In the ringed resource topos, the **universal implementation incidence** $\mathscr{Z}$ and its **feasibility image** $\mathscr{F}$ are

$$\mathscr{Z} = \Delta_{\mathfrak{C}_{\mathrm{Ev}}} P \times_{\bar{K}} \mathscr{M}, \qquad \mathscr{F} = \mathrm{im}(\mathscr{Z} \to \Delta_{\mathfrak{C}_{\mathrm{Ev}}} V). \tag{4.19}$$

Definition 4.23 (The sheaf of feasible implementations)**.** For $v : V$, the **sheaf of feasible implementations** $\mathscr{W}_v$ is the pullback of $\mathscr{Z}$ along $\Delta\{v\} \to \Delta V$. Its image in the terminal sheaf is the **feasible open** $\mathcal{U}_v$.

This applies Proposition 4.21 in $\mathscr{S}$. The cartesian square selects the implementations meeting the universal budget, and the regular image records their feasible denotations:

$$\begin{array}{ccc} \mathscr{Z} & \longrightarrow & \mathscr{M} \\ \downarrow & \lrcorner & \downarrow \\ \Delta P & \xrightarrow[\Delta\Phi]{} & \bar{K} \\ {\scriptstyle \Delta p}\downarrow & & \\ \Delta V & & \end{array} \qquad \mathscr{Z} \twoheadrightarrow \mathscr{F} \hookrightarrow \Delta V. \tag{4.20}$$

Definition 4.24 (Feasible denotations at a budget)**.** At an ideal point I, the **implementation witnesses** Z_I and **feasible denotations** $\mathcal{C}_{\mathrm{cost}}(I)$ are

$$Z_I = I^*\mathscr{Z} = P \times_K M_I, \qquad \mathcal{C}_{\mathrm{cost}}(I) = I^*\mathscr{F} = \mathrm{im}(Z_I \to V). \tag{4.21}$$

This defines semantic budget fibres; restricting I to $J_\Sigma L$ gives the admitted computational instances.

Theorem 4.25 (Classified computational upper bounds)**.** *The incidence and its image are the budget diagrams*

$$\mathscr{Z}(a) = \{e : P : \kappa(e) \le a\}, \qquad \mathscr{F}(a) = \{v : V : \exists e \in P_v,\ \kappa(e) \le a\}.$$

The open-valued classifier of $\mathscr{F} \hookrightarrow \Delta V$ is

$$\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)}. \tag{4.22}$$

At every ideal point,

$$I^*\mathscr{F} = \mathcal{C}_{\mathrm{cost}}(I) = \{v : \exists e \in P_v,\ \kappa(e) \in I\}. \tag{4.23}$$

Regular-epimorphic refinements of implementation covers preserving denotation and measured cost leave this image unchanged.

Proof.

(1) **Compute incidence and feasibility at each budget.** Apply Theorem 3.8 to the pullback in (4.19). Images are computed at each budget, giving the two diagrams. Membership of e has extent $U_{\kappa(e)}$; existential quantification along p gives (4.22).

(2) **Compute the fibres.** Geometric inverse image preserves pullbacks and regular images, proving the fibre formula.

(3) **Prove invariance under presentation refinement.** For a refinement $r : P' \twoheadrightarrow P$, $\exists_r \top = \top$; composition of images proves the invariance.

□

The budget fibres satisfy $\mathcal{C}_{\mathrm{cost}}(I) = \bigvee_{a \in I} \mathcal{C}_{\mathrm{cost}}(\downarrow a)$ by interchanging their two existential quantifiers. Their incidence has type $J_\Sigma L \times V \to \Omega_{\mathcal{E}}$, with the separately specified positive observations of Section 2.6.1.

4.9 Complexity judgments as subobject adjunctions

An upper bound asks for an implementation whose cost meets the bound. A lower bound constrains every implementation of the same denotation. The adjunctions $\exists_p \dashv p^* \dashv \forall_p$ express these different quantifiers over the fixed implementation family. Their classifiers are $\Omega_{\mathcal{E}}$-valued; positive observations retain the separately specified meaning of Section 2.6.1.

Definition 4.26 (Upper, lower, and tight complexity judgments)**.** For a nonnegative bound g, form

$$A_g = \{e : P : \Phi(e) \in Bg\}, \qquad D_g = \{e : P : g \in B\Phi(e)\},$$

The **upper**, **lower**, and **tight complexity judgments** are

$$\mathrm{Upper}_g = \exists_p A_g, \qquad \mathrm{Lower}_g = \forall_p D_g, \qquad \mathrm{Tight}_g = \mathrm{Upper}_g \wedge \mathrm{Lower}_g\,. \tag{4.24}$$

Their internal formulas are respectively $\exists e \in P_v\, \Phi(e) \in O(g)$ and $\forall e \in P_v\, \Phi(e) \in \Omega(g)$. With the environment displayed, the first is

$$\mathrm{Upper}_g^{\theta}(v) \iff \exists e \in P_{\theta,v}\ \Phi_\theta(e) \in Bg,$$

and the lower and tight judgments use that same $P_{\theta,v}$. The lower-bound constants and eventual extents occur inside $\forall e$; the variant with uniform constants is a different dependent-quantifier expression.

These are internal judgments retaining their witness transformations. For conventional worst-case lower bounds, the exact classical predicate is obtained by implementationwise double-negation closure before $\forall_p$, as constructed and proved in Construction 6.9 and Theorem 6.10.

Proposition 4.27 (Complexity bounds as inclusions of feasibility opens)**.** *For $g \in K^+$ and $a = [g]$, the upper, lower and tight judgments have the geometric descriptions*

$$\begin{aligned} \mathrm{Upper}_g(v) &\iff U_a \subseteq \mathcal{U}_v, \\ \mathrm{Lower}_g(v) &\iff \mathcal{U}_v \subseteq U_a, \\ \mathrm{Tight}_g(v) &\iff \mathcal{U}_v = U_a. \end{aligned} \tag{4.25}$$

These equivalences are internal and uniform in the original denotation, bound and computational-environment parameters.

Proof.

(1) **Reflect growth order through principal opens.** The principal opens satisfy $U_b \subseteq U_a \iff a \le b$: evaluate an inclusion at b for one direction, and use upperness for the other.

(2) **Characterize the upper-bound judgment.** If $\kappa(e) \le a$, then $U_a \subseteq U_{\kappa(e)} \subseteq \mathcal{U}_v$. Conversely, evaluation of $U_a \subseteq \mathcal{U}_v$ at a and (4.22) give $\exists e \in P_v\ \kappa(e) \le a$, precisely the upper judgment. Equivalently, supercompactness of U_a extracts a member of the canonical implementation cover.

(3) **Characterize the lower and tight judgments.** For the lower judgment,

$$\begin{aligned} \mathcal{U}_v \subseteq U_a &\iff \forall e \in P_v\ U_{\kappa(e)} \subseteq U_a \\ &\iff \forall e \in P_v\ a \le \kappa(e) \iff \mathrm{Lower}_g(v). \end{aligned}$$

Combining the two comparisons proves the tight statement.

□

The feasibility open therefore determines every principal upper, lower and tight judgment. A tight bound is the assertion that this open is the corresponding principal open. The implementation cover will identify principality with an attained optimal growth class in Theorems 5.3 and 5.4. The lower statement in (4.25) is an inclusion between opens; its proof retains the universal implementation quantifier and its individual domination witnesses.

Proposition 4.28 (Matching bounds and shared witnesses)**.** *A tight bound supplies an implementation with cost in $\Theta(g)$. On the eventually nonzero comparison locus, a lower bound excludes an implementation in $o(g)$. An existential implementation in $\Theta(g)$ does not imply the universal lower bound. For two incidence subobjects $A, D \hookrightarrow P$, the shared-witness classification is $\exists_p(A \wedge D)$, which lies below $(\exists_p A) \wedge (\exists_p D)$; equality is not automatic.*

Proof.

(1) **Combine bounds using a shared implementation.** Frobenius reciprocity and the counit of $p^* \dashv \forall_p$ give

$$(\exists_p A_g) \wedge (\forall_p D_g) = \exists_p (A_g \wedge p^* \forall_p D_g) \leq \exists_p (A_g \wedge D_g).$$

Mutual domination gives $\Theta(g)$.

(2) **Exclude simultaneous smallness and a nonzero lower bound.** If a small implementation also satisfies $g \preceq \Phi(e)$ with constant a, smallness at $2a$ gives eventually $2|g| \leq |g|$, contradicting the eventually nonzero extent and properness of the filter.

(3) **Distinguish an implementation cost from a universal bound.** A constant-output function has both an immediate-return implementation of bounded cost and one scanning every input letter, of cost $\Theta(|x| + 1)$. The latter does not make that bound a universal lower bound.

(4) **Establish the shared-witness inclusion.** Finally monotonicity of images gives the shared-witness inclusion; taking the two images separately permits different implementations.

□

All these predicates are defined in $\mathcal{E}$ with their displayed parameters. Constant-diagram transport preserves their particular quantifications: a map of constant objects has the same fibre at every budget and Weil stage, and a constant predicate has the same universal clause on that fibre. Parameter substitution inside a topos obeys the dependent-product Beck–Chevalley law, by its right-adjoint universal property. For a cover refinement $r : P' \twoheadrightarrow P$, both $\exists_r r^*$ and $\forall_r r^*$ are the identity on subobjects of P, so pulled-back upper and lower judgments are invariant. This follows from the inhabited fibres of the regular epi.

The open $\mathcal{U}_v$ is Scott open by its principal cover. Universal resource assertions need not be: in the cofinite numerical fibre put $g_k(n) = (n+1)^k$ and $I_m = {\downarrow}\,[g_m]$. The assertion $\forall k\, [g_k] \in I$ holds at $\bigcup_m I_m$ but at no I_m, since $[g_{m+1}] \not\leq [g_m]$. Universal judgments therefore retain their dependent-product type instead of being identified with feasibility opens.

4.10 The implementation sheaf and compiler morphisms

Proposition 4.29 (Substitution of fixed environments)**.** *For a parameter substitution $s : \mathsf{Env}' \to \mathsf{Env}$, reindexing the execution construction gives the same measured evaluator as interpreting its substituted descriptions. It carries P, V, p, Φ, their implementation incidence and feasibility image, and their upper and lower judgments to the corresponding constructions over Env'. The implementation-sheaf, compiler, and optimality constructions below are likewise performed in this fixed-environment family.*

Proof.

(1) **Transport interpretation through environment substitution.** The interpretation statement is the finite substitution equation of Proposition 4.15 and Theorem 4.16.

(2) **Identify totality, denotations, and cost functions.** Pullback between parameter slices preserves finite limits and regular images. The dependent-product Beck–Chevalley equation for the input projection identifies the totality subobjects; exponentiation in the slice identifies their transposed denotations and cost functions. The quotient and image comparisons then identify p, V, Φ.

(3) **Transport feasibility and lower judgments.** The same pullback and image equations identify incidence and upper feasibility. Beck–Chevalley for p identifies the universal lower judgment.

(4) **Verify parameter coherence of the remaining constructions.** All budget and coefficient constructions retain these parameters; the sheaf constructions below have the same pointwise formulas in each slice. In particular an implementation quantifier is never replaced by an existential over environments. The displayed comparisons satisfy identity and composition by their universal properties.

□

In particular the denotation objects and their implementation covers form a cartesian square under $s : \mathsf{Env}' \to \mathsf{Env}$:

$$\begin{array}{ccc} P_{\mathsf{Env}'} & \longrightarrow & P_{\mathsf{Env}} \\ {\scriptstyle p_{\mathsf{Env}'}}\downarrow & \lrcorner & \downarrow{\scriptstyle p_{\mathsf{Env}}} \\ V_{\mathsf{Env}'} & \longrightarrow & V_{\mathsf{Env}}. \end{array}$$

Pullback of the measured cost leg gives the compatible incidence square; Beck–Chevalley identifies its existential and universal judgments.

Fixing a denotation in the universal incidence leaves a family of implementations indexed by budgets. We now describe this witness sheaf explicitly. Its image is the feasible open, while its sections retain the choices of implementation needed for the later optimality criterion.

Theorem 4.30 (The universal sheaf of feasible implementations)**.** *For $v : V$, the sheaf $\mathscr{W}_v$ is the dependent coproduct*

$$\mathscr{W}_v \cong \coprod_{e \in P_v} U_{\kappa(e)} \qquad \text{in } \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}). \tag{4.26}$$

Its image over $\mathfrak{C}_{\mathrm{Ev}}$ is $\mathcal{U}_v$, and its restriction to that open is a regular epimorphism $\mathscr{W}_v \twoheadrightarrow \mathcal{U}_v$. By the sheaf–local-homeomorphism correspondence, this constructs an internal étale locale $E_v \to \mathfrak{C}_{\mathrm{Ev}}$ whose fibre over a resource ideal I is $\{e \in P_v : \kappa(e) \in I\}$.

Proof.

(1) **Identify the implementation sheaf by membership.** Over an implementation $e : P_v$, the defining pullback $\mathscr{Z} = \Delta P \times_{\Delta K} \mathscr{M}$ restricts to the membership subterminal $[\![\Phi(e) \in \mathscr{M}]\!] = U_{\kappa(e)}$. Dependent disjoint union over P_v gives (4.26).

(2) **Compute feasibility and the regular epimorphism.** The image of its projection is the join of these membership opens, exactly $\mathcal{U}_v$ by Theorem 4.25. Factoring through the image gives the regular epi.

(3) **Construct the associated locale and its fibres.** The standard internal sheaf–étale-locale equivalence for a localic sheaf topos (Vickers 1989; Mac Lane and Moerdijk 1992) constructs E_v; pullback along I computes the displayed fibre.

□

Definition 4.31 (The implementation diagram)**.** The **implementation diagram** is

$$\mathsf{Impl}_v : L \to \mathcal{E}, \qquad \mathsf{Impl}_v(a) = \{e \in P_v : \kappa(e) \leq a\},$$

whose transition maps retain the same implementation as the budget increases. Its ideal fibre is the directed colimit over $a \in I$.

Each implementation is available at its own cost and at every larger budget. Extending this generating data along the cost map produces the entire witness diagram. The following universal property says that a transformation out of the diagram is determined by what it does to those generators.

Proposition 4.32 (The free budget extension of an implementation family)**.** *Write $\kappa_v : \mathrm{Disc}(P_v) \to L$ for the cost functor from the discrete internal category of implementations. The witness diagram is the left Kan extension of its terminal diagram:*

$$\mathsf{Impl}_v \cong \mathrm{Lan}_{\kappa_v} \mathbf{1}, \qquad \mathsf{Impl}_v(a) \cong \coprod_{e \in P_v} L(\kappa_v(e), a).$$

Here $L(b, a)$ is the subterminal object expressing $b \leq a$. For every internal diagram $D : L \to \mathcal{E}$ there is a natural isomorphism

$$\mathrm{Nat}(\mathsf{Impl}_v, D) \cong \prod_{e \in P_v} D(\kappa_v(e)). \tag{4.27}$$

Proof.

(1) **Identify the budget-indexed implementation family.** At a, the displayed dependent coproduct has exactly the elements $e \in P_v$ with $\kappa_v(e) \leq a$, and its transition maps retain e.

(2) **Extract data at each implementation's cost.** For a transformation α, evaluate α at the element e of $\mathsf{Impl}_v(\kappa_v(e))$ to obtain $d_e \in D(\kappa_v(e))$.

(3) **Construct the inverse and prove the universal property.** Conversely transport d_e along $D(\kappa_v(e) \leq a)$ whenever e is feasible at a. These operations are inverse by naturality. They are internal dependent operations, so require no selection from unrelated fibres. This proves the Kan-extension universal property.

□

The Kan-extension formula retains each implementation across all budgets that admit it; its regular image is the feasible open.

4.11 A problem probes the locale by an open-valued observable

The witness sheaf describes implementations at each budget. Its support answers the corresponding question about the problem: on which budget families does an implementation exist? For $v : V$, use the fibre P_v of the measured span (4.17), with witness diagram $\mathrm{Lan}_{\kappa_v} \mathbf{1}$. Its support is

$$\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)} \quad \in \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}). \tag{4.28}$$

Thus $I \models \mathcal{U}_v$ if and only if $\exists e \in P_v\ \Phi(e) \in M_I$. This is the same cost incidence as in Theorem 4.25, with neither names nor implementation costs quotiented by denotation.

Proposition 4.33 (The functor-of-points feasibility probe)**.** *For each represented denotation $v : V$, there is a locale map $\chi_v : \mathfrak{C}_{\mathrm{Ev}} \to \mathbb{S}$ to the Sierpiński locale, uniquely classified by $\mathcal{U}_v$. On every test locale Z it induces a natural transformation*

$$h_{\mathfrak{C}_{\mathrm{Ev}}}(Z) \longrightarrow \mathcal{O}(Z), \qquad b \longmapsto b^* \mathcal{U}_v. \tag{4.29}$$

For the budget family $\mathscr{M}_b \subseteq \Delta_Z K$, the open $b^*\mathcal{U}_v$ is the regular-image locus of the pullback

$$\Delta_Z P_v \times_{\Delta_Z K} \mathscr{M}_b \longrightarrow 1_Z, \tag{4.30}$$

where the first map to $\Delta_Z K$ is $\Delta_Z \Phi$. Consequently the functor of points of the open sublocale $\mathcal{U}_v \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$ is the subfunctor

$$h_{\mathcal{U}_v}(Z) \cong \{b : Z \to \mathfrak{C}_{\mathrm{Ev}} : b^*\mathcal{U}_v = \top_Z\} \hookrightarrow h_{\mathfrak{C}_{\mathrm{Ev}}}(Z). \tag{4.31}$$

For an ideal point I, the specialization is

$$\chi_v(I) = \top \iff v \in \mathcal{C}_{\mathrm{cost}}(I) \iff \exists e \in P_v\ \kappa(e) \in I.$$

Proof.

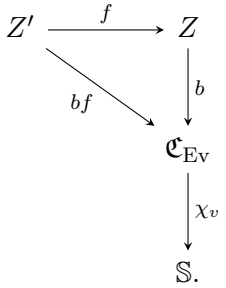
(1) **Classify the feasibility open and verify naturality.** The Sierpiński classification of opens gives the unique χ_v. Under the natural identification $\mathrm{Loc}_{\mathcal{E}}(Z, \mathbb{S}) \cong \mathcal{O}(Z)$, composition with χ_v is exactly (4.29). For a map $f : Z' \to Z$, $(bf)^*\mathcal{U}_v = f^*(b^*\mathcal{U}_v)$, proving naturality.

(2) **Pull back the witness family and its image.** The universal incidence is the pullback $\mathscr{Z} = \Delta P \times_{\bar{K}} \mathscr{M}$. Pulling its v-fibre back along b gives (4.30). Geometric inverse image preserves pullbacks and regular images, so the image open is $b^*\mathcal{U}_v$.

(3) **Characterize feasible factorization and ideal fibres.** An open immersion $\mathcal{U}_v \hookrightarrow \mathfrak{C}_{\mathrm{Ev}}$ has the asserted factorization property for locale maps, proving (4.31). The last equivalences are the fibre formula in Theorem 4.25.

□

For $f : Z' \to Z$, the problem observable respects the parameter triangle



Its open is $(bf)^*\mathcal{U}_v = f^*(b^*\mathcal{U}_v)$. The pullback witness family supplies its local implementation witnesses.

Equivalently, the Yoneda correspondence identifies the probe with

$$\mathrm{Nat}(h_{\mathfrak{C}_{\mathrm{Ev}}}, h_{\mathbb{S}}) \;\cong\; \mathrm{Loc}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}, \mathbb{S}) \;\cong\; \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}),$$

where $h_{\mathbb{S}}(Z) = \mathcal{O}(Z)$. In this precise sense *problems probe the complexity locale*: the generalized points are budget families, while a problem gives an open-valued observable on those families. By Theorem 5.1, the same observable is the point $c(v)$ of the exponential locale $\mathfrak{F} = \mathbb{S}^{\mathfrak{C}_{\mathrm{Ev}}}$, with $c : \mathrm{Disc}(V) \to \mathfrak{F}$ natural in the denotation parameter.

The witness sheaf remains distinct from its feasible open. By Theorem 4.30, it has the presentation

$$\mathscr{W}_v \cong \coprod_{e \in P_v} U_{\kappa(e)}, \qquad \mathscr{W}_v \twoheadrightarrow \mathcal{U}_v.$$

After pullback along $b : Z \to \mathfrak{C}_{\mathrm{Ev}}$, the equation $b^*\mathcal{U}_v = \top_Z$ says that the witness projection is locally inhabited over Z. A section supplies the stronger datum of compatible implementation witnesses throughout the family. The distinction between these two statements will govern the optimality result.

4.12 A constant problem and redundant programs

The distinction between a program's cost and a problem's complexity already appears for a constant function. Work on $X = \mathbb{N}$ with its cofinite filter. Select an implemented unit-valued idle primitive a that leaves the result unchanged, and count its recorded invocations with $w(\mathsf{a}) = 1$ and empty-word cost 0. This is the history-fold measurement of Propositions 4.2 and 4.11.

Let $v : \mathbb{N} \to \mathbb{N}$ be the constant-zero function. The two descriptions are given in Algorithm 4.1; $\mathsf{Idle}(\mathsf{a})$ invokes the selected primitive once. This measurement counts those invocations. The pure constant-return description e_0 has an empty event word on every input, and therefore

$$p(e_0) = v, \qquad H_{e_0}(n) = [], \qquad T_{e_0}(n) = 0, \qquad \kappa(e_0) = [0].$$

Algorithm 4.1 Two implementations of the constant-zero function

Input: $n \in \mathbb{N}$; the selected idle primitive a

```
1: procedure e_0(n)
2:     return 0
3: end procedure
4: procedure e_1(n)
5:     for j = 1, ..., n do
6:         IDLE(a)
7:     end for
8:     return 0
9: end procedure
```

The loop is NNO iteration of the implemented primitive. The log equation and finite induction give

$$p(e_1) = v, \qquad H_{e_1}(n) = \mathsf{a}^n, \qquad T_{e_1}(n) = n, \qquad \kappa(e_1) = [n].$$

Here a^n is the finite word of charged idle events.

The first program contributes $U_{[0]} = \top$ to the feasibility open. Since every growth ideal contains its bottom element, it follows that

$$\mathcal{U}_v = \top_{\mathfrak{C}_{\mathrm{Ev}}}. \tag{4.32}$$

The additional contribution $U_{[n]} \subseteq U_{[0]}$ leaves this open unchanged. Thus the problem is feasible at the zero-cost budget although e_1 has linear event cost. For the corresponding bounded and linear machine-update example, see Proposition 4.28.

We next compare witness families by translating their implementations. The Kan-extension property reduces the semantic comparison to the cost of each translated generator. A finite compiler description then supplies the implementation of that translation.

4.13 Implemented natural transformations

Proposition 4.34 (Lax cost squares classify witness transformations)**.** *For a monotone budget transformation $r : L_v \to L_w$ there is a natural isomorphism*

$$\mathrm{Nat}(\mathsf{Impl}_v, \mathsf{Impl}_w \circ r) \cong \{\gamma : P_v \to P_w : \kappa_w \gamma \leq r \kappa_v\}. \tag{4.33}$$

These correspondences preserve identities and composition.

Proof.

(1) **Apply the Kan-extension classification.** Apply Equation (4.27) with $D = \mathsf{Impl}_w \circ r$. Its right side is the dependent product of the objects $\{d \in P_w : \kappa_w(d) \leq r(\kappa_v(e))\}$, which is precisely the object of displayed maps γ.

(2) **Identify the action and its composition laws.** The transformation sends e at every feasible budget to $\gamma(e)$. The two descriptions therefore have the same identity and composition.

□

The compiler's cost bound is the inequality in the lax square

$$\begin{array}{ccc} P_v & \xrightarrow{\gamma} & P_w \\ {\scriptstyle \kappa_v}\downarrow & \leq & \downarrow{\scriptstyle \kappa_w} \\ L_v & \xrightarrow[r]{} & L_w. \end{array} \qquad \kappa_w \gamma \leq r \kappa_v. \tag{4.34}$$

Monotonicity transports the bound to every larger budget. The resulting components send the same implementation to the same compiled code, which proves naturality. We now retain finite-code realizations of these maps.

Definition 4.35 (Measured compiler comparisons)**.** A **measured compiler comparison** from an implementation family P_v to P_w is a pair (γ, r) consisting of a map $\gamma : P_v \to P_w$ and a monotone map $r : L_v \to L_w$ with $\kappa_w \gamma \leq r \kappa_v$. The map γ is required to lie in the regular image of the interpretation of finite transformation descriptions that return a member of P_w on every member of P_v. This image records local implementation witnesses. The construction below forms the comparison object and its category operations.

Construction 4.36 (The category of measured compiler comparisons)**.**

(1) **Fix the comparison context.** Work at the fixed environment of Construction 4.13. Source and target signatures may share that environment; a comparison using an implemented change of environment retains the specified map and is formed after the corresponding pullback. Its execution cost is obtained from that change's implementation description.

(2) **Construct transformation descriptions.** Use the data extension of the implementation signature in Section 4.5, retaining its code carriers as sorts as in Construction D.5. For two code carriers E_v, E_w, let $T_{v,w}$ be their carrier of well-typed finite transformation descriptions, with interpreted partial map

$$\mathrm{ev}^{\mathrm{tr}}_{v,w} : T_{v,w} \times E_v \to L_\Sigma E_w.$$

This interpretation is the finite-tree construction of Lemma D.6; its identity and sequential-composition constructors give descriptions of the identity and composite code maps. In the numerical instance $T_{v,w} = \mathbb{N}$ with its partial numerical evaluator on program codes; the identity and composition wrappers are those of Lemma D.6 and Proposition D.9 and Construction 13.2.

(3) **Restrict to returning transformations of implementations.** For the measured fibre inclusions $i_v : P_v \hookrightarrow E_v$, $i_w : P_w \hookrightarrow E_w$, form the subobject

$$\mathrm{CompCode}(P_v, P_w) = \{t : T_{v,w} : \forall e : P_v \; \exists d : P_w,\ \mathrm{ev}^{\mathrm{tr}}_{v,w}(t, i_v e) = \eta_{E_w}(i_w d)\}.$$

The condition means that the partial domain is true and its returned value belongs to P_w; it is an $\Omega_{\mathcal{E}}$-valued condition. Restriction to this true domain and transposition construct

$$\gamma : \mathrm{CompCode}(P_v, P_w) \to (P_w)^{P_v}.$$

(4) **Impose the resource bound and take the image.** Define the bound subobject

$$R(v,w) = \left\{ (t,r) : \begin{array}{l} t \in \mathrm{CompCode}(P_v, P_w), \quad r \in \mathrm{Mon}(L_v, L_w), \\ \forall e : P_v, \quad \kappa_w(\gamma_t e) \le r(\kappa_v(e)) \end{array} \right\}.$$

The hom object of measured comparisons is its regular image

$$\mathrm{MComp}(v,w) = \mathrm{im}\left(R(v,w) \xrightarrow{(\gamma,\pi_2)} (P_w)^{P_v} \times \mathrm{Mon}(L_v, L_w) \right). \tag{4.35}$$

Thus a comparison retains a code-output map and its resource transformation, with local evidence of a finite implementation.

(5) **Construct identities and composition.** Its identity and composition are

$$1_v = (\mathrm{id}_{P_v}, \mathrm{id}_{L_v}), \qquad (\delta, s) \circ (\gamma, r) = (\delta\gamma, sr).$$

They land in the image because the finite constructors implement identity and composition. Their bounds follow from

$$\kappa_z(\delta\gamma e) \le s(\kappa_w(\gamma e)) \le sr(\kappa_v(e)).$$

(6) **Verify the category laws and retained equality.** Associativity and units are equations of interpreted functions and monotone maps, proving the internal category laws for every retained internal family of objects. The codes themselves remain a presentation of this category; syntactic composition expressions need not be equal as syntax. Nor are two maps identified merely because their outputs have equal denotations: distinct codes in P_w can have different costs. The quotient in (4.35) uses equality in $(P_w)^{P_v}$, not equality after $p_w : P_w \to V_w$.

Theorem 4.37 (Compiler action on the implementation moduli)**.** *For a measured comparison $(\gamma, r) : v \to w$ in* MComp*, the implementation map $\gamma : P_v \to P_w$ and monotone resource transformation $r : L_v \to L_w$ satisfy*

$$\kappa_w(\gamma e) \le r(\kappa_v(e)) \qquad (e \in P_v). \tag{4.36}$$

The growth transformation induces a locale map $s_r : \mathfrak{C}_v \to \mathfrak{C}_w$, acting on ideals by $I \mapsto\!\downarrow r(I)$. The compiler induces a morphism of witness sheaves over s_r,

$$\mathscr{W}_v \longrightarrow s_r^* \mathscr{W}_w,$$

and its regular image gives

$$\mathcal{U}_v \le s_r^* \mathcal{U}_w. \tag{4.37}$$

Identity and composed comparisons in MComp *act by identity and composition of these morphisms. This action is independent of the choice of finite code representing the same interpreted comparison in Construction 4.36.*

Proof.

(1) **Construct the compiler transformation.** Under Theorem 3.8, s_r^* is precomposition with r. The lax comparison therefore gives the required witness transformation by Proposition 4.34. Its explicit component at budget a sends e to γe, with

$$\kappa_w(\gamma e) \le r(\kappa_v(e)) \le r(a),$$

so this is a natural transformation $\mathsf{Impl}_v \to \mathsf{Impl}_w \circ r$.

(2) **Transport feasibility and verify coherence.** Taking images proves (4.37). Both composites send an implementation to the same composite code and use the same composed monotone bound, proving identity, composition, and independence of the finite description of the interpreted map.

□

For a generalized budget $b : Z \to \mathfrak{C}_v$, the same compiler therefore gives

$$b^*\mathscr{W}_v \longrightarrow (s_r b)^*\mathscr{W}_w, \qquad b^*\mathcal{U}_v \le (s_r b)^*\mathcal{U}_w.$$

These comparisons commute with further substitution of parameter locales because geometric inverse image preserves the witness map and its image. The family statement retains the code map γ; an inclusion of feasibility opens alone supplies no compiler.

Over the locale $\mathfrak{R}_r$ of (3.12), this comparison further gives $\pi_1^*\mathscr{W}_v \to \pi_2^*\mathscr{W}_w$. Indeed $\kappa_v(e) \in I$ implies $\kappa_w(\gamma e) \in\!\downarrow r(I) \subseteq J$. The map still retains γe; it is not obtained by choosing a witness from the target feasibility image. Observation transport is the equation $\chi_{J_\Sigma r(I)}(u) = \chi_I(u \circ r)$ of Proposition 2.31, and implemented name maps for γ are supplied by the finite compiler descriptions above. The budget-name operation is a separate implementation, as follows.

Proposition 4.38 (Named transport of admitted budgets)**.** *The observed map $F_\Sigma r$ has admitted-point map $J_\Sigma r$. For name covers $C_L \twoheadrightarrow J_\Sigma L$ and $C_{L'} \twoheadrightarrow J_\Sigma L'$, its canonical covered-span presentation is*

$$C_L \xleftarrow[\text{epi}]{p} \widehat{P}_r \xrightarrow{v} C_{L'}, \qquad q_{L'} v = J_\Sigma r\, q_L p. \tag{4.38}$$

It is uniform in a named family of monotone maps r. The left leg covers the entire input name carrier, and effective descent recovers exactly $J_\Sigma r$. The compiler code for γ and this covered budget operation jointly present the semantic action on named implementations and admitted budgets.

Proof.

(1) **Construct the uniform semantic operation.** For $R = \mathrm{Mon}(L, L')$, Proposition 2.31 gives $(r, I) \mapsto J_\Sigma r(I)$ and $\chi_{J_\Sigma r(I)}(u) = \chi_I(u \circ r)$ uniformly in r, I, u.

(2) **Construct the name cover and its tracked projections.** Take the canonical covers $C_R \twoheadrightarrow R$, $C_L \twoheadrightarrow J_\Sigma L$, and $C_{L'} \twoheadrightarrow J_\Sigma L'$. Pull back $C_{L'}$ along the displayed operation on $C_R \times C_L$, then take its canonical partitioned cover. Its projections p, v satisfy the family version of (4.38), and p is regular epic. Specialization gives $\widehat{P}_r$.

$$\begin{array}{ccc} \widehat{P}_r & \xrightarrow{\;v\;} & C_{L'} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q_{L'}} \\ C_L & \xrightarrow[J_\Sigma r\, q_L]{} & J_\Sigma L' \end{array}$$

(3) **Prove exact descent and coherence.** The displayed equation makes $q_{L'}v$ constant on the kernel pair of p, so its unique descent is $J_\Sigma r\, q_L$. Composing the spans by pullback and partitioned refinement denotes $J_\Sigma(sr)$, and the identity span denotes $J_\Sigma\,\mathrm{id}$. Uniqueness of effective descent proves their semantic equations; the positive evaluation equation is substitution of tests.

□

The cover retains the output-name witness rather than choosing a section over the original input carrier. An actual budget program is recorded by its finite implementation description and measured history, as for the other operations. The semantic parameter r and the covered-span equation alone assign no runtime to such a program and do not compute r from the compiler code γ.

4.14 Measured changes of input representation

Keep X_0, Y_0 and the environment fixed. Use two input presentations $q_X : C_X \twoheadrightarrow X_0$ and $q'_X : C'_X \twoheadrightarrow X_0$. Form the object of conversion descriptions whose total returned-name map $r : C'_X \to C_X$ satisfies $q_X r = q'_X$. Its defining predicates are formed as subobjects, just as for CompCode; its generic element carries its history H_r and selected measurement. For resource transport, take the subobject on which the two supplied proper-filter contexts make r a filtered-input map of Section 3.9. This retains the eventual-substitution witness needed to form $r^* : K_X \to K_{C'_X}$.

Proposition 4.39 (Conversion overhead from recorded histories)**.** *Sequential composition constructs an implementation e^r on C'_X with the same extensional denotation as e. Its domain is the pullback of the original domain along r. In the writer interpretation its history and running-time function are*

$$H_{e^r}(c') = H_r(c') +\!\!+ H_e(r(c')), \qquad T_{e^r} = T_r + T_e \circ r. \tag{4.39}$$

On the total-implementation subobject, its cost germ is $\Phi(e^r) = h + r^*\Phi(e)$, where $h = [T_r]_{\mathrm{Ev}}$. If $\varphi_r : L_X \to L_{C'_X}$ is the induced growth map, the monotone budget transformation

$$a \longmapsto [h] \vee \varphi_r(a)$$

constructs the corresponding measured compiler comparison.

Proof.

(1) **Identify the composite domain and output.** The code-composition constructor first runs the converter and then e. Totality of the converter and the bind domain equation give the stated domain. The two value-adequacy squares and $q_X r = q'_X$ identify the extensional output.

(2) **Compute the history and cost.** Writer bind concatenates the two histories, proving the first equation; the additive fold proves the second. Passing to germs gives the equality.

(3) **Derive the growth bound and its implementation witness.** For nonnegative functions, the growth class of a sum is the join of their classes. Thus the cost of e^r has class $[h] \vee \varphi_r(\kappa(e))$, and monotonicity gives the required compiler bound at every larger budget. The conversion and composition descriptions provide an implementation witnessing this bound.

□

For a different history valuation the formula is $\nu(H_r +\!\!+ H_e \circ r)$; no additive identity is inferred. In a numerical compilation every dispatch and simulation instruction belongs to the compiled history. Its bound is computed from that full history, including any overhead of realizing the writer-level composition. Opposite representation conversions give opposite measured comparisons with their own overheads. This supplies the quantitative comparison between presentations; equality of their unmeasured denotations alone does not identify their cost functions.

The measured span now supplies a witness sheaf and a feasible open for each represented problem, together with the action of implemented compilers. The next section studies the feasible open itself. Its covering properties will say whether one implementation, or a finite portfolio, accounts for all feasible budgets.

5 Feasibility moduli and optimality

A problem determines an open of feasible budgets. We first classify these open-valued specifications and then compare their covers with the implementation witnesses that generate them. One witness accounting for the whole open is an optimal implementation; finitely many such witnesses form a dominating portfolio. The same construction applies to joint resources, provided every constraint is imposed on one shared implementation.

5.1 The classifying locale of feasibility specifications

The environment parameter remains fixed throughout this section: V, P_v mean $V_\theta, P_{\theta,v}$. Its classifying maps and optimality statements are families over that parameter, with the substitution comparisons of Proposition 4.29. Let $\mathbb{S}$ denote the Sierpiński locale, which classifies opens, and let $K_c(\mathfrak{C}_{\mathrm{Ev}})$ be the internal lattice of compact opens. Its elements are precisely finite unions of principal opens: compactness extracts a finite subcover from the principal decomposition, and finite unions of compact opens are compact. Finite unions and intersections make $K_c(\mathfrak{C}_{\mathrm{Ev}})$ a bounded distributive lattice.

Theorem 5.1 (Feasibility specifications are classified)**.** *The locale $\mathfrak{F}$ with frame $\mathcal{O}(\mathfrak{F}) = \mathrm{Up}(K_c(\mathfrak{C}_{\mathrm{Ev}}))$ has the natural universal property*

$$\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{F}) \cong \mathcal{O}(Z \times \mathfrak{C}_{\mathrm{Ev}}). \tag{5.1}$$

Thus it constructs the exponential $\mathbb{S}^{\mathfrak{C}_{\mathrm{Ev}}}$. Its universal open is

$$\mathscr{U} = \bigvee_{a \in L} V_{U_a} \times U_a \ \in \mathcal{O}(\mathfrak{F} \times \mathfrak{C}_{\mathrm{Ev}}), \qquad V_k = \{\ell \in K_c(\mathfrak{C}_{\mathrm{Ev}}) : k \subseteq \ell\}.$$

The measured incidence determines a canonical map $c : \mathrm{Disc}(V) \to \mathfrak{F}$ classifying $v \mapsto \mathcal{U}_v$.

Proof.

(1) **Describe opens of the product by monotone functions.** For $H = \mathcal{O}(Z)$, the frame of the product $Z \times \mathfrak{C}_{\mathrm{Ev}}$ is naturally the frame of monotone maps $w : L \to H$. Indeed a rectangle $z \times U_a$ corresponds to the function $b \mapsto \bigvee_{a \le b} z$, and every monotone function is $\bigvee_a w(a) \times U_a$.

(2) **Verify the frame-coproduct universal property.** For frame maps $f : H \to H'$ and $h : \mathrm{Up}(L) \to H'$, the unique extension is $w \mapsto \bigvee_a f(w(a)) \wedge h(U_a)$. Finite meets

are preserved using $U_a \cap U_b = U_{a\vee b}$, and joins and top are preserved by principal decomposition. This proves the frame-coproduct universal property, including the inverse formulas.

(3) **Pass from classifying ideals to monotone functions.** By Theorem 3.3, a map $Z \to \mathfrak{F}$ is an H-valued ideal j on $K_c(\mathfrak{C}_{\mathrm{Ev}})$. It determines $w(a) = j(U_a)$.

(4) **Construct the inverse ideal and verify independence.** Conversely, for monotone w, put

$$j\left(\bigcup_{i=1}^{n} U_{a_i}\right) = \bigwedge_{i=1}^{n} w(a_i).$$

This is independent of the finite presentation: if $U_a \subseteq \bigcup_i U_{b_i}$, supercompactness gives internally an i with $b_i \leq a$, hence $\bigwedge_i w(b_i) \leq w(a)$. Applying this to each generator proves order reversal and independence; the empty union has value $\top$, and unions become meets.

(5) **Identify the universal open and the classification map.** The constructions are inverse and natural in H. Their composition proves (5.1), and sends the universal ideal to the displayed open $\mathscr{U}$. Apply it to the open relation (4.22) over $\mathrm{Disc}(V)$ to obtain c.

□

Let $\chi_{\mathscr{F}}$ classify the universal measured feasibility family on $\mathrm{Disc}(V) \times \mathfrak{C}_{\mathrm{Ev}}$. The exponential transpose c is characterized by

$$\begin{array}{ccc} \mathrm{Disc}(V) \times \mathfrak{C}_{\mathrm{Ev}} & \xrightarrow{c\times \mathrm{id}} & \mathfrak{F} \times \mathfrak{C}_{\mathrm{Ev}} \\ & \searrow^{\chi_{\mathscr{F}}} & \downarrow{\scriptstyle \mathrm{ev}} \\ & & \mathbb{S}. \end{array}$$

Here ev classifies the universal open $\mathscr{U}$. This triangle relates the problem-indexed observable to the representing locale of feasibility specifications.

The point $c(v)$ records the feasible specification. Its measured instance also carries the epimorphism $\mathscr{W}_v \twoheadrightarrow \mathcal{U}_v$, which records the implementations. Selecting a section of that epimorphism is a further question. Effective names and positive observations retain the structures already constructed; the classification alone assigns neither to an arbitrary point of $\mathfrak{F}$. The principal opens contributed by implementations cover the feasible open. If one member covers the whole, its implementation is optimal. If a finite subfamily covers it, those implementations suffice as a portfolio. Supercompactness and compactness express these two possibilities intrinsically.

5.2 Optimization and topological compactness

The implementation comparison is with abstract complexity measures (Blum 1967); finite portfolios have also been studied as algorithm-selection objects (Gomes and Selman 2001). Here the relevant properties will be expressed by compactness of a feasibility open. Fix a denotation $v : V$ and write $P_v = \{e : P : p(e) = v\}$. The universal cost open of Theorem 4.25 is $\mathcal{U}_v = \bigvee_{e\in P_v} U_{\kappa(e)}$.

Definition 5.2 (Optimal implementations and dominating portfolios)**.** An **optimal implementation** of v is an $e_0 \in P_v$ such that $\kappa(e_0) \leq \kappa(e)$ for every $e \in P_v$. A **finite dominating portfolio** is a finite list $e_1, \ldots, e_n \in P_v$ such that every $e \in P_v$ has some $k \in \{1, \ldots, n\}$ with $\kappa(e_k) \leq \kappa(e)$. The empty list is allowed.

We use compactness and supercompactness as in Definition 3.2.

Theorem 5.3 (Optimal algorithms and finite portfolios)**.** *The classified feasibility open satisfies*

$$\mathcal{U}_v \text{ is supercompact} \iff \exists e_0 \in P_v\ \forall e \in P_v\ \kappa(e_0) \le \kappa(e). \tag{5.2}$$

It is compact precisely when a finite asymptotically dominating portfolio exists:

$$\begin{aligned} \mathcal{U}_v \text{ is compact} \iff \exists n \in \mathbb{N}\, \exists e_1, \dots, e_n \in P_v\ \forall e \in P_v \\ \exists k \in \{1, \dots, n\}\ \kappa(e_k) \le \kappa(e). \end{aligned} \tag{5.3}$$

In particular the empty portfolio accounts for the empty feasibility open. Both equivalences are internal and stable under substitution of denotation parameters.

Proof.

(1) **Recall principal-open order reflection.** By Proposition 4.27,

$$U_b \le U_a \iff a \le b. \tag{5.4}$$

(2) **Characterize supercompactness.** An optimal e_0 gives $\mathcal{U}_v = U_{\kappa(e_0)}$, which is supercompact. Conversely, supercompactness applied to the canonical cover $\mathcal{U}_v = \bigvee_{e \in P_v} U_{\kappa(e)}$ supplies e_0 with $\mathcal{U}_v \le U_{\kappa(e_0)}$. Order reflection gives $\kappa(e_0) \le \kappa(e)$ for every e.

(3) **Characterize compactness.** Compactness applied to the same cover supplies finitely many $e_1, \dots, e_n$. For each e, supercompactness of $U_{\kappa(e)}$ gives a k with $U_{\kappa(e)} \le U_{\kappa(e_k)}$, equivalently $\kappa(e_k) \le \kappa(e)$. Conversely, such a portfolio gives $\mathcal{U}_v = \bigvee_k U_{\kappa(e_k)}$, a finite join of compact opens. This includes the empty join. The argument is internal and uses only finite lists and existential elimination.

□

Theorem 5.4 (Optimal implementations are sections)**.** *For $v : V$, there is a natural isomorphism of internal $\mathcal{E}$-objects*

$$\mathrm{Sect}(\mathscr{W}_v \to \mathcal{U}_v) \cong \{e_0 \in P_v : \forall e \in P_v,\ \kappa(e_0) \le \kappa(e)\}. \tag{5.5}$$

Here Sect *represents sections in all parameter contexts, equivalently $p_{\mathfrak{C}_{\mathrm{Ev}^*}}(\mathscr{W}_v^{\mathcal{U}_v})$. Consequently the witness epimorphism splits internally exactly when $\mathcal{U}_v$ is supercompact.*

Proof.

(1) **Show that a section selects one common implementation.** In Theorem 3.8, a section assigns a feasible implementation e_a to every feasible budget a, with $e_a = e_b$ when $a \le b$. Any two feasible budgets have the feasible common upper bound $a \vee b$; hence all selected implementations agree. Since $v \in V = \mathrm{im}(p_0)$, the fibre P_v is internally inhabited. Thus there is a unique common implementation e_0; unique existence descends through the inhabited parameter cover.

(2) **Prove optimality of the selected implementation.** At $a = \kappa(e)$ feasibility gives $\kappa(e_0) \le \kappa(e)$.

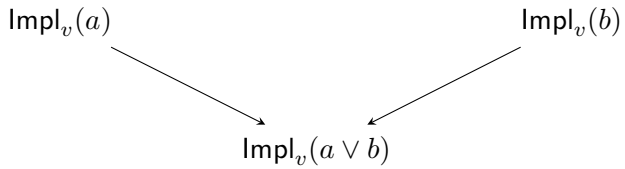
(3) **Construct the inverse section and verify naturality.** Conversely such an e_0 is available at every feasible budget and gives a constant compatible section. The two operations are inverse and commute with parameters. The final assertion is Theorem 5.3.

□

An optimal implementation is therefore exactly a splitting

$$\mathscr{W}_v \underset{s}{\overset{\pi}{\rightleftarrows}} \mathcal{U}_v \qquad \pi s = \mathrm{id}_{\mathcal{U}_v}\,.$$

A section makes compatible choices at all feasible budgets. In the diagram presentation, the common upper budget compares the choices made at a and b:



Both transition maps retain the implementation code, so compatibility forces those choices to agree. This is why a section supplies one optimal witness throughout the feasible open.

The optimality criterion can also be evaluated after allowing chosen overheads. Pulling the witness family to the gauge sublocale gives the same questions relative to the localized notion of resource.

5.3 Gauge-relative feasibility

Corollary 5.5 (Soft optimality and portfolios)**.** *Let $\mathcal{U}_v^{(P)}$ be the restriction of the measured feasibility open to $\mathfrak{C}_{\mathrm{Ev}}^{(P)}$. Then*

$$\mathcal{U}_v^{(P)} = \bigvee_{e \in P_v} U^{(P)}_{q_P(\kappa(e))}.$$

It is supercompact precisely when an implementation e_0 satisfies $\kappa(e_0) \preceq_P \kappa(e)$ for all $e \in P_v$. It is compact precisely when finitely many implementations dominate every implementation in this gauge-relative preorder. For $P = P_\lambda$, this is optimality and finite-portfolio completeness for soft-O comparison.

Proof.

(1) **Compute feasibility in the localized growth order.** Under (3.16), membership of a cost in a saturated budget is membership of its q_P-class in the corresponding ideal of L/P. This gives the displayed open formula.

(2) **Characterize soft optimality and finite portfolios.** Principal opens of L/P are supercompact; applying the proof of Theorem 5.3 to that principal cover proves both equivalences. The bitlength gauge identifies $\preceq_{P_\lambda}$ with soft domination by Section 2.7.

□

5.4 Joint-resource moduli and Pareto portfolios

Separate resource bounds may be witnessed by different programs. Joint feasibility instead requires one program meeting every bound. Let L_T, L_S be growth lattices for time and space measurements of the same implementation object, with classifying locales $\mathfrak{C}_T, \mathfrak{C}_S$. The product construction below retains that common implementation while varying both budgets.

Theorem 5.6 (Product moduli and joint feasibility)**.** *There is a canonical frame isomorphism*

$$\mathcal{O}(\mathfrak{C}_T \times \mathfrak{C}_S) \cong \mathrm{Up}(L_T \times L_S). \tag{5.6}$$

Each principal rectangle $U_a^T \times U_b^S$ is supercompact. For joint measured costs $\kappa_T : P_v \to L_T$, $\kappa_S : P_v \to L_S$, the simultaneous feasibility open is

$$\mathcal{U}_v^{T,S} = \bigvee_{e \in P_v} (U^T_{\kappa_T(e)} \times U^S_{\kappa_S(e)}). \tag{5.7}$$

It is supercompact exactly when one implementation is simultaneously asymptotically optimal in both costs. It is compact exactly when finitely many implementations jointly dominate every implementation: for each e there is a member e_i of the finite list satisfying

$$\kappa_T(e_i) \le \kappa_T(e), \qquad \kappa_S(e_i) \le \kappa_S(e).$$

The witness sheaf is the coproduct of the principal rectangles in (5.7); hence both budget tests refer to the same implementation.

Proof.

(1) **Construct the frame map on principal rectangles.** Every upper predicate on $L_T \times L_S$ is the union of principal rectangles. Given frame maps $h_T : \mathrm{Up}(L_T) \to H$ and $h_S : \mathrm{Up}(L_S) \to H$, define

$$h(W) = \bigvee_{(a,b) \in W} (h_T(U_a^T) \wedge h_S(U_b^S)).$$

(2) **Verify the product universal property.** The identity $(U_a^T \times U_b^S) \wedge (U_c^T \times U_d^S) = U^T_{a \vee c} \times U^S_{b \vee d}$ and frame distributivity prove that h preserves finite meets; principal decomposition proves join preservation. Restriction along the two projections gives the inverse bijection. This is the coproduct universal property of the frame of the localic product, proving (5.6).

(3) **Prove supercompactness of principal rectangles.** Evaluating a covering family at (a, b) proves supercompactness of each principal rectangle as in Theorem 3.3.

(4) **Construct joint feasibility and its witness sheaf.** The simultaneous incidence is the pullback of the two membership predicates along the diagonal implementation map, followed by the regular image to the budget product. The image is exactly (5.7); taking its implementation fibre gives the stated witness sheaf.

(5) **Characterize joint optimality and portfolios.** The supercompactness and compactness proofs now repeat Theorem 5.3 with the product order and principal rectangles. Both comparisons are coordinatewise, so they yield exactly the two displayed cost inequalities.

□

In particular, intersecting the two separately existential feasibility predicates may use different implementations. Equation (5.7) is the geometric construction of simultaneous time–space feasibility.

The section characterization of Theorem 5.4 also applies to the product budget order: a section of the joint witness sheaf is exactly one implementation optimal in every displayed cost. Its proof uses the componentwise join as the common upper budget.

5.5 A compact two-resource portfolio

The next example calculates a compact feasible open for which neither implementation is optimal in both resources. We use a selected two-description presentation and two event-count valuations, so the entire implementation fibre can be inspected. Work on $X = \mathbb{N}$ with its cofinite filter, and fix two implemented unit-valued idle primitives with distinct tags a, b and identical extensional effect. Define the product valuation by

$$w(\mathsf{a}) = (1, 0), \qquad w(\mathsf{b}) = (0, 1), \qquad w([]) = (0, 0),$$

and extend it to words by componentwise addition. These are two points of the history-weight object of Proposition 4.11. They count the selected events; total machine time and peak-space occupancy are separate measurements.

Specializing Algorithm 5.1 at $s = A$ or $s = B$ gives the fixed total descriptions e_A, e_B. Both return $v(n) = n$; the selected event counts give their respective logged histories

$$H_{e_A}(n) = \mathsf{a}^n\, \mathsf{b}^{n^2}, \qquad H_{e_B}(n) = \mathsf{a}^{n^2}\, \mathsf{b}^n. \tag{5.8}$$

Algorithm 5.1 The two-description resource portfolio

Input: $n \in \mathbb{N}$; fixed choice $s \in \{A, B\}$

$(k_a, k_b) \leftarrow (n, n^2)$
if $s = B$ **then**
    $(k_a, k_b) \leftarrow (k_b, k_a)$
end if
for $j = 1, \ldots, k_a$ **do**
    IDLE(a)
end for
for $j = 1, \ldots, k_b$ **do**
    IDLE(b)
end for
return n

The loops use the specified finite arithmetic and iteration constructors. The history fold gives cost pairs

$$(T_{1,A}(n), T_{2,A}(n)) = (n, n^2), \qquad (T_{1,B}(n), T_{2,B}(n)) = (n^2, n).$$

Put $a = [n + 1]$ and $b = [(n + 1)^2]$ in the numerical growth lattice. Then $a < b$: the inequality $a \le b$ holds, while $b \not\le a$ is witnessed by arbitrarily large numerical inputs. The growth pairs of the two implementations are (a, b) and (b, a), which are incomparable in the coordinatewise order.

Take the two-description subpresentation $E_{\mathrm{port}} = \{e_A, e_B\}$. Its evaluator is the copairing of the two total evaluations, its denotation image contains just v, and its implementation object consists of the two displayed members. The measured-image and incidence constructions therefore give its complete joint feasible open:

$$\mathcal{U}_v^{\mathrm{port}} = (U_a^{(1)} \times U_b^{(2)}) \;\vee\; (U_b^{(1)} \times U_a^{(2)}) \quad \in \mathcal{O}(\mathfrak{C}_{\mathrm{Ev}}^{(1)} \times \mathfrak{C}_{\mathrm{Ev}}^{(2)}). \tag{5.9}$$

The budget points $(\downarrow a, \downarrow b)$ and $(\downarrow b, \downarrow a)$ are feasible, whereas $(\downarrow a, \downarrow a)$ is not feasible in this presentation. Each program meets one of the smaller bounds, but neither meets both. Choosing different witnesses for the two resources would therefore give the wrong joint judgment.

For clarity, put $I_a =\downarrow a$ and $I_b =\downarrow b$. Direct evaluation of the two principal rectangles gives the following complete table for the displayed portfolio:

joint budget	e_A feasible	e_B feasible	some implementation feasible
(I_a, I_a)	no	no	no
(I_a, I_b)	yes	no	yes
(I_b, I_a)	no	yes	yes
(I_b, I_b)	yes	yes	yes

Each entry tests both resource components of one of the two measured cost pairs. The table evaluates the complete open of this finite presentation.

The open in (5.9) is compact, being a union of two supercompact principal rectangles. It is not supercompact. Indeed, if one of the two rectangles contained the union, it would contain the other rectangle. The budget point (I_b, I_a) belongs to the second rectangle but not the first, while (I_a, I_b) belongs to the first but not the second. Thus the displayed two-rectangle cover has no single covering member. By Theorems 5.3 and 5.6, the two codes form a finite dominating portfolio, whereas neither code is simultaneously optimal in both measurements *within this selected subpresentation*. This computes both sides of the compactness criterion on an implemented instance.

In the full description language, this calculated open is a subopen of $\mathcal{U}_v^{(1,2)}$. Other implementations can enlarge it: an uncharged pure identity description makes the full open top for these idle-event counters. The example thus establishes the portfolio criterion for its specified two-description fibre. The product theorem also applies to the time and occupancy measurements of Construction 13.2; identifying a particular frontier there additionally requires the corresponding universal lower-bound analysis.

The classifier now describes budget families, their feasible opens, and their implementation witnesses. We first instantiate this construction for conventional worst-case complexity, then vary the valuations of fixed histories to obtain differential resource families.

6 Conventional worst-case complexity

The measured-span construction now gives a concrete instance for deterministic computation on finite binary words. Finite maxima over word-length fibres construct its worst-case costs, and the resulting resource predicates recover the conventional time and workspace upper classes. The polynomial budget is identified with **P** by explicit clocks and tape simulations. All finite syntax, numerical recursions, images, and quotients are constructed in the fixed effective topos $\mathcal{E}$.

The external classes below use ordinary finite machine descriptions and represented numerical bounds. Their upper classes are the global instances of the internal upper predicates. Universal lower predicates retain the distinction between effective witness transformations and classical truth; Section 6.4 constructs the exact classical reading. We write $\mathrm{DTIME}_{\mathcal{T}}(O(b))$ to make the multiplicative-constant convention explicit, retaining the machine model when fine bounds are at issue.

6.1 Finite length fibres and worst-case aggregation

Write $X = \{0,1\}^*$ and $\ell : X \to \mathbb{N}$ for word length.

Construction 6.1 (Canonical length fibres and their maxima)**.**

(1) **Enumerate the fibre without choice.** Let E_0 be the one-element list containing the empty word. Define E_{n+1} by prefixing 0 to each member of E_n, followed by the list

obtained by prefixing 1. Induction proves that E_n lists each word of length n exactly once, has length 2^n, and is inhabited.

(2) **Fold by maximum.** For $a : X \to \mathbb{Q}_{\geq 0}$, define

$$(Wa)(n) = \max_{x \in E_n} a(x).$$

Rational order is decidable; a finite fold constructs this value and the first word attaining it. Consequently $W : (\mathbb{Q}_{\geq 0})^X \to (\mathbb{Q}_{\geq 0})^{\mathbb{N}}$ is a constructed arrow. The same fold gives $W : \mathbb{N}^X \to \mathbb{N}^{\mathbb{N}}$. For a parameterized $a : Z \times X \to \mathbb{Q}_{\geq 0}$, apply the identical fold while retaining z.

For functions use pointwise order. Equip X with its length filter and $\mathbb{N}$ with its cofinite filter from Definition 2.4. Let $K_X^+, K_{\mathbb{N}}^+$ be the nonnegative eventual-germ objects, and $L_X, L_{\mathbb{N}}$ their Landau growth posets, abbreviating the corresponding L_{Ev} of Definition 2.5. Equivalently, $a \preceq b$ means that $a \leq cb$ on an eventual extent for a positive rational constant c: rounding c up gives an integer constant in the defining relation. As before, the growth poset identifies mutual domination. We write $[a]$ for a growth class where no ambiguity results.

Theorem 6.2 (Worst-case aggregation is left adjoint to length pullback)**.** *Precomposition $\ell^* b = b \circ \ell$ and the finite maximum satisfy*

$$W \dashv \ell^*, \qquad W\ell^* = \mathrm{id}, \qquad a \leq \ell^* W a$$

on nonnegative functions. They descend to nonnegative eventual germs and to an adjunction of growth posets

$$W_L : L_X \rightleftarrows L_{\mathbb{N}} : \ell^*, \qquad W_L[a] \leq [b] \iff [a] \leq \ell^*[b].$$

The right adjoint is fully faithful. The constructions commute with substitution of parameters.

Proof.

(1) **Prove the finite-fibre adjunction.** For each n, the inequality $\max_{x \in E_n} a(x) \leq b(n)$ is equivalent to $a(x) \leq b(n)$ for every $x \in E_n$. Quantifying over n gives $Wa \leq b$ if and only if $a \leq \ell^* b$. Nonemptiness of each fibre gives $W\ell^* b = b$. The inequality $a(x) \leq Wa(|x|)$ is the unit. These equations establish the adjunction and its triangle identities in the posetal categories.

(2) **Descend through eventual equality.** If $a = a'$ on all words of length at least N, the two maxima agree for all $n \geq N$. If $b = b'$ for all $n \geq N$, their pullbacks agree on that length extent. Thus both operations coequalize the relevant kernel pairs and descend through the effective quotients. For the eventual order on germs, the same fibrewise equivalence on a common tail proves the descended adjunction.

(3) **Descend the adjunction through Landau domination.** For each $c > 0$ and N,

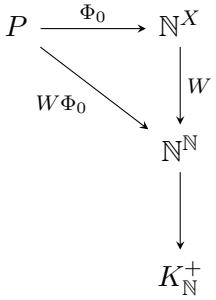
$$\forall n \geq N\ Wa(n) \leq cb(n) \quad \iff \quad \forall x\ (|x| \geq N \Rightarrow a(x) \leq cb(|x|)).$$

The same witnesses c, N work in both directions. Existentially quantifying them proves the displayed adjunction on growth classes. Monotonicity of finite maximum also preserves domination between two input functions, so both quotient maps are well-defined.

(4) **Retain the parameter context.** Every fold, inequality, cutoff, and witness transformation above is performed on the displayed finite fibre with the parameter unchanged. The quotient equations are uniform. Substitution therefore commutes with the operations and their unit and counit.

□

For a measured cost map $\Phi_0 : P \to \mathbb{N}^X$ as in Proposition 4.19, the construction gives the diagram



and its semantic input bound is controlled by the unit $a \leq \ell^* W a$. This is a finite-fibre left Kan extension in the posetal sense.

Proposition 6.3 (Algebraic scope of maximum)**.** *Worst-case maximum preserves zero, binary joins, and multiplication by pulled-back nonnegative coefficients. It satisfies*

$$W(a + a') \leq Wa + Wa', \qquad W(aa') \leq (Wa)(Wa').$$

It is not an additive or multiplicative homomorphism in general.

Proof.

(1) **Prove the stated operations fibrewise.** For a fixed finite nonempty fibre, maximum commutes with a binary maximum. A scalar $b(n) \geq 0$ is constant on that fibre and can be pulled outside its maximum. Each summand and factor is bounded by its respective maximum, giving the two inequalities.

(2) **Exhibit failure of ring preservation.** On nonempty words let a indicate first bit 0 and a' indicate first bit 1; give both value zero on the empty word. For $n \geq 1$,

$$Wa(n) = Wa'(n) = 1, \quad W(a + a')(n) = 1, \quad W(aa')(n) = 0.$$

Thus the failures persist after eventual equality. The construction is used on measured costs, not as a ring map transporting arbitrary coefficient algebra.

□

We use maxima at exactly length n. The running maximum over lengths at most n need not preserve eventual equality: a single exceptional early value can remain the running maximum forever. No such operation is used in the descent proof above.

6.2 A measured Turing-machine instance

Construction 6.4 (Finite descriptions, configurations, and execution)**.**

(1) **Construct the program object.** A program is a finite deterministic transition table with a finite state set, a finite tape alphabet containing the binary input symbols and a blank, and a finite positive number of work tapes. There is one read-only input tape with endmarkers. A transition reads the scanned symbols, updates the state and work symbols, and moves each head by at most one position. Decision programs have two halting states; for function computation an append-only output tape is also used. Finite-list parsing and finite checks construct the object of valid tables. Invalid numerical codes denote a fixed loop. Thus no validity oracle is used. We write $\mathcal{T}$ for this family of machines.

(2) **Construct configurations and transitions.** Represent a work tape by its scanned symbol and two finite lists, with nearest cells first, extending an empty list by blank on demand. A right move pushes the written symbol onto the left list and pops the right list, or reads blank if that list is empty; a left move is symmetric. The input word and its bounded head position determine the input symbol by finite lookup. Finite table lookup constructs a total primitive-recursive map s_e on configurations, an initialization map $\mathrm{init}(e, x)$, and a Boolean halting map h_e. Halted configurations are made absorbing. For decisions, the return map reads the halting-state bit.

(3) **Apply measured finite return.** At a nonhalted configuration emit one transition event and continue with $s_e(c)$; at a halted configuration return the output. This is a finite case-defined body for Theorem 4.6; Theorem 4.9 records its finite history. The history is the ordered list of simulated transitions. Its length is the measured time $T_e(x)$ on the return domain.

(4) **Construct numerical clocks.** Primitive recursion gives $s_e^t(\mathrm{init}(e, x))$. Put

$$\alpha(e, x, t) = h_e(s_e^t(\mathrm{init}(e, x))).$$

This is a primitive-recursive Boolean predicate on finite codes, words, and numerical t. Its interpretation counts target-machine transitions; it does not identify the target time with the time taken by a numerical interpreter computing α.

Proposition 6.5 (Exact operational and numerical cost adequacy)**.** *The constructed evaluator has exactly the ordinary domain and output of the encoded machine. Moreover*

$$T_e(x) = t \iff \alpha(e, x, t) = 1 \ \wedge \ \forall j < t \ \alpha(e, x, j) = 0, \qquad \mathrm{Within}(e, x, t) = \alpha(e, x, t).$$

Both displayed Boolean tests are primitive recursive. The time map is partial recursive and has the same domain as machine evaluation.

Proof.

(1) **Identify every transition.** Induction on t identifies $s_e^t(\mathrm{init}(e, x))$ with the ordinary configuration after t updates, including the absorbing convention. The list-stack equations in Construction 6.4 implement exactly one symbol update and one permitted head displacement.

(2) **Identify first return and cost.** A least successful clock stage has exactly that many preceding nonhalted updates. Each contributes one event. Its return value is the ordinary halted value, so the finite-return evaluator has the same domain, output, and time. Bounded conjunction gives the first-hit test.

(3) **Construct the total implementation subobject.** For decision programs set

$$P = \{e : \forall x \in \{0, 1\}^* \ \exists t \ \alpha(e, x, t) = 1\}.$$

This is an $\Omega_{\mathcal{E}}$-classified subobject of codes. Restricted to $P \times \{0,1\}^*$, evaluation and time are total, and transpose to $p_0 : P \to 2^{\{0,1\}^*}$ and $\Phi_0 : P \to \mathbb{N}^{\{0,1\}^*}$. As in Proposition 4.19, let $P \twoheadrightarrow V \hookrightarrow 2^{\{0,1\}^*}$ be the image factorization of p_0. These letters denote the fixed Turing-machine instance throughout this section. No algorithm recognizing P has been used or asserted.

□

These are the two numerical measure properties used by Blum (1967), now proved for the bit-input instance. Universal interpretation and machine simulation are separate measured algorithms; quantitative simulation is addressed in Section 6.6.

Proposition 6.6 (Uniform execution of the worst-case profile)**.** *Define the primitive-recursive predicate*

$$\beta(e, n, t) = \bigwedge_{x \in E_n} \alpha(e, x, t).$$

Its first-hit search has domain precisely the program–length pairs on which all inputs of length n halt. On this domain it returns

$$W_e(n) = \max_{|x|=n} T_e(x).$$

In particular $e \mapsto W_e$ is a total arrow $P \to \mathbb{N}^{\mathbb{N}}$.

Proof.

(1) **Identify the return domain.** A successful stage supplies a halting bound for every word in E_n. Conversely, finitely many halting witnesses have a maximum by finite induction, which makes β true. This uses finite existential elimination, not an axiom of choice.

(2) **Identify the first successful stage.** Absorption of halted states gives $\beta(e, n, t) = 1$ exactly when all the individual first-hit times are at most t. Their maximum is attained and is therefore the least such t, including the case $t = 0$.

(3) **Retain totality and parameters.** For $e \in P$ the domain statement holds for every n, so restriction and transposition yield the total profile map. The program searches successive t and evaluates the finite Boolean conjunction; it does not first decide whether e belongs to P. All equations are uniform in additional parameters.

□

The value $W_e(n)$ is the maximum of target execution times, not the running time of the procedure calculating that maximum. The latter may enumerate exponentially many inputs. No efficient cost-analysis oracle is asserted by Proposition 6.6.

6.3 Worst-case resource predicates

Let $B_{\mathbb{N}} \hookrightarrow K_{\mathbb{N}}$ be the bounded numerical germ ring and its ambient germ algebra. Define the new measured span using the same implementations and denotations:

$$V \xleftarrow{p} P \xrightarrow{\Phi^{\mathrm{wc}}} K_{\mathbb{N}}, \qquad \Phi^{\mathrm{wc}}(e) = [W_e]_{\mathrm{ev}}.$$

Write $\kappa^{\mathrm{wc}}(e) = [W_e] \in L_{\mathbb{N}}$.

Theorem 6.7 (Conventional worst-case predicates)**.** *For a numerical nonnegative bound b, put $I_b = {\downarrow}\,[b]$ and $M_b = B_{\mathbb{N}}[b]_{\mathrm{ev}}$. Then*

$$\begin{aligned} Z_b^{\mathrm{wc}} &= P \times_{K_{\mathbb{N}}} M_b, \\ \mathcal{C}^{\mathrm{wc}}(I_b) &= \mathrm{im}(Z_b^{\mathrm{wc}} \to V), \\ v \in \mathcal{C}^{\mathrm{wc}}(I_b) &\iff \exists e \in P_v\ W_e = O(b). \end{aligned}$$

For the Turing-machine instance and Boolean denotations, its external class is exactly $\mathrm{DTIME}_{\mathcal{T}}(O(b))$*. Its lower and tight predicates are, respectively,*

$$\forall e \in P_v\ b = O(W_e), \qquad (\exists e \in P_v\ W_e = O(b)) \ \wedge\ (\forall e \in P_v\ b = O(W_e)).$$

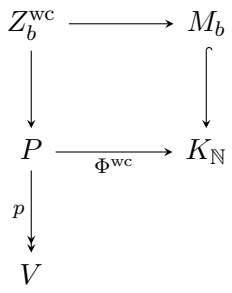
The associated feasibility open is

$$\mathcal{U}_v^{\mathrm{wc}} = \bigvee_{e \in P_v} U_{[W_e]}.$$

Internally, Proposition 4.27 and Theorem 5.3 give its exact characterizations of upper bounds, lower bounds, optimal implementations, and finite dominating portfolios, with κ^{wc} in place of the inputwise cost map.

Proof.

(1) **Construct witnesses before the feasibility image.** The pullback selects precisely the codes with $\Phi^{\mathrm{wc}}(e) \in M_b$. By Lemma 2.14, membership is exactly $W_e = O(b)$. The image over p is the displayed existential predicate. In diagram form,



(2) **Identify the conventional class.** By Proposition 6.5, the costs are exactly the chosen machine's step counts. By Theorem 6.2, $W_e = O(b)$ is exactly the uniform asymptotic bound on all strings of each sufficiently large length. The index e ranges over all total decision programs in that model. These are precisely the defining quantifiers of $\mathrm{DTIME}_{\mathcal{T}}(O(b))$. Conventional definitions are recorded, for example, in Barak (2023, Chapter 13).

(3) **Construct lower bounds using the same implementation fibre.** Apply $\forall_p$ to the subobject defined by $b = O(W_e)$. Its formula is the displayed lower predicate. Intersect this subobject of V with the upper subobject for the tight predicate. No uniform lower constant across different implementations is introduced.

(4) **Identify the locale and its optimization predicates.** Apply the universal budget-family construction to the displayed span. Each implementation contributes $U_{[W_e]}$; taking its image gives the indicated join. Principal-open order reflection $U_c \subseteq U_a \iff a \leq c$ proves the upper and lower inclusion formulas. Supercompactness of this join is equivalent to one member covering it, hence to an O-optimal implementation. Compactness is equivalent to a finite dominating subfamily, by the same principal-open argument. All constructions use pullbacks and images and retain parameters.

□

Proposition 6.8 (The quantifiers in a worst-case lower bound)**.** *The preceding lower predicate is exactly*

$$\forall e \in P_v \ \exists c \in \mathbb{N}_{>0} \ \exists N \ \forall n \geq N \ \exists x \in E_n : \quad b(n) \leq cT_e(x).$$

It does not require the inequality on every input of length n. The infinitely-often version is obtained by replacing $\exists N \, \forall n \geq N$ with $\forall N \, \exists n \geq N$, while retaining the implementation quantifier and its constant. These equivalences hold internally with witnesses, as well as for the displayed formulas read externally in classical logic. They do not identify classical truth with realizability of a universally quantified formula; Section 6.4 supplies that separate exact identification.

Proof.

(1) **Extract a maximizer constructively.** The finite fold in Construction 6.1 returns the first maximizing word. Therefore $b(n) \leq cW_e(n)$ implies the displayed inequality for that word; the reverse implication follows from $T_e(x) \leq W_e(n)$. Positive rational constants can be rounded up to positive integers. The transformations retain e, c, N, n.

(2) **Check the distinction from inputwise lower bounds.** For $n \geq 1$, the profile equal to $n+1$ on words beginning with 0 and to $(n+1)^2$ on words beginning with 1 has worst-case profile $(n+1)^2$, but has no inputwise quadratic lower bound. This distinction also occurs for actual Turing-machine time: scan the input, remember its first bit, and halt immediately after the scan on the 0 branch; on the 1 branch first construct the unary clock of $(n+1)^2$ marks given in Theorem 6.12, then halt. The first branch takes $\Theta(n+1)$ time and the second $\Theta((n+1)^2)$, the latter lower bound following from the number of marks written. The empty word is handled by a fixed halting branch. Thus even conventional time has distinct inputwise and worst-case lower predicates.

(3) **Retain the infinitely-often quantifiers.** The equivalence at each e, c, n permits the indicated quantifier substitution. Externally, classical failure of an $O(b)$ upper bound has formula $\forall c \forall N \, \exists n \geq N : W_e(n) > cb(n)$. Internally, we retain direct witness formulas rather than using this classical quantifier-negation rule. None of these statements is silently interchanged with an eventual lower bound.

□

6.4 The classical reading of universal lower bounds

The realizability clauses for universal and existential quantification retain a procedure that produces witnesses, as in Kleene (1945, Section 5). Thus the internal formula $\forall e \in P_v \, \exists c, N \, \forall n \geq N \ b(n) \leq cW_e(n)$ includes effective dependence of c, N on a name of the implementation and its membership evidence. Conventional classical lower bounds assert existence separately for each implementation. Neither notion asks for one constant shared by all implementations.

Construction 6.9 (Close the relation before quantifying)**.**

(1) **Construct the implementationwise relation.** For the displayed numerical bound b, form the subobject

$$R_b = \{e \in P : b = O(W_e)\} \hookrightarrow P.$$

Its characteristic arrow is an ordinary $\Omega_{\mathcal{E}}$-valued arrow.

(2) **Apply the existing double-negation closure.** Put $j(q) = \neg\neg q$ on $\Omega_{\mathcal{E}}$ and let $j_P R_b$ be the subobject classified by $j\chi_{R_b}$. The intuitionistic identities $q \Rightarrow j(q)$, $j(j(q)) = j(q)$, and $j(q \wedge q') = j(q) \wedge j(q')$ make this the double-negation Lawvere–Tierney closure. It is defined in the existing topos; no principle $j(q) \Rightarrow q$ is added. The two characteristic pullbacks are

$$\begin{array}{ccccc} R_b & \longrightarrow & 1 & \longleftarrow & j_P R_b \\ \downarrow & & \downarrow{\scriptstyle \top} & & \downarrow \\ P & \xrightarrow[\chi_{R_b}]{} & \Omega_{\mathcal{E}} & \xleftarrow[j\chi_{R_b}]{} & P \end{array}$$

See also Hyland (1982, Proposition 4.4) for the classical sheaf interpretation.

(3) **Quantify over implementations afterwards.** Define the subobject of denotations

$$\mathsf{Low}_b^{\mathrm{cl}} = \forall_p(j_P R_b), \qquad \mathsf{Low}_b^{\mathrm{cl}}(v) \equiv \forall e \in P_v \; \neg\neg(b = O(W_e)).$$

Intersect this subobject with the previously constructed upper subobject for the classical tight predicate. The closure is deliberately inside the universal quantifier; $j(\forall_p R_b)$ is a different expression.

Theorem 6.10 (Exact classical lower predicate)**.** *For a global denotation v and global represented bound b, global validity of $\mathsf{Low}_b^{\mathrm{cl}}(v)$ is equivalent to the ordinary classical statement*

$$\forall \text{ total machine codes } e \text{ computing } v \quad \exists c \in \mathbb{N}_{>0} \; \exists N \; \forall n \geq N : b(n) \leq cW_e(n).$$

The original predicate $\forall_p R_b$ retains the stronger requirement of an effective witness transformation. Upper-class membership needs no such closure. The same construction identifies the classical infinitely-often lower predicate exactly.

Proof.

(1) **Identify the pointwise realizability relation.** Fix an actual total program e. Its membership in P is realized by the program that searches its halting clock on each input. For fixed c, N satisfying the displayed inequality, evaluation of b and the total profile W_e realizes every numerical inequality on the tail, uniformly in n. Conversely any realizer of $R_b(e)$ supplies such c, N and valid inequalities. Thus $R_b(e)$ has a realizer exactly when the classical eventual bound holds. Membership in P_v is similarly exactly extensional equality with the given denotation, together with totality; no decision procedure for that membership is inferred.

(2) **Calculate double negation on truth-value realizers.** In the Kleene realizability presentation, a truth value is represented by a set $A \subseteq \mathbb{N}$. A realizer of $A \Rightarrow \varnothing$ is a partial-recursive index that sends every member of A to an element of the empty set. Consequently, in the classical metatheory,

$$\neg A = \begin{cases} \mathbb{N}, & A = \varnothing, \\ \varnothing, & A \neq \varnothing, \end{cases} \qquad jA = \begin{cases} \varnothing, & A = \varnothing, \\ \mathbb{N}, & A \neq \varnothing. \end{cases}$$

These are calculations in the realizability model, not an internal double-negation-elimination rule. Relative subobject predicates also retain the given realizer of their ambient element.

(3) **Prove both directions of the external characterization.** If every code in the classical fibre satisfies the eventual bound, step (1) makes each relevant realizer set nonempty. Step (2) gives one fixed realizer for its double negation. A constant procedure, retaining the ambient implementation name where necessary, therefore realizes the universally quantified closed predicate. Conversely a realizer of that predicate gives an inhabited $jR_b(e)$ at every actual implementation. By step (2), $R_b(e)$ is inhabited, and step (1) gives the classical eventual bound. No procedure choosing its witnesses from e has been asserted in this converse.

(4) **Check upper and infinitely-often predicates.** For a principal upper-class witness there is one code e and one pair c, N. A correct total decider supplies its totality realizer by clock search; numerical bounds with these witnesses have realizers by computation. Conversely a realizer supplies the code and bounds. Hence the existing upper image has exactly the classical upper class. For an infinitely-often lower relation, fix its constant c and search $n \geq N$ until the decidable inequality holds. Classical truth guarantees termination for each N, so this realizes $\forall N\, \exists n \geq N$. The implementationwise relation again has a realizer exactly when its classical statement holds; steps (2)–(3) apply unchanged.

□

Proposition 6.11 (A computable family with no effective cutoff selector)**.** *There is a primitive-recursive family of positive numerical profiles $a_e(n)$ such that, classically, $n+1 = O(a_e(n))$ for every e, but no total recursive procedure returns a valid domination constant and cutoff uniformly in e. Thus the distinction between pointwise classical existence and a realized universal statement cannot be discarded for general measured profiles.*

Proof.

(1) **Construct a single possible exceptional length.** Let $H(e, n)$ be the bounded Boolean test that machine e, on empty input, halts for the first time at update n. Set

$$a_e(n) = \begin{cases} 1, & H(e,n) = 1, \\ n+1, & H(e,n) = 0. \end{cases}$$

This is primitive recursive in e, n. At most one length is exceptional. For each e the classical eventual bound therefore holds with constant 1, and with cutoff $h+1$ if its halting time is h, or cutoff 0 if it never halts.

(2) **Derive a halting decider from a proposed selector.** Suppose a total recursive procedure supplied valid positive integer c_e and cutoff N_e. Put $M_e = \max(c_e, N_e)$. A first halting time $h \geq M_e$ would give $h+1 > c_e a_e(h) = c_e$ at a length $h \geq N_e$, a contradiction. Thus the machine halts if and only if it halts before M_e updates. Computing M_e and checking these finitely many updates would decide the halting problem. No such selector exists.

(3) **Identify the scope of the example.** These are uniformly computable numerical profiles. They can also be realized as tagged-event valuations by computing $a_e(|x|)$ and then emitting that many counted events. They are not asserted to be the complete running times of that computing procedure. The example establishes the logical obstruction for the general resource framework, while Theorem 6.10 already gives the exact classical predicate for the actual Turing-time instance.

□

The same placement of closure handles ordinary external optimality: for a candidate e_0, use $\forall e \in P_v \neg\neg(W_{e_0} = O(W_e))$. For a displayed finite portfolio, close the relation $\exists i\,(W_{e_i} = O(W_e))$ before quantifying over e. A true instance has a finite index and numerical witnesses, so the same proof applies. Theorem 5.3 continues to characterize the original internal witness-preserving predicates; one must not replace their internal quantifiers by classical ones without this explicit interpretation.

6.5 Polynomial budgets and clocked programs

Set $b_d(n) = (n+1)^d$ and form the ascending union

$$I_{\mathrm{poly}} = \bigcup_{d\in\mathbb{N}} \downarrow [b_d], \qquad M_{\mathrm{poly}} = \bigcup_{d\in\mathbb{N}} B_{\mathbb{N}}[b_d]_{\mathrm{ev}}.$$

The explicit inclusions $b_d \leq b_{d+1}$ construct an inhabited directed ideal and its resource module.

Theorem 6.12 (Exact polynomial-time class)**.** *In the Turing-machine instance,*

$$\mathcal{C}^{\mathrm{wc}}(I_{\mathrm{poly}}) = \mathbf{P}_{\mathcal{T}}.$$

The polynomial-time denotations have an explicit numerical presentation by clocked finite machine descriptions.

Proof.

(1) **Expand the ideal membership.** The left-hand side has formula

$$\exists e \in P_v\ \exists d\ \exists c \in \mathbb{N}_{>0}\ \exists N\ \forall n \geq N: \quad W_e(n) \leq c(n+1)^d.$$

These are exactly the polynomial-time quantifiers for the fixed model. The exponent and constant may depend on the implementation, not on the input length.

(2) **Absorb finite exceptions by a finite computation.** Given witnesses e, d, c, N, compute the finitely many $W_e(n)$ for $n < N$ by Proposition 6.6. Let c' be the maximum of $1, c$ and these values, and set $d' = \max(1, d)$. Since $(n+1)^{d'} \geq 1$,

$$\forall n: \quad W_e(n) \leq c'(n+1)^{d'}.$$

This constructs a global polynomial bound from an asymptotic bound with its witnesses; it does not search for such witnesses from an arbitrary program.

(3) **Construct a polynomial clock.** For fixed positive integers c, d, copy unary $n+1$ onto d counter tapes and enumerate the d-tuples of positions in these tapes in lexicographic order. Write c tally marks per tuple. This produces a unary clock of length $c(n+1)^d$. Advancing the last counter costs one move; on a carry it is reset by a scan of $n+1$ cells. The total reset work over all carries is bounded by a constant depending on c, d times $(n+1)^d$: at depth j there are at most $(n+1)^{d-j}$ resets of length $n+1$, and the finite sum of these terms has that bound. Initialization costs $O(d(n+1))$. Thus the clock is constructed in $O(c(n+1)^d)$ time for its fixed description. Rewind the clock to its first mark and the input to its initial position; this costs $O(c(n+1)^d + n + 1)$ and is absorbed by the same bound because $c, d \geq 1$.

(4) **Compile a total clocked decision program.** For any program e, produce a program $\mathrm{Clock}(e, c, d)$ which builds this clock and then executes at most $c(n+1)^d$ target transitions, consuming one clock mark per transition. It checks for halting also after the last

permitted transition. It returns the target's Boolean value on timely halting, and returns 0 on timeout. The transition table of e is incorporated into the compiled finite control, with an extra clock tape, so the simulation overhead per target transition is bounded by a constant for that compiled program. Code generation is effective and total. Every such clocked program halts in polynomial time, including when e is invalid or partial. The degree belongs to the fixed compiled description; this does not assert one polynomial degree for an interpreter taking d as input.

(5) **Prove surjectivity onto the class.** Every clocked description denotes a member of $\mathbf{P}_{\mathcal{T}}$. For every polynomial-time implementation, step (2) supplies clock parameters for which timeout never changes its result. Thus the denotation map from triples (e, c, d) has image exactly that class. Pairing the positive parameters gives a numerical presentation. The construction makes no assertion that equality of its denotation with an arbitrary supplied program is semidecidable.

□

The clocks supply concrete algorithms, rather than an additional "time-constructibility" premise for these polynomial bounds. For other explicitly specified clock programs, the same construction records their actual generation cost; an arbitrary computable bound is not declared to be computable within its own value.

6.6 Machine models, simulations, and encodings

The numerical execution instance of Construction 13.2 uses increment, zero-test with decrement, jumps, and halt. Computability adequacy of this set does not establish binary-input polynomial-time adequacy. The obstruction can be proved directly for its usual numeric input convention.

Proposition 6.13 (Numeric counters do not supply binary-time adequacy)**.** *Let a counter program start with $R_0 = N$ and every other register zero, use only the four instructions just listed, and return the number in a designated register. Every total program returning $N \bmod 2$ takes at least N updates on input $N \geq 1$. In contrast, a binary-input Turing machine computes the same function in linear time in the bit length. Consequently no polynomial-overhead compiler from binary-input Turing machines to this numeric counter convention can preserve these inputs and outputs.*

Proof.

(1) **Couple adjacent inputs before the smaller counter reaches zero.** Suppose the run on N halts after $t < N$ updates. Throughout these updates its R_0 is at least $N - t > 0$. The runs on N and $N + 1$ have the same control state and the same other registers; their R_0 values differ by one. Induction proves this invariant because every zero test on R_0 takes its nonzero branch in both runs, and all other tests have identical operands.

(2) **Contradict Boolean parity output.** The second run therefore halts at the same time. If the output register is different from R_0, the outputs agree, contradicting parity. If it is R_0, the second output is at least two, contradicting Boolean output. Thus $t < N$ is impossible.

(3) **Read the binary input directly.** A Turing machine scans a nonempty binary numeral to its right end, retains the last bit in its finite control, and returns it. This uses $O(m+1)$ updates for an m-bit numeral. Inputs $N = 2^{m-1}$ already force at least 2^{m-1} updates in the counter model. No polynomial in m can bound those runs.

□

The input convention is part of this proposition, not a claim about counter machines with additional bit-reading primitives or input tapes. The repair for conventional complexity is to construct the bit-input machine instance, while retaining the numeric counter instance for its own stated measurement. General machine-independent structural theorems do not identify the complexity of an individual function across arbitrary encodings; this distinction is already explicit in Blum (1967).

Proposition 6.14 (Constructed polynomial-overhead tape simulation)**.** *Every fixed multitape decision machine e has an effectively constructed single-work-tape simulator $S(e)$, preserving its input, return domain, and Boolean result, with*

$$T_{S(e)}(x) \leq C_e(|x| + T_e(x) + 1)^2$$

on the return domain. A one-work-tape machine is already a special case of the multitape family. Consequently their polynomial-time classes are equal.

Proof.

(1) **Encode the finitely many tapes and heads.** Use one finite block per simulated work tape, separated by delimiters, with one marked cell for its head and explicit boundary blanks. The simulated state lies in finite control. The input head can be retained on the simulator's read-only input tape. After t transitions the encoded work blocks have length $O_e(t + 1)$; copying input onto work storage in a single-total-tape convention instead gives $O_e(|x| + t + 1)$, which is also covered by the stated bound.

(2) **Implement a simulated transition by sweeps.** A sweep reads the finitely many marked symbols into finite control. The compiled transition table determines the new symbols and moves. A further sweep updates marks and symbols. If a block needs a new boundary cell, shift the finite suffix once. A fixed number of sweeps and shifts costs at most $C_e(|x| + t + 1)$ moves. Every such operation is a finite transition-table construction.

(3) **Sum the bound and prove adequacy.** Induction shows that the encoded configuration after each complete sweep cycle is the target configuration. Initialization costs $O_e(|x| + 1)$. Summing the sweep bounds for $t < T_e(x)$ gives the displayed quadratic bound. A nonhalting target never reaches a simulated halting configuration. If T_e is polynomial in input length, so is the simulator time. The reverse inclusion uses the literal one-work-tape inclusion.

□

The sharper two-tape construction of Hennie and Stearns (1966) gives a logarithmic rather than quadratic simulation overhead; it can be cited when that sharper quantitative statement is needed. The explicit quadratic construction above suffices to identify the polynomial budget with conventional **P**. Neither statement identifies fine bounds such as linear time on different tape models.

Proposition 6.15 (Finite-alphabet and bit-length encodings)**.** *A finite input alphabet A, specified by a list of distinct symbols, has a fixed-block binary encoding, with linear-time encoder, decoder, and image test. Under this encoding its polynomial-time decision class is exactly the binary class restricted to encoded words. The encoding $x \mapsto \mathrm{bin}^{-1}(1x)$ of binary words as positive integers has bit length $|x| + 1$; the bit-string conversions are linear.*

Proof.

(1) **Construct the block code.** Choose the least positive d with $2^d \geq \max(1, |A|)$ and send the ith symbol to the d-bit numeral for i. The given finite list determines this assignment without choice. A finite-state parser checks block boundaries and unused codes. It rejects invalid strings and otherwise decodes one symbol per block. The output length is $d|x|$, and both conversions take linear time.

(2) **Compose the machines with their conversions.** Encoding then running a binary machine, or decoding then running an A-machine, preserves polynomial time by the linear length relation and the composition construction in Proposition 6.16. Invalid binary encodings can be rejected in linear time. These give both directions of the class statement.

(3) **Separate bit encoding from numeric counter access.** Prefixing a 1 gives the standard binary numeral of an integer $N \in [2^{|x|}, 2^{|x|+1})$; removing its first bit is the inverse. As string operations they are linear. Loading the value N into a unit increment/decrement input register gives the different interface of Proposition 6.13; the length identity does not make that interface a polynomial-time bit reader.

□

6.7 Composition and conventional reductions

For function programs use an append-only binary output tape, initially empty, with at most one bit written per target transition. Therefore $|f(x)| \leq T_f(x)$ on return. Define $\overline{W}_g(m) = \max_{j \leq m} W_g(j)$ on actual total cost functions. This running maximum is used before quotienting; no descent to eventual-equality germs is claimed for it.

Proposition 6.16 (A quantitative composition compiler)**.** *There is an effective code constructor for $g \circ f$ with*

$$W_{g \circ f}(n) \leq C_{f,g}(n + 1 + W_f(n) + \overline{W}_g(W_f(n))).$$

In particular, polynomial-time function programs are closed under composition. Polynomial-time many-one reductions preserve membership in the polynomial-time decision class.

Proof.

(1) **Compile the sequential execution.** Rename and separate the finitely many work tapes of the two programs. Run f and record its output on a work tape. Rewind that tape and use it as the simulated input for g. The fixed transition tables are incorporated into the compiled finite control. This costs at most a fixed constant times $|x| + 1 + T_f(x) + |f(x)| + T_g(f(x))$. Its domain and returned value are exactly those of the composite.

(2) **Take the appropriate maxima.** The output-length bound gives $|f(x)| \leq W_f(n)$ for $|x| = n$. Hence $T_g(f(x)) \leq \overline{W}_g(W_f(n))$. Taking the finite maximum in the preceding estimate gives the stated inequality.

(3) **Construct polynomial bounds and reductions.** For global nondecreasing polynomial bounds p, q, the right side is bounded by $C_{f,g}(n + 1 + p(n) + q(p(n)))$. Such global bounds are obtained from asymptotic witnesses by the finite-exception procedure in Theorem 6.12. If r is a polynomial-time many-one reduction from a language A to B, compose its machine with a polynomial-time decider for B. The defining equality $\chi_A = \chi_B \circ r$ proves correctness, and the bound proves polynomial time.

□

The final equality is semantic correctness; the displayed machine construction supplies the implementation and its overhead. In particular, an arbitrary semantic arrow is never declared to be a polynomial-time compiler merely because it has a tracker. This same separation between names, denotations, and measured execution is central to Kawamura and Cook (2012).

6.8 Workspace and simultaneous constraints

Construction 6.17 (Conventional workspace and its worst-case profile)**.**

(1) **Record precisely the counted cells.** Use the read-only-input, work-tape decision convention from Construction 6.4. For each work tape, retain the current integer head position and the least and greatest positions visited. Initially they are all zero. A transition updates these three integers by addition, minimum, and maximum. Since heads move by at most one, the visited cells form the interval between the two extrema. The number of work cells visited is the sum of the interval lengths over the finitely many work tapes. The input tape is excluded. These counters describe the history measurement; they are not additional work tapes charged to the simulated program.

(2) **Fold the returning history.** On a finite halted history, the final visited-cell count is the peak count because the visited sets only grow. Define this value to be $S_e(x)$. It is a total finite fold of the history and is defined on exactly the return domain. On total implementations it gives $P \to \mathbb{N}^X$. Apply W to obtain $S_e^{\mathrm{wc}}(n) = \max_{|x|=n} S_e(x)$.

(3) **Keep one implementation witness for joint bounds.** Use the paired measurement $(\Phi_T^{\mathrm{wc}}, \Phi_S^{\mathrm{wc}}) : P \to K_{\mathbb{N}} \times K_{\mathbb{N}}$. For budgets M_b, M_s, first form $P \times_{K_{\mathbb{N}} \times K_{\mathbb{N}}} (M_b \times M_s)$, then take its image under p. This imposes both bounds on the same program.

Theorem 6.18 (Exact deterministic space and joint-resource classes)**.** *The space-only feasibility image of M_s is exactly* $\mathrm{DSPACE}_{\mathcal{T}}(O(s))$*. The joint image is exactly the class of problems with one total decision program satisfying both the worst-case time bound $O(b)$ and the worst-case work-space bound $O(s)$. In particular its defining predicate retains a single implementation witness for both resources.*

Proof.

(1) **Identify the measurement with conventional workspace.** Induction on the history identifies the recorded extrema with the actual visited positions. Unit head moves make each visited set an interval, so its cardinality is exactly the recorded length. This proves equality with the declared conventional work-space count.

(2) **Apply finite maxima and bounded division.** The finite-maximum adjunction gives $S_e = O(s \circ \ell)$ if and only if $S_e^{\mathrm{wc}} = O(s)$. Membership in M_s is exactly this condition. Existential image over p gives the stated space class.

(3) **Retain the shared witness.** The paired pullback has formula $\exists e \in P_v\, [W_e = O(b) \wedge S_e^{\mathrm{wc}} = O(s)]$. Separate images instead give $(\exists e \in P_v\, W_e = O(b)) \wedge (\exists e' \in P_v\, S_{e'}^{\mathrm{wc}} = O(s))$. The construction has the former formula; it makes no inference interchanging these quantifiers.

□

6.9 Finite portfolios in the conventional time instance

Theorem 5.3 applies to arbitrary fixed measurements. In the usual time setting there is an additional construction, finite racing. To account explicitly for input copying, let e^+ be the program that scans the input to its end and back before running e. Put $\tau_e = T_{e^+}$ and $\widehat{W}_e = W(\tau_e)$. This constructs constants independent of x with

$$|x| + 1 + T_e(x) \le C\tau_e(x), \qquad \tau_e(x) \le C(|x| + 1 + T_e(x)).$$

An extra fixed transition handles the empty word. This scan-prefixed instance has the same polynomial-time class. It also has the same $O(b)$ time class whenever $b(n) \ge n+1$ eventually. These assertions follow directly from the two inequalities.

Proposition 6.19 (Finite racing and optimality)**.** *For a nonempty finite list of total programs $e_1, \dots, e_k$ with the same Boolean denotation, an effective scan-prefixed racing program r has that denotation and satisfies*

$$\widehat{W}_r(n) \le C_{e_1,\dots,e_k} \min_i \widehat{W}_{e_i}(n).$$

For a computable denotation in this scan-prefixed multitape time instance, a finite asymptotically dominating portfolio exists if and only if an asymptotically optimal program exists. Its feasibility open is therefore compact if and only if it is supercompact.

Proof.

(1) **Construct the finite parallel machine.** Allocate separate work-tape blocks and an input copy for each of the finitely many machines. Copying the input costs $O_k(n+1)$. Their finite control states form a finite product. Execute one transition of each machine per round and stop when the first halts. Because the list and its tape counts are fixed in this compiled machine, each round has constant cost depending only on the list. Every returned bit is correct, since all components have the same denotation.

(2) **Bound the race before taking worst cases.** The cost is at most $C(n{+}1{+}\min_i \tau_{e_i}(x))$. Every scan-prefixed competitor already pays for $n+1$, so a further scan prefix for the race is absorbed into the same constant. Thus $\tau_r(x) \le C' \min_i \tau_{e_i}(x)$. For each i this is at most $C'\tau_{e_i}(x)$; taking maxima gives $\widehat{W}_r(n) \le C'\widehat{W}_{e_i}(n)$ for every i, hence the displayed finite minimum bound.

(3) **Collapse a finite dominating portfolio to one program.** For any competing program e, domination supplies some i with $\widehat{W}_{e_i} = O(\widehat{W}_e)$. The racing bound then gives $\widehat{W}_r = O(\widehat{W}_e)$. Hence r is optimal. An optimal program gives a one-element dominating portfolio. Applying the principal-open characterizations in Theorem 6.7 proves the compactness statement. For a computable denotation the portfolio cannot be empty; an empty implementation fibre is a separate case.

□

This proposition is specific to the constructed time instance. It does not assert that racing preserves the best space usage, that arbitrary event valuations support this inequality, or that two-tape and unrestricted-multitape optimality agree up to a constant factor. The polynomial-overhead simulation theorem is sufficient for **P**, but its overhead cannot be discarded in fine optimality questions.

These constructions retain the logical distinction of Proposition 2.30. Finite clock tests concern the displayed executions, and worst-case profiles are formed before taking eventual germs. Totality and asymptotic membership remain $\Omega_{\mathcal{E}}$-valued; the cofinite germ quotient

acquires no nonconstant positive resource test. The finite code constructors and first-return searches above therefore instantiate the existing foundation without an additional validity oracle, representative selection, or logical axiom. We now return to varying the valuations of fixed histories.

7 Differential calculus of resource constraints

The measured span separates a program's denotation from its resource valuation. Keeping its execution history fixed while varying event weights produces a family of resource values. We begin with this family and its budget equation, then construct the finite coefficient calculus that differentiates them.

7.1 Measured valuation families and budget variations

For a fixed total implementation e with finitely many measured event types, let $N_{e,i}$ be its event-count functions. Additive rational weights give the measured family

$$F_e(w) = \sum_i N_{e,i} w_i.$$

This follows by folding the returning history (Theorem 4.9). Passing to germs and extending coefficients gives the B-linear map $F_e : B^k \to K$ with the same formula. A polynomial parametrization $w = w(z)$ therefore gives a polynomial resource law. Its finitely many coefficients lie in the principal resource generated by the sum of their absolute values, so it is calculated on a finite-projective resource cell.

For an embedded budget $M \subseteq K$, put $q_M : K \to K/M$. Feasibility is the equation $q_M F_e(z) = 0$. Over the dual numbers, polynomial expansion gives

$$q_M F_e(z + \varepsilon v) = q_M F_e(z) + \varepsilon q_M DF_{e,z}(v).$$

At a feasible point, the fixed-budget tangent condition is therefore $q_M DF_{e,z}(v) = 0$. If a finite-projective budget also varies, its tangent class is $\theta \in \mathrm{Hom}_B(M, K/M)$ and the condition becomes

$$q_M DF_{e,z}(v) = \theta(F_e(z)).$$

Theorems 8.2 and 8.7 construct the budget tangent and prove this equation by comparing coefficients.

Theorem 7.1 (Measured Taylor coefficients and geometric jets)**.** *Let H_e be the returning history of an implementation, with finitely many measured event types and polynomial weights $w_i(z) \in \mathbb{Q}[z_1, \dots, z_m]$. Write*

$$\tau_r : \mathbb{Q}[z] \longrightarrow \mathbb{Q}[z, \varepsilon_1, \dots, \varepsilon_m]/(\varepsilon_1, \dots, \varepsilon_m)^{r+1}, \qquad p(z) \longmapsto p(z + \varepsilon).$$

Folding H_e with the weights $\tau_r w_i$ gives the Taylor coefficients of its measured cost through order r:

$$\mathrm{fold}(\tau_r w, H_e) = \tau_r \,\mathrm{fold}(w, H_e). \tag{7.1}$$

The two executions have the same return domain and returned value. For a total implementation on the specified input carrier, taking germs identifies these coefficients with the order-r geometric jet of $F_e(z) = \sum_i N_{e,i} w_i(z)$. The comparisons commute with truncation, polynomial parameter substitution, and the specified environment maps. Their finite coefficient operations have uniform implementations in the order and the displayed polynomial data.

Proof.

(1) **Commute Taylor substitution with execution.** The algebra homomorphism τ_r preserves the additive monoid of weights. Thus Theorems 4.7 and 4.9 give (7.1), including iteration. Changing the valuation preserves the return domain and value by Theorems 4.4 and 4.6.

(2) **Pass to germs and identify the jet.** The equality is coefficientwise in finite rational polynomials, so descends through the germ quotient. The finitely many coefficients of F_e lie in the principal cell generated by their absolute sum. On this cell, Theorems 7.24 and 12.5 and Proposition 12.8 identify substitution by $z + \varepsilon$ with the geometric jet.

(3) **Verify the comparisons.** Truncation discards higher monomials. Polynomial parameter maps compose with Taylor substitution by their full Taylor law, and history substitution gives the environment comparison.

(4) **Implement finite coefficients.** Monomial enumeration, rational arithmetic, and finite folds are uniform NNO recursions in the order and polynomial data. The numerical covers and arithmetic trackers of Proposition D.3 implement them. The resulting Taylor coefficients may be signed, and lie in K.

□

The theorem supplies the execution interpretation of the differential calculus. With $C_r = \mathbb{Q}[z, \varepsilon]/(\varepsilon)^{r+1}$ and the finite alphabet A of measured event types, read the returning history as its word of measured event types. The valuation comparison is

$$\begin{array}{ccc} A^* & \xrightarrow{\overline{w}} & \mathbb{Q}[z] \\ & \searrow^{\overline{\tau_r w}} & \downarrow{\scriptstyle \tau_r} \\ & & C_r. \end{array}$$

Both paths evaluate the same returning history. The finite coefficient calculation is Algorithm 7.1; compatible truncations give the formal families used below.

Algorithm 7.1 Measured Taylor coefficients on a returning history

Input: Returning word H; polynomial weights $(w_a)_{a \in A}$; order r
Output: The cost polynomial in $C_r = \mathbb{Q}[z, \varepsilon]/(\varepsilon)^{r+1}$

for each event type $a \in A$ **do**
    $v_a \leftarrow \mathrm{trunc}_{\deg_\varepsilon \le r}\, w_a(z + \varepsilon)$
end for
$c \leftarrow 0$
for each event a of H, in order **do**
    $c \leftarrow c + v_a$
end for
return c

Example 7.2 (A measured polynomial resource family)**.** Work on $\mathbb{N}$ with the cofinite filter and the unit-valued idle events a, b of Section 5.5. The fixed total program in Algorithm 7.2 uses finite iteration. Assign weights 1 and z^2 to a, b, respectively.

Algorithm 7.2 The measured quadratic resource family

Input: $n \in \mathbb{N}$

```
IDLE(a)
for j = 1, ..., (n+1)^2 do
    IDLE(b)
end for
return n
```

The history fold gives the nonnegative rational cost

$$T_z(n) = 1 + (n+1)^2 z^2 \qquad (z \in \mathbb{Q}).$$

Writing $t(n) = 1/(n+1)$, its polynomial coefficient extension is

$$F : B \longrightarrow K, \qquad F(z) = 1 + t^{-2} z^2. \tag{7.2}$$

The base value $F(0) = 1$ belongs to B. At first order, $F(d) = 1$ in $K[d]/d^2$. At second order the additional coefficient is t^{-2}, whose class in K/B is nonzero. We will use this same family to calculate fixed, moving, and coupled-budget obstructions in Examples 8.12, 8.18 and 8.21.

Free commutative algebras represent the polynomial expressions above. Their universal derivatives retain the coefficients. Based Weil stages then represent finite coefficient variations, and module realization carries the budget equation into differential geometry.

7.2 Polynomial algebras and universal differentials

Let R be a commutative rational algebra in $\mathcal{E}$; the applications use $R = B$ for relative resource calculus and $R = \mathbb{Q}$ for absolute differentials. All modules and algebras in this subsection are internal.

Theorem 7.3 (Free commutative algebra)**.** *The forgetful functor* $\mathrm{CAlg}_R \to \mathrm{Mod}_R$ *has left adjoint* $T = \mathrm{Sym}_R$*. Its unit inserts linear generators and its multiplication substitutes polynomials.*

Proof.

(1) **Construct the polynomial quotient.** Form finite commutative polynomial expressions with coefficients in R and symbols $[m]$ for $m \in M$, and take the effective quotient by the algebra equations and

$$[m + m'] = [m] + [m'], \qquad [rm] = r[m], \qquad [0] = 0.$$

(2) **Prove the universal property.** Evaluation of expressions extends each R-linear map $M \to A$ uniquely to an R-algebra map $\mathrm{Sym}_R M \to A$. Conversely restriction to the symbols recovers that linear map. This proves the adjunction.

(3) **Verify the monad and algebra equations.** Insertion and substitution satisfy the monad laws because both sides agree on generators. Algebra multiplication and its unit are natural, and substitution preserves them.

□

Definition 7.4 (Square-zero extensions)**.** For an R-algebra A and an A-module N, its **square-zero extension** is

$$A \ltimes N = A \oplus N, \qquad (a, n)(b, n') = (ab, an' + bn), \qquad 1 = (1, 0). \tag{7.3}$$

The module laws prove the algebra equations.

Theorem 7.5 (Universal derivative)**.** *There is a natural R-linear map*

$$d_M : TM \longrightarrow TM \otimes_R M$$

representing R-derivations of TM. It satisfies the constant, Leibniz, linear, chain, and interchange identities.

Proof.

(1) **Construct the universal derivative.** The R-linear map $m \mapsto (\eta_M m, 1 \otimes m)$ into $TM \ltimes (TM \otimes_R M)$ extends uniquely to an algebra map. Its first component is id_{TM} by freeness. Its second component d_M satisfies

$$d_M(1) = 0, \qquad d_M(ab) = a\, d_M b + b\, d_M a, \qquad d_M(\eta_M m) = 1 \otimes m. \tag{7.4}$$

(2) **Prove universality and naturality.** More generally, for a TM-module N, any R-linear $v : M \to N$ gives the algebra map with generator $(\eta_M m, v(m))$. Thus

$$\mathrm{Der}_R(TM, N) \cong \mathrm{Hom}_R(M, N) \cong \mathrm{Hom}_{TM}(TM \otimes_R M, N),$$

which proves universality. For $f : M \to N$, the derivations $(Tf \otimes f)d_M$ and $d_N Tf$ agree on each generator, proving naturality.

(3) **Prove the chain identity.** Write $\mu_M : TTM \to TM$ for substitution and $m_M : TM \otimes_R TM \to TM$ for multiplication. The chain identity is

$$d_M \mu_M = (m_M \otimes 1_M)(\mu_M \otimes d_M) d_{TM}. \tag{7.5}$$

Both sides are derivations along μ_M and send a generator $p \in TM$ to $d_M p$, so universality proves the equation.

(4) **Prove interchange on finite polynomials.** On a finite monomial, writing m_i also for its inserted generator,

$$d_M(m_1 \cdots m_n) = \sum_i \left(\prod_{j \neq i} m_j \right) \otimes m_i. \tag{7.6}$$

Applying d_M to the first factor sums over ordered pairs of distinct positions. Exchanging those positions preserves the sum, proving that $(d_M \otimes 1_M) d_M$ is invariant under switching its last two factors. Every element is represented by a finite polynomial, so this proves interchange on the whole algebra.

□

Definition 7.6 (Derivations and Kähler differentials)**.** An **R-derivation** from an R-algebra A to an A-module N is an R-linear map satisfying $d(ab) = a\, db + b\, da$. The **module of Kähler differentials** $\Omega^1_{A/R}$ is the A-module on symbols da, subject to additivity, R-linearity, $d1 = 0$, and this Leibniz relation.

Proposition 7.7 (Universal differentials)**.** *For an R-algebra A, the module $\Omega^1_{A/R}$ represents R-derivations. For any A-module N,*

$$\mathrm{Hom}_A(\Omega^1_{A/R}, N) \cong \mathrm{Der}_R(A, N) \cong (\mathrm{CAlg}_R)_{/A}(A, A \ltimes N).$$

Proof.

(1) **Construct the differential module.** Take the free A-module on symbols da, $a \in A$, and quotient by

$$d(a+b) = da + db, \qquad d(ra) = r\,da, \qquad d1 = 0, \qquad d(ab) = a\,db + b\,da.$$

(2) **Identify derivations and square-zero sections.** Its maps to N are precisely R-linear derivations. Such a derivation is equivalently an algebra section $a \mapsto (a, da)$ by (7.3).

(3) **Retain relative structure and parameters.** Relative differentials for a specified map $A_0 \to A$ impose the additional equations $d(a_0) = 0$. All quotients are effective and all constructions retain their parameters, so the universal maps commute with substitution.

□

7.3 Finite-projective stages for resource geometry

7.3.1 Rational stages and finite-projective closure

Definition 7.8 (Rational Weil stages)**.** A **rational Weil presentation** is a finite-dimensional augmented commutative $\mathbb{Q}$-algebra $W = \mathbb{Q} \oplus \mathfrak{n}$, specified by a rational basis, a multiplication table, and r with $\mathfrak{n}^{r+1} = 0$. Let $C_W = B \otimes_{\mathbb{Q}} W$. The **rational-stage category** $\mathsf{W}_B^{\mathrm{rat}}$ has these objects and all B-algebra maps between them, encoded by finite lists of images satisfying their multiplication equations.

Tensor product gives coproducts of these stages, but the category with all B-algebra maps is not, in general, closed under equalizers.

Example 7.9 (A coefficient-dependent rational equalizer)**.** In the cofinite numerical fibre, let e be the parity idempotent, which is neither zero nor one. On $C = B[\varepsilon]/\varepsilon^2$ the maps $u(\varepsilon) = \varepsilon$ and $v(\varepsilon) = e\varepsilon$ have equalizer in B-algebras

$$\mathrm{Eq}(u, v) = B \oplus Be\varepsilon.$$

Its nilpotent summand is rank one on the support of e and rank zero on the complementary support. It is finite projective over B, but is not $B \otimes_{\mathbb{Q}} W$ for a fixed rational Weil algebra W: on the two idempotent components the augmentation ranks of such a rational extension would agree. Thus the full rational-stage category cannot serve as the stage category for all coefficient-dependent finite Weil limits.

Definition 7.10 (Finite-projective Weil stages)**.** A **finite-projective Weil stage** is an augmented commutative B-algebra $C = B \oplus N$ with a displayed finite-free retract presentation of N and a finite bound $N^{r+1} = 0$. The category $\mathsf{W}_B^{\mathrm{proj}}$ has these stages and all B-algebra maps between them.

Theorem 7.11 (Finite-projective Weil-stage closure)**.** *There is a small internal category $\mathsf{W}_B^{\mathrm{proj}}$ whose objects are augmented commutative B-algebras $C = B \oplus N$ with $N^{r+1} = 0$ and a displayed finite-free retract (idempotent-matrix presentation) of N. It contains $\mathsf{W}_B^{\mathrm{rat}}$ as a full subcategory and is closed under finite limits and Weil tensor coproducts. Each stage is finite projective in this explicit sense and flat over B. These presentations and closure operations are uniform in parameters.*

Proof.

(1) **Construct the internal category of Weil stages.** A stage code consists of an integer n, an idempotent $p \in M_n(B)$, a finite bilinear multiplication table on $N = \mathrm{im}(p)$, and a positive integer r witnessing $N^{r+1} = 0$. Commutativity and associativity are finite matrix equations. A stage morphism is similarly a finite matrix representing a multiplication-preserving map between these augmentation modules. Matrix identity, substitution, and composition construct the internal category. The validity conditions are matrix equations over B, interpreted as in Section 2.6.1. The ring is $B \oplus N$, with multiplication $(b, x)(c, y) = (bc, by + cx + xy)$.

(2) **Construct coproducts and finite products.** The tensor coproduct of $B \oplus N$ and $B \oplus N'$ has underlying module

$$B \oplus N \oplus N' \oplus (N \otimes_B N'),$$

a finite-free retract. The augmentation ideal is nilpotent: a sufficiently long product contains either more than r factors from the first augmentation ideal or more than r' from the second. The terminal stage is B. Products in the augmented-algebra category are fibre products over B, with augmentation ideal $N \oplus N'$.

(3) **Split the image of the difference map.** For equalizers take two stage maps $u, v : C \rightrightarrows C'$. Every such map preserves augmentation: its composite with $C' \to B$ kills the nilpotent ideal of C, because B is reduced. Thus $d = u - v$ restricts to a B-linear map $d : N \to N'$. Its image $J \subseteq N'$ is generated by images of the displayed finite generators of N. Embed N' as a displayed summand of a finite free module. Apply Theorem 2.27 to the corestriction $d : N \twoheadrightarrow J$, with those images and their specified preimages in N. It constructs a section $s : J \to N$ satisfying $ds = \mathrm{id}_J$.

(4) **Construct the equalizer idempotent.** The endomorphism

$$p_N = \mathrm{id}_N - sd : N \longrightarrow N$$

satisfies

$$p_N^2 = \mathrm{id}_N - 2sd + s(ds)d = p_N, \qquad dp_N = d - dsd = 0.$$

For $x \in \ker d$ one has $p_N x = x$, so $\mathrm{im}(p_N) = \ker d$. Therefore the equalizer in B-algebras is

$$\mathrm{Eq}(u, v) = B \oplus \mathrm{im}(p_N),$$

a displayed finite-free retract of C as a module.

(5) **Verify algebra closure and finite-limit stability.** It is closed under multiplication because u and v are algebra maps; its augmentation ideal inherits nilpotence from N. Finite limits follow from products and equalizers, with the idempotent presentations retained. All calculations are finite, compatible with parameter substitution, and use the displayed lifts.

□

Definition 7.12 (Based Weil geometry and module realization)**.** The **based Weil geometry** uses $\mathsf{W}_B = \mathsf{W}_B^{\mathrm{proj}}$ and

$$\mathcal{G}_B = [\mathsf{W}_B, \mathcal{E}].$$

The functor topos is internal and covariant, equivalently the presheaf topos on the opposite stage category; finite limits, colimits, and images are computed stagewise. Its **resource**

line, **module realization**, and **representable infinitesimal probes** are respectively

$$\mathcal{R}_B(C) = C, \quad j_B M(C) = C \otimes_B M, \quad D_A(C) = \mathrm{Hom}_{B\text{-alg}}(A, C) \qquad (A, C \in \mathsf{W}_B).$$

The resource line is a ring object and $j_B M$ an $\mathcal{R}_B$-module. Each D_A is a representable infinitesimal object.

A budget family and an infinitesimal coefficient stage can vary together. The next construction combines these two parameters, so each budget carries its realized resource module at every Weil stage.

7.4 Relative geometry of budgets and Weil coefficients

Theorem 7.13 (Budget–Weil presentation)**.** *For the finite-matrix stage presentations constructed above,*

$$\mathsf{W}_{\bar{B}}^{\mathrm{proj}} \cong \Delta_{\mathfrak{C}_{\mathrm{Ev}}} \mathsf{W}_B^{\mathrm{proj}}, \qquad \mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}} := [\mathsf{W}_{\bar{B}}^{\mathrm{proj}}, \mathscr{S}] \simeq [L \times \mathsf{W}_B^{\mathrm{proj}}, \mathcal{E}].$$

At a probe (a, A) the structure ring and universal resource are

$$\mathcal{R}_{\bar{B}}(a, A) = A, \qquad \mathfrak{M}(a, A) = A \otimes_B M_a. \tag{7.7}$$

For $a \leq b$ and $u : A \to A'$, the transition sends $c \otimes m$ to $u(c) \otimes m$ using $M_a \hookrightarrow M_b$.

Proof.

(1) **Transport the finite stage presentations.** A stage code comprises natural numbers, finite coefficient lists, idempotent and multiplication matrices, and finite equations. The nilpotence bound is a displayed natural number and its equations are checked on the finite generators. The object of codes and the object of arrows are therefore built from finite limits and natural-number-indexed finite lists of coefficients. The constant-diagram functor preserves these constructions; in particular it preserves their equality subobjects. This identifies both stage categories, including identities and composition.

(2) **Identify budget diagrams of Weil diagrams.** An internal diagram over the resulting constant category is equivalently a budget diagram of Weil diagrams: the two actions commute. Currying gives the asserted equivalence.

(3) **Compute the probe formulas.** Tensor and module operations are computed at each probe, giving (7.7).

□

Write $i_W : \mathscr{S} \to \mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}}$ for constancy in the Weil variable and $\Gamma_W Z(a) = Z(a, B)$. The restricted based-Weil adjunctions are $i_W \dashv \Gamma_W \dashv i_W \dashv \Gamma_W$. They calculate reduction on fixed-base probes at each budget. The finite-hit dominance of $\mathscr{S}$ is $\Delta_{\mathfrak{C}_{\mathrm{Ev}}}\Sigma$: constant diagrams preserve $2^{\mathbb{N}}$ and its finite-hit image, and composition of the classified monos is checked at each budget. Applying the same construction in the Weil variable gives $\Sigma_{\mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}}} = i_W \Delta_{\mathfrak{C}_{\mathrm{Ev}}}\Sigma$. For every positive test $u : Z \to \Sigma_{\mathcal{G}_{\mathfrak{C}_{\mathrm{Ev}}}}$,

$$u = i_W(\Gamma_W u)\eta_Z. \tag{7.8}$$

This follows from $\Gamma_W \dashv i_W$ exactly as in Proposition 7.15. Budget locality is retained; infinitesimal coefficients are invisible to these positive tests. For a name cover $q_D : C_D \twoheadrightarrow$

$\mathrm{Pt}_\Sigma D$ and an implemented map $\widehat{F} : C_D \to C_{D'}$, let $k = i_W \Delta_{\mathfrak{C}_{\mathrm{Ev}}}$. The transported cover remains a regular epimorphism and

$$(kq_{D'})(k\widehat{F}) = k(\mathrm{Pt}_\Sigma F)(kq_D), \qquad k\,\mathrm{Obs}_D : k\,\mathrm{Pt}_\Sigma D \times k\mathcal{O}_D \to k\Sigma.$$

These equations are evaluated at each probe as the original name-realization and observation equations; pointwise finite limits and images preserve their types. For a partial implementation the same argument preserves its Σ-classified domain span and its value map. Thus the relative geometry retains both domains and returned values.

7.5 Reduction, observations, and Weil coefficient calculus

7.5.1 Reduction and infinitesimal shape

Every B-algebra map $C = B \oplus N \to B$ kills the nilpotent augmentation ideal because B is reduced. It therefore equals the augmentation. Thus B is terminal as well as initial in the full projective Weil-stage category W_B. Write $i : \mathcal{E} \to \mathcal{G}_B$ for constant objects and $\Gamma(F) = F(B)$.

Theorem 7.14 (The ambidextrous based-Weil restriction)**.** *On the based-Weil functor topos, the functor i is fully faithful and there is an ambidextrous adjoint quadruple*

$$i \dashv \Gamma \dashv i \dashv \Gamma.$$

The endofunctor $\mathfrak{I}_W = i\Gamma$ is left exact and idempotent, with natural maps

$$\epsilon_X : \mathfrak{I}_W X \to X, \qquad \eta_X : X \to \mathfrak{I}_W X, \qquad \eta_X \epsilon_X = \mathrm{id}_{\mathfrak{I}_W X}.$$

It carries the induced reduction comonad and infinitesimal-shape monad structures. The associated triple of adjoint idempotent endofunctors has coincident underlying functor $\mathfrak{I}_W$.

Proof.

(1) **Construct the adjunction from initiality.** Initiality of B gives

$$\mathrm{Nat}(iA, F) \cong \mathrm{Hom}_{\mathcal{E}}(A, F(B)) :$$

a map at B extends at C through $F(B \to C)$.

(2) **Construct the adjunction from terminality.** Terminality gives

$$\mathrm{Nat}(F, iA) \cong \mathrm{Hom}_{\mathcal{E}}(F(B), A) :$$

a map at B extends at C through $F(C \to B)$.

(3) **Verify the adjunction quadruple and full faithfulness.** Both inverse equations are naturality along those stage maps. These two adjunctions repeat to give the quadruple. Since $\Gamma i = \mathrm{id}$, i is fully faithful.

(4) **Identify the units, counits, and modal structures.** At C the maps ϵ_X, η_X are respectively $X(B \to C)$ and $X(C \to B)$. Their composite on $X(B)$ is the identity. Evaluation and constant formation preserve finite limits, giving left exactness; $\Gamma i = \mathrm{id}$ gives idempotence. The units, counits, and their triangle identities give the stated monad, comonad, and adjoint structures.

□

The ambidexterity here has a specific source: both endpoints of a based probe reduce to the fixed B-point. It describes the based-Weil restriction of the primary étale endpoint geometry of Theorem 9.5. The comparison in Theorem 10.6 places the following Weil coefficient and microlinearity calculations within that larger geometry. For the line,

$$\Im_W \mathcal{R}_B = iB, \qquad \eta_{\mathcal{R}_B}(C) : C \to B$$

is augmentation; for every representable probe, $\Im_W D_A = iD_A(B) = 1$. Thus reduction extracts the reduced point of each infinitesimal probe.

The reduction $B \to B/\mathfrak{m}$ has a different type: it removes numerical errors vanishing along the eventual filter. The map η_X removes formal nilpotent directions. Both are used with their displayed coefficient objects.

7.5.2 Computational observations inside the cohesive geometry

The functor $i : \mathcal{E} \to \mathcal{G}_B$ preserves finite limits, images, and the NNO stagewise. Between constant objects it preserves exponentials:

$$(iY)^{iX} \cong i(Y^X).$$

Indeed a natural map $h_C \times iX \to iY$ is determined by its value on (id_C, x); naturality along $u : C \to C'$ forces the value on (u, x) to be the same. This proves the isomorphism in every parameter context.

Proposition 7.15 (Positive observations factor through reduction)**.** *The canonical finite-hit Rosolini truth-value object of $\mathcal{G}_B$ is*

$$\Sigma_{\mathcal{G}_B} \cong i\Sigma.$$

It is a dominance. Every positive computational test has the unique factorization

$$u : Z \to \Sigma_{\mathcal{G}_B}, \qquad u = i(\Gamma u)\eta_Z : Z \to i\Gamma Z \to i\Sigma.$$

Consequently its pullbacks along the two projections of $Z \times_{\Im_W Z} Z$ are equal.

Proof.

(1) **Construct the geometric positive truth-value object.** Constant-subobject classification gives a monomorphism $i\Omega_{\mathcal{E}} \to \Omega_{\mathcal{G}_B}$: at each stage, evaluation of the classified natural family on its identity arrow recovers the original truth value. The preceding exponential calculation gives $(i2)^{i\mathbb{N}} \cong i(2^{\mathbb{N}})$. The finite-hit predicate is the same natural-number existential at every stage; its classifier is the constant finite-hit classifier followed by this monomorphism. Images are stagewise, so its image is $i\Sigma$.

(2) **Factor positive observations through reduction.** The adjunction $\Gamma \dashv i$ identifies a map $Z \to i\Sigma$ with its component $\Gamma Z \to \Sigma$ and gives the displayed factorization. A subobject so classified is the pullback along η_Z of its reduced classified subobject.

(3) **Verify dominance and formal-disk invariance.** For two composable such subobjects, left exactness of Γ identifies the reduced composite; dominance in $\mathcal{E}$ classifies it, and pullback along η_Z recovers the original composite. Truth and pullback stability follow by the same argument. Thus $i\Sigma$ is a dominance. The two formal-disk projections have the same composite with η_Z, proving the last assertion.

□

For $j_B M$, reduction is augmentation $C \otimes_B M \to M$, since $\Gamma j_B M = M$. Canonical positive tests therefore observe the reduced value and are constant along its formal disk. Infinitesimal coefficients are instead read by the coefficient calculations below.

Computational names, natural numbers, and denotations are constant objects of this geometry. The coefficient calculation in Theorem 7.16 also expresses their infinitesimal discreteness as $(iX)^{D_A} \cong iX$. Resource values lie in the nonconstant modules $j_B M$. These two roles allow an implementation witness to stay fixed while its resource constraint varies infinitesimally.

7.5.3 Weil coefficient identity and comparison of stages

Theorem 7.16 (Exact realization and Weil coefficients)**.** *The functor $j_B : \mathrm{Mod}_B(\mathcal{E}) \to \mathrm{Mod}_{\mathcal{R}_B}(\mathcal{G}_B)$ is exact, fully faithful, strong symmetric monoidal, and reflects exactness and isomorphisms. For every $F \in \mathcal{G}_B$ and projective stage A,*

$$F^{D_A}(C) \cong F(C \otimes_B A),$$

and therefore

$$(j_B M)^{D_A}(C) \cong (C \otimes_B A) \otimes_B M \cong (C \otimes_B M) \otimes_B A. \tag{7.9}$$

All comparisons are natural in the stages, their maps, and the module, and commute with augmentation and Weil substitution.

Proof.

(1) **Compute exponentials by Weil disks.** Writing $h_C = \mathrm{Hom}_{B\text{-alg}}(C, -)$, tensor coproduct and Yoneda give

$$h_C \times D_A \cong h_{C \otimes_B A}, \qquad F^{D_A}(C) = \mathrm{Nat}(h_C \times D_A, F) \cong F(C \otimes_B A).$$

This also proves the displayed coefficient formula.

(2) **Prove exactness of realization.** Every stage is finite projective and hence flat, so each $C \otimes_B -$ is exact, making j_B exact stagewise.

(3) **Prove full faithfulness and reflection.** A natural $\mathcal{R}_B$-linear map $j_B M \to j_B N$ is determined by its component at B: naturality along $B \to C$ determines its values on $1 \otimes m$, and C-linearity determines all $c \otimes m$. Conversely any B-linear map gives such a natural transformation by scalar extension, proving full faithfulness. Evaluation at B reflects exactness and isomorphisms.

(4) **Verify strong monoidality and coherence.** The associativity of balanced tensor gives $j_B(M \otimes_B N) \cong j_B M \otimes_{\mathcal{R}_B} j_B N$ and $j_B B \cong \mathcal{R}_B$; uniqueness on elementary tensors proves unit, symmetry, and associativity coherence.

□

Proposition 7.17 (Restriction to rational Weil probes)**.** *The full inclusion $k : \mathsf{W}_B^{\mathrm{rat}} \hookrightarrow \mathsf{W}_B^{\mathrm{proj}}$ induces restriction $k^* : \mathcal{G}_B \to [\mathsf{W}_B^{\mathrm{rat}}, \mathcal{E}]$ with coherent isomorphisms*

$$k^* j_B^{\mathrm{proj}} M \cong j_B^{\mathrm{rat}} M, \qquad k^* D_{C_W}^{\mathrm{proj}} \cong D_{C_W}^{\mathrm{rat}}, \qquad k^* i^{\mathrm{proj}} X \cong i^{\mathrm{rat}} X.$$

It preserves the rational Weil coefficient and exponential identities and commutes with evaluation at B and the resulting infinitesimal-shape operation.

Proof.

(1) **Compare restriction and Weil exponentiation.** All three functors have the displayed stagewise formulas. For rational stages C_W, C_V, their tensor coproduct $C_{W\otimes V}$ is again rational; applying the exponential calculation of Theorem 7.16 before or after restriction therefore gives the same expression $F(C_W \otimes_B C_V)$.

(2) **Compare reduction and verify coherence.** Evaluation at B commutes with restriction, and constant functors do so stagewise. These identities give the stated shape comparison, while equality of the elementary-tensor formulas gives all coherence diagrams.

□

7.5.4 Finite Kock–Lawvere identities and microlinearity

Definition 7.18 (Microlinearity for a specified class of Weil diagrams)**.** An object Y is **microlinear** for a specified class of finite augmented Weil diagrams when the functor $A \mapsto (Y^{D_A} \to Y)$ preserves their limits in the slice over Y. The arrow to Y is induced by augmentation. The diagram class is stated explicitly when rational stages and general finite-projective stages are compared.

For $A = B[\varepsilon]/\varepsilon^2$, put $D = D_A$. Yoneda identifies $D \subseteq \mathcal{R}_B$ with the square-zero elements. Formula (7.9) gives

$$\mathcal{R}_B^D \cong \mathcal{R}_B \times \mathcal{R}_B, \qquad (j_B M)^D \cong j_B M \times j_B M,$$

where (a, b) corresponds to the map $d \mapsto a + db$. The same formula for every finite-projective probe is the finite Weil–Kock–Lawvere coefficient identity (Kock 2006).

Theorem 7.19 (Rational and projective microlinearity)**.** *For every B-module M, the augmented coefficient functor*

$$W \longmapsto ((j_B M)^{D_{C_W}} \to j_B M)$$

preserves finite limits of rational Weil diagrams with rational structure maps. For every embedded resource module $M \subseteq K$, the functor

$$A \longmapsto ((j_B M)^{D_A} \to j_B M)$$

preserves finite limits of the full finite-projective Weil-stage category. In both statements the codomain is $\mathcal{G}_B/j_B M$ and the displayed maps are induced by augmentation.

Proof.

(1) **Preserve rational Weil limits.** Finite limits of augmented rational Weil algebras are computed by products over $\mathbb{Q}$ and kernels of linear differences. Gaussian elimination splits their finite-dimensional exact sequences, so tensoring with any $\mathbb{Q}$-module preserves them. By Theorem 7.16, the stage-C coefficient functor is $W \mapsto (C \otimes_B M) \otimes_{\mathbb{Q}} W$, augmented to $C \otimes_B M$. This proves the rational assertion.

(2) **Preserve projective Weil limits for resources.** For an embedded resource M, Theorem 2.24 makes M flat; the finite-projective stage C is also flat. Hence $T = C \otimes_B M$ is flat. The coefficient functor $A \mapsto T \otimes_B A$ preserves the underlying module fibre products and kernels computing the augmented Weil limits of Theorem 7.11. These stagewise comparisons are natural and preserve augmentation, giving the isomorphisms in $\mathcal{G}_B/j_B M$.

□

Proposition 7.20 (Failure for arbitrary coefficient modules)**.** *For the numerical cofinite ring B and regular nonunit $t(n) = 1/(n+1)$, the augmented coefficient functor of Theorem 7.19 need not preserve projective Weil equalizers for $M = B/tB$.*

Proof.

(1) **Compute the original equalizer.** On $A = B[\varepsilon]/\varepsilon^2$, take $u(\varepsilon) = t\varepsilon$ and $v(\varepsilon) = 0$. Regularity of t gives $\mathrm{Eq}(u, v) = B$.

(2) **Exhibit failure of the equalizer comparison.** After tensoring with $M = B/tB$, the two maps coincide and their equalizer becomes $M \oplus M\varepsilon$, whereas tensoring the original equalizer gives only M. The two are not isomorphic as the indicated augmented equalizer comparison: its image is the constant-coefficient summand and misses ε. The embedded resources of Theorem 7.19 do not have this obstruction because they are flat.

□

At $C = B[\varepsilon]/\varepsilon^2$ the generic infinitesimal ε is nonzero: coefficient extraction would otherwise force $1 = 0$ in B, contradicting properness of the filter. Thus the model has nontrivial infinitesimal directions.

7.5.5 Coefficient change and reduced-base cohesion

Finite Weil coefficient realization commutes with a ring map $\beta : B \to B'$ on rational probes: for any rational Weil W,

$$(B' \otimes_{\mathbb{Q}} W) \otimes_{B'} (B' \otimes_B M) \cong B' \otimes_B ((B \otimes_{\mathbb{Q}} W) \otimes_B M).$$

Both sides send an elementary coefficient–Weil–module tensor to the same expression; the unit and substitution diagrams follow from tensor coherence. The adjoint quadruple of Theorem 7.14 has the further property that the coefficient stage is terminal, which is a consequence of reducedness rather than of tensor coherence alone.

Proposition 7.21 (A coefficient reduction producing nilpotents)**.** *In the cofinite numerical fibre, let $t(n) = 1/(n+1)$ and $z(n) = 1/(1 + \lfloor\sqrt{n+1}\rfloor)$. Then $z \notin tB$ but $z^2 \in tB$. Thus $B' = B/tB$ has a nonzero square-zero element, and its coefficient stage is not terminal in the category of B'-Weil stages.*

Proof.

(1) **Construct a nonzero square-zero coefficient.** Write $m = \lfloor\sqrt{n+1}\rfloor$. The inequalities

$$\frac{z(n)^2}{t(n)} = \frac{n+1}{(1+m)^2} \le 1, \qquad \frac{z(n)}{t(n)} = \frac{n+1}{1+m} \ge \frac{m^2}{1+m}$$

show that z^2/t is bounded but z/t is unbounded along square inputs. Hence the class $\bar{z} = z + tB$ is nonzero and satisfies $\bar{z}^{\,2} = 0$.

(2) **Show failure of terminality.** The two B'-algebra maps

$$B'[\varepsilon]/\varepsilon^2 \rightrightarrows B', \qquad \varepsilon \longmapsto 0, \quad \varepsilon \longmapsto \bar{z}$$

are distinct, so B' is not terminal among those stages. This explicitly separates coefficient-tensor comparisons from the reduced-base cohesion calculation.

□

7.5.6 Explicit linear adjoints

Let $\mathscr{M}_B = \mathrm{Mod}_{\mathcal{R}_B}(\mathcal{G}_B)$ and $\Gamma_{\mathrm{lin}}(F) = F(B)$ with its B-module structure.

Theorem 7.22 (Linear realization adjoints)**.** *There is an adjoint triple*

$$\Lambda_{\mathrm{lin}} \dashv j_B \dashv \Gamma_{\mathrm{lin}}, \qquad \Lambda_{\mathrm{lin}} j_B \cong \mathrm{id} \cong \Gamma_{\mathrm{lin}} j_B.$$

Proof.

(1) **Construct the right-adjoint correspondence.** A B-linear $v : M \to F(B)$ extends at C by $c \otimes m \mapsto cF(B \to C)(v(m))$. Naturality and linearity prove it is inverse to evaluation, giving $j_B \dashv \Gamma_{\mathrm{lin}}$.

(2) **Construct the coend candidate.** For the left adjoint put $C^\vee = \mathrm{Hom}_B(C, B)$ with right C-action $(\phi c)(x) = \phi(cx)$, and construct

$$\Lambda_{\mathrm{lin}} F = \mathrm{coeq}\left(\bigoplus_{u : C \to C'} (C')^\vee \otimes_C F(C) \rightrightarrows \bigoplus_C C^\vee \otimes_C F(C) \right). \tag{7.10}$$

(3) **Define the balanced coequalizer maps.** One map sends $\phi \otimes x$ to $(\phi u) \otimes x$ in the C-summand; the other to $\phi \otimes F(u)x$ in the C'-summand. The right C-action on $(C')^\vee$ uses u, so both maps are balanced. The indexing objects are the object and arrow objects of the internal stage category; internal free modules and quotients construct these sums and this coequalizer.

(4) **Establish the stagewise duality correspondence.** Finite projectivity gives

$$\mathrm{Hom}_C(F(C), C \otimes_B M) \cong \mathrm{Hom}_B(C^\vee \otimes_C F(C), M).$$

It sends v to $\phi \otimes x \mapsto (\phi \otimes 1)v(x)$. For a dual basis (c_i, ϕ_i) the inverse sends h to $x \mapsto \sum_i c_i \otimes h(\phi_i \otimes x)$. The dual-basis identities and balancing prove the inverse equations and C-linearity.

(5) **Prove the left adjunction and composite identities.** Naturality of the family v is exactly equalization of the two maps in (7.10). The coequalizer universal property proves $\Lambda_{\mathrm{lin}} \dashv j_B$. Full faithfulness of j_B gives the two composite identities.

□

The coefficient model now lets us express resource operations as geometric maps. We first identify the coordinate algebra of a finite-projective module. Its dual appears because coordinates are linear functions on that module. Universal differentials then describe the first-order change of those functions.

7.6 Affine resource coordinates and universal differentials

7.6.1 Dual coordinate modules

Proposition 7.23 (Finite-cell differential comparison)**.** *Let M be a finite projective B-module and set $M^\vee = \mathrm{Hom}_B(M, B)$. Its affine coordinate algebra is $\mathrm{Sym}_B(M^\vee)$, and for every B-algebra C,*

$$\mathrm{Hom}_{B\text{-}alg}(\mathrm{Sym}_B(M^\vee), C) \cong \mathrm{Hom}_B(M^\vee, C) \cong C \otimes_B M.$$

Under this identification, the universal derivation gives the coefficient of ε in evaluation on $C[\varepsilon]/\varepsilon^2$.

Proof.

(1) **Establish the finite-duality bijections.** The first bijection is the free-algebra universal property. For a finite dual basis (m_i, λ_i), the inverse of the second sends a functional $f : M^\vee \to C$ to $\sum_i f(\lambda_i) \otimes m_i$; the dual-basis identity proves both inverse equations and independence of the presentation.

(2) **Extract the dual-number derivative.** A polynomial coordinate p satisfies

$$p(x + \varepsilon v) = p(x) + \varepsilon\, dp_x(v)$$

because multiplying factors chooses either no ε or exactly one ε. This is precisely formula (7.6). Addition and relations preserve the equality. Thus the induced map of Kähler modules, transposed to tangent vectors, agrees with the dual-number coefficient.

□

The support isomorphism $Bg \cong Be_g$ makes each principal resource a displayed finite-projective cell.

For a general resource $M = \bigcup_g Bg$, a finite tuple of inputs lies in a common finite projective cell. A finite polynomial resource operation lands in the finite principal cell generated by the sum of the absolute values of its finitely many coefficient-budget products. The comparison above applies on each such cell. Cell inclusions commute with evaluation and derivation by naturality, so these comparisons assemble over the directed resource presentation. This proves the derivative comparison for resource arithmetic and polynomial overhead cells on this directed presentation.

Theorem 7.24 (Canonical polynomial lifting to Weil geometry)**.** *Let E, N be finite-projective B-modules. A polynomial resource map $p : E \to N$, specified by its coordinate-algebra map*

$$p^* : \mathrm{Sym}_B(N^\vee) \longrightarrow \mathrm{Sym}_B(E^\vee),$$

induces canonically a geometric map $\widehat{p} : j_B E \to j_B N$. It preserves identities and polynomial composition, and

$$D\widehat{p}(x, v) = dp_x(v).$$

These maps and derivatives commute with finite-projective resource cell inclusions whenever the polynomial expressions and their coefficient budgets are transported by those inclusions.

Proof.

(1) **Construct the stagewise polynomial map.** By Proposition 7.23, $C \otimes_B E$ and $C \otimes_B N$ are the C-points of their symmetric coordinate algebras. Precomposition with p^* gives $\widehat{p}_C$, natural in the Weil stage. Composition of algebra maps proves identity and composition preservation.

(2) **Identify the derivative.** The dual-number comparison of Proposition 7.23 identifies the coefficient of ε in $p(x + \varepsilon v)$ with $dp_x(v)$. Finite duality detects equality in the target module, giving $D\widehat{p}(x, v) = dp_x(v)$.

(3) **Compare resource cells.** A finite polynomial expression uses finitely many arguments and coefficient budgets. Their absolute sum supplies a common finite-projective principal cell by Theorem 2.22. Naturality of coordinate evaluation and of the differential comparison proves compatibility with the stated cell inclusions.

□

Proposition 7.25 (Finite polynomial laws on arbitrary modules)**.** *For B-modules H, N, a finite family of B-multilinear maps $m_r : H^{\otimes_B r} \to N$ for $0 \le r \le d$ (with $m_0 \in N$) constructs the polynomial law $p(x) = \sum_{r=0}^{d} m_r(x, \dots, x)$ and a canonical natural geometric lift $\widehat{p} : j_B H \to j_B N$, satisfying*

$$D\widehat{p}(x, v) = \sum_{r=1}^{d} \sum_{j=1}^{r} \widetilde{m}_r(x, \dots, x, \underset{j}{v}, x, \dots, x),$$

where $\widetilde{m}_r$ is the scalar-extended multilinear map at the generalized Weil stage. On finite-projective modules this agrees with the coordinate-algebra construction of Theorem 7.24.

Proof.

(1) **Extend the multilinear coefficients to Weil stages.** For each Weil stage C, balanced tensor extension of m_r produces a C-multilinear map $(C \otimes_B H)^{\otimes_C r} \to C \otimes_B N$. Evaluating the finite sum diagonally defines $\widehat{p}_C : C \otimes_B H \to C \otimes_B N$.

(2) **Verify naturality and extract the derivative.** Tensor functoriality makes these maps natural in every B-algebra stage morphism. Evaluating at $x + \varepsilon v$, the square-zero multiplication removes all terms with two or more occurrences of ε. The terms with exactly one occurrence give the displayed formula, by coefficient uniqueness.

(3) **Compare with the finite-duality construction.** For finite projective H, N, both stagewise constructions are polynomial evaluations with the same coordinate values, so finite duality identifies them.

□

An arbitrary numerical value family also has a canonical, but different, realization. The constant-object/evaluation adjunction of Theorem 7.14 gives

$$\mathcal{G}_B(iX, j_B K) \cong \mathcal{E}(X, K), \qquad f \longmapsto (x \mapsto 1 \otimes f(x)).$$

The source iX is infinitesimally discrete: $(iX)^D \cong iX$. This realization therefore preserves the values of an arbitrary numerical family without differentiating its discrete input. Polynomial resource variables have a different role: the lifting theorem realizes their polynomial maps geometrically and identifies their derivatives.

7.6.2 Relative differential modules

For a commutative B-algebra A, Proposition 7.7 constructs $\Omega^1_{A/B}$ by generators and relations and identifies its maps with B-derivations, equivalently with algebra sections of $A \ltimes N \to A$. For a finite projective coordinate cell this gives

$$\Omega^1_{\mathrm{Sym}_B(M^\vee)/B} \cong \mathrm{Sym}_B(M^\vee) \otimes_B M^\vee.$$

Dual bases identify its evaluation pairing with the tangent coefficient of Proposition 7.23.

The absolute module $\Omega^1_{B/\mathbb{Q}}$ is obtained by fixing only the rational constants in Proposition 7.7. These constructions retain their coefficient maps. For instance every relative derivation over B kills a scale $t \in B$, whereas its absolute differential dt is the specified generator in $\Omega^1_{B/\mathbb{Q}}$.

Localization is compatible with relative differentials:

$$\Omega^1_{A_s/B} \cong A_s \otimes_A \Omega^1_{A/B}.$$

To prove this, a derivation extends uniquely by $d(s^{-1}) = -s^{-2}ds$ and the quotient rule $d(a/s^r) = s^{-r}da - ras^{-r-1}ds$. Clearing denominators verifies independence of the fraction, and the Leibniz law verifies the extension. Representability proves the isomorphism. The gluing equalizer of Proposition 3.23 therefore also glues these differential modules and their maps.

7.7 Finite tangent calculus

Proposition 7.26 (Differentiation in the realized geometry)**.** *Every map $F : j_B M \to j_B N$ in $\mathcal{G}_B$ has a unique first-order expansion*

$$F(x + \varepsilon v) = F(x) + \varepsilon\, DF(x, v).$$

Its derivative is linear in v, obeys the chain rule, and agrees with Proposition 7.23 on polynomial resource cells.

Proof.

(1) **Obtain existence and uniqueness of the derivative.** The dual-number coefficient identity gives existence and uniqueness.

(2) **Prove additivity in the tangent direction.** For addition in v, use the Weil algebra with two generators $\varepsilon_1, \varepsilon_2$ and all their pairwise products zero. Apply F to $x+\varepsilon_1 v+\varepsilon_2 w$. Projections to each dual-number axis identify its two linear coefficients as $DF(x, v)$ and $DF(x, w)$. The algebra map sending both ε_1 and ε_2 to ε then proves additivity.

(3) **Prove scalar linearity.** The stage map $\varepsilon \mapsto a\varepsilon$ proves scalar linearity, including generalized coefficients $a \in C$.

(4) **Prove the chain rule.** Composing two expansions proves the chain rule by uniqueness of coefficients.

(5) **Compare universal derivations and geometric derivatives.** The finite polynomial calculation in Proposition 7.23 gives the comparison.

(6) **Obtain higher symmetry and Taylor expansions.** Iterated probes and permutation of their generators give the higher symmetry identities; truncated probes give finite Taylor expansions with the rational factorials available.

□

For a linear map L, its coefficient formula is $DL(x, v) = L(v)$. For addition and multiplication in the resource line it gives

$$D(+)((x, y), (v, w)) = v + w, \qquad D(\cdot)((x, y), (v, w)) = yv + xw.$$

For a composite the chain rule is $D(GF)(x, v) = DG(Fx, DF(x, v))$. The tangent functor is $T_{\tan}X = X^D$ with projection induced by $0 : 1 \to D$. On realized modules it is the usual pair of base point and direction. Its zero section, addition of directions, and scalar action are the coefficient maps just constructed.

Iterated tangent uses $\mathbb{Q}[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2)$. Its coefficient of $\varepsilon_1\varepsilon_2$ is invariant under interchanging the two generators. This proves symmetry of the second base-point derivative. Iteration and adjacent transpositions prove symmetry of every higher base-point derivative, with directions held fixed. These are equations of natural maps, so include generalized coefficients at every stage.

We can now apply the differential calculus to resource relations. The multiplier descriptions of O, o, Θ, and $\sim$ use multiplier witnesses. Realizing those multiplication maps first retains the witnesses needed to differentiate the constraints.

7.8 Realization of the Landau incidences

Multiplication makes $j_B K$ an $\mathcal{R}_B$-algebra. Write $U = \mathcal{R}_B^\times$ and $\mathcal{M} = j_B\mathfrak{m}$. The subobject $1 + \mathcal{M}$ is a subgroup of U. To see this at $C = B \oplus N$, augment $a \in C \otimes_B \mathfrak{m}$ to $a_0 \in \mathfrak{m}$. Then $1 + a_0$ is a unit of B, while $a - a_0$ lies in the nilpotent augmentation ideal. A finite geometric inverse makes $1 + a$ a unit of C, and $(1 + a)^{-1} - 1 = -a(1 + a)^{-1}$ lies in the same extended ideal.

Definition 7.27 (Geometric Landau relations)**.** The **geometric Landau relations** are the images of

$$\begin{aligned}
\mathcal{R}_B \times j_B K &\longrightarrow j_B K \times j_B K, \quad (b, g) \longmapsto (bg, g),\\
\mathcal{M} \times j_B K &\longrightarrow j_B K \times j_B K, \quad (a, g) \longmapsto (ag, g),\\
U \times j_B K &\longrightarrow j_B K \times j_B K, \quad (u, g) \longmapsto (ug, g),\\
\mathcal{M} \times j_B K &\longrightarrow j_B K \times j_B K, \quad (a, g) \longmapsto ((1 + a)g, g).
\end{aligned}$$

They realize respectively $O, o, \Theta, \sim$. Reversing the first two gives the realizations of Ω, ω. These formulas specify the geometric relations also on nonconstant points.

Theorem 7.28 (Recovery of Landau relations)**.** *Pullback of each of these six geometric relations along $i(K \times K) \to j_B K \times j_B K$ is the corresponding constant Landau relation. This statement is uniform in computational parameters.*

Proof.

(1) **Extend multiplier witnesses to all Weil stages.** A multiplier, unit, or small multiplier extends to every stage and witnesses the forward inclusion by Theorem 2.20.

(2) **Reflect membership through augmentation.** Conversely, augment a stagewise witness to B. Augmentation takes an arbitrary scalar to a bounded coefficient, a unit to a unit, and an element of $C \otimes_B \mathfrak{m}$ to an element of $\mathfrak{m}$. Its equation becomes the original equation on f, g.

(3) **Identify incidence and the reversed comparisons.** Thus membership is reflected. Images and the resulting pullbacks are computed stagewise. Reversal gives the two lower-bound cases.

□

Differentiating a witnessed incidence varies both its value and its multiplier. The resulting equations describe how the resource relation is preserved to first order; taking its image recovers the corresponding relation on values.

7.9 Differentials of the Landau incidences

Proposition 7.29 (Tangents of multiplier orbits)**.** *For a reference germ g, let $\widehat{\Theta}_g$ be the image of $U = \mathcal{R}_B^\times \to j_B K$, $u \mapsto ug$, and put $\widehat{\sim}_g = g + j_B(\mathfrak{m}g)$. At the common point g,*

$$T_g\widehat{\Theta}_g \cong j_B(Bg), \qquad T_g\widehat{\sim}_g \cong j_B(\mathfrak{m}g). \tag{7.11}$$

Proof.

(1) **Extract a tangent direction from a unit witness.** A tangent in the first image at stage C is a point over $C[\varepsilon]/(\varepsilon^2)$ reducing to g. By the regular-image presentation, after a regular-epimorphic refinement it has a unit witness $u = u_0 + \varepsilon u_1$. Its value is $ug = u_0 g + \varepsilon u_1 g$. Reduction gives $u_0 g = g$, and coefficient extraction places the direction $u_1 g$ in $C \otimes_B Bg$.

(2) **Descend the direction through the witness cover.** This membership descends because $C \otimes_B Bg \hookrightarrow C \otimes_B K$ is monic and subobject inclusions are reflected by regular-epimorphic covers.

(3) **Realize every supported tangent direction.** Conversely, for every $b \in C$, the element $1+\varepsilon b$ is a unit with inverse $1-\varepsilon b$ and has orbit value $g+\varepsilon bg$. Thus the tangent directions at every stage are exactly $C \otimes_B Bg$, naturally in C.

(4) **Compute the small-multiplier tangent.** For the second locus, translate $j_B(\mathfrak{m}g)$ by g and use the dual-number coefficient identity; its directions are exactly $C \otimes_B \mathfrak{m}g$. The construction and inverse are natural in the stage and reflect equalities of tangent coefficients.

□

Lower-bound witnesses also have a differential equation. In the incidence $g = af$, with a a bounded multiplier, the tangent equation is

$$\delta g = a\,\delta f + f\,\delta a.$$

For the strict lower-bound incidence the multiplier and its variation lie in $j_B\mathfrak{m}$. This equation is the derivative of the witnessed multiplication constraint, whose image realizes the lower-bound relation. Thus the multiplier witness is retained while the differential equation is formed.

7.10 The universal differential resource family

Theorem 7.30 (Universal differential resource family)**.** *The functor $j_{\bar{B}} : \mathrm{Mod}_{\bar{B}}(\mathscr{S}) \to \mathrm{Mod}_{\mathcal{R}_{\bar{B}}}(\mathcal{G}_{\mathfrak{E}_{\mathrm{Ev}}})$, given at a stage A by $A \otimes_{\bar{B}} -$, is exact, fully faithful, and strong symmetric monoidal. The module $\mathfrak{M} = j_{\bar{B}}\mathscr{M}$ is microlinear for all finite-projective Weil diagrams. At every generalized ideal point I, with stage codes reindexed along inverse image,*

$$I^*\mathfrak{M} \cong j_B M_I. \tag{7.12}$$

Moreover $\mathfrak{M}^D \cong \mathfrak{M} \times \mathfrak{M}$. For a polynomial resource map $F : j_{\bar{B}}\bar{H} \to j_{\bar{B}}\bar{K}$, the universal constrained locus $S_F = F^{-1}(\mathfrak{M})$ has tangent

$$T_x S_F = \ker(j_{\bar{B}}(q)DF_x), \qquad q : \bar{K} \to \bar{K}/\mathscr{M}. \tag{7.13}$$

These comparisons specialize to the corresponding budget fibres.

Proof.

(1) **Realize the universal family and prove microlinearity.** The Yoneda and finite-dual-basis proof of Theorem 7.16 is internal and applies in $\mathscr{S}$, giving the functorial properties and the Weil coefficient identities. The flatness of $\mathscr{M}$ from Theorem 3.6 lets the proof of Theorem 7.19 apply to the full projective stages.

(2) **Compute ideal fibres.** Under Theorem 7.13, inverse image at I gives

$$\varinjlim_{a \in I}(A \otimes_B M_a) \cong A \otimes_B \varinjlim_{a \in I} M_a \cong A \otimes_B M_I.$$

(3) **Verify parameter and stage coherence.** For a locally varying stage code, the same isomorphism follows by reindexing its finite matrices; inverse image preserves the idempotent equations and the balanced tensor presentation. Both comparison paths along a stage map send $c \otimes m$ to the same substituted elementary tensor. This proves the parameter and stage coherence of (7.12).

(4) **Compute tangent coefficients and the normal kernel.** The dual-number basis gives the exponential formula. Finally, exactness identifies $\mathfrak{M}$ with the kernel of $j_{\bar{B}}(q)$; expanding $qF(x+\varepsilon v)$ at a feasible point gives $\varepsilon qDF_x(v)$. Coefficient uniqueness proves (7.13). Inverse image preserves this kernel and its finite coefficient calculation.

□

7.11 Conservative realization of measured feasibility

Realization preserves the measured incidence and its feasibility image.

Theorem 7.31 (Conservative computational comparison)**.** *Cost incidence and its Weil realization classify the same measured implementations. At a budget I, the natural map $iK \to j_B K$ sends f to $1 \otimes f$, and*

$$iP \times_{j_B K} j_B M_I \cong iZ_I, \qquad \operatorname{im}(iZ_I \to iV) \cong i\mathcal{C}_{\mathrm{cost}}(I).$$

These comparisons also hold relatively over $\mathscr{S}$ for $i_W \mathscr{Z} \cong i_W \Delta P \times_{j_{\bar{B}} \bar{K}} \mathfrak{M}$.

Proof.

(1) **Preserve and reflect resource membership.** The inclusion $K^+ \hookrightarrow K$ retains each measured cost and its membership in the marked submodule $M_I \hookrightarrow K$. At a Weil stage A, membership of $1 \otimes \Phi(e)$ in $A \otimes_B M_I$ follows from membership in M_I. Conversely augmentation $A \to B$ recovers $\Phi(e) \in M_I$ from a stage witness.

(2) **Identify the incidence subobject and its image.** Flatness of A makes the scalar-extended marking monic. Thus the incidence is the same constant subobject at each stage; images are stagewise. The identical calculation at each budget a proves the relative assertion.

□

Equivalently, the realization square

$$\begin{array}{ccc} iM_I & \xrightarrow{1\otimes -} & j_B M_I \\ \downarrow & \lrcorner & \downarrow \\ iK & \xrightarrow[1\otimes -]{} & j_B K \end{array} \tag{7.14}$$

is cartesian.

Corollary 7.32 (Preservation of implemented feasibility and optimality)**.** *Formal principal parts and de Rham coefficient constructions preserve the original implementation diagram's feasible opens in $\mathfrak{C}_{\mathrm{Ev}}$ and their compactness and supercompactness. Thus the optimal-witness and finite-portfolio equivalences of Section 5 apply with the same implementation fibres and computational environments.*

Proof.

(1) **Recover measured incidence and feasibility by reduction.** The coefficient constructions act on the resource-coordinate variable. Reduction recovers the measured incidence by Theorem 7.31; its image and the lattice of opens of $\mathfrak{C}_{\mathrm{Ev}}$ are unchanged.

□

The comparison uses the split Weil extension. Arbitrary coefficient changes behave differently: tensoring $B \hookrightarrow K$ with B/tB gives a nonzero source and zero target.

The finite calculus now differentiates the measured resource families and their budget constraints. We next classify variations of the embedded budgets and calculate the successive feasibility obstructions.

8 Deformations of resource-constrained computation

Ordinary budget families have already been classified by $\mathfrak{C}_{\mathrm{Ev}}$. We now ask how an embedded budget can vary over a nilpotent coefficient extension. This is a different moduli problem. The earlier point object $\mathrm{Pt}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Idl}(L)$ has constant realization, with $(i\,\mathrm{Idl}(L))^D \cong i\,\mathrm{Idl}(L)$ by Section 7.5; it carries no infinitesimal variation. We therefore define deformations directly by extending the coefficient ring and varying the marked submodule of the extended fraction algebra. Tangents and lifting obstructions will describe this functor.

8.1 The marked deformation functor

For a Weil stage $A = B \oplus \mathfrak{n}_A$ put $K_A = K \otimes_B A$ and $\mathfrak{n}_{K,A} = K \otimes_B \mathfrak{n}_A$. The split augmentation gives this kernel formula. For an idempotent e, the multiplicative group $e + e\mathfrak{n}_{K,A}$ is formed in the corner ring eK_A; inverses are finite geometric series.

Definition 8.1 (The principal embedded-resource deformation functor)**.** Fix $M = Bg \hookrightarrow K$ and its support $e = e_g$. The **principal embedded-resource deformation functor** assigns to a finite-projective Weil stage A the object $\mathrm{Def}^{\mathrm{pr}}_M(A)$ of marked images $Ax \hookrightarrow K_A$ with $x \in eK_A$ reducing to g. Generators determine the same point precisely when their embedded images agree. An augmented Weil map $A \to A'$ sends this point to $A'x' \hookrightarrow K_{A'}$, where x' is the substituted generator.

To verify functoriality, write $x = g(e + z)$ for a unique $z \in e\mathfrak{n}_{K,A}$, since $gg^\dagger = e$. The multiplier $e + z$ is invertible in the corner, and the finite-projective stage A is flat over B. Multiplication by x therefore identifies the displayed image with eA. If $Ax = Ay$, each generator is a scalar multiple of the other; substitution preserves these two equations, hence preserves equality of images. Identity and composition hold on generators and descend to the image quotient. Tight replacement of g gives the same marked functor, as proved in Corollary 8.5.

The functor thus classifies nilpotent deformations of an embedded budget, while $h_{\mathfrak{C}_{\mathrm{Ev}}}(Z)$ classifies ordinary budget families over locales. We first calculate the deformation functor at first order, then use that tangent module to describe its formal classifying disk.

8.2 First-order deformations of finite-projective cells

At first order, a deformation changes the embedding of a module while reducing to its original marking. Let $A_1 = B[\varepsilon]/(\varepsilon^2)$ and $K_1 = K \otimes_B A_1 \cong K[\varepsilon]/(\varepsilon^2)$. For a displayed finite-projective $i_M : M \hookrightarrow K$, a framed deformation is an A_1-linear monomorphism $\widetilde{i} : M \otimes_B A_1 \hookrightarrow K_1$ reducing to i_M. The frame is the displayed source $M \otimes_B A_1$. Forgetting it identifies monomorphisms with the same embedded image. The theorem calculates this image quotient.

Theorem 8.2 (The embedded resource tangent module)**.** *Every framed first-order embedded deformation has the unique form*

$$i_h(m + \varepsilon n) = m + \varepsilon(n + h(m)), \qquad h \in \mathrm{Hom}_B(M, K). \tag{8.1}$$

Two deformations have the same image if and only if $h - h'$ factors through $M \hookrightarrow K$. Consequently the object of image classes is canonically

$$T_M^{\mathrm{emb}} \cong \mathrm{Hom}_B(M,K)/\mathrm{Hom}_B(M,M) \cong \mathrm{Hom}_B(M,K/M). \tag{8.2}$$

For $M = Bg \cong Be_g$ this reduces to

$$T_{Bg}^{\mathrm{emb}} \cong e_gK/Bg; \qquad T_B^{\mathrm{emb}} \cong K/B. \tag{8.3}$$

These isomorphisms are natural under marked isomorphisms of finite-projective resource modules and substitution of parameters.

Proof.

(1) **Classify framed deformations.** In $A_1 = B \oplus \varepsilon B$, an A_1-linear map reducing to i_M has the unique form (8.1): its value on m determines $h(m)$ and its value on εn is forced. If its output is zero, the constant coefficient gives $m = 0$ and then the ε-coefficient gives $n = 0$; hence it is monic.

(2) **Identify changes of framing.** Equal images give an A_1-linear automorphism of $M \otimes_B A_1$ reducing to the identity. It is necessarily $1 + \varepsilon u$ for a B-linear $u : M \to M$, with inverse $1 - \varepsilon u$. The equation $i_{h'} = i_h(1 + \varepsilon u)$ is $h' - h = i_M u$. Conversely this equation supplies the image-preserving automorphism, proving the first quotient in (8.2).

(3) **Compute the quotient.** The displayed finite-free retract makes $\mathrm{Hom}_B(M,-) \cong M^\vee \otimes_B -$ exact. Applying it to $0 \to M \to K \to K/M \to 0$ identifies the quotient with $\mathrm{Hom}_B(M,K/M)$.

(4) **Specialize and verify naturality.** For $M \cong Be_g$, evaluation at e_g gives $\mathrm{Hom}_B(M,K/M) \cong e_g(K/M) = e_gK/Bg$. These descriptions use module operations and quotient universal properties, so commute with marked isomorphisms and parameters.

□

Change of frame and its effective quotient. The tangent calculation can be organized by the changes of frame that leave the image fixed. Write $T_M = T_M^{\mathrm{emb}} \cong \mathrm{Hom}_B(M,K/M)$, $\mathcal{F}_M = \mathrm{Hom}_B(M,K)$, and $\mathcal{G}_M = \mathrm{End}_B(M)$. The additive action is $u \cdot h = h + i_M u$. Its compatibility with the embedding is

$$\begin{array}{ccc} M \otimes_B A_1 & \xrightarrow[\cong]{1+\varepsilon u} & M \otimes_B A_1 \\ {\scriptstyle i_{h'}}\downarrow & & \downarrow{\scriptstyle i_h} \\ K_1 & = & K_1. \end{array} \qquad h' = h + i_M u.$$

The action is free since i_M is monic. Its effective quotient is the marked-image object, presented by the coequalizer

$$\mathcal{G}_M \times \mathcal{F}_M \underset{\pi_2}{\overset{\mathrm{act}}{\rightrightarrows}} \mathcal{F}_M \twoheadrightarrow T_M. \tag{8.4}$$

Exactness of $\mathrm{Hom}_B(M,-)$ gives the short exact sequence

$$0 \longrightarrow \mathrm{End}_B(M) \xrightarrow{i_M \circ -} \mathrm{Hom}_B(M,K) \xrightarrow{q \circ -} \mathrm{Hom}_B(M,K/M) \longrightarrow 0.$$

The cokernel therefore removes exactly the infinitesimal changes of frame. Monicity of the marking makes the action free, and the effective quotient records its image.

For a principal resource $M = Bg$, the first-order image classes in Definition 8.1 coincide with those just computed. A frame sends g to $g + \varepsilon h(g)$; B-linearity makes $h(g)$ supported on e_g. Conversely each supported generator reducing to g gives a frame on $M \otimes_B A_1 \cong e_g A_1$. Equality of images is the same equivalence in both descriptions. Thus (8.3) is the first-order value of the already defined deformation functor.

For cofinite numerical input the class of $t^{-1} = n + 1$ is nonzero in K/B; the deformation

$$B[\varepsilon](1 + \varepsilon t^{-1}) \subset K[\varepsilon]$$

therefore differs from the constant deformation $B[\varepsilon] \subset K[\varepsilon]$ while reducing to the same embedded module $B \subset K$. This exhibits a nonzero moduli tangent already on the rank-one resource cell.

8.3 Formal resource cells and their classifying disks

The first-order quotient identifies the intrinsic tangent module. For a principal resource, finite logarithm and exponential extend this description to every nilpotence order. We now construct the quotient and representing formal disk of Definition 8.1.

Theorem 8.3 (The formal classifier of a principal resource cell)**.** *For $M = Bg \hookrightarrow K$, with $e = e_g$, the image classes of principal A-submodules generated by an element reducing to g and supported in eK_A form the functor*

$$\mathrm{Def}_M^{\mathrm{pr}}(A) = \frac{e + e\mathfrak{n}_{K,A}}{e + e\mathfrak{n}_A}, \qquad [u] \longmapsto A(gu) \subseteq K_A. \tag{8.5}$$

Put $T_M = \mathrm{Hom}_B(M, K/M)$. There are canonical natural isomorphisms

$$eK/eB \xrightarrow{\cong} T_M, \qquad \mathrm{Def}_M^{\mathrm{pr}}(A) \cong \mathfrak{n}_A \otimes_B T_M. \tag{8.6}$$

The first map sends a multiplier class $[z]$ to $(m \mapsto zm \bmod M)$. With $0 : 1 \to iT_M$, the formal disk

$$\mathfrak{D}_M = 1 \times_{iT_M} j_B T_M \tag{8.7}$$

represents the functor: its value at A, equivalently the internal object of maps $D_A \to \mathfrak{D}_M$, is $\mathrm{Def}_M^{\mathrm{pr}}(A)$.

Proof.

(1) **Classify supported deformations by unit cosets.** The support inverse $g^\dagger = e/g$ satisfies $gg^\dagger = e$. Thus every supported generator reducing to g is uniquely gu with $u \in e + e\mathfrak{n}_{K,A}$. Multiplication by gu embeds eA in eK_A: the inclusion $A \hookrightarrow K_A$ is monic since A is a finite-free retract and regular denominators act injectively on it. Two such images agree precisely when $u^{-1}v$ is a unit of eA reducing to e, namely an element of $e + e\mathfrak{n}_A$. This proves the quotient classification and its functoriality.

(2) **Construct finite logarithm and exponential.** Choose the displayed bound r with $\mathfrak{n}_A^{r+1} = 0$. The ideal $e\mathfrak{n}_{K,A}$ has the same nilpotence bound. For x in that ideal define in the corner

$$\log(e + x) = \sum_{k=1}^{r} \frac{(-1)^{k+1}}{k} x^k, \qquad \exp(x) = e + \sum_{k=1}^{r} \frac{x^k}{k!}.$$

(3) **Identify the multiplicative and additive quotients.** The rational polynomial identities, reduced modulo terms of degree greater than r, prove that these are inverse and that logarithm takes products to sums. Increasing the displayed bound changes neither map. They carry the subgroup over eA onto $e\mathfrak{n}_A$, so they induce

$$\mathrm{Def}_M^{\mathrm{pr}}(A) \cong (eK \otimes_B \mathfrak{n}_A)/(eB \otimes_B \mathfrak{n}_A) \cong \mathfrak{n}_A \otimes_B (eK/eB).$$

(4) **Identify the tangent module intrinsically.** Here $\mathfrak{n}_A$ is a displayed finite-free retract, hence flat. Multiplication by g identifies eK/eB with eK/Bg; the dual-basis calculation of Theorem 8.2 identifies this with T_M. Its intrinsic formula is the multiplier map in the statement, so it is independent of the chosen generator.

(5) **Identify the representing formal disk.** Finally, augmentation on $A \otimes_B T_M$ has kernel $\mathfrak{n}_A \otimes_B T_M$ by the split decomposition of A. This calculates (8.7); the internal Yoneda lemma gives its representing property. All polynomial identities and quotients commute with Weil maps and parameters.

□

The representing disk is the fibre over zero of augmentation:

$$\begin{array}{ccc} \mathfrak{D}_M & \longrightarrow & j_B T_M \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \mathrm{aug}} \\ 1 & \xrightarrow[0]{} & iT_M. \end{array}$$

At a Weil stage A, the right-hand map is induced by $A \to B$.

At every stage the logarithm is an isomorphism of the quotient presentations. In the corner supported on e, it gives

$$\begin{array}{ccccc} e + e\mathfrak{n}_A & \hookrightarrow & e + e\mathfrak{n}_{K,A} & \twoheadrightarrow & \mathrm{Def}_M^{\mathrm{pr}}(A) \\ \cong\downarrow{\scriptstyle \log} & & \cong\downarrow{\scriptstyle \log} & & \downarrow\cong \\ e\mathfrak{n}_A & \hookrightarrow & e\mathfrak{n}_{K,A} & \twoheadrightarrow & \mathfrak{n}_A \otimes_B T_M. \end{array} \tag{8.8}$$

The upper row is a quotient of multiplicative groups and the lower row is its additive cokernel presentation. Naturality of the finite logarithm and exponential makes this a diagram of Weil-stage functors.

Corollary 8.4 (Universal images and finite lifts)**.** *For $z \in \mathfrak{n}_A \otimes_B T_M$, choose a representative $\widetilde{z} \in e\mathfrak{n}_{K,A}$ locally along the quotient epimorphism. The image $A(g \exp \widetilde{z})$ is independent of that representative and constructs the universal marked family over $\mathfrak{D}_M$. Surjective Weil-stage maps induce internal epimorphisms on these deformation objects. For $A_r = B[d]/d^{r+1}$ and $M = B$,*

$$\mathrm{Def}_B^{\mathrm{pr}}(A_r) \cong \frac{1 + dK[d]/d^{r+1}}{1 + dB[d]/d^{r+1}} \cong \bigoplus_{j=1}^{r} d^j(K/B). \tag{8.9}$$

Reduction to order $r-1$ has kernel K/B.

Proof.

(1) **Descend the universal image through the logarithmic quotient.** Changing $\widetilde{z}$ by $e\mathfrak{n}_A$ multiplies its exponential by a unit of eA, hence preserves the image. These equal images descend across the quotient epi and commute with stage maps, giving a submodule of K_A over the formal classifier.

(2) **Construct lifts along surjective stage maps.** A surjective augmented algebra map is surjective on augmentation ideals: subtract the augmentation of a lift. Tensoring preserves that epimorphism. The last assertion follows by the displayed direct-sum description.

□

Corollary 8.5 (Invariance under tight replacement)**.** *If $g' \in \Theta(g)$, then $Bg' = Bg$ as marked modules and their formal classifiers, universal images, and tangent comparisons are canonically identical.*

Proof.

(1) **Identify the deformation classifier under tight replacement.** Write $g' = vg$ with $v \in B^\times$ by Theorem 2.20. The support is unchanged and $A(g'u) = A(gu)$. The multiplier description of T_M and logarithm identify the same quotient and representing disk at every stage.

□

Logarithm and exponential use finite rational arithmetic at the displayed nilpotence bound, so they act on the coefficient presentations. Subgroup membership and equality of image classes remain the associated quotient predicates, with their separately specified observation structure. A deformation of a resource need not preserve a particular resource constraint. We now vary a polynomial resource value and its budget together. At first order, the change of the value modulo the budget must match the change of the budget at that value.

8.4 Deformation of incidence with a moving budget

Let $M \hookrightarrow K$ be a displayed finite-projective resource cell, $q : K \to Q = K/M$, and let $F : j_B H \to j_B K$ be one of the polynomial resource maps constructed in Theorem 7.24 and Proposition 7.25. A reduced feasible point $x : 1 \to H$ has $F(x) \in M$. The displayed generators of M supply the principal generator $g = \sum_i |i_M r(e_i)|$, where $i_M : M \hookrightarrow K$ is the marking, $r : B^n \to M$ is its displayed retraction, and (e_i) is the standard basis. Thus $M = Bg$ by bounded division, and the preceding formal classifier applies. Its independence of the generator was proved in Corollary 8.5.

Definition 8.6 (Incidence deformations)**.** The **incidence deformation functor** is

$$\mathsf{Inc}_{F,x,M}(A) = \left\{ (x_A, N_A) : \begin{array}{ll} x_A \in A \otimes_B H, & (\epsilon_A \otimes 1)x_A = x, \\ N_A \in \mathrm{Def}_M^{\mathrm{pr}}(A), & F_A(x_A) \in N_A \end{array} \right\}. \tag{8.10}$$

Here $\epsilon_A : A \to B$ is the augmentation and $N_A \hookrightarrow K_A$ is the universal marked image of Corollary 8.4. Weil substitution acts on both entries and preserves membership by naturality of F and of that image. This constructs a subfunctor of the formal disk of $j_B H$ at x times $\mathfrak{D}_M$. Source frames are forgotten by equality of embedded images. Its first-order object is denoted $T_{(x,M)}\mathsf{Inc}_F := \mathsf{Inc}_{F,x,M}(B[\varepsilon]/\varepsilon^2)$.

Theorem 8.7 (The tangent of resource incidence)**.** *The first-order incidence deformations are the kernel of the B-linear map*

$$H \oplus T_M \longrightarrow Q, \qquad (v, \theta) \longmapsto qDF_x(v) - \theta(F(x)).$$

Equivalently,

$$T_{(x,M)}\mathsf{Inc}_F = \{(v, \theta) : qDF_x(v) = \theta(F(x))\}. \tag{8.11}$$

For a fixed budget this gives

$$T_x F^{-1}(j_B M) = \ker(qDF_x). \tag{8.12}$$

Proof.

(1) **Represent the varying point and budget.** A first-order point is $x+\varepsilon v$ and its value is $F(x)+\varepsilon DF_x(v)$ by the Weil coefficient identity. By Theorem 8.2, a budget deformation is locally represented by $h : M \to K$, with image of $m + \varepsilon n \mapsto m + \varepsilon(n + h(m))$ and class $\theta = qh$.

(2) **Derive the incidence equation in the quotient.** Membership of the value is exactly $DF_x(v) - h(F(x)) \in M$. Passing to Q gives (8.11).

(3) **Recover membership and descend the calculation.** Conversely that equation supplies the unique element of M representing the displayed difference, and hence the required membership. Replacing h by a map with the same class changes neither condition. The calculations therefore descend from the quotient cover, naturally in all parameters. The fixed-budget slice is $\theta = 0$.

□

The incidence deformation complexes. The same linearized map governs first-order solutions and higher-order corrections. We record the fixed and moving versions as two-term complexes of B-modules in degrees $0, 1$:

$$\begin{aligned} \mathsf{C}^{\mathrm{fix}}_{F,x,M} &= [\, H \xrightarrow{qDF_x} Q \,], \\ \mathsf{C}^{\mathrm{mov}}_{F,x,M} &= [\, H \oplus T_M \xrightarrow{L^{\mathrm{mov}}_{(x,M)}} Q \,], \\ L^{\mathrm{mov}}_{(x,M)}(v, \theta) &= qDF_x(v) - \theta(F(x)). \end{aligned} \tag{8.13}$$

In each complex, H^0 is the kernel and H^1 the cokernel. The kernel gives the first-order incidence deformations. At higher orders, a residual can be cancelled precisely when its class in the cokernel vanishes; the lift theorems below then identify the inhabited lift object as a torsor under H^0. Allowing the budget to vary gives the comparison

$$\begin{array}{ccc} H & \xrightarrow{qDF_x} & Q \\ {\scriptstyle v \mapsto (v,0)}\big\downarrow & & \big\| \\ H \oplus T_M & \xrightarrow[L^{\mathrm{mov}}_{(x,M)}]{} & Q. \end{array} \tag{8.14}$$

This compares the tangent and obstruction modules for the two problems: the vertical inclusion allows the additional budget direction while preserving the value residual.

Proposition 8.8 (Normal directions for a fixed marked budget)**.** *For any embedded resource M and geometric valuation family $F : j_B H \to j_B K$, the tangent of $S_F = F^{-1}(j_B M)$ at a feasible point is* $\ker(j_B(q)DF_x)$.

Proof.

(1) **Realize the fixed resource as a kernel.** Exactness of j_B identifies $j_B M$ with the kernel of $j_B(q)$. Exponentiation by the dual-number object preserves this pullback.

(2) **Compute the tangent kernel at a feasible point.** Its coefficient formula gives $(qF(x), qDF_x(v))$, whose zero fibre at a feasible point is the asserted kernel. This proves (8.12) for arbitrary embedded budgets.

□

Corollary 8.9 (Rigidity of a tight budget containing a fixed cost)**.** *For a fixed implemented cost $c \in K$, the only principal formal deformation of Bc containing the constant cost c is the constant deformation. Its first-order version is $\{\theta \in T_{Bc} : \theta(c) = 0\} = 0$.*

Proof.

(1) **Translate containment of the fixed cost into a unit equation.** Write $e = e_c$. A deformation is $A(cu)$ for $u \in e + e\mathfrak{n}_{K,A}$. If $c \in A(cu)$, multiplication by $c^\dagger$ yields $e = au$ for some $a \in eA$.

(2) **Prove triviality of the deformation.** Thus $a = u^{-1}$ reduces to e and is a nilpotent perturbation of the unit in eA. Its finite geometric inverse belongs to eA, so $u \in e + e\mathfrak{n}_A$ and its class is trivial.

(3) **Deduce first-order rigidity.** At first order, c generates Bc, so a linear map vanishing on c is zero.

□

8.5 Higher-order feasibility and lifting obstructions

A feasible deformation through order r need not extend to order $r+1$. Its lower-order coefficients determine the next failure of the constraint, and a new coefficient changes that failure through the linearized incidence map. We calculate whether this residual can be cancelled.

Definition 8.10 (Fixed-budget feasible jets and residuals)**.** Use the fixed budget M and polynomial F above, and put $L_x = qDF_x : H \to Q$. An **order-r feasible jet** is

$$x_r = x + \sum_{j=1}^{r} d^j v_j \quad \text{over } B[d]/d^{r+1}, \qquad qF(x_r) = 0.$$

View the same displayed polynomial over $B[d]/d^{r+2}$ and define

$$R_{r+1} = [d^{r+1}]qF(x_r) \in Q.$$

The coefficients v_j are uniquely determined by the jet, so this is its **order-$(r+1)$ residual**.

Theorem 8.11 (The obstruction and the object of lifts)**.** *The obstruction to lifting x_r by one order is*

$$\mathrm{ob}_{r+1}(x_r) = [R_{r+1}]_{\operatorname{coker} L_x} \in \operatorname{coker} L_x. \tag{8.15}$$

It vanishes exactly when the internal object of feasible lifts is inhabited. That object is the fibre $\{w : H : L_x(w) = -R_{r+1}\}$; when inhabited, it is a torsor under $\ker L_x$. The construction is natural in the polynomial data and reduced parameters, and applies relatively over the structured resource space.

Proof.

(1) **Compute the next-order correction equation.** A lift has the unique form $x_r+d^{r+1}w$. The polynomial Taylor identity gives, modulo d^{r+2},

$$qF(x_r + d^{r+1}w) = d^{r+1}(R_{r+1} + L_x(w)).$$

Indeed terms quadratic in the added correction have degree at least $r + 2$, and the positive-degree part of $DF_{x_r} - DF_x$ also has that property after multiplication by d^{r+1}. For $r = 0$, the same identity is the dual-number formula.

(2) **Identify lifts with the linear fibre.** Equality of polynomial coefficients proves the fibre assertion.

(3) **Characterize existence by the obstruction class.** The kernel of $Q \to \operatorname{coker} L_x$ is the regular image of L_x, so zero obstruction means exactly internal existence of w.

(4) **Identify the lift torsor and parameter stability.** Two lifts differ by a unique element of $\ker L_x$, and addition by such an element preserves a lift. This proves the torsor assertion. All maps are coefficient operations, kernels, and cokernels; their internal constructions and identities apply in $\mathscr{S}$ as well.

□

Example 8.12 (The second-order obstruction of the measured family)**.** For the family $F(z) = 1 + t^{-2}z^2$ of Example 7.2, take $H = B$, $M = B$, and $x = 0$. Here $L_0 = qDF_0 = 0$, and the first-order jet $z_1 = d$ is feasible. Its next residual is $R_2 = [t^{-2}]_{K/B} \in K/B$, which is nonzero because $(n + 1)^2$ is unbounded. Every lift $d + d^2w$ has the same square modulo d^3, so no parameter correction removes the obstruction. Under the coordinate isomorphism $B \to t^{-1}B$, $z \mapsto b = t^{-1}z$, this is equivalently the family $1 + b^2$ with first-order direction t^{-1}.

8.6 Complete principal-resource moduli

The principal-resource classifier of Theorem 8.3 applies at every finite nilpotence order. We now assemble its image classes into a compatible formal family. The classification is taken at the level of those classes, so compatible generators need not be chosen.

Theorem 8.13 (Completed principal deformation functor)**.** *For a principal marked resource $M = Bg \hookrightarrow K$ put $T_M = \operatorname{Hom}_B(M, K/M)$ and $A_r = B[d]/d^{r+1}$. Then*

$$\widehat{\operatorname{Def}}^{\mathrm{pr}}_M := \varprojlim_r \operatorname{Def}^{\mathrm{pr}}_M(A_r) \;\cong\; d\,T_M[[d]] = \prod_{j\geq 1} d^j T_M. \tag{8.16}$$

The transition maps are projections onto initial coefficient segments. Every finite-order class admits a compatible extension to all higher orders, and the displayed comparison is natural in marked principal-resource isomorphisms and parameters.

Proof.

(1) **Compute each finite deformation stage.** The augmentation ideal of A_r is the displayed free module $\mathfrak{n}_r = dB \oplus \cdots \oplus d^r B$. Applying the finite logarithm classification of Theorem 8.3 yields

$$\operatorname{Def}^{\mathrm{pr}}_M(A_r) \cong \mathfrak{n}_r \otimes_B T_M \cong \bigoplus_{1\leq j\leq r} d^j T_M.$$

(2) **Identify the inverse limit with coefficient sequences.** Its naturality in Weil-stage maps identifies truncation with deletion of the highest coefficient. A point of the inverse limit is therefore exactly a compatible finite-prefix family of T_M-coefficients, equivalently an internal sequence $(z_j)_{j\geq 1}$. This gives (8.16).

(3) **Construct an extension of every finite prefix.** Extend a given coefficient prefix by zero coefficients to obtain an explicit compatible sequence of deformation *classes*.

□

Principal-resource image deformations are formally unobstructed by coefficientwise zero extension. The following calculation describes the quotient that supplies their coefficients.

Proposition 8.14 (B-linear nonsplitting of the normal quotient)**.** *In the cofinite numerical fibre, the quotient $q : K \twoheadrightarrow K/B$ does not admit a B-linear section.*

Proof.

(1) **Exhibit nonzero torsion in the normal quotient.** Let $t(n) = 1/(n+1) \in B$. It is regular in B and invertible in K, while $t^{-1} = n + 1 \notin B$. Therefore $\xi = q(t^{-1}) \in K/B$ is nonzero but $t\xi = q(1) = 0$.

(2) **Derive a contradiction from a linear section.** If a B-linear section $s : K/B \to K$ existed, then

$$t\, s(\xi) = s(t\xi) = 0.$$

Multiplication by t is invertible on K, so $s(\xi) = 0$, contradicting $q(s(\xi)) = \xi \neq 0$.

□

We now combine these two calculations. The budget itself admits formal extension, but its extension must still be compatible with the resource value. Allowing a budget correction enlarges the linearized incidence operator and gives the moving-budget obstruction.

8.7 Higher-order incidence with a moving principal budget

Let H be a displayed finite-projective B-module, let $F : H \to K$ be a specified polynomial resource map, and let $x \in H$ satisfy $F(x) \in M = Bg$. Write

$$Q = K/M, \quad q : K \twoheadrightarrow Q, \quad T_M = \mathrm{Hom}_B(M, Q).$$

The first-order moving-incidence operator of Section 8 is

$$L^{\mathrm{mov}}_{(x,M)} : H \oplus T_M \longrightarrow Q, \qquad (v, \theta) \longmapsto qDF_x(v) - \theta(F(x)). \tag{8.17}$$

This map is intrinsic to the marking $M \hookrightarrow K$.

Definition 8.15 (Feasible moving formal jet)**.** At order r, a **moving formal jet** is a valuation coefficient $x_r = x + \sum_{j=1}^{r} d^j v_j \in H \otimes_B A_r$ together with a principal deformation image $M_r \in \mathrm{Def}^{\mathrm{pr}}_M(A_r)$. It is **feasible** when the image of $F(x_r)$ in $(K \otimes_B A_r)/M_r$ is zero. The object of these pairs is formed as a pullback of the corresponding polynomial evaluation and marked-incidence subobjects. All its equalities are $\Omega_{\mathcal{E}}$-valued.

Theorem 8.16 (Moving-budget lifting obstruction)**.** *For a feasible moving jet (x_r, M_r), there is a canonical class*

$$\mathrm{ob}^{\mathrm{mov}}_{r+1}(x_r, M_r) \in \mathrm{coker}\, L^{\mathrm{mov}}_{(x,M)}. \tag{8.18}$$

Its vanishing is equivalent to internal inhabitation of the object of feasible lifts to order $r+1$. When inhabited, that object is a torsor under $\ker L^{\mathrm{mov}}_{(x,M)}$. The constructions are natural in polynomial resource maps, marked principal modules, and all supplied parameters. The fixed-budget theorem Theorem 8.11 is the specialization in which the budget tangent component is zero.

Proof.

(1) **Write the local feasibility equation.** Put $e = e_g$. By Theorem 8.3, locally along the representative epimorphism the budget has a logarithm $z_r(d) = \sum_{j=1}^{r} d^j z_j \in eK \otimes_B dA_r$. The ambient unit

$$U_r = (1-e) + \exp_e(z_r), \qquad U_r^{-1} = (1-e) + \exp_e(-z_r)$$

gives $M_r = U_r(M \otimes_B A_r)$ and feasibility $q(U_r^{-1}F(x_r)) = 0 \pmod{d^{r+1}}$. Zero-extend z_r, x_r to order $r+1$, evaluate the exponential there, and set

$$R_{r+1} = [d^{r+1}]q(U_r^{-1}F(x_r)).$$

(2) **Descend the residual.** Two representatives differ by $b(d) \in deB[d]/d^{r+1}$. After zero extension their ambient units satisfy

$$U_r' = U_r h, \qquad h = (1-e) + \exp_e(b) \in (B[d]/d^{r+2})^{\times}, \quad h(0) = 1.$$

The residual series is multiplied by h^{-1}. Its coefficients through degree r vanish in Q, so its next coefficient is unchanged. This kernel-pair equality descends R_{r+1} through the effective quotient.

(3) **Compute the correction equation.** A lift adds $d^{r+1}v$ to x_r and a budget coefficient $\theta \in T_M$. Locally represent the latter by $\zeta \in eK$, with $\theta(m) = q(\zeta m)$. Modulo d^{r+2} the Taylor formula gives

$$F(x_r + d^{r+1}v) = F(x_r) + d^{r+1}DF_x(v), \qquad U_{r+1}^{-1} = U_r^{-1}(1 - d^{r+1}\zeta).$$

Thus the next feasibility coefficient is

$$R_{r+1} + qDF_x(v) - \theta(F(x)) = R_{r+1} + L^{\mathrm{mov}}_{(x,M)}(v,\theta).$$

Changing ζ by eB leaves it unchanged because $F(x) \in M$.

(4) **Identify the obstruction and lift torsor.** The lift object is the fibre of $L^{\mathrm{mov}}_{(x,M)}$ at $-R_{r+1}$. As in Theorem 8.11, it is inhabited precisely when the cokernel class of R_{r+1} vanishes; subtraction and addition identify it as a torsor under the kernel. Finite coefficient operations and quotient descent give parameter naturality. For the constant budget $U_r = 1$, restricting to $\theta = 0$ recovers the fixed-budget calculation.

□

Corollary 8.17 (Comparison of fixed and moving obstructions)**.** *For a fixed-budget feasible jet, regarded as a moving jet with constant budget, the morphism (8.14) carries its fixed obstruction to its moving obstruction. Explicitly the comparison square is*

$$\begin{array}{ccc} Q & \twoheadrightarrow & H^1(\mathsf{C}^{\mathrm{fix}}_{F,x,M}) \\ \| & & \downarrow \\ Q & \twoheadrightarrow & H^1(\mathsf{C}^{\mathrm{mov}}_{F,x,M}). \end{array}$$

Proof.

(1) **Identify the residual for a constant budget.** The constant budget is represented by $U_r = 1$. Both residual constructions therefore give $[d^{r+1}]qF(x_r)$ in Q.

(2) **Construct the comparison of obstruction classes.** The inclusion $\mathrm{im}(qDF_x) \subseteq \mathrm{im}(L^{\mathrm{mov}}_{(x,M)})$ induces the displayed map of cokernels and proves the assertion.

□

Example 8.18 (The measured family with a moving budget)**.** For Examples 7.2 and 8.12, choose

$$M_2 = (1 + d^2t^{-2})B[d]/d^3.$$

Then $F(d) = 1 + d^2t^{-2} \in M_2$. Here $T_B = K/B$ and $L^{\mathrm{mov}}_{(0,B)}(v, \theta) = -\theta(1)$ is surjective, so the moving obstruction vanishes. The displayed generator supplies the feasible second-order budget deformation.

The measured linear budget. Multiplication by the full-support germ $g = t^{-1}$ transports this example to the budget of the doubling program in Section 13.4. In fact

$$K/B \xrightarrow{\cong} K/Bg, \quad [a] \longmapsto [ga], \qquad F_g(b) = g(1 + b^2).$$

The map is well-defined and invertible because g is a unit of K, and it carries B onto Bg. Thus the fixed second-order residual is $[gt^{-2}]$, and the moving budget $(1 + d^2t^{-2})Bg[d]/d^3$ contains $F_g(dt^{-1})$.

8.8 Infinitesimal compatibility of budget diagrams

Proposition 8.19 (The tangent of a resource flag)**.** *For displayed finite-projective embedded cells $M_0 \subseteq M_1$, write $q_{01} : K/M_0 \to K/M_1$. The first-order deformations of this flag are the pairs*

$$(\theta_0, \theta_1) \in \mathrm{Hom}_B(M_0, K/M_0) \oplus \mathrm{Hom}_B(M_1, K/M_1) \quad \textit{with} \quad q_{01}\theta_0 = \theta_1|_{M_0}.$$

For a finite diagram, impose this equation on every inclusion.

Proof.

(1) **Derive the first-order inclusion equation.** Locally lift the classes to $h_i : M_i \to K$ using the specified finite dual bases. The deformed image of M_0 lies in that of M_1 exactly when $h_0(m) - h_1(m) \in M_1$ for every $m \in M_0$. This is the displayed equation.

(2) **Recover the inclusion and descend the condition.** Its converse constructs the inclusion by the unique correction in M_1; changes of lifts preserve it. The condition therefore descends and is necessary and sufficient.

(3) **Extend the calculation to finite diagrams.** Imposing it for finitely many edges gives the diagram statement, with commutativity following from the common ambient marking. This is the formal incidence problem lying over the classifier of Theorem 3.12, not the tangent of its infinitesimally discrete constant point object.

□

8.9 Complete finite budget-diagram equations

A system of resource constraints may include inclusions between several budgets. Deforming it requires those inclusions to persist. For a finite diagram $M_i \subseteq M_j$ of principal marked resources, write $M_i = Bg_i$, put $e_i = e_{g_i}$ and choose local supported logarithms $z_i(d)$. Use the ambient units

$$U_i(d) = (1 - e_i) + \exp_{e_i}(z_i(d)) \in K[d]/d^{r+2}, \qquad U_i(0) = 1.$$

Their products with M_i represent the deformations. Extending by $1 - e_i$ makes every inverse below an inverse in the common ambient coefficient algebra. An edge remains an inclusion precisely when

$$q_j(U_j(d)^{-1}U_i(d)m) = 0 \quad (m \in M_i), \qquad q_j : K \to K/M_j.$$

It is enough to test a displayed finite family of generators. The coefficient at first order is the existing tangent flag equation $q_{ij}\theta_i = \theta_j|_{M_i}$. At degree $r+1$ the new coefficients occur through this same linear operator, while all lower coefficients determine an explicit residual.

Intrinsically the finite diagram has the two-term complex

$$\mathsf{C}^{\bullet}_{\mathrm{flag}} = \left[\bigoplus_i T_{M_i} \xrightarrow{\partial_{\mathrm{flag}}} \prod_{(i \to j)} \mathrm{Hom}_B(M_i, K/M_j) \right], \tag{8.19}$$

where $\partial_{\mathrm{flag}}(\theta)_{ij} = q_{ij}\theta_i - \theta_j|_{M_i}$. The products and sums here are finite. Each target is an intrinsic edge quotient module; a displayed generator identifies it with the corresponding supported quotient used in the coefficient calculation. The vanishing of one edge component is the commutative square

$$\begin{array}{ccc} M_i & \hookrightarrow & M_j \\ {\scriptstyle\theta_i}\downarrow & & \downarrow{\scriptstyle\theta_j} \\ K/M_i & \xrightarrow[q_{ij}]{} & K/M_j. \end{array}$$

Use this differential as L_{flag} below. The degree-$(r+1)$ residual on an edge is the B-linear map $m \mapsto [d^{r+1}]q_j(U_j^{-1}U_i m)$, so it has precisely the target in (8.19). Changes of next-order coefficients act through ∂_{flag}. Thus H^0 organizes compatible tangent directions and H^1 the finite-diagram obstructions.

Proposition 8.20 (Formal obstruction for a finite budget flag)**.** *A finite diagram of principal embedded-resource inclusions has a linear first-order compatibility map L_{flag} into the finite product of its edge quotient modules. Every compatible order-r moving diagram has a canonical order-$(r+1)$ obstruction in* $\operatorname{coker} L_{\mathrm{flag}}$*. Its vanishing is internal inhabitation of the compatible lift object; when inhabited that object is a torsor under* $\ker L_{\mathrm{flag}}$*. Adding polynomial valuation incidences enlarges the same finite operator by their first-order rows and gives the analogous obstruction theorem.*

Proof.

(1) **Extract the residual on each inclusion edge.** For each inclusion edge, conjugate the deformed inclusion by the target supported unit as displayed above. Its expression has zero coefficients through degree r. Extract the next coefficient.

(2) **Prove independence of representatives.** Changing logarithmic representatives gives $U_i' = U_i h_i$, where h_i is an ambient $B[d]$-unit with constant term 1 and preserves M_i. For an edge $i \to j$, commutativity of the coefficient algebra gives

$$q_j((U_j')^{-1}U_i' m) = h_j^{-1} h_i \, q_j(U_j^{-1} U_i m).$$

The quotient is a $B[d]$-module, and the original expression has zero coefficients through order r. Since $(h_j^{-1}h_i)(0) = 1$, the order-$(r+1)$ residual is unchanged. This proves descent on both the source and target representative kernel pairs.

(3) **Identify the flag correction map.** New coefficients of source and target units enter linearly and give precisely the first-order flag-tangent map.

(4) **Assemble the obstruction and lift torsor.** The finite graph of edges allows all residual coefficients and maps to be assembled into one finite product. The lift equation is therefore $L_{\mathrm{flag}}(w) = -R_{r+1}$. Its image, cokernel, and kernel supply the obstruction, inhabited fibre, and torsor statements exactly as in the preceding theorem. Adding valuation rows uses the same polynomial Taylor calculation.

□

The next example combines two measured incidences with a budget inclusion. Its obstruction will then be compared under changes of the allowed overheads.

Example 8.21 (An obstruction to jointly moving two budgets)**.** In the cofinite numerical fibre put $t(n) = 1/(n+1)$ and take

$$M_0 = B \subset M_1 = t^{-1}B, \qquad F_0(z) = 1 + t^{-2}z^2, \qquad F_1(z) = t^{-1}.$$

The first valuation is Example 7.2; the second is measured by Algorithm 8.1, assigning unit weight to the selected idle event.

Algorithm 8.1 The measured linear resource in the coupled-budget example

Input: $n \in \mathbb{N}$
1: **for** $j = 1, \dots, n+1$ **do**
2: IDLE
3: **end for**
4: **return** n

At order one, $z = d$ and the two constant budgets satisfy both incidences and their inclusion.

Each incidence separately extends to order two: the inner budget may be $(1 + d^2t^{-2})B[d]/d^3$, while the outer budget remains constant. For a joint lift, put $Q_i = K/M_i$, let $q_{01} : Q_0 \to Q_1$, and write $a = \theta_0(1) \in Q_0$, $b = \theta_1(t^{-1}) \in Q_1$ for the new budget coefficients. A parameter correction $d + d^2v$ has square d^2 modulo d^3. Consequently the two incidence equations and the inclusion equation require

$$a = [t^{-2}]_{Q_0}, \qquad b = 0, \qquad q_{01}a - tb = 0.$$

The last equation uses $\theta_1(1) = t\theta_1(t^{-1})$. It would imply $t^{-2} \in t^{-1}B$, contrary to the unboundedness of t^{-1}. Thus no joint lift exists.

The full obstruction calculation identifies its intrinsic quotient. The correction operator is

$$L : B \oplus Q_0 \oplus Q_1 \longrightarrow Q_0 \oplus Q_1 \oplus Q_1, \qquad L(v, a, b) = (-a, -b, q_{01}a - tb).$$

The surjection $(x, y, z) \mapsto q_{01}x - ty + z$ has kernel $\operatorname{im} L$: if its value is zero, the triple is $L(0, -x, -y)$. Hence $\operatorname{coker} L \cong Q_1$, and the residual $([t^{-2}]_{Q_0}, 0, 0)$ represents

$$o = [t^{-2}]_{K/t^{-1}B} \in K/t^{-1}B, \qquad \operatorname{Ann}_B(o) = tB. \tag{8.20}$$

Indeed $bo = 0$ means $bt^{-2} \in t^{-1}B$, equivalently $b \in tB$. The obstruction records compatibility of the budget system even though both individual moving-incidence problems are solvable.

8.10 Gauge change of infinitesimal resources

Proposition 8.22 (Localization of finite-cell deformation tangents)**.** *For a multiplicative overhead ideal P containing* [1] *and a finite-projective resource module $M \subseteq K$, localization induces a canonical isomorphism*

$$B_P \otimes_B T_M^{\mathrm{emb}} \cong \operatorname{Hom}_{B_P}(B_P \otimes_B M,\, K/B_P M). \tag{8.21}$$

For $M = B$ it sends the tangent quotient K/B to K/B_P. For a principal cell, finite logarithm equips its formal-deformation objects with their canonical additive module structures. With $C = B_P$, $A_C = C \otimes_B A$, and $M_C = C \otimes_B M$, there is a natural isomorphism

$$C \otimes_B \operatorname{Def}_M^{\mathrm{pr}}(A) \cong \operatorname{Def}_{M_C}^{\mathrm{pr}}(A_C). \tag{8.22}$$

It is induced by coefficient substitution on the defining unit-group quotients.

Proof.

(1) **Localize the ambient module and normal quotient.** The coefficient ring B_P is a localization of B at regular denominators, and K is already the total quotient ring. Hence $B_P \otimes_B K \cong K$. Flat localization sends the quotient K/M to $K/B_P M$.

(2) **Commute tangent Hom objects with localization.** A finite-projective module has a finite dual basis, so its internal hom commutes with scalar extension:

$$B_P \otimes_B \operatorname{Hom}_B(M, K/M) \cong \operatorname{Hom}_{B_P}(B_P \otimes_B M, K/B_P M).$$

Apply Theorem 8.2 to obtain (8.21).

(3) **Identify the localized formal-deformation objects.** The split augmentation gives $\mathfrak{n}_{A_C} = C \otimes_B \mathfrak{n}_A$. The tangent isomorphism above gives $T_{M_C} \cong C \otimes_B T_M$. The finite logarithm of Theorem 8.3 identifies each unit-group quotient with its additive module, so

$$\begin{aligned} \operatorname{Def}_{M_C}^{\mathrm{pr}}(A_C) &\cong \mathfrak{n}_{A_C} \otimes_C T_{M_C} \\ &\cong (C \otimes_B \mathfrak{n}_A) \otimes_C (C \otimes_B T_M) \\ &\cong C \otimes_B (\mathfrak{n}_A \otimes_B T_M) \\ &\cong C \otimes_B \operatorname{Def}_M^{\mathrm{pr}}(A). \end{aligned}$$

These are (8.22). Finite logarithm and exponential are rational polynomials on nilpotent ideals; they commute with coefficient substitution and every augmented Weil map. Hence the isomorphism is induced by the original unit-group quotients and is natural in stages and parameters.

□

Theorem 8.23 (Overhead localization of resource obstructions)**.** *For the fixed, moving, or finite-diagram incidence complexes constructed above, write* $C^\bullet = [E^0 \xrightarrow{L} E^1]$*. Let* $B_P = S_P^{-1}B$ *be an overhead localization from Theorem 2.37. There are natural identifications*

$$C_P^\bullet \cong B_P \otimes_B C^\bullet, \qquad H^i(C_P^\bullet) \cong B_P \otimes_B H^i(C^\bullet) \quad (i = 0, 1).$$

At every finite order, the obstruction o *of a feasible jet maps to* $o_P = 1 \otimes o$*, and*

$$o_P = 0 \iff \exists s \in S_P \; so = 0 \iff S_P \cap \mathrm{Ann}_B(o) \text{ is inhabited.} \tag{8.23}$$

The existential and the lift objects are interpreted internally.

Proof.

(1) **Compare the localized deformation complexes.** Localization identifies $B_P \otimes_B K$ with K and carries K/M to K/B_PM. Every Hom object in the complexes has a displayed finite-projective source. Its finite dual basis therefore gives the scalar-extension identification of Proposition 8.22. The finite sums, edge products, derivatives, and evaluation maps in L commute with these identifications, proving the complex comparison.

(2) **Preserve tangent and obstruction modules.** Flatness of localization preserves its kernel and cokernel.

(3) **Identify the localized obstruction class.** Residuals are obtained by finite coefficient extraction from polynomial valuations and finite supported exponentials. Those operations commute with scalar extension. Their equality on representative kernel pairs therefore identifies the localized obstruction with $1 \otimes o$.

(4) **Characterize vanishing by denominator annihilation.** By the fraction-equivalence relation, $o/1 = 0$ in $S_P^{-1}H^1(C)$ exactly when some $s \in S_P$ annihilates o. This gives (8.23); the localized obstruction theorem identifies its vanishing with the inhabited lift object.

□

For Example 8.21, allowing the polynomial gauge $B[t^{-1}]$ removes the obstruction, since $to = 0$. Polylogarithmic overheads do not: writing $\lambda(n) = 1 + \mathrm{bitlength}(n+1)$, no λ^{-k} belongs to tB. On $n = 2^m$ the ratio $t^{-1}\lambda^{-k}$ is unbounded, as the binomial lower bound for 2^m of degree $k+1$ shows. Thus the same finite diagram has no lift after polylogarithmic localization but has a lift after polynomial localization. The annihilator gives the overhead criterion.

The resource deformation problem is now expressed by tangent modules, incidence complexes, and their successive lift objects. The following sections construct the geometric function objects and transport structures associated with these families.

9 Geometric function objects and differential cohesion

The preceding calculus describes variations of measured resource values and embedded budgets. To compare functions on their infinitesimal neighbourhoods with differential forms and transport, we now represent the coordinate spaces themselves. A nilpotent arrow records both a family and its reduced boundary. Its two endpoints construct the adjunctions below, while affine representability identifies its function objects. Restriction to based probes recovers the coefficient calculations used in the resource deformation theory.

9.1 The primary infinitesimal-arrow geometry and its étale topology

A nilpotent thickening has a smaller boundary and a larger infinitesimal space. Keeping both endpoints produces the primary geometry. Its restriction to the based stages of Section 7.3 will recover their resource coefficient calculus.

Definition 9.1 (Affine tests and infinitesimal thickenings)**.** The **category of affine tests** $\mathsf{Aff}_B^{\mathrm{fp}}$ is the opposite of the internal category of finitely presented commutative B-algebras. A presentation is a quotient $B[x_1,\dots,x_n]/(f_1,\dots,f_m)$ by a displayed finite list; its equations, homomorphisms, and changes of presentation are formed by internal finite lists, quotients, and equality in the presented algebras. Write $\operatorname{Spec} A$ for the corresponding object of $\mathsf{Aff}_B^{\mathrm{fp}}$. The **infinitesimal-arrow category** I_B has as objects the **infinitesimal thickenings** given by surjective maps of those algebras $S \twoheadrightarrow R$ with a nilpotent kernel, written

$$j = (U = \operatorname{Spec} R \hookrightarrow T = \operatorname{Spec} S).$$

Its arrows are *all* commutative squares of the corresponding affine arrows. The nilpotence condition is an existential finite power equation, $\exists n \geq 1\ (\ker(S \to R))^n = 0$; witnesses enter the arguments by existential elimination.

Finite polynomial codes and their interpretations define these categories internally. Surjectivity and nilpotence are bounded power-object predicates. Finite polynomial and finite-projective presentations carry names for coefficient calculations by Proposition D.3; their validity predicates remain $\Omega_{\mathcal{E}}$-valued.

Lemma 9.2 (The two affine endpoints)**.** *There are functors $c, d : \mathsf{I}_B \to \mathsf{Aff}_B^{\mathrm{fp}}$ and $\iota : \mathsf{Aff}_B^{\mathrm{fp}} \to \mathsf{I}_B$ with*

$$c(j) = T, \qquad d(j) = U, \qquad \iota(V) = (V \xrightarrow{\mathrm{id}} V), \qquad c \dashv \iota \dashv d, \quad c\iota = d\iota = \mathrm{id}\,.$$

Proof.

(1) **Identify maps into the identity arrow.** A square $j \to \iota(V)$ is uniquely specified by its larger endpoint map $T \to V$; the smaller map must be its composite with $U \to T$.

(2) **Identify maps out of the identity arrow.** A square $\iota(V) \to j$ is uniquely specified by a map $V \to U$; its larger map is forced.

(3) **Verify naturality and the endpoint identities.** These give mutually inverse internal hom-object bijections, natural in both variables; the two identities follow on identity arrows.

□

The two endpoints have different universal properties. Their adjunctions are expressed by

$$\mathrm{Hom}_{\mathsf{Aff}_B^{\mathrm{fp}}}(c(j), V) \cong \mathrm{Hom}_{\mathsf{I}_B}(j, \iota V), \qquad \mathrm{Hom}_{\mathsf{I}_B}(\iota V, j) \cong \mathrm{Hom}_{\mathsf{Aff}_B^{\mathrm{fp}}}(V, d(j)).$$

Their canonical arrows are the commutative squares

$$\begin{array}{ccc} U & \xrightarrow{\ j\ } & T \\ {\scriptstyle j}\downarrow & & \| \\ T & = & T \end{array} \qquad \begin{array}{ccc} U & = & U \\ \| & & \downarrow{\scriptstyle j} \\ U & \xrightarrow[\ j\ ]{} & T. \end{array}$$

These are respectively the arrows $j \to \iota c(j)$ and $\iota d(j) \to j$. Their opposite directions will determine the endpoint adjunctions on sheaves.

Definition 9.3 (The affine and infinitesimal étale sites)**.** On $\mathsf{Aff}_B^{\mathrm{fp}}$, use the **big finitely presented étale site**: a basic covering family consists of a finite presented family of finitely presented étale maps $T_\lambda \to T$ with an effective jointly covering image. Such finite families generate the topology on the finitely presented affine site; covering diagrams are indexed internal families. The **infinitesimal-arrow topology** on I_B is generated by the covering families

$$U_\lambda = U \times_T T_\lambda \hookrightarrow T_\lambda \quad (T_\lambda \to T \text{ an étale covering}).$$

Lemma 9.4 (Étale lift of finite nilpotent covering diagrams)**.** *For a finite nilpotent thickening $U \hookrightarrow T$, base change induces an equivalence between the categories of finitely presented étale T-schemes and étale U-schemes. It preserves finite intersections and jointly covering families, and the comparison is natural in commutative squares of thickenings. Consequently the larger- and smaller-endpoint constructions carry the stated étale covers to étale covers.*

Proof.

(1) **Construct lifts across square-zero and nilpotent ideals.** For a square-zero ideal the standard-étale Jacobian presentation gives uniqueness and existence of lifts: the invertible Jacobian solves the correction equation by finite linear algebra. Iteration over the powers of a nilpotent ideal provides the finite-order statement.

(2) **Glue local lifts by uniqueness.** An arbitrary étale presentation is locally standard étale; uniqueness on overlaps and ordinary étale descent glue the local lifts and maps. This is the infinitesimal invariance of the étale site (Stacks Project Authors 2026b, Theorems 41.15.2 and 41.16.1; Section 59.45).

(3) **Preserve covers, refinements, and descent diagrams.** The equivalence identifies covering categories, refinements, and Čech diagrams. Surjectivity of a family is preserved since a nilpotent closed immersion induces the same underlying open-support lattice. The proof is finite in each nilpotence witness.

□

Theorem 9.5 (Differential cohesion over the Landau ring)**.** *The internal étale sheaf categories*

$$\mathscr{X}_{B,\mathrm{et}} = \mathrm{Sh}_{\mathrm{et}}(\mathsf{Aff}_B^{\mathrm{fp}}; \mathcal{E}), \qquad \mathscr{H}_{B,\mathrm{et}} = \mathrm{Sh}_{\mathrm{et}}(\mathsf{I}_B; \mathcal{E})$$

carry four adjoints

$$\iota_! \dashv \iota^* \dashv \iota_* \dashv \iota^!. \tag{9.1}$$

Both endpoint embeddings are fully faithful, and

$$(\iota_! F)(U \hookrightarrow T) = F(T), \qquad (\iota_* F)(U \hookrightarrow T) = F(U).$$

The first three functors preserve all internal limits and colimits; $\iota^!$ preserves all internal limits. The associated idempotent modalities

$$\mathfrak{R} = \iota_! \iota^*, \qquad \mathfrak{I}_{\mathrm{inf}} = \iota_* \iota^*, \qquad \mathfrak{C} = \iota_* \iota^!$$

obey $\mathfrak{R} \dashv \mathfrak{I}_{\mathrm{inf}} \dashv \mathfrak{C}$. The canonical $\iota_! \to \iota_*$ and $\mathfrak{R} \to \mathfrak{I}_{\mathrm{inf}}$ are not isomorphisms.

Proof.

(1) **Identify the presheaf adjoints and full faithfulness.** On presheaves, left and right Kan extension along ι^{op} are evaluation at c and d, respectively, by Lemma 9.2. Their restrictions to the identity-arrow objects recover the input functor, proving full faithfulness.

(2) **Compare the endpoint covering diagrams.** We spell out the sheafification comparison. At an arrow $j = (U \hookrightarrow T)$, basic covers and their refinements are precisely the finite étale covering diagrams of T. For the larger endpoint c this is the defining identification. For the smaller endpoint d, Lemma 9.4 identifies that category, including its intersections and refinements, with the corresponding category over U. At an identity arrow, the same diagrams are identity-arrow covers.

(3) **Commute endpoint functors with sheafification.** In each case, matching families for a precomposed presheaf are therefore the corresponding endpoint matching families. Taking their equivalence classes under refinement gives natural isomorphisms

$$(F \circ c)^+ \cong F^+ \circ c, \qquad (F \circ d)^+ \cong F^+ \circ d, \qquad (H \circ \iota)^+ \cong H^+ \circ \iota.$$

Applying the construction twice gives the same comparisons for sheafification. Thus endpoint precomposition and restriction preserve sheaves and commute with sheafification.

(4) **Verify preservation of limits and colimits.** Presheaf limits are pointwise, and sheaf colimits are sheafifications of pointwise colimits; this proves all the asserted limit and colimit preservation for $\iota_!, \iota^*, \iota_*$.

(5) **Construct the remaining adjoint.** For $H \in \mathscr{H}_{B,\mathrm{et}}$, first define the presheaf

$$Q_H(T) = \mathrm{Nat}_{\mathscr{H}_{B,\mathrm{et}}}(\iota_* a y_T, H),$$

where ay_T is the sheafification of the representable presheaf of T. The internal natural-transformation object is an end, constructed from products and equalizers in $\mathcal{E}$. For $f : T' \to T$, precomposition with $\iota_* a y_f$ gives the restriction $Q_H(T) \to Q_H(T')$.

(6) **Verify descent for the candidate.** For a basic étale cover $(T_\lambda \to T)_\lambda$, put $T_{\lambda\mu} = T_\lambda \times_T T_\mu$. In $\mathscr{X}_{B,\mathrm{et}}$, the covering family gives the coequalizer

$$\coprod_{\lambda,\mu} a y_{T_{\lambda\mu}} \rightrightarrows \coprod_{\lambda} a y_{T_\lambda} \longrightarrow a y_T.$$

The preceding colimit-preservation calculation shows that ι_* preserves this coequalizer and its coproducts. Applying $\mathrm{Nat}_{\mathscr{H}_{B,\mathrm{et}}}(-, H)$ therefore gives the equalizer

$$Q_H(T) \longrightarrow \prod_{\lambda} Q_H(T_\lambda) \rightrightarrows \prod_{\lambda,\mu} Q_H(T_{\lambda\mu}).$$

All constructions are internal and hold in every parameter context. Thus Q_H satisfies the sheaf condition for the generating covers and defines an object of $\mathscr{X}_{B,\mathrm{et}}$.

(7) **Prove the right adjunction.** Internal Yoneda density writes a sheaf F as $F \cong \int^{T \in \mathsf{Aff}_B^{\mathrm{fp}}} F(T) \cdot a y_T$, where the dot denotes the internal copower. Preservation of internal colimits by ι_* and the defining end for Q_H give

$$\mathrm{Nat}_{\mathscr{H}_{B,\mathrm{et}}}(\iota_* F, H) \cong \mathrm{Nat}_{\mathscr{X}_{B,\mathrm{et}}}(F, Q_H).$$

The isomorphism is natural in F and H and in all internal parameters. Set $\iota^! H = Q_H$. The unit and counit are the images of identity maps under this isomorphism and its inverse; naturality gives both triangle identities. This constructs $\iota_* \dashv \iota^!$.

(8) **Distinguish the two endpoint embeddings.** Take the affine-line sheaf $F(T) = \Gamma(T, \mathcal{O}_T)$ and $j = (\operatorname{Spec} B \hookrightarrow \operatorname{Spec}(B[\varepsilon]/\varepsilon^2))$. At j the two endpoint embeddings have components $B[\varepsilon]/\varepsilon^2$ and B, with canonical map given by reduction. This is not an isomorphism, since coefficient extraction sends ε to its nonzero coefficient 1.

(9) **Distinguish the modalities and verify their laws.** Apply the two associated modalities to $\iota_! F$ to see that their comparison is also not an isomorphism. Finally all adjunction and idempotency equations follow from the endpoint adjunctions and their full faithfulness.

□

Proposition 9.6 (Smooth automorphisms under nilpotent reduction)**.** *Over $S = B[\varepsilon]/\varepsilon^2 \twoheadrightarrow R = B$, the distinct smooth S-automorphisms of the affine line $x \mapsto x$ and $x \mapsto x + \varepsilon$ become identical after reduction. Thus the functor on categories of smooth schemes induced by a nilpotent thickening is not faithful; the proof of Theorem 9.5 cannot be obtained by substituting "smooth" for "étale".*

Proof.

(1) **Exhibit a nontrivial automorphism with identity reduction.** The inverse of $x \mapsto x + \varepsilon$ is $x \mapsto x - \varepsilon$; its reduction is the identity. The two maps differ at x by the nonzero element ε. The conclusion follows.

□

9.2 Recovery of algebraic synthetic differential geometry

The primary site gives an affine calculation of function objects. We use it to show that the coordinate line supports the algebraic SDG identities needed by the subsequent differential and formal constructions. Here *algebraic* refers to finitely presented commutative algebras and their Weil extensions, rather than a specified C^∞-ring structure.

Definition 9.7 (Primary affine objects and the coordinate line)**.** Let $\Delta_{\mathrm{inf}} : \mathcal{E} \to \mathscr{H}_{B,\mathrm{et}}$ be structural inverse image. For a finitely presented B-algebra A, its **affine sheaf** F_A, **primary affine object** X_A, and the **coordinate line** $\mathcal{O}_{\mathrm{inf}}$ are

$$F_A(\operatorname{Spec} S) = \operatorname{Hom}_{B\text{-alg}}(A, S), \qquad X_A = \iota_! F_A, \qquad \mathcal{O}_{\mathrm{inf}} = X_{B[z]}.$$

At $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$, these have values $\operatorname{Hom}_{B\text{-alg}}(A, S)$ and S, respectively. A square from j to $\iota(\operatorname{Spec} A)$ is determined by its larger-endpoint map, so $X_A = y\iota(\operatorname{Spec} A)$. These representables are sheaves: algebra maps at both endpoints glue along the stated étale covers, and their commutative-square equation is local. In the tensor formulas below, scalar extension is along $\Delta_{\mathrm{inf}} B \to \mathcal{O}_{\mathrm{inf}}$.

Definition 9.8 (Full and zero-boundary Weil probes)**.** For a rational Weil algebra W, put $W_B = B \otimes_{\mathbb{Q}} W$. Its **full Weil probe** and **zero-boundary Weil probe** are

$$D_W^{\mathrm{all}} = X_{W_B}, \qquad D_W^0 = y(\operatorname{Spec} B \hookrightarrow \operatorname{Spec} W_B).$$

For $W = \mathbb{Q}[\varepsilon]/\varepsilon^2$, write D^{all} for the full dual-number probe.

Theorem 9.9 (Primary algebraic SDG recovery)**.** *For every finitely presented B-algebra A, canonical evaluation induces an isomorphism of $\mathcal{O}_{\mathrm{inf}}$-algebras*

$$\mathcal{O}_{\mathrm{inf}} \otimes_{\Delta_{\mathrm{inf}} B} \Delta_{\mathrm{inf}} A \xrightarrow{\cong} \mathcal{O}_{\mathrm{inf}}^{X_A}. \tag{9.2}$$

Its value at j is $S \otimes_B A$. Consequently, for every internal commutative $\mathcal{O}_{\mathrm{inf}}$-algebra C, coordinate evaluation gives the natural isomorphism

$$\underline{\mathrm{Hom}}_{\mathcal{O}_{\mathrm{inf}}\text{-}alg}(\mathcal{O}_{\mathrm{inf}}^{X_A}, C) \;\cong\; \mathrm{Spec}_C(A), \tag{9.3}$$

where $\mathrm{Spec}_C(A)$ is the object of solutions in C of a displayed finite presentation of A. In particular, taking $A = B \otimes_{\mathbb{Q}} A_0$ for finitely presented rational A_0 proves the comprehensive algebraic SDG axiom over $\mathbb{Q}$.

For every rational Weil algebra W, the full probe satisfies

$$\mathcal{O}_{\mathrm{inf}}^{D_W^{\mathrm{all}}} \cong \mathcal{O}_{\mathrm{inf}} \otimes_{\mathbb{Q}} W. \tag{9.4}$$

For $W = \mathbb{Q}[\varepsilon]/\varepsilon^2$, the probe is $D^{\mathrm{all}} = \{d \in \mathcal{O}_{\mathrm{inf}} : d^2 = 0\}$ and the comparison is the Kock–Lawvere isomorphism

$$\mathcal{O}_{\mathrm{inf}} \times \mathcal{O}_{\mathrm{inf}} \xrightarrow{\cong} \mathcal{O}_{\mathrm{inf}}^{D^{\mathrm{all}}}, \qquad (a, b) \longmapsto (d \mapsto a + db). \tag{9.5}$$

This infinitesimal object is nontrivial. The line and every displayed finite-projective affine module cell are microlinear for rational Weil diagrams: their augmented Weil-prolongation functors preserve finite limits in the slice over the cell. All comparisons are internal, natural in the displayed algebras and their maps, and uniform in parameters. The construction therefore also applies in $\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$ with coefficient ring $\bar{B}$.

Proof.

(1) **Compute the function algebra by Yoneda.** The product $j \times \iota(\mathrm{Spec}\, A)$ is

$$\mathrm{Spec}(R \otimes_B A) \hookrightarrow \mathrm{Spec}(S \otimes_B A).$$

Its endpoints are finitely presented, and its quotient kernel is generated by the image of $\ker(S \to R)$, with the same nilpotence bound. Products of represented sheaves and internal Yoneda give

$$(\mathcal{O}_{\mathrm{inf}}^{X_A})(j) \cong \mathcal{O}_{\mathrm{inf}}(j \times \iota(\mathrm{Spec}\, A)) = S \otimes_B A.$$

Evaluation sends $s \otimes a$ to $(\alpha \mapsto s\alpha(a))$, so preserves the algebra structure and is natural in both tests and coordinate algebras.

(2) **Identify scalar extension and affine duality.** For $A = B[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$, internal scalar extension is the sheafification of this finite presentation over S. The preceding calculation identifies it with the already sheaf-valued function algebra, proving (9.2). Maps from it to C are precisely tuples in C^n satisfying the f_i. The same calculation in every slice proves (9.3), independently of presentation.

(3) **Recover the infinitesimal coefficient identities.** Take $A = W_B$ to obtain (9.4). The dual-number basis $1, \varepsilon$ gives (9.5). At the dual-number test, $\varepsilon \neq 0$ by coefficient extraction and $1 \neq 0$ in B; hence the square-zero probe is nontrivial.

(4) **Prove microlinearity of module cells.** For $X_E = X_{\mathrm{Sym}_B(E^\vee)}$, finite duality and the same product calculation give

$$X_E(j) = S \otimes_B E, \qquad (X_E^{D_W^{\mathrm{all}}})(j) \cong (S \otimes_B E) \otimes_{\mathbb{Q}} W.$$

The split rational-linear limit argument of Theorem 7.19 therefore preserves finite augmented Weil limits in the slice over X_E. The line is the rank-one case.

(5) **Retain internal parameters.** Finite presentations, quotients, and these universal properties are internal and parameter-uniform. They give the stated natural comparisons and the same construction in $\mathscr{S}$ over $\bar{B}$.

□

This establishes an algebraic SDG model over the Landau coefficient ring, in the sense of the comprehensive and Weil axioms of (Kock 2006, Sections I.12, I.16, III.1 and Appendix D). It supplies the function objects used for formal completion and jets.

Lemma 9.10 (Full and zero-boundary infinitesimal probes)**.** *For every rational Weil algebra W, there is a canonical monomorphism $D_W^0 \hookrightarrow D_W^{\mathrm{all}}$. It induces isomorphisms*

$$\mathcal{O}_{\mathrm{inf}}^{D_W^{\mathrm{all}}} \xrightarrow{\cong} \mathcal{O}_{\mathrm{inf}}^{D_W^0}, \qquad X_E^{D_W^{\mathrm{all}}} \xrightarrow{\cong} X_E^{D_W^0} \tag{9.6}$$

for displayed finite-projective affine module cells X_E. The two probes themselves need not be isomorphic. For dual numbers, D^0 consists of square-zero coordinates whose reduction at the smaller endpoint is zero.

Proof.

(1) **Identify the zero-boundary condition.** At j, a W_B-point of S belongs to D_W^0 precisely when its composite to R equals the augmentation followed by $B \to R$. This is a subobject of the full set of W_B-points.

(2) **Construct the zero-boundary product test.** The product of j with the representing zero-boundary arrow has larger algebra $S \otimes_B W_B$ and smaller algebra R. Its quotient kernel is the sum of the two extended nilpotent ideals, which is nilpotent by a finite binomial bound.

(3) **Compare the two function objects.** Yoneda therefore computes the same larger-endpoint function algebra as in Theorem 9.9. The map between the products is the identity on that larger algebra, proving the first isomorphism; the identical argument with $S \otimes_B E$ proves the second.

(4) **Distinguish the full and zero-boundary probes.** At the identity test of $B[\varepsilon]/\varepsilon^2$, the element ε belongs to the full square-zero probe and has nonzero reduction, so it does not belong to D^0.

□

The zero-boundary probes describe displacements relative to the smaller endpoint of a thickening. They are the probes used by the shape relation and its boundary principal parts below.

9.3 Comparison with the based resource calculus

Proposition 9.11 (Based recovery of the primary coefficient identities)**.** *Restriction to $j_C = (\operatorname{Spec} B \hookrightarrow \operatorname{Spec} C)$, $C \in \mathsf{W}_B$, defines a functor $\mathrm{res}_W : \mathscr{H}_{B,\mathrm{et}} \to \mathcal{G}_B$ with natural identifications*

$$\mathrm{res}_W \mathcal{O}_{\mathrm{inf}} \cong \mathcal{R}_B, \qquad \mathrm{res}_W X_E \cong j_B E, \qquad \mathrm{res}_W D_W^{\mathrm{all}} \cong \mathrm{res}_W D_W^0 \cong D_{C_W}.$$

For the line and these affine module cells, restriction also identifies the exponentials in Theorem 9.9 and Lemma 9.10 with the corresponding based Weil exponentials of Theorem 7.16. These comparisons preserve augmentation, Weil substitution and polynomial coordinate maps.

Proof.

(1) **Evaluate the primary objects at based tests.** Evaluation at j_C gives C, $C \otimes_B E$ and $\mathrm{Hom}_{B\text{-alg}}(C_W, C)$, respectively.

(2) **Identify the two probes on based stages.** Every B-algebra map from a Weil stage to the reduced ring B kills its nilpotent ideal. The zero-boundary condition is therefore automatic at j_C, identifying the two probes there.

(3) **Recover the coefficient identities and coherence.** Evaluating their primary function objects gives $C \otimes_{\mathbb{Q}} W$, or $(C \otimes_B E) \otimes_{\mathbb{Q}} W$ for a module cell. These are exactly the expressions obtained from Theorem 7.16 in $\mathcal{G}_B$. Every map is the same coefficient substitution or elementary-tensor map, proving the stated coherence.

□

Proposition 9.12 (Budget-indexed endpoint geometry)**.** *The constructions of Lemma 9.2 and Theorem 9.5 may be performed in the budget-diagram category $\mathscr{S} = [L, \mathcal{E}]$, yielding endpoint adjunctions on the associated indexed internal étale sheaves. Their defining formulas are pointwise in the budget parameter and compatible with $M_a \hookrightarrow M_b$. Transport of a name-cover and tracked operation into this indexed geometry preserves its equality of denotations, its Σ-classified domain and its original positive-observation evaluation.*

Proof.

(1) **Reindex the endpoint geometry over budgets.** A diagram $a \mapsto M_a$ reindexes every polynomial, nilpotent-arrow, and endpoint formula using the same ring B; finite ring operations and naturality commute with the inclusions.

(2) **Preserve classified computation and its witnesses.** The named computation equations are originally equations of maps and classified spans in $\mathcal{E}$. Applying the left-exact sheaf realization to their finite diagrams preserves equality, pullbacks and regular images; the stored classifier is transported with the span. Thus the original Σ-valued observation remains typed and its finite witnesses remain tracked.

□

The primary function objects now supply coordinate algebras for the finite neighbourhoods used above. Completing their diagonal ideals assembles the neighbourhoods of every order.

10 Formal completion of differential resource geometry

The finite calculus measures resource variation to a specified order. Completion assembles compatible orders and exposes a further resource question: whether one budget contains every coefficient of the assembled family. The measured tower in Example 10.14 has a polynomial budget at each finite order and no common polynomial growth degree. It will also supply the de Rham and crystalline example.

We construct formal neighbourhoods by powers of the diagonal ideal, first over B in $\mathcal{E}$ and then over the resource parameters in $\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$. Principal parts retain both endpoints and the full Taylor action of coordinate changes. Boundary principal parts represent pairs with equal smaller-endpoint value on the primary site; their inverse-limit function objects identify the completed infinitesimal nerve. Uniform finite computations on coefficient names are then obtained from the same coefficient tower.

10.1 The diagonal and its finite principal parts

An order-r neighbourhood retains differences between two points through degree r. Its coordinate algebra is obtained by quotienting the corresponding power of the diagonal ideal. Let E be a displayed finite-projective B-module, let $V = E^\vee$, and write $A_E = \mathrm{Sym}_B(V)$ for its coordinate algebra, with the variance of Proposition 7.23. Put

$$D_E = A_E \otimes_B A_E, \qquad \mu_E : D_E \longrightarrow A_E, \qquad I_E = \ker \mu_E.$$

Both tensor-factor inclusions will be retained; they give the source and target actions on every principal-parts object.

Definition 10.1 (Formal diagonal filtration)**.** For $r \in \mathbb{N}$, the **order-r principal-parts algebra** is

$$P^r_{E/B} = D_E / I_E^{r+1}.$$

The transition maps $P^{r+1}_{E/B} \twoheadrightarrow P^r_{E/B}$ define the **formal diagonal pro-algebra** $\boldsymbol{P}_E = (P^r_{E/B})_r$ and its **completed principal-parts algebra**

$$\widehat{P}_{E/B} = \varprojlim_r P^r_{E/B}.$$

The pro-algebra is retained along with its realization. A B-algebra map $P^r_{E/B} \to C$ is an **order-r formal neighbourhood point** of the diagonal.

Definition 10.2 (Formal coefficient families)**.** A **formal coefficient family** is a compatible family of finite stages in a displayed inverse tower of coefficient objects. A computable name for the family is additional data; compatibility alone does not supply such a name.

Theorem 10.3 (Polynomial calculation of formal principal parts)**.** *There are natural filtered A_E-algebra isomorphisms*

$$\begin{aligned} P^r_{E/B} &\cong \bigoplus_{0 \le j \le r} A_E \otimes_B \mathrm{Sym}^j_B(V), \\ \widehat{P}_{E/B} &\cong \prod_{j \ge 0} A_E \otimes_B \mathrm{Sym}^j_B(V), \\ \mathrm{gr}_{I_E}(D_E) &\cong \mathrm{Sym}_{A_E}(\Omega^1_{A_E/B}), \qquad I_E/I_E^2 \cong \Omega^1_{A_E/B}. \end{aligned} \tag{10.1}$$

Each finite degree is a finite-projective A_E-module; multiplication on the first line is truncated by total degree and on the second is coefficientwise finite convolution. The filtered isomorphisms are natural under polynomial coordinate substitution with its full Taylor transport: for $y = p(x)$ the displacement is sent to $p(x+u) - p(x)$. On the associated graded this transport is the symmetric algebra of the cotangent map. All comparisons also commute with changes of displayed finite-projective presentation.

Proof.

(1) **Introduce displacement coordinates.** Freeness gives $D_E \cong \mathrm{Sym}_B(V \oplus V)$. The invertible change $(v_0, v_1) \mapsto (v_0, v_1 - v_0)$ makes μ_E evaluation at displacement zero, hence

$$D_E \cong A_E \otimes_B \mathrm{Sym}_B(V), \qquad I_E^k = \bigoplus_{j \ge k} A_E \otimes_B \mathrm{Sym}^j_B(V).$$

Taking quotients and their inverse limit gives the first two formulas, with finite convolution in each coefficient.

(2) **Identify the finite coefficients and associated graded.** The displayed retract for V makes $V^{\otimes j}$ finite-projective. The averaging idempotent $\frac{1}{j!}\sum_{\sigma\in S_j}\sigma$ exhibits $\mathrm{Sym}^j_B(V)$ as its retract. The universal differential formula $\Omega^1_{A_E/B} \cong A_E \otimes_B V$ identifies I_E/I_E^2 and the symmetric multiplication on the associated graded.

(3) **Compute polynomial coordinate transport.** The two-vertex map for $y = p(x)$ sends (y_0, y_1) to $(p(x_0), p(x_1))$, so in displacement coordinates it is

$$y = p(x), \qquad v = p(x+u) - p(x). \tag{10.2}$$

The displacement has zero constant term, so preserves every filtration power and induces compatible maps on quotients and limits.

(4) **Verify coherence and graded transport.** For successive maps p, q, substitution gives

$$q(p(x) + [p(x+u) - p(x)]) - q(p(x)) = q(p(x+u)) - q(p(x)).$$

This proves composition coherence; the identity gives the unit. The linear part is $dp_x(u)$, giving the symmetric cotangent map on the associated graded. All formulas commute with displayed linear retractions by their two-vertex universal property. For example, $p(x) = x^2$ sends v to $2xu + u^2$; its graded transport keeps only $2xu$.

□

The coordinate diagram below fixes the two actions and reduction.

$$\begin{array}{ccccc} A_E & \overset{s^\sharp}{\underset{t^\sharp}{\rightrightarrows}} & A_E \otimes_B A_E & \longrightarrow & P^r_{E/B} \\ & & & \searrow^{\mu_E} & \downarrow{\scriptstyle \mathrm{aug}_r} \\ & & & & A_E. \end{array} \tag{10.3}$$

The augmentation ideal of $P^r_{E/B}$ has $(r+1)$-st power zero. The quotient $P^0_{E/B} = A_E$, so this filtration begins at the ordinary resource-coordinate space and adds no independently specified infinitesimals.

10.2 Finite formal neighbourhoods on based Weil probes

Principal parts describe a neighbourhood over the whole polynomial cell: their reduction is A_E. They need not be finite-projective over B and hence need not be B-Weil stages themselves. We realize them by evaluating their functor of points on the already constructed category $\mathsf{W}_B^{\mathrm{proj}}$:

$$\mathcal{D}_E^{\leq r}(C) = \mathrm{Hom}_{B\text{-alg}}(P^r_{E/B}, C), \qquad C \in \mathsf{W}_B^{\mathrm{proj}}.$$

The quotient map makes this a subobject of $j_B E \times j_B E$.

Theorem 10.4 (Completion–Weil comparison)**.** *There is a canonical isomorphism in $\mathcal{G}_B$,*

$$\varinjlim_r \mathcal{D}_E^{\leq r} \;\cong\; j_B E \times_{\mathfrak{I}_W(j_B E)} j_B E. \tag{10.4}$$

It is natural in finite-projective polynomial coordinate maps and is compatible with the source and target projections. For every finite ordinal $[q]$, completion of the higher diagonal $A_E^{\otimes_B(q+1)} \to A_E$ gives a functor of points naturally identified with the $(q+1)$-fold fibre product of $j_B E$ over $\mathfrak{I}_W(j_B E)$. These isomorphisms respect every simplicial operator.

Proof.

(1) **Factor equal-reduction pairs through finite principal parts.** Fix a stage $C = B \oplus N$, with the supplied finite nilpotence bound $N^{s+1} = 0$. By finite duality, a pair of points of $j_B E(C)$ is a pair of B-algebra maps $A_E \to C$. It belongs to the right-hand side of (10.4) precisely when its two reductions to B agree. Equivalently the difference of the two evaluations of every $v \in V$ belongs to N. The induced map $A_E \otimes_B A_E \to C$ therefore sends I_E to N and kills I_E^{s+1}; it factors through $P^s_{E/B}$.

(2) **Recover equal reductions from finite factorization.** Conversely, if the map factors through $P^r_{E/B}$, the images of the coordinate differences are nilpotent. Their reductions in B are zero because B is reduced, by Proposition 2.15. Duality then implies equality of the reduced E-points. The implications hold internally, uniformly in C and its nilpotence datum. They identify the subobjects at every stage, and hence the corresponding functors.

(3) **Extend the comparison to higher diagonals.** For $q+1$ points use differences with the first point, or successive differences; both generate the kernel of the higher multiplication map. The same finite nilpotence argument identifies its completed formal points with points having equal reductions.

(4) **Verify simplicial coherence.** Deleting, repeating, or permuting vertices acts by substitution of these differences and preserves all powers of the higher diagonal ideal. The stagewise identifications therefore commute with faces, degeneracies, and permutations.

□

10.3 Formal completion in the primary geometry

The based calculation records displacements at fixed-base probes. We now identify the full relation on the primary site, where both endpoints of a thickening remain visible. For $A = \mathrm{Sym}_B(E^\vee)$, retain the affine sheaf $F_A \in \mathscr{X}_{B,\mathrm{et}}$ of Section 9.2, with $F_A(\operatorname{Spec} S) = \mathrm{Hom}_{B\text{-alg}}(A, S)$, and put

$$X_A = \iota_! F_A \in \mathscr{H}_{B,\mathrm{et}}, \qquad \mathfrak{I}_{\mathrm{inf}} X_A = \iota_* F_A.$$

The shape unit $\eta_{X_A} : X_A \to \mathfrak{I}_{\mathrm{inf}} X_A$ evaluated at $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ is the restriction map $\mathrm{Hom}_{B\text{-alg}}(A, S) \to \mathrm{Hom}_{B\text{-alg}}(A, R)$.

Definition 10.5 (Finite boundary principal parts)**.** The **order-r boundary principal parts** form the subobject of $X_A \times X_A$ given at $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ by

$$\mathcal{D}^{\le r}_{A,\mathrm{et}}(j) = \mathrm{Hom}_{B\text{-alg}}(P^r_{A/B}, S) \underset{\mathrm{Hom}_{B\text{-alg}}(A\otimes_B A, R)}{\times} \mathrm{Hom}_{B\text{-alg}}(A, R), \tag{10.5}$$

where the last arrow is the diagonal and the first arrow reduces both vertices modulo $S \to R$.

Theorem 10.6 (Étale completion of the full infinitesimal relation)**.** *The boundary principal parts are étale sheaves, they form an increasing union, and the comparison*

$$\varinjlim_r \mathcal{D}^{\le r}_{A,\mathrm{et}} \xrightarrow{\cong} \mathcal{D}_{X_A} := X_A \times_{\mathfrak{I}_{\mathrm{inf}} X_A} X_A \tag{10.6}$$

is an isomorphism of internal étale sheaves. It is natural under maps of displayed polynomial coordinate algebras, retains both vertex maps, and identifies completed higher diagonals with the complete infinitesimal Čech nerve of the shape unit, including all face and degeneracy maps.

Restriction to the based probe $\operatorname{Spec} B \hookrightarrow \operatorname{Spec} C$, $C \in \mathsf{W}_B^{\mathrm{proj}}$, *identifies* (10.5) *with the previously constructed* $\mathrm{Hom}_{B\text{-}alg}(P^r_{A/B}, C)$, *and* (10.6) *with* (10.4).

Proof.

(1) **Factor equal-boundary pairs through finite principal parts.** At j, a pair of A-points of S lies in the right side exactly when its reductions in R agree. Its algebra map $A \otimes_B A \to S$ therefore takes the diagonal ideal I into $J = \ker(S \to R)$. For a nilpotence witness $J^N = 0$, it annihilates I^N, so the pair comes from P^{N-1}.

(2) **Recover boundary equality and identify the sheaves.** Conversely every element of (10.5) has equal reduced vertices by its displayed fibre-product equation. There is no chosen global N: these implications prove equality of subobjects by existential elimination in $\mathcal{E}$. Pullback of nilpotent quotients along étale maps preserves their power filtration, so the formulas define sheaves and the stagewise isomorphism is natural in coverings and refinement.

(3) **Extend the comparison to the completed nerve.** For $q+1$ vertices use the kernel of $A^{\otimes_B(q+1)} \to A$. Equality of all reductions places every difference into J, and the same nilpotence argument applies. Face, degeneracy, and permutation maps are algebra substitutions preserving powers of this ideal; therefore the comparison is simplicial.

(4) **Recover the based-Weil formula.** On the based arrow $B \hookrightarrow C = B \oplus N$, the image of every difference under a P^r-point is nilpotent. The reduced ring B has no nonzero nilpotents, so the boundary equality is automatic. These identities give the based-Weil formula directly at every stage.

□

The completed relation is the kernel pair of the primary shape unit:

$$\begin{array}{ccc} \mathcal{D}_{X_A} & \xrightarrow{\ t\ } & X_A \\ {\scriptstyle s}\downarrow & \lrcorner & \downarrow{\scriptstyle \eta_{X_A}} \\ X_A & \xrightarrow[\eta_{X_A}]{} & \Im_{\inf} X_A. \end{array}$$

The theorem identifies its finite boundary filtration with $\varinjlim_r \mathcal{D}^{\le r}_{A,\mathrm{et}}$ inside this cartesian square. Higher fibre products give its full nerve.

10.4 Represented principal parts and formal functions

Identifying a formal relation is not yet a calculation of its functions. Representability supplies that calculation: finite neighbourhoods are primary tests, so Yoneda evaluates their function objects; compatible truncations then give the functions on the formal union. Retain the coordinate-line ring object

$$\mathcal{O}_{\inf} = \iota_! F_{B[z]}, \qquad \mathcal{O}_{\inf}(\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S) = S.$$

For $q \ge 0$ put

$$P^r_q = A^{\otimes_B(q+1)}/I^{r+1}_q, \qquad I_q = \ker(A^{\otimes_B(q+1)} \to A), \qquad k_{q,r} = (\operatorname{Spec} A \hookrightarrow \operatorname{Spec} P^r_q).$$

The finite-projective polynomial presentation supplies finitely many generators for I_q and for each of its finite powers. Thus P^r_q is finitely presented over B, and $k_{q,r}$ is an object of I_B. Write $\mathcal{D}^{\le r}_{A,q}$ for the higher-vertex version of (10.5), and $\mathcal{D}^{(q)}_{X_A} = X_A \times_{\Im_{\inf} X_A} \cdots \times_{\Im_{\inf} X_A} X_A$ with $q+1$ factors.

Lemma 10.7 (Primary representability of finite boundary principal parts)**.** *There are natural isomorphisms of primary sheaves*

$$yk_{q,r} \cong \mathcal{D}^{\leq r}_{A,q}, \qquad \varinjlim_r yk_{q,r} \cong \mathcal{D}^{(q)}_{X_A}.$$

They preserve all vertex maps, truncations, and polynomial coordinate substitutions. The representing sheaves are already étale sheaves.

Proof.

(1) **Identify the representing boundary square.** A square from $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ to $k_{q,r}$ is precisely a pair of algebra maps $P^r_q \to S$, $A \to R$ whose composites from $A^{\otimes_B(q+1)}$ agree. This is the displayed boundary fibre product, including equality of all reduced vertices.

(2) **Verify the sheaf condition.** Algebra maps into an étale covering diagram glue uniquely at both endpoints; the square equation glues because equality is local. Hence the representable is a sheaf.

(3) **Identify the completed union and coherence.** The union assertion is the higher-diagonal case of Theorem 10.6. Every comparison is induced by the same algebra substitution on the two endpoints, which proves the stated coherence.

□

Theorem 10.8 (Functions on the completed infinitesimal nerve)**.** *Let* $\underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}$ *denote the internal* $\mathcal{E}$*-object of sheaf maps, and define the ring objects*

$$\mathcal{C}^q_r = \mathcal{O}_{\mathrm{inf}}^{\mathcal{D}^{\leq r}_{A,q}}, \qquad \widehat{\mathcal{C}}^{\,q} = \mathcal{O}_{\mathrm{inf}}^{\mathcal{D}^{(q)}_{X_A}}.$$

Then

$$\underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}^{\leq r}_{A,q}, \mathcal{O}_{\mathrm{inf}}) \cong P^r_q, \tag{10.7}$$

$$\underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}^{(q)}_{X_A}, \mathcal{O}_{\mathrm{inf}}) \cong \varprojlim_r P^r_q, \tag{10.8}$$

$$\widehat{\mathcal{C}}^{\,q} \cong \varprojlim_r \mathcal{C}^q_r. \tag{10.9}$$

At $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$*, these coefficient sheaves have the explicit values*

$$\mathcal{C}^q_r(j) \cong S \otimes_B P^r_q, \qquad \widehat{\mathcal{C}}^{\,q}(j) \cong \varprojlim_r (S \otimes_B P^r_q).$$

All comparisons respect the cosimplicial maps, algebra operations, and polynomial coordinate substitutions with their full Taylor law.

Proof.

(1) **Compute finite and completed internal function objects.** Internal Yoneda and Lemma 10.7 give $\underline{\mathrm{Hom}}(yk_{q,r}, \mathcal{O}_{\mathrm{inf}}) = \mathcal{O}_{\mathrm{inf}}(k_{q,r}) = P^r_q$. Mapping out of the internally indexed union gives the inverse-limit formula, uniformly in any additional $\mathcal{E}$-parameter.

(2) **Compute the completed coefficient sheaf.** In the sheaf topos, product with a fixed object preserves colimits. Applying this fact to the exponential adjunction proves (10.9).

(3) **Construct the product test.** The product $j \times k_{q,r}$ has larger-endpoint algebra $S \otimes_B P_q^r$ and smaller-endpoint algebra $R \otimes_B A$. The kernel of its quotient is generated by the two extended nilpotent kernels, whose sum is nilpotent by a finite binomial bound. Both endpoints are finitely presented, so this is a product in I_B.

(4) **Compute stagewise coefficients and their limit.** Yoneda therefore calculates

$$(\mathcal{O}_{\mathrm{inf}}^{yk_{q,r}})(j) = \underline{\mathrm{Hom}}(yj \times yk_{q,r}, \mathcal{O}_{\mathrm{inf}}) = S \otimes_B P_q^r.$$

Sheaf limits are computed pointwise, giving the completed value.

(5) **Recover internal maps and verify coherence.** At the terminal test $\iota(\operatorname{Spec} B)$ this recovers the two internal map objects above. Algebra substitutions represent the vertex and coordinate maps, so Yoneda identifies every associated coherence diagram, including multiplication of functions.

□

Restriction of functions along a finite-neighbourhood inclusion is the truncation map under Yoneda:

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(yk_{q,r+1}, \mathcal{O}_{\mathrm{inf}}) & \longrightarrow & \underline{\mathrm{Hom}}(yk_{q,r}, \mathcal{O}_{\mathrm{inf}}) \\ \cong \downarrow & & \downarrow \cong \\ P_q^{r+1} & \twoheadrightarrow & P_q^r. \end{array}$$

Here the hom objects are those of $\mathscr{H}_{B,\mathrm{et}}$ internal to $\mathcal{E}$. Taking the inverse limit of this natural diagram gives the completed function comparison, including its cosimplicial structure.

The function calculation must also respect the endpoint structure. We next compare the endpoint functors with inverse limits of the coefficient sheaves, including the units, counits, and maps of the infinitesimal nerve.

10.5 The endpoint modalities through formal completion

Theorem 10.9 (Endpoint adjunctions on formal towers and their realizations)**.** *The endpoint quadruple of Theorem 9.5 acts levelwise on internally $\mathbb{N}^{\mathrm{op}}$-indexed towers, with the same four adjunctions, full-faithfulness equations, and idempotent modalities. Every endpoint functor commutes with realization of these towers by internal inverse limits. In particular*

$$\mathfrak{M}(\varprojlim_r H_r) \cong \varprojlim_r \mathfrak{M}(H_r), \qquad \mathfrak{M} \in \{\mathfrak{R}, \mathfrak{I}_{\mathrm{inf}}, \mathfrak{C}\}. \tag{10.10}$$

These comparisons commute with all adjunction units and counits, the modality multiplication and comultiplication, and maps of towers. They apply to the completed coefficient sheaves $\widehat{\mathcal{C}}^q$ of Theorem 10.8, including their cosimplicial operations and normalized additive cochains. Rings and modules are transported over the corresponding transported coefficient rings.

Proof.

(1) **Lift the adjunctions and limits to towers.** By Theorem 9.5, all four endpoint functors preserve internal limits. Their componentwise adjunction bijections restrict to maps of towers by naturality, with triangle and full-faithfulness equations checked componentwise. Composing their limit comparisons gives (10.10).

(2) **Verify compatibility with structure maps.** For a unit, counit, modality structure map, or tower morphism, the two comparison paths agree after every limit projection. The limit universal property therefore identifies them.

(3) **Transport coefficients and cochains.** Ring and module operations are finite-product diagrams; normalization in each cochain degree is a finite intersection of degeneracy kernels. The limit-preserving functors retain these structures. Apply the comparisons to (10.9); its cosimplicial and coordinate maps are maps of the finite towers.

□

For the shape modality, compatibility with the unit is the square

$$\begin{array}{ccc} \varprojlim_r H_r & \xrightarrow{\eta} & \Im_{\inf}(\varprojlim_r H_r) \\ \| & & \downarrow \cong \\ \varprojlim_r H_r & \xrightarrow[\varprojlim_r \eta_{H_r}]{} & \varprojlim_r \Im_{\inf} H_r. \end{array}$$

The analogous counit squares and their mates express the compatibility for the other endpoint modalities.

There are two related limiting operations here. Formal spaces are unions of finite represented neighbourhoods, identified in Theorem 10.6; their function objects are inverse limits. The theorem applies to the inverse limits of coefficient sheaves.

Proposition 10.10 (Completion of geometric coefficients)**.** *For an internal coefficient tower P_r, scalar extension has a canonical comparison*

$$S \otimes_B \varprojlim_r P_r \longrightarrow \varprojlim_r (S \otimes_B P_r).$$

It is not always an isomorphism, even for a finitely presented polynomial algebra S. The completed coefficient sheaf in Theorem 10.8 is the right-hand construction.

Proof.

(1) **Construct the coefficient-completion comparison.** Tensor the limit projections and use the target limit universal property to construct the comparison.

(2) **Exhibit the polynomial-series comparison.** Take $S = B[z]$ and $P_r = B[u]/u^{r+1}$. It becomes

$$B[[u]][z] \longrightarrow B[z][[u]].$$

(3) **Prove injectivity and failure of surjectivity.** It is injective by coefficient extraction. Every element of the source has a finite bound on its z-degree, whereas $\sum_{m\geq 0} z^m u^m$ has none: its coefficient at any larger u-degree has that larger z-degree, and $1 \neq 0$ in B. Thus the comparison is not surjective.

(4) **Recover the stagewise sheaf calculation.** The sheaf formula is the pointwise calculation already proved.

□

Proposition 10.11 (Polynomial lifting, effective descent, and a singular counterexample)**.** *For a displayed finite-projective E, the unit $\eta_{X_A} : X_A \to \Im_{\inf} X_A$ is an effective (epimorphic) descent morphism of $\mathscr{H}_{B,\mathrm{et}}$. This assertion fails for general finitely presented affine algebras: the unit of $X = \iota_! \operatorname{Spec}(B[x]/x^2)$ is not an epimorphism.*

Proof.

(1) **Prove effective lifting for finite-projective affine cells.** The functor of points of $A = \mathrm{Sym}_B(E^\vee)$ is $S \mapsto E \otimes_B S$. A displayed finite-free retract for E exhibits $E \otimes_B S \to E \otimes_B R$ as a retract of a finite power of the surjection $S \to R$; hence it is a regular epimorphism in $\mathcal{E}$, uniformly in j. The sheaf unit is thus locally surjective, hence effective epic. In an elementary topos every epimorphism is an effective descent morphism for objects, by the regular kernel-pair and descent construction.

(2) **Construct the singular lifting problem.** For the counterexample use $S = B[t]/(t^3) \twoheadrightarrow R = B[t]/(t^2)$. The element $t \in R$ is a point of $\mathrm{Spec}(B[x]/x^2)$.

(3) **Show that the lift fails locally.** Any lift to S would have the form $t + at^2$, but $(t + at^2)^2 = t^2 \neq 0$ in S; the equation cannot hold even after a faithfully flat étale extension of S. Thus the associated component of the unit is not locally surjective. The equation has no solution in the coefficient algebra, independently of the chosen implementation names.

□

Proposition 10.12 (A natural nonpolynomial Weil transformation)**.** *The reduction–constant-extension transformation $R = \epsilon_{\mathcal{R}_B}\eta_{\mathcal{R}_B} : \mathcal{R}_B \to \mathcal{R}_B$ is natural in projective Weil stages, satisfies $R_B = \mathrm{id}_B$, and annihilates every nilpotent coefficient. It is not represented by any polynomial in $B[x]$. Hence the polynomial higher-diagonal cochains form a proper subcalculus of all maps of Weil functors.*

Proof.

(1) **Identify the natural reduction map.** At $C = B \oplus N$, $R_C(b + n) = b$; augmented stage maps preserve this formula. It fixes B and kills N.

(2) **Exclude a polynomial representative.** If $p \in B[x]$ represented R, then $p(b) = b$ for every $b \in B$. Choose a degree bound $d \geq 1$ for $p - x$ and evaluate at $d + 1$ distinct rational constants. The Vandermonde determinant is a product of nonzero rationals, hence a unit of B. Thus $p = x$. But at $B[\varepsilon]/\varepsilon^2$, reduction sends ε to zero, whereas $p(\varepsilon) = \varepsilon \neq 0$, a contradiction.

(3) **Identify the scope of the example.** This is a nonpolynomial map in the based-Weil geometry. In the primary geometry, Theorem 10.8 instead identifies the completed coordinate cochains with all $\mathcal{O}_{\mathrm{inf}}$-valued functions on the represented formal nerve.

□

At each finite order, the coefficient constructions commute with the stated gauge and input changes. Passing to an infinite coefficient object raises a further question: can one denominator serve all orders? The next comparison records precisely where finite-order transport and completed transport differ.

10.6 Formal-order coherence and coefficient localization

Proposition 10.13 (Finite-order coefficient change)**.** *For a localization $B \to B_P$ constructed by an overhead gauge, put $A_P = B_P \otimes_B A_E$. For every r, the canonical map*

$$B_P \otimes_B P^r_{E/B} \xrightarrow{\cong} P^r_{A_P/B_P}$$

is an isomorphism of augmented algebras. It commutes with the source and target actions, the differential comparison, truncation, and all higher-diagonal substitution maps.

Proof.

(1) **Localize the diagonal ideal and its powers.** Localization is flat by Theorem 2.37. It preserves the kernel of multiplication μ_E, and hence carries its ideal to the kernel of the base-changed multiplication. Products of a finite number of ideal elements commute with tensor extension, so it carries I_E^{r+1} onto the $(r+1)$-st power of the new ideal.

(2) **Compare finite principal parts and their structure maps.** Flatness preserves the resulting quotient sequence. Every map in the comparison is induced by the universal algebra maps and finite multiplication, so the stated diagrams commute.

□

For the completed algebra the finite comparisons induce

$$B_P \otimes_B \widehat{P}_{E/B} \longrightarrow \varprojlim_r (B_P \otimes_B P^r_{E/B}). \tag{10.11}$$

The following measured family identifies the resource distinction between its two sides.

Example 10.14 (Measured finite programs and a completed resource family)**.** Work in the cofinite numerical instance and put $t(n) = 1/(n+1)$, $q = t^{-1}$, and $R = B[t^{-1}]$. Use an idle invocation carrying a natural-number tag; it leaves the returned input unchanged, and its tag is retained in the history. The finite iterations in Algorithm 10.1 construct e_r; specializing the finite description at r constructs the map $r \mapsto e_r$. The variable m records the repetition count $(n+1)^j$ at the start of the j-th pass.

Algorithm 10.1 A finite program for the order-r measured family

Input: $n, r \in \mathbb{N}$

```
m ← 1
for j = 0, ..., r do
    for k = 1, ..., m do
        IdleTag(j)
    end for
    m ← (n + 1)m
end for
return n
```

Give event j the weight z^j and other invocation events weight zero. For each nonnegative rational z, the measured cost and its germ polynomial are

$$T_{r,z}(n) = \sum_{j=0}^{r} (n+1)^j z^j, \qquad F_r(z) = \sum_{j=0}^{r} q^j z^j. \tag{10.12}$$

The polynomial coefficient extension is calculated in the germ algebra K. The histories have the same returned value, and Theorem 7.1 identifies their measured coefficients with geometric Taylor coefficients. Every coefficient of F_r belongs to $t^{-r}B$, since $q^j = t^{-r}t^{r-j}$ for $j \le r$. Truncation gives $F_{r+1} \bmod z^{r+1} = F_r$, so these polynomials determine

$$F(z) = \sum_{j \ge 0} q^j z^j \in R[[z]], \qquad (1 - qz)F(z) = 1.$$

The measured objects are the finite programs e_r and their costs; F is the compatible formal generating function of their coefficients.

Proposition 10.15 (Finite polynomial budgets without a common degree)**.** *The canonical map*

$$B[[z]][t^{-1}] \longrightarrow B[t^{-1}][[z]] \tag{10.13}$$

is injective. Its image consists exactly of coefficient families contained in one budget $t^{-N}B$ for some $N \in \mathbb{N}$. The measured formal family F of Example 10.14 lies outside this image, although every finite truncation lies in it.

Proof.

(1) **Characterize series with a common denominator.** A fraction $t^{-N}b(z)$ has all its coefficients in $t^{-N}B$. Conversely, a coefficient family with that common budget becomes a unique B-coefficient series after multiplication by t^N, giving the required fraction.

(2) **Prove injectivity and exclude the displayed series.** Regularity of t proves injectivity coefficientwise. If F had such a denominator, its coefficient at $j = N + 1$ would give $t^N q^{N+1} = t^{-1} \in B$, a contradiction.

(3) **Recover the finite-order budget bounds.** The finite budget calculation in Example 10.14 proves the last assertion.

□

Thus localization and completion distinguish an order-dependent polynomial growth degree from a single degree valid for every coefficient. The constants in each coefficient's Landau estimate may depend on that coefficient. Formal order is also distinct from merely quotienting by asymptotically vanishing errors, as the next calculation shows.

Proposition 10.16 (The Landau vanishing ideal is idempotent)**.** *For every proper-filter rational germ context,*

$$\mathfrak{m}^2 = \mathfrak{m}, \qquad \varprojlim_r B/\mathfrak{m}^{r+1} \cong B/\mathfrak{m}.$$

Consequently completion at the original small-error ideal cannot replace the diagonal-adic formal-order tower.

Proof.

(1) **Construct a rational square-root bound.** Let $f \in \mathfrak{m}$. There is a uniform rational operation $q \mapsto s(q)$ on nonnegative rationals satisfying

$$s(0) = 0, \qquad q \le s(q)^2 \le 4q \quad (q > 0),$$

obtained by locating a positive rational between successive powers of two using finite rational comparison and doubling.

(2) **Factor a vanishing germ.** Apply it pointwise to $|f|$ and write $g = s(|f|)$. Define $h = f/g$ on the nonzero support, zero elsewhere. These are rational-valued functions and respect eventual equality. One has $f = gh$ and $|h| \le g$ pointwise.

(3) **Prove that both factors vanish.** For every $k \ge 1$, the smallness bound $4k^2|f| \le 1$ on an eventual extent yields $kg \le 1$ there, and hence also $k|h| \le 1$. Thus $g, h \in o(1) = \mathfrak{m}$. So $f \in \mathfrak{m}^2$.

(4) **Deduce idempotence and the constant quotient tower.** The opposite inclusion follows from ideality. All positive powers of the idempotent ideal equal $\mathfrak{m}$, and the inverse system of quotients is constant at $B/\mathfrak{m}$.

□

10.7 Formal coordinates on indexed resource cells

We now apply formal coordinates to the universal resource family. The budget classifier remains $\mathfrak{C}_{\mathrm{Ev}}$; the new objects are neighbourhoods of its resource cells. At a principal budget $a \in L$, the module is $\mathscr{M}(a) = M_a$. Over the regular-epimorphic cover supplying a representative g, it is the displayed finite-projective module $Bg \cong Be_g$. Construct $A_a = \mathrm{Sym}_B(M_a^\vee)$ and its principal parts locally, then descend along the representative kernel pair. An inclusion $M_a \hookrightarrow M_b$ induces the contravariant coordinate map

$$A_b \longrightarrow A_a.$$

Free algebras, diagonal ideals, finite quotients, and completed higher diagonals are functorial in these coordinate maps. The coordinate algebras therefore vary contravariantly with the budget order, while their represented formal spaces vary covariantly. This gives the geometric budget diagram with the required variance.

The descent depends only on the marked module. Equal growth classes give the same embedded image $Bg \subseteq K$, and finite-projective duality and principal parts are intrinsic to that image. Their local presentations consequently agree on kernel pairs and descend by effective quotients in $\mathcal{E}$.

Proposition 10.17 (Preservation of the global resource classifiers)**.** *The functors and formal systems of this section leave unchanged the classifying equivalence* $\mathrm{Loc}_{\mathcal{E}}(Z, \mathfrak{C}_{\mathrm{Ev}}) \cong \mathrm{Sub}_{\Delta_Z B}(\Delta_Z K)$*, the universal marked module, all finite resource-diagram classifiers, and the feasibility classifier* $\mathfrak{F} = \mathbb{S}^{\mathfrak{C}_{\mathrm{Ev}}}$*. The formal differential objects are constructed* over *their already classified parameters.*

Proof.

(1) **Preserve the indexed classifying data under completion.** The classifiers are defined from B, K, L and their marked subobjects. Formal completion acts on the polynomial coordinate algebras over these parameters. The finite module and algebra constructions commute with parameter pullback, so the original classifying maps and universal properties continue to classify the indexed data.

□

10.8 Observed formal towers and their effective points

We equip a compatible family of formal coefficients with the observation structure generated by its finite stages and apply Section 1.3. The generator test of Lemma 1.14 identifies its effective-point object. For cofinite numerical input, positive tests on these coefficient modules are constant by Proposition 2.30. The numerical covers of Proposition D.3 retain the finite computations on coefficient names; their evaluation equations are specified below.

Theorem 10.18 (Effective points of an observed inverse tower)**.** *For an internal tower* $(D_r, \mathcal{O}_r)$ *in* $\mathbf{EDcpo}_\Sigma$*, form the underlying compatible-family poset* $D_\infty = \varprojlim_r D_r$*. Its pointwise directed suprema and the observation structure*

$$\mathcal{O}_\infty = \mathrm{cl}_\Sigma\{\pi_r^* U : r \in \mathbb{N},\ U \in \mathcal{O}_r\} \tag{10.14}$$

construct its inverse limit in $\mathbf{EDcpo}_\Sigma$*. There is a canonical isomorphism over* D_∞*,*

$$\mathrm{Pt}_\Sigma D_\infty \cong \varprojlim_r \mathrm{Pt}_\Sigma D_r. \tag{10.15}$$

It is uniform in parameters and natural in maps of observed towers. On the generating observations its positive evaluation is

$$\mathrm{Obs}_{D_\infty}(x, \pi_r^* U) = \mathrm{Obs}_{D_r}(\pi_r x, U).$$

Proof.

(1) **Construct the observed limit.** Take directed suprema coordinatewise in the compatible-family poset. Scott continuity of the transition maps preserves compatibility, and the coordinatewise universal property gives the supremum. Projection observations are Scott open; their Σ-observation closure remains so by Section 1.3. Every observed cone preserves these generators and their closure, proving the limit property in $\mathbf{EDcpo}_{\Sigma}$.

(2) **Identify its effective points.** Apply Lemma 1.14 to the projection-generated observations. Explicitly, a parameterized compatible family of effective points has the single positive evaluation

$$(r, U) \longmapsto \mathrm{Obs}_{D_r}(x_r, U).$$

If two generators have the same pullback, their values agree in $\Omega_{\mathcal{E}}$, hence in Σ by monicity. The evaluation descends to the generator image and extends to its observation closure. This constructs the inverse to the effective-projection map in (10.15).

(3) **Verify naturality.** Both maps lie over the same compatible-family object. Uniqueness in the effective-point pullback makes them inverse and proves naturality in tower maps and parameters.

□

Construction 10.19 (Canonical observations on coefficient towers)**.**

(1) **Realize each coefficient object discretely.** For a coefficient object C of $\mathcal{E}$, set $\mathsf{D}(C) = F_{\Sigma}(C_{\mathrm{disc}})$. By the discrete free-completion calculation in Section 1.3, its underlying dcpo is C, its observations are the maps $C \to \Sigma$ viewed in $\Omega_{\mathcal{E}}$, and

$$\mathrm{Pt}_{\Sigma}\, \mathsf{D}(C) \cong C.$$

(2) **Transport coefficient maps.** Precomposition makes every arrow $f : C \to C'$ observation-preserving, defining $\mathsf{D}(f)$.

(3) **Assemble the observed inverse tower.** Apply this construction to $C = P_q^r$ and its truncation maps, and then use Theorem 10.18. It constructs an observed dcpo $\widehat{\mathsf{D}}^{\,q}$ with

$$\mathrm{Pt}_{\Sigma}\, \widehat{\mathsf{D}}^{\,q} \cong \varprojlim_r \mathrm{Pt}_{\Sigma}\, \mathsf{D}(P_q^r) \cong \varprojlim_r P_q^r \cong \underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}_{X_A}^{(q)}, \mathcal{O}_{\mathrm{inf}}). \tag{10.16}$$

The same construction applies to finite normalized coefficient objects and their compatible-family limit. The final isomorphism is Theorem 10.8.

(4) **Identify semantic tests and construct named observations.** In the cofinite numerical fibre these objects and their limits are B-modules, so their maps to Σ are constant by Proposition 2.30. Thus the effective-point identifications above do not assert nonconstant tests of coefficient germs. Use the numerical covers $q_r : N_r \twoheadrightarrow P_q^r$ and $q_\infty : N_\infty \twoheadrightarrow \varprojlim_r P_q^r$ of Proposition D.3. Their coordinate maps satisfy $q_r p_r = \pi_r q_\infty$. A finite numerical test on $p_r(n)$ is a test on the supplied coefficient name. It descends to a semantic positive test exactly when it coequalizes the name kernel pair; every such descended test is constant in this fibre.

Proposition 10.20 (Finite-output operations on coefficient names)**.** *An internal map between coefficient towers, or between their limits whose order-r output is given uniformly by a finite coefficient map at an order $b(r)$ computed from r, preserves the generated observations. It therefore induces the corresponding map of effective points. Finite arithmetic, polynomial*

Taylor substitution, truncation, and the coface maps have this property; the de Rham differentials and completed homotopies constructed below have the same property by their stated finite coefficient formulas.

In $\mathcal{E}$, the numerical covers of Proposition D.3 carry uniform trackers for these finite formulas. If $q_\infty : N_\infty \twoheadrightarrow C_\infty$ presents the input limit and $q'_r : N'_r \twoheadrightarrow C'_r$ presents output order r, the constructed operations satisfy

$$q'_r \, \widehat{f}_r \, p_{b(r)} = \pi'_r f \, q_\infty.$$

The order and the supplied parameter names are inputs to one numerical operation. Finite output tests are made on these returned names; the displayed equation gives their exact coefficient denotations.

Proof.

(1) **Transport finite-output observations.** For a generating output observation $\pi_r^* U$, its pullback is $\pi_{b(r)}^*(U \circ f_r)$, where f_r is the displayed finite coefficient map. Precomposition is an allowed observation in the canonical observation structure on coefficients. For a map of general observed towers the same assertion is its finite-stage observation-preservation equation. Lemma 1.14 extends the result to all generated observations; its effective-point functor then supplies the claimed map.

(2) **Compose the specified arithmetic name maps.** Each finite formula is a composition of arithmetic, finite-list, rational-coefficient, and matrix operations. Their numerical covers and tracked lifts are constructed in Proposition D.3; their implementations compose by Lemma D.6.

(3) **Construct one uniform operation on finite outputs.** The order bound $b(r)$ and the finite lists of coefficient operations are constructed by NNO recursion, so the index is an input to one uniform operation.

(4) **Verify products, descent, and dependent name transport.** Use the finite product of numerical coefficient covers for several inputs. Coordinate evaluation followed by the finite tracker gives $\widehat{f}_r p_{b(r)}$; composing with q'_r gives $\pi'_r f q_\infty$ by the defining value equations. For dependent inputs retain the parameter cover and pull back the coefficient presentations as in Proposition D.3. These equations descend, proving exact denotation and parameter coherence.

□

The limit classifies compatible effective points with the generated observations. Its numerical name cover and uniform finite evaluations are constructed in Proposition D.3. For cofinite coefficients, positive semantic tests are constant; the finite computation remains recorded on its names.

We can now use the formal neighbourhoods and their identified function objects to build cochains on the infinitesimal nerve. The next section compares those cochains with relative differential forms.

11 De Rham resource geometry

Resource variations have differential coefficients, and compatible changes of those coefficients require transport. De Rham forms express the infinitesimal data; curvature tests the compatibility of a proposed connection, while a stratification records coherent transport at every formal order. The event instrumentations of Proposition 11.12 exhibit this compatibility

test. The completed measured family of Example 10.14 provides a nonzero logarithmic class and an explicit flat transport (Proposition 11.15).

Functions on the completed infinitesimal nerve and Kähler forms give two descriptions of this differential information. We construct a chain-homotopy equivalence between them over the resource coefficients. The geometric function calculation is Theorems 9.9 and 10.8; its uniform finite computations on coefficient names are constructed in Construction 10.19 and Proposition 10.20. The de Rham and crystalline precedents are (Grothendieck 1966; Berthelot and Ogus 1978), and the finite rational contraction uses a bar–Koszul calculation related to (Hochschild et al. 1962).

Relative forms differentiate valuation parameters while fixing B. Absolute forms over $\mathbb{Q}$ also differentiate resource coefficients; the logarithmic connections on $t^{-k}B$ in Proposition 13.15 use that second calculus. The primary site supplies the function objects for the comparison, and based-Weil restriction recovers their finite polynomial coefficient formulas.

11.1 Completed higher-diagonal cochains

A cochain evaluates a function on a finite sequence of infinitesimally related points. The completed higher diagonals provide these configurations and their coordinate algebras.

Definition 11.1 (Completed diagonal cochains)**.** Fix E finite-projective, $V = E^{\vee}$, and $A = \mathrm{Sym}_B(V)$. The **completed diagonal cochains** in degree $q \geq 0$ are

$$D_q = A^{\otimes_B(q+1)}, \quad I_q = \ker(D_q \xrightarrow{\mu_q} A), \quad \widehat{C}^q(A/B) = \varprojlim_r D_q/I_q^{r+1}.$$

By Theorem 10.8 there is a natural cosimplicial algebra identification

$$\widehat{C}^q(A/B) \cong \underline{\mathrm{Hom}}_{\mathscr{H}_{B,\mathrm{et}}}(\mathcal{D}^{(q)}_{X_A}, \mathcal{O}_{\mathrm{inf}}).$$

These limits are taken in internal B-modules or algebras; (10.9) gives their coefficient-sheaf realization. To calculate their maps, write the successive coordinate differences as $u_i = x_i - x_{i-1}$, $1 \leq i \leq q$. The finite-projective change of variables gives

$$\widehat{C}^q(A/B) \cong A[[u_1, \ldots, u_q]], \tag{11.1}$$

where each u_i denotes a block of variables dual to E and the completion is in the total difference degree. The degree $q = 0$ algebra is A.

A simplicial map of the higher diagonal sends a vertex to a selected or repeated vertex, hence sends each difference to a finite sum of successive differences. This preserves the powers of I_q and constructs the cosimplicial algebra

$$\widehat{C}^{\bullet}(A/B) : \Delta \longrightarrow \mathrm{CAlg}_B(\mathcal{E}).$$

Definition 11.2 (Normalized infinitesimal cochains)**.** The **normalized infinitesimal cochains** and their differential are

$$N^q_{\mathrm{inf}}(A/B) = \bigcap_{i=0}^{q-1} \ker(s_i : \widehat{C}^q \to \widehat{C}^{q-1}), \qquad \delta = \sum_{i=0}^{q+1} (-1)^i d_i. \tag{11.2}$$

By the cosimplicial identities $\delta^2 = 0$ and $\delta N^q \subseteq N^{q+1}$. These are equations of maps, proved by pairing the two terms corresponding to any two deleted vertices.

Proposition 11.3 (Explicit completed coface formula)**.** *For a normalized q-cochain $f(x; u_1, \dots, u_q)$,*

$$\begin{aligned}\delta f(x; u_1, \dots, u_{q+1}) = {} & f(x + u_1; u_2, \dots, u_{q+1}) \\ & + \sum_{i=1}^{q} (-1)^i f(x; u_1, \dots, u_i + u_{i+1}, \dots, u_{q+1}) \\ & + (-1)^{q+1} f(x; u_1, \dots, u_q). \end{aligned} \tag{11.3}$$

Normalization consists exactly of series having positive degree in each successive difference block u_i. All displayed operations are defined by finite polynomial substitution at each fixed homogeneous degree and commute with truncation.

Proof.

(1) **Compute the coface substitutions.** Deleting the first vertex changes the base point from x to $x+u_1$; deleting the i-th interior vertex replaces the two adjacent differences by their sum; deleting the last vertex removes the last difference. This proves the formula.

(2) **Identify normalized cochains.** Repeating one vertex sets one of its adjacent differences to zero. Being in all codegeneracy kernels therefore means that every monomial contains a positive contribution from each difference block.

(3) **Extend the formulas to completed cochains.** The substitutions respect the total-degree filtration, and each resulting coefficient is a finite sum, proving the completion assertion.

□

The normalized-completed comparison is summarized by the commuting diagram

$$\begin{array}{ccc} N^{\bullet}_{\mathrm{inf}}(A/B) & \xrightarrow{\ \rho\ } & \Omega^{\bullet}_{A/B} \\ \Big\downarrow{\scriptstyle \delta} & & \Big\downarrow{\scriptstyle d} \\ N^{\bullet+1}_{\mathrm{inf}}(A/B) & \xrightarrow[\ \rho\ ]{} & \Omega^{\bullet+1}_{A/B}. \end{array} \tag{11.4}$$

The maps will be constructed and compared in Theorem 11.6.

The normalized complex has an Alexander–Whitney cup product,

$$(f \smile g)(x; u_1, \dots, u_{p+q}) = f(x; u_1, \dots, u_p) g(x + u_1 + \cdots + u_p; u_{p+1}, \dots, u_{p+q}). \tag{11.5}$$

Associativity is reassociation of the same ordered vertices. The coface calculation gives $\delta(f \smile g) = \delta f \smile g + (-1)^p f \smile \delta g$.

11.2 Relative forms and the infinitesimal comparison

We now compare the normalized cochains with the algebraic de Rham complex.

Definition 11.4 (The relative de Rham complex)**.** Here **algebraic de Rham complex** means the complex formed from Kähler differentials of the specified B-algebra. The module of Section 7.6 supplies its degree-one forms. Define

$$\Omega^p_{A/B} = \bigwedge_A^p \Omega^1_{A/B}, \qquad d : A \to \Omega^1_{A/B}$$

and extend d as the unique graded derivation of degree one with $d(da) = 0$.

The alternating Leibniz identities give $d^2 = 0$. For a polynomial resource cell,

$$\Omega^{\bullet}_{A/B} \cong A \otimes_B \bigwedge_B^{\bullet} V. \tag{11.6}$$

This construction is internal to the B-algebras in $\mathcal{E}$.

The comparison extracts the first-order contribution in each successive displacement and alternates it. For a normalized q-cochain f, a temporary finite free presentation gives the coefficient formula

$$\rho_q(f) = \sum_{a_1,\dots,a_q} [u_1^{a_1} \cdots u_q^{a_q}] f \; dx^{a_1} \wedge \cdots \wedge dx^{a_q}. \tag{11.7}$$

The exterior quotient makes this alternating tensor formula independent of the temporary basis. At degree zero it is $\rho_0 = \mathrm{id}_A$. The theorem proves that this map retains the cohomological information of the completed cochains.

Lemma 11.5 (Homogeneous additive-cochain calculation and natural contraction)**.** *Let V be a displayed finite-projective module over the rational commutative ring B. In homogeneous difference degree m, put*

$$C_0^{q,m}(V) = \bigoplus_{\substack{m_1+\cdots+m_q=m \\ m_i \geq 1}} \mathrm{Sym}_B^{m_1}(V) \otimes_B \cdots \otimes_B \mathrm{Sym}_B^{m_q}(V)$$

with $C_0^{0,0} = B$ and all other degree-zero summands absent. The degree-preserving component δ_0 of (11.3) has cohomology

$$H^q(C_0^{\bullet,m}(V)) \cong \begin{cases} \bigwedge_B^m V, & q = m, \\ 0, & q \neq m. \end{cases}$$

There are B-linear homogeneous operators i_m, p_m, h_m of cochain degrees $0, 0, -1$, respectively, with target complex $\bigwedge_B^m V[-m]$ (and $B[0]$ at $m = 0$), such that

$$\begin{gathered} p_m i_m = 1, \quad \delta_0 i_m = 0, \quad p_m \delta_0 = 0, \\ \delta_0 h_m + h_m \delta_0 = 1 - i_m p_m, \qquad h_m^2 = h_m i_m = p_m h_m = 0. \end{gathered} \tag{11.8}$$

They are natural under linear maps of displayed finite-projective modules, and their coefficients are finite rational expressions constructed uniformly in m.

Proof.

(1) **Identify the homogeneous bar–cobar complex.** The differential is the reduced cobar differential of the graded symmetric coalgebra on V, with coproduct induced by $u \mapsto u \otimes 1 + 1 \otimes u$. In homogeneous degree m this is a finite complex. Its graded dual is the reduced bar complex of $\mathrm{Sym}_B(V^\vee)$: the divided-power dual is identified with the symmetric algebra using invertible factorials.

(2) **Compute its cohomology by the bar–Koszul comparison.** The Koszul resolution on a displayed free basis of $V^\vee$ resolves its augmentation B; exactness follows by tensoring the one-variable free resolutions. The normalized bar resolution resolves the same augmentation by its extra-degeneracy contraction. Their antisymmetrization comparison is a quasi-isomorphism of free resolutions (the usual bar–Koszul calculation (Hochschild et al. 1962)); tensoring with B and taking finite graded duals gives the stated cohomology, concentrated at $q = m$.

(3) **Present the finite permutation complexes.** We construct the homotopy on finite permutation modules. Give the set of ordered block decompositions of $\{1,\dots,m\}$ its decidable finite permutation presentation. In degree q the induced permutation modules

$$P_m^q = \bigoplus_{\substack{m_1+\cdots+m_q=m \\ m_i\geq 1}} \mathbb{Q}[S_m] \otimes_{\mathbb{Q}[S_{m_1}\times\cdots\times S_{m_q}]} \mathbb{Q}$$

carry the bar–cobar differential by splitting adjacent blocks into their shuffles with the signs of δ_0. With right/left permutation conventions matched, symmetrization gives a natural isomorphism

$$C_0^{q,m}(V) \cong V^{\otimes_B m} \otimes_{\mathbb{Q}[S_m]} P_m^q.$$

The alternating representation sgn_m gives the corresponding presentation of $\bigwedge_B^m V$. All these representations are finite-dimensional rational vector spaces with specified permutation bases.

(4) **Identify the alternating cohomology representation.** To identify its cohomology equivariantly, evaluate the free-module bar–Koszul calculation at $V = \mathbb{Q}^m$ and take the multidegree $(1,\dots,1)$ in its m labelled generators. The resulting summand of $C_0^{\bullet,m}(V)$ is precisely $P_m^\bullet$, and the corresponding summand of $\bigwedge^m V$ is the sign representation sgn_m. Thus the calculation identifies the cohomology of $P_m^\bullet$ as sgn_m in degree m and zero elsewhere, equivariantly under permutations.

(5) **Construct the projection and antisymmetrization.** Take $p : P_m^\bullet \to \mathrm{sgn}_m[-m]$ to be the exterior quotient and take i to be normalized antisymmetrization:

$$i(v_1 \wedge \cdots \wedge v_m) = \frac{1}{m!} \sum_{\sigma\in S_m} \mathrm{sgn}(\sigma) v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(m)}.$$

These equivariant chain maps satisfy $pi = 1$. The cohomology calculation above shows that $K_m^\bullet = \ker p$ is acyclic.

(6) **Construct equivariant splittings of the acyclic kernel.** Write $B_m^q = \mathrm{im}(\delta_0 : K_m^{q-1} \to K_m^q)$. In every degree there is an exact sequence of finite rational representations

$$0 \longrightarrow B_m^q \longrightarrow K_m^q \xrightarrow{\delta_0} B_m^{q+1} \longrightarrow 0.$$

Algorithm 11.1 constructs a rational section $s_q : B_m^{q+1} \to K_m^q$ and its equivariant average $\bar{s}_q$. Since δ_0 is equivariant, $\delta_0\bar{s}_q = 1$ still holds. Consequently $K_m^q = B_m^q \oplus \bar{s}_q(B_m^{q+1})$ is an equivariant direct-sum decomposition.

(7) **Define the contracting homotopy.** For $z \in P_m^q$ set

$$h_q(z) = \bar{s}_{q-1}(z - ipz - \bar{s}_q\delta_0 z).$$

The argument of $\bar{s}_{q-1}$ belongs to B_m^q: it lies in $\ker p$ and its differential is zero.

(8) **Verify the homotopy equation and side conditions.** Now $\delta_0 h_q z = z - ipz - \bar{s}_q\delta_0 z$, while $h_{q+1}\delta_0 z = \bar{s}_q\delta_0 z$. Thus $\delta_0 h + h\delta_0 = 1 - ip$. The image of h_q is the complementary summand $\bar{s}_{q-1}(B_m^q)$, on which the next h vanishes; also $hi = 0$ and $ph = 0$. This proves all the side conditions in (11.8). The $m = 0$ case is $i = p = 1$, $h = 0$.

(9) **Transport the contraction naturally to coefficient modules.** Finally apply $V^{\otimes m} \otimes_{\mathbb{Q}[S_m]} (-)$. The homotopy equations remain equations of finite matrices, valid over every rational B; all operations are natural in V because S_m acts by permuting tensor factors. If V is a retract of a finite free module, the same functorial maps restrict along its displayed idempotent.

(10) **Construct the family uniformly in the degree.** Algorithm 11.1 uses the finite permutation lists and rational matrices supplied above. Its loops and Gaussian elimination give one NNO-recursive construction for all m.

Algorithm 11.1 Equivariant contraction of the finite permutation complex

Input: Degree m; rational matrices for $P_m^\bullet, \delta_0, i, p$ and the S_m-action

if $m = 0$ **then**
  return $(i, p, h) = (1, 1, 0)$
end if
for $q = 0, \dots, m$ **do**
  Compute $K_m^q = \ker p$ and $B_m^q = \delta_0(K_m^{q-1})$
end for
for $q = 0, \dots, m$ **do**
  Compute a section s_q of $\delta_0 : K_m^q \to B_m^{q+1}$
  by rational Gaussian elimination in lexicographic order
  $\bar{s}_q \leftarrow \frac{1}{m!} \sum_{\sigma \in S_m} \sigma s_q \sigma^{-1}$
end for
for $q = 0, \dots, m$ **do**
  $h_q \leftarrow \bar{s}_{q-1}(1 - ip - \bar{s}_q \delta_0)$
end for
return (i, p, h)

Here complexes and sections are zero outside their displayed degrees.

□

Theorem 11.6 (Completed infinitesimal de Rham comparison)**.** *For every displayed finite-projective B-module E, the maps* (11.7) *assemble into a natural cochain map*

$$\rho : N_{\inf}^\bullet(\mathrm{Sym}_B(E^\vee)/B) \longrightarrow \Omega^\bullet_{\mathrm{Sym}_B(E^\vee)/B}.$$

It is a chain-homotopy equivalence*, with a parameter-uniform homotopy constructed on compatible truncations. In particular it is a quasi-isomorphism, natural under polynomial substitution and in internal resource parameters. Through Theorem 10.8, this is the comparison from normalized functions on the completed infinitesimal nerve to the relative de Rham complex. It carries the Alexander–Whitney product to the exterior product and induces an isomorphism of graded cohomology rings.*

Proof.

(1) **Decompose normalized completed cochains by degree.** Put $V = E^\vee$, $A = \mathrm{Sym}_B(V)$ and use the total difference-degree filtration. Normalization and the finite symmetric-power decomposition of the previous lemma identify

$$N_{\inf}^q(A/B) \cong \prod_{m \geq q} A \otimes_B C_0^{q,m}(V).$$

(2) **Separate the homogeneous differential and perturbation.** Split the coface differential as $\delta = \delta_0 + \Delta$. The first part keeps the base point x fixed and has homogeneous difference degree zero. The perturbation Δ comes only from expanding the coefficients of $f(x + u_1; \dots)$ around x: it raises total difference degree by at least one.

(3) **Assemble the homogeneous contraction.** The contraction operators of Lemma 11.5, extended over A and assembled as product maps i, p, h, obey

$$pi = 1, \quad \delta_0 h + h\delta_0 = 1 - ip, \quad h^2 = hi = ph = 0, \quad p = \rho.$$

The target is $\Omega^{\bullet}_{A/B}$ with zero homogeneous differential before perturbation.

(4) **Truncate both sides of the contraction.** For $r \in \mathbb{N}$, retain difference degrees at most r on the source and form degrees at most r on the target:

$$\begin{aligned} N^q_{\leq r} &= \bigoplus_{q \leq m \leq r} A \otimes_B C_0^{q,m}(V), \\ T^q_r &= \begin{cases} \Omega^q_{A/B}, & 0 \leq q \leq r, \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \tag{11.9}$$

The differential on T_r is initially zero. Since h and δ_0 preserve difference degree, while i inserts difference degree equal to form degree and p extracts it, the homogeneous contraction descends to $N_{\leq r}$ and T_r. Write its operators as $i_r, p_r, h_r, \delta_{0,r}$. The perturbation also descends, giving Δ_r and $\delta_r = \delta_{0,r} + \Delta_r$.

(5) **Construct the finite perturbation maps.** Both $h_r\Delta_r$ and $\Delta_r h_r$ raise difference degree, so their $(r+1)$st powers vanish. Define the finite sums

$$\begin{aligned} J_r &= \sum_{k=0}^{r} (-h_r\Delta_r)^k i_r, \\ H_r &= \sum_{k=0}^{r} (-1)^k h_r (\Delta_r h_r)^k, \\ p'_r &= p_r \sum_{k=0}^{r} (-\Delta_r h_r)^k, \qquad d'_r = p_r \Delta_r J_r. \end{aligned}$$

(6) **Verify the perturbation identities coefficientwise.** These are the finite-geometric-series formulas of the basic perturbation lemma, applied to the contraction onto T_r (Miller and Rahmati 2024). Substitution into $(\delta_{0,r} + \Delta_r)^2 = 0$ and the homogeneous contraction equations gives

$$\begin{aligned} (d'_r)^2 &= 0, \qquad \delta_r J_r = J_r d'_r, \qquad p'_r \delta_r = d'_r p'_r, \\ p'_r J_r &= 1_{T_r}, \qquad \delta_r H_r + H_r \delta_r = 1_{N_{\leq r}} - J_r p'_r. \end{aligned}$$

Deleting difference degree $r+1$ on the source and form degree $r+1$ on the target commutes with all these operators. In the finite sums, the additional terms vanish after this deletion. Thus the contractions form a compatible inverse system.

(7) **Identify the transferred projection.** Filtration identifies their transferred terms exactly. The map p extracts difference degree equal to cochain degree; h preserves difference degree but lowers cochain degree, so $p\Delta h = 0$. The same degree calculation holds at each finite stage. Thus $p'_r = p_r = \rho_r$, where ρ_r is the induced map $N_{\leq r} \to T_r$.

(8) **Identify the transferred de Rham differential.** Terms in $p\Delta(h\Delta)^k i$ with $k \geq 1$ have difference degree strictly above target cochain degree, and hence vanish. Only $p_r\Delta_r i_r$ survives in $d'_r = p_r\Delta_r J_r$. For a polynomial coefficient a, the component of degree

one of $a(x+u)-a(x)$ is da; applying alternating projection to the remaining linear difference coefficients gives

$$(d'_r)^q = d_r^q = \begin{cases} d_{\mathrm{dR}}^q, & 0 \le q < r, \\ 0, & \text{otherwise.} \end{cases}$$

Consequently (T_r, d_r) is the brutal truncation of the de Rham complex in form degrees at most r, with zero differential out of its top degree.

(9) **Pass to the compatible completed maps.** The source inverse limit is the completed normalized complex, and the target inverse limit is the full de Rham complex:

$$\varprojlim_r N^\bullet_{\le r} \cong N^\bullet_{\mathrm{inf}}(A/B), \qquad \varprojlim_r (T_r, d_r) \cong (\Omega^\bullet_{A/B}, d_{\mathrm{dR}}).$$

The second identification is degreewise eventually constant. The compatible finite formulas therefore define internal maps

$$\begin{aligned} J &= (1+h\Delta)^{-1} i = \sum_{k\ge 0} (-h\Delta)^k i, \\ H &= h(1+\Delta h)^{-1} = \sum_{k\ge 0} (-1)^k h(\Delta h)^k, \\ p' &= p(1+\Delta h)^{-1} = \rho, \qquad d' = p\Delta J = d_{\mathrm{dR}}. \end{aligned} \tag{11.10}$$

Every displayed sum is finite in each specified output difference degree. Taking inverse limits of the explicit operator identities proves the corresponding completed identities; no exactness assertion about inverse limits is used.

(10) **Establish the completed homotopy equivalence.** Consequently the completed maps satisfy

$$\begin{gathered} N^\bullet_{\mathrm{inf}}(A/B) \underset{J}{\overset{\rho}{\rightleftarrows}} \Omega^\bullet_{A/B} \\ \rho J = 1, \qquad \delta H + H\delta = 1 - J\rho, \\ \delta J = J d_{\mathrm{dR}}, \qquad \rho\delta = d_{\mathrm{dR}}\rho. \end{gathered} \tag{11.11}$$

This proves the asserted homotopy equivalence, including injectivity and surjectivity on all cohomology objects.

(11) **Verify finite-output construction and naturality.** All operators in a fixed output difference degree depend on only finitely many lower coefficients. Their defining algorithms use finite rational matrices and sums and commute with parameter substitution. In particular the compatible infinite corrections are produced by the NNO and the product universal property. The comparison ρ commutes with polynomial coordinate substitutions by the chain rule and the universal alternating quotient. The auxiliary maps H, J have the linear naturality established by their finite coefficient construction.

(12) **Verify compatibility with cup and wedge products.** Finally the degree-$(p+q)$ component of (11.5) linear in every difference block is the concatenation of those of f and g. Alternation gives $\rho(f \smile g) = \rho(f) \wedge \rho(g)$, as claimed.

□

For a polynomial coordinate map $p : E \to F$, naturality of the comparison itself is the commutative square of complexes

$$\begin{array}{ccc} N^{\bullet}_{\mathrm{inf}}(A_F/B) & \xrightarrow{\rho_F} & \Omega^{\bullet}_{A_F/B} \\ \Big\downarrow{\scriptstyle p^*} & & \Big\downarrow{\scriptstyle p^*} \\ N^{\bullet}_{\mathrm{inf}}(A_E/B) & \xrightarrow[\rho_E]{} & \Omega^{\bullet}_{A_E/B}. \end{array}$$

The finite homogeneous contraction constructs its homotopy equivalence; the full Taylor substitution supplies the vertical cochain map.

11.3 Relative formal Poincaré contraction

The preceding comparison relates two descriptions of forms. A second calculation checks the local formal neighbourhood itself by contracting its relative displacement directions.

Definition 11.7 (Continuous relative forms)**.** For n finite free formal variables put $R = A[[u_1, \dots, u_n]]$ and define the **continuous relative forms**

$$\widehat{\Omega}^{p}_{R/A} = R \otimes_A \bigwedge_A^{p} \Big(\bigoplus_{i=1}^{n} A\, du_i \Big).$$

The differential is termwise formal differentiation, which is well-defined because each output coefficient is a finite sum.

Theorem 11.8 (Formal Poincaré lemma)**.** *The inclusion $A \to \widehat{\Omega}^{\bullet}_{R/A}$ is a chain-homotopy equivalence. Its relative de Rham cohomology is A in degree zero and zero in positive degrees. The result is natural in finite-projective formal displacement modules.*

Proof.

(1) **Compute the Euler contraction identity.** Let $\mathsf{E} = \sum_i u_i \partial/\partial u_i$ and let ι_{E} be contraction. The graded Leibniz rule gives

$$d\iota_{\mathsf{E}} + \iota_{\mathsf{E}} d = \mathcal{L}_{\mathsf{E}}.$$

(2) **Define the contraction on positive weights.** On homogeneous coefficient degree m and form degree p the Lie derivative is multiplication by $m + p$. For $m + p > 0$ define $h = (m + p)^{-1} \iota_{\mathsf{E}}$, and set $h = 0$ on degree-zero constants. Rational invertibility and coefficientwise finite arithmetic make this an internal operator on the completed product.

(3) **Verify the deformation retraction and projective descent.** Its identity is

$$dh + hd = \mathrm{id} - \mathrm{const}\, \mathrm{ev}_0 \,.$$

This exhibits the required deformation retraction. The Euler operator and the homogeneous decomposition are intrinsic to symmetric powers and commute with finite-projective retractions.

□

11.4 Crystalline stratifications and integrable resource connections

Differential coefficients can also be transported between infinitesimally related points. A stratification records that transport and its composition law; an integrable connection records its first-order differential data. We construct their comparison for a displayed finite-projective module N over $A = \mathrm{Sym}_B(E^\vee)$. Pull N back along the two coefficient actions on principal parts:

$$s_r^* N = P^r_{A/B} \otimes_{A,s_r} N, \qquad t_r^* N = P^r_{A/B} \otimes_{A,t_r} N.$$

Definition 11.9 (Formal resource stratification)**.** With $s_r(a) = a(x)$ and $t_r(a) = a(x+u)$, a **formal stratification** in the source-to-target convention is a family of $P^r_{A/B}$-linear isomorphisms

$$\varepsilon_r : s_r^* N \xrightarrow{\cong} t_r^* N$$

compatible with truncation and equal to the identity on the reduced diagonal. At every finite completed triple diagonal, writing ε_{xy} for transport from the fibre over x to the fibre over y, its cocycle equation is

$$\varepsilon_{yz}\, \varepsilon_{xy} = \varepsilon_{xz}.$$

The morphisms are A-linear maps commuting with this family.

Definition 11.10 (Connections and curvature)**.** An **integrable relative connection** is a B-linear map

$$\nabla : N \to N \otimes_A \Omega^1_{A/B}$$

satisfying the Leibniz identity $\nabla(an) = a\nabla(n) + n \otimes da$ and whose graded extension satisfies $\nabla^2 = 0$. More generally, a **relative connection** satisfies the displayed Leibniz identity; its **curvature** is the square of the graded extension.

Curvature is an A-module map.

Theorem 11.11 (Connections are formal stratifications)**.** *For displayed finite-projective E and finite-projective $A = \mathrm{Sym}_B(E^\vee)$-modules N, integrable B-connections on N and formal stratifications of N form naturally equivalent categories. The equivalence commutes with polynomial substitution, parameter substitution, and finite-order coefficient base change.*

Proof.

(1) **Derive the iterated covariant Leibniz formula.** Write $P^r = P^r_{A/B}$ and $u = y - x$. We first construct the transport in the direction forced by its semilinearity. In a finite-free coordinate presentation write $\nabla n = \sum_i \nabla_i(n) \otimes dx^i$. The Leibniz identity and flatness say

$$\nabla_i(an) = \partial_i(a)n + a\nabla_i(n), \qquad [\nabla_i, \nabla_j] = 0.$$

The finite multi-index Leibniz formula follows by induction:

$$\nabla^\alpha(an) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} \partial^\beta(a) \nabla^{\alpha-\beta}(n).$$

(2) **Construct the finite Taylor transport.** Define a P^r-linear map by its values on t_r-generators,

$$T_r : t_r^* N \longrightarrow s_r^* N,$$
$$T_r(1 \otimes_{t_r} n) = \sum_{|\alpha| \le r} \frac{u^\alpha}{\alpha!} (1 \otimes_{s_r} \nabla^\alpha n). \tag{11.12}$$

(3) **Verify balancing and reduction.** Indeed finite reindexing of the Leibniz sum gives

$$T_r(1 \otimes_{t_r} an) = a(x+u)T_r(1 \otimes_{t_r} n) = t_r(a)T_r(1 \otimes_{t_r} n),$$

so the defining formula respects the t_r-balanced relation. Its reduction modulo I is the identity.

(4) **Prove the transport cocycle.** On the finite completed triple diagonal use vertices x, y, z, with differences $u = y - x$ and $v = z - y$. Expanding the iterated covariant derivatives and using their commutation, the ordinary multinomial identity gives

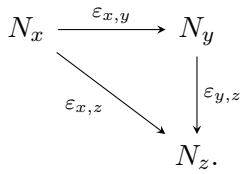
$$T_{x \leftarrow y}\, T_{y \leftarrow z} = T_{x \leftarrow z}, \qquad T_{x \leftarrow x} = 1.$$

Each coefficient is a finite sum, since total difference degree is truncated.

(5) **Construct the inverse stratification.** Swapping the vertices gives the reverse transport; the cocycle with $z = x$ proves that T_r is invertible. Define

$$\varepsilon_r = T_r^{-1} : s_r^* N \xrightarrow{\cong} t_r^* N.$$

The identity for T yields the source-to-target cocycle. Writing N_x, N_y, N_z for the three pullbacks on the completed triple diagonal, it is the commutative triangle



Reduction and truncation follow from the same properties of T.

(6) **Extract the connection from first-order transport.** Conversely, given ε, set $T = \varepsilon^{-1}$. Working in $s_1^* N$, define ∇n by

$$T_1(1 \otimes_{t_1} n) - (1 \otimes_{s_1} n) \mod I^2 \ \in\ (I/I^2) \otimes_A N \cong \Omega^1_{A/B} \otimes_A N.$$

(7) **Verify Leibniz and zero curvature.** The equality $T_1(1 \otimes_{t_1} an) = t_1(a)T_1(1 \otimes_{t_1} n)$ gives $\nabla(an) = a\nabla(n) + n \otimes da$ after taking the first difference coefficient. The alternating coefficient of $u_i v_j$ in the triple cocycle is $[\nabla_i, \nabla_j]$, so the cocycle gives zero curvature.

(8) **Recover every higher Taylor coefficient.** Differentiating the same triple cocycle once in the final nilpotent displacement recursively gives the higher coefficient rule $T_{\alpha+e_i} = (\alpha_i + 1)^{-1} \nabla_i T_\alpha$. It recovers uniquely the finite Taylor polynomial (11.12) because all positive integers are invertible in B.

(9) **Descend the construction and verify naturality.** For a displayed projective retract, replace the temporary free basis by a finite dual basis of E; constant directional derivations commute, and their symmetric tensor expressions factor through the displayed idempotent. Every identity above is an identity of finite symmetric-power morphisms. The construction therefore descends to the finite-projective cell, commutes with coordinate substitutions and coefficient base change, and is functorial on connection-preserving module maps.

(10) **Verify that the constructions are inverse.** The two procedures are mutually inverse by the first-order coefficient and the unique higher-coefficient recurrence.

□

11.5 Measured instrumentation, curvature, and coherent transport

Two changes of event measurement can preserve returned values while their order changes the resource data. Use the idle events a, b and let $N = (N_a, N_b)^{\mathsf{T}}$ be the corresponding count functions of a specified program. For natural numbers u, v, the finite substitutions

$$\sigma_u(a) = a, \quad \sigma_u(b) = ba^u, \qquad \tau_v(a) = ab^v, \quad \tau_v(b) = b$$

are implemented by the finite description transformation in Algorithm 11.2. It preserves parameter and variable leaves and replaces only the selected idle invocations.

Algorithm 11.2 Instrumentation by finite description substitution

Input: Well-typed finite description c; substitution $\chi = \sigma_u$ or τ_v

```
procedure INSTRUMENT(c, χ)
    if c is an invocation of a or b then
        return the sequential idle block χ(tag(c))
    else if c is a parameter, variable, or other primitive leaf then
        return c
    else
        for each child c_i of c do
            d_i ← INSTRUMENT(c_i, χ)
        end for
        return the same typed constructor with children (d_i)
    end if
end procedure
```

Since the inserted blocks are finite and leave the state unchanged, the finite return comparison identifies the return domains and values. The returning histories are the displayed free-monoid substitutions.

Writing count vectors as columns, their actions are

$$U_u = I + uP, \qquad V_v = I + vQ, \qquad P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

For instance σ_u changes (N_a, N_b) to $(N_a + uN_b, N_b)$. These polynomial laws extend to resource coefficients. The count functions lie in a common principal cell, so the two-component calculation is realized on a finite-projective resource bundle.

Proposition 11.12 (Instrumentation curvature and ordered transport)**.** *The two instrumentations have mixed-jet discrepancy*

$$U_\varepsilon V_\delta - V_\delta U_\varepsilon = \varepsilon\delta[P, Q], \qquad [P, Q] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{11.13}$$

over $B[\varepsilon, \delta]/(\varepsilon^2, \delta^2)$. On the coefficient bundle $B[u, v]^2$, the connection prescribing these constant infinitesimal updates,

$$\nabla_0 = d - P\,du - Q\,dv,$$

has curvature $[P, Q]\,du \wedge dv$ and admits no formal stratification inducing it. The ordered family $G(u, v) = V_v U_u$ instead constructs the integrable connection

$$\nabla_G = d - (dG)G^{-1}$$

and the formal transport $\varepsilon_{x,y} = G(y)G(x)^{-1}$. Its associated crystalline coaction is given by Proposition 12.9.

Proof.

(1) **Compute the measured mixed coefficient.** Expand the two products and compare the coefficient of $\varepsilon\delta$. The signed difference is (11.13). These are measured coefficients by the count formulas and Theorem 7.1.

(2) **Compute curvature and exclude a constant-rule stratification.** The differential of $-P\,du - Q\,dv$ is zero, and its exterior square is $[P, Q]\,du \wedge dv$, which is nonzero in the free coefficient bundle. The connection–stratification equivalence therefore rules out a stratification with these constant update rules.

(3) **Construct the polynomial frame and flat connection.** Since $P^2 = Q^2 = 0$, the inverse of G is $(I - uP)(I - vQ)$, a polynomial matrix. For a column h, $\nabla_G(Gh) = G\,dh$; applying the graded connection again gives zero curvature.

(4) **Verify the stratification cocycle.** The transport $G(y)G(x)^{-1}$ is the identity on the diagonal, is compatible with every truncation, and satisfies

$$G(z)G(y)^{-1}\,G(y)G(x)^{-1} = G(z)G(x)^{-1}.$$

It therefore gives the required stratification.

(5) **Identify the ordered update rules.** Its v-update is Q, while its u-update is $V_v P V_v^{-1}$; at the origin these are the original two updates. The conjugation records their specified order and supplies the compatible transport.

□

This example distinguishes the mixed response of two implemented instrumentations from their common returned computation. It also exhibits, on the same resource coefficients, a curvature obstruction and a flat transport. Formal stratifications act on the primary infinitesimal relation by Proposition 12.9; based restriction gives the displayed Taylor matrices.

11.6 Global resource coefficients, absolute forms, and a nonzero class

The construction is uniform in E, its displayed idempotent, and its coefficient maps. It therefore applies at each budget to $A_a = \mathrm{Sym}_B(M_a^\vee)$. Budget transitions act by polynomial coordinate restriction, and Theorem 11.6 identifies the induced maps on cochains and forms. This assembles the relative complexes and crystalline transport over the resource cells. The parameter space is still $\mathfrak{C}_{\mathrm{Ev}}$, with its previously specified points and positive observations.

Theorem 11.13 (Formal principal parts and de Rham comparison after localization)**.** *Let $A = \mathrm{Sym}_B(V)$ for a displayed finite-projective V, let $f \in A$, and put $A_f = A[f^{-1}]$. Write $D_q = A^{\otimes_B(q+1)}$ and $I_q = \ker(D_q \to A)$, with analogous notation $I_{q,f}$ over A_f. At every q, r there are natural isomorphisms*

$$\frac{A_f^{\otimes_B(q+1)}}{I_{q,f}^{r+1}} \;\cong\; A_f \otimes_A \frac{D_q}{I_q^{r+1}}. \tag{11.14}$$

They respect every vertex and truncation map. Consequently normalized completed coordinate cochains of A_f are chain-homotopy equivalent to $\Omega^\bullet_{A_f/B}$. The connection–stratification equivalence and finite Taylor transport extend to finite-projective A_f-modules.

Proof.

(1) **Construct inverses at every vertex.** Write f_i for the image of f at vertex i in the higher-diagonal algebra. Modulo I_q^{r+1}, the element $f_i - f_0$ is nilpotent of exponent $r+1$. After inverting f_0, the inverse of every other f_i is constructed by

$$f_i^{-1} = f_0^{-1} \sum_{k=0}^{r} \left(-\frac{f_i - f_0}{f_0} \right)^k \pmod{I_q^{r+1}}.$$

(2) **Identify the localized principal-parts algebra.** Thus the universal algebra obtained by inverting all the f_i at finite order is precisely the one obtained by inverting f_0. Since localization commutes with quotients and with finite powers of their defining ideals, the two universal properties give the inverse maps of (11.14).

(3) **Verify simplicial and truncation coherence.** The finite inverse is unique, so the comparison commutes with all simplicial vertex maps and truncation.

(4) **Extend the homogeneous contraction over the localization.** The normalized completed complex is consequently the inverse limit of its finite localizations $A_f \otimes_A (D_q/I_q^{r+1})$. In difference coordinates its homogeneous additive complex is $A_f \otimes_B C_0^{\bullet,m}(V)$. The maps of Equation (11.8) extend over A_f.

(5) **Construct the localized de Rham retraction.** Expanding the inverse $f(x+u)^{-1}$ by the displayed finite geometric formula produces a perturbation strictly raising formal difference degree. The same finite-order perturbation construction (11.10) therefore gives a deformation retraction onto $\Omega^{\bullet}_{A_f/B}$, whose transferred differential is the universal localized derivation $d(f^{-1}) = -f^{-2}df$. Localization is performed at each finite diagonal order before taking the limit.

(6) **Extend connection transport over the localization.** The Taylor transport of Theorem 11.11 uses the same finite inverse identities; it therefore extends and satisfies its unit and cocycle equations over A_f.

□

Proposition 11.14 (A logarithmic relative de Rham class)**.** *On $A = B[x, x^{-1}]$, relative to B,*

$$H^0_{\mathrm{dR}}(A/B) \cong B, \qquad H^1_{\mathrm{dR}}(A/B) \cong B \cdot [d\log x],$$

and the higher cohomology objects vanish. The formal diagonal has $P^r_{A/B} \cong A[u]/(u^{r+1})$, and $d\log x = dx/x$ is represented by the completed normalized cocycle

$$\ell(x,y) = \log\left(1 + \frac{y-x}{x}\right) = \sum_{j\geq 1} \frac{(-1)^{j+1}}{j} \left(\frac{y-x}{x}\right)^j.$$

Proof.

(1) **Compute the Laurent de Rham cohomology.** Termwise relative differentiation gives $d(b_j x^j) = j b_j x^{j-1} dx$. Every $j \neq 0$ is invertible in B and the derivative has no $x^{-1}dx$ coefficient. This computes the two nonzero cohomology objects; the cotangent module is free of rank one, so higher forms vanish.

(2) **Compute the finite and completed principal parts.** Put $y = x + u$. At finite order y has the explicitly constructed inverse

$$y^{-1} = x^{-1} \sum_{k=0}^{r} (-u/x)^k \pmod{u^{r+1}}.$$

It follows from Theorem 11.13 that $P^r_{A/B} \cong A[u]/(u^{r+1})$ and $\widehat{P}_{A/B} \cong A[[u]]$.

(3) **Construct the logarithmic cocycle and identify its class.** The formal logarithm $\ell(x, y)$ is therefore a legitimate completed coordinate cochain. Its finite logarithm identity $\ell(x, y) + \ell(y, z) = \ell(x, z)$ holds modulo each higher diagonal power, so $\delta\ell = 0$. Its first difference coefficient is x^{-1}; therefore $\rho_1(\ell) = dx/x$, as claimed.

□

Proposition 11.15 (A measured logarithmic class and its transport)**.** *For the measured family of Example 10.14, retain $R = B[t^{-1}]$, $q = t^{-1}$, and put*

$$A = R[z, (1 - qz)^{-1}], \qquad f(z) = (1 - qz)^{-1}, \qquad \omega = \frac{q}{1 - qz}\, dz.$$

Relative to R, the form ω is closed and has nonzero de Rham class. Its completed diagonal representative is

$$\ell(z, z + u) = \log \frac{1 - qz}{1 - q(z + u)} = \sum_{j \geq 1} \frac{1}{j} \left(\frac{qu}{1 - qz} \right)^j. \tag{11.15}$$

The rank-one connection $\nabla = d - \omega$ is integrable, f is horizontal, and its source-to-target stratification is multiplication by

$$\tau(z, z') = \frac{f(z')}{f(z)} = \frac{1 - qz}{1 - qz'}. \tag{11.16}$$

The Taylor coefficients of f at zero are exactly the measured coefficients q^j of the family F_r.

Proof.

(1) **Identify the nonzero logarithmic de Rham class.** Since q is a unit of R, $x = 1 - qz$ identifies A with $R[x, x^{-1}]$ and ω with $-dx/x$. Proposition 11.14, applied over R, identifies its nonzero class. The same relative calculation is available over B: localization gives $\Omega^1_{R/B} = 0$. Alternatively $A = B[z, (t(t - z))^{-1}]$, a principal open of a polynomial B-cell.

(2) **Construct the measured logarithmic cocycle.** At every finite diagonal order, $qu/(1 - qz)$ is nilpotent, so (11.15) is a finite logarithm there. Its truncations are compatible. Multiplication of the ratios in (11.16) gives the additive cocycle identity for their formal logarithms, and the first difference coefficient is $q/(1 - qz)$. Thus the de Rham comparison sends ℓ to ω.

(3) **Verify integrability and the horizontal frame.** The relative exterior square vanishes in this one-coordinate algebra, and $d\omega = 0$, giving integrability. Direct differentiation gives $df = \omega f$. The unit f is a horizontal frame, so transport of its scalar coefficients is multiplication by $f(z')/f(z)$.

(4) **Verify transport coherence and the measured expansion.** These ratios satisfy the identity and triple cocycle equations at every finite order. Expanding $(1 - qz)^{-1}$ at zero gives q^j, which agrees with (10.12).

□

The de Rham class concerns the absence of an additive primitive for ω in the localized coordinate algebra. The displayed multiplicative transport exists and will give the crystalline jet coaction in Example 12.12. Its horizontal generating function retains the complete coefficient family whose resource bounds were calculated in Proposition 10.15.

The absolute algebraic de Rham complex on these cells has an Euler contraction.

Theorem 11.16 (Absolute Euler contraction and the Laurent class)**.** *For every displayed finite-projective V over the rational ring B, inclusion and zero-section augmentation construct a natural chain-homotopy equivalence*

$$\Omega^{\bullet}_{B/\mathbb{Q}} \;\simeq\; \Omega^{\bullet}_{\mathrm{Sym}_B(V)/\mathbb{Q}}.$$

For $L = B[x, x^{-1}]$, there is a natural chain-homotopy equivalence

$$\Omega^{\bullet}_{L/\mathbb{Q}} \;\simeq\; \Omega^{\bullet}_{B/\mathbb{Q}} \oplus \Omega^{\bullet-1}_{B/\mathbb{Q}} \cdot d\log x,$$

where the shifted summand has the corresponding shifted differential. Consequently

$$H^k_{\mathrm{dR}}(L/\mathbb{Q}) \cong H^k_{\mathrm{dR}}(B/\mathbb{Q}) \oplus H^{k-1}_{\mathrm{dR}}(B/\mathbb{Q})\,[d\log x].$$

Proof.

(1) **Decompose the absolute de Rham complex by weight.** The symmetric algebra has its canonical weight grading $A = \bigoplus_{m\geq 0} \mathrm{Sym}^m_B(V)$. Give the universal differential da the same weight as homogeneous a. The relations defining $\Omega^1_{A/\mathbb{Q}}$ and its exterior powers are homogeneous, so the absolute de Rham complex is a direct sum by nonnegative weight. Its weight-zero subcomplex is exactly $\Omega^{\bullet}_{B/\mathbb{Q}}$, with inclusion and projection induced by $B \to A \to B$.

(2) **Construct the Euler contraction.** The Euler derivation E is zero on B and acts as multiplication by m on $\mathrm{Sym}^m_B(V)$. Its interior contraction on forms obeys the Cartan identity

$$d\iota_{\mathsf{E}} + \iota_{\mathsf{E}} d = \mathcal{L}_{\mathsf{E}}.$$

(3) **Contract the positive-weight summands.** On weight $m > 0$ the right side equals m. Define $h = m^{-1}\iota_{\mathsf{E}}$ on this weight and zero on weight zero. Since each form has finitely many weights, h is a map of internal B-modules. It gives $dh + hd = 1 - \iota_B \epsilon_B$, proving the first homotopy equivalence. The grading and Euler map are intrinsic and commute with finite-projective retractions.

(4) **Contract the nonzero Laurent weights.** For $L = B[x, x^{-1}]$, give x weight 1, x^{-1} weight -1, and dx weight 1. The same Euler contraction kills every nonzero integer weight, since every nonzero integer is a unit of B. The weight-zero complex is

$$\Omega^{\bullet}_{B/\mathbb{Q}} \oplus \Omega^{\bullet-1}_{B/\mathbb{Q}} \wedge \frac{dx}{x}.$$

(5) **Identify the remaining complex and its cohomology.** The absolute differential preserves this direct sum and $d(dx/x) = 0$. More explicitly, a chain isomorphism from the standard cochain shift (whose differential is $-d_B$) sends α of degree $k-1$ to $(-1)^{k-1}\alpha \wedge d\log x$ in degree k. Hence the same Cartan identity supplies the asserted deformation retraction and its cohomology calculation.

□

Relative formal cochains use B as base. The absolute retractions above use $\mathbb{Q}$ and apply to the numerical scale differentials and logarithmic connections of Section 13.9.

11.7 Artin stacks presented by smooth groupoids

A finite-projective resource-coordinate module may be described in local frames. Changes of frame act on its coordinate expression and on every polynomial resource map defined on it. To organize these isomorphisms and their stabilizers, we use algebraic stacks in groupoids on the same big étale site. An internal groupoid-valued presheaf is constructed as a finite-limit category object in $\mathcal{E}$. Descent requires its values on an étale Čech nerve to give the groupoid of compatible objects and arrows, and stackification glues these finite covering data. These groupoid-valued stacks are considered alongside the $\mathcal{E}$-valued sheaf categories of Theorem 9.5.

Definition 11.17 (Smooth Artin quotients)**.** For a displayed smooth affine groupoid $R \rightrightarrows U$ of finitely presented B-schemes with smooth source and target, the **smooth Artin quotient** $[U/R]$ is the stack of local U-objects and their R-valued change-of-trivialization arrows, glued along étale covering diagrams. Equivalently, on a test T its objects are groupoid-principal R-bundles $P \to T$ with equivariant map $P \to U$, and its arrows are equivariant isomorphisms.

Construction 11.18 (Smooth Artin quotients)**.**

(1) **Construct the atlas.** For the quotient in Definition 11.17, the atlas $U \to [U/R]$ has kernel-pair groupoid R and is smooth and surjective in the ordinary algebraic-stack sense.

(2) **Construct the full simplicial nerve.** The construction includes the full simplicial nerve $U_\bullet$ with $U_0 = U$, $U_1 = R$ and $U_q = R \times_U \cdots \times_U R$.

Proof.

(1) **Verify descent for the torsor quotient.** The torsor cocycle on a covering family is finite diagram data; refining two coverings compares them by their pullbacks. The sheaf condition for U and R supplies effectiveness of object and arrow descent, and torsor descent makes the displayed assignment a stack.

(2) **Recover the groupoid nerve and smooth atlas.** Pullback of the universal torsor along $U \to [U/R]$ reconstructs the groupoid arrows and its full nerve. Smoothness and surjectivity of the atlas follow by pulling back to a torsor trivializing cover; this is the standard ordinary smooth-groupoid presentation theorem (Stacks Project Authors 2026c, Sections 94.12 and 94.17).

□

Theorem 11.19 (Endpoint cohesion on groupoid-valued étale stacks)**.** *Let* $\mathrm{St}_{\mathrm{et}}^{(1)}(\mathsf{Aff}_B^{\mathrm{fp}}; \mathcal{E})$ *and* $\mathrm{St}_{\mathrm{et}}^{(1)}(\mathsf{I}_B; \mathcal{E})$ *denote the 2-categories of internal groupoid-valued étale stacks, with pseudonatural maps and modifications. The endpoint construction lifts to a string of 2-adjoints*

$$\iota_! \dashv \iota^* \dashv \iota_* \dashv \iota^!,$$

with the same larger- and smaller-endpoint formulas as (9.1)*. On a representable affine* T*, the rightmost functor is determined by*

$$(\iota^! \mathcal{H})(T) = \mathrm{PsNat}(\iota_* y_T, \mathcal{H}),$$

the groupoid of pseudonatural transformations and their invertible modifications. The two endpoint embeddings remain nonisomorphic, witnessed by the discrete affine-line stack. The full Artin quotient stacks from Construction 11.18 occur as objects of the base stack 2-category.

Proof.

(1) **Lift the endpoint adjunctions to groupoid-valued presheaves.** The endpoint universal properties of Lemma 9.2 apply to pseudonatural transformations of groupoid-valued presheaves, yielding the first three 2-adjunctions and full faithfulness by the two identities $c\iota = d\iota = 1$.

(2) **Compare endpoint descent and stackification.** The étale lift of Lemma 9.4 identifies not only covering families but their complete descent groupoids; therefore endpoint precomposition commutes with stackification by gluing effective descent data. It preserves the 2-colimits of stack diagrams and consequently has a right 2-adjoint.

(3) **Construct the remaining right 2-adjoint.** To construct it, express a groupoid-valued presheaf by the Grothendieck construction on its representable objects and arrows, with their specified invertible coherence. Its 2-colimit is the original presheaf by the bicategorical Yoneda calculation. Applying ι_* to that presentation and mapping to $\mathcal{H}$ identifies the resulting groupoid of maps with the pseudonatural transformations into the displayed rightmost formula.

(4) **Descend the adjunction and distinguish the embeddings.** Stackification gives the same adjunction on étale stacks. The affine-line example in Theorem 9.5 consists of discrete groupoids, so the endpoint embeddings are also distinct here.

□

Proposition 11.20 (Infinitesimal symmetries of resource frames)**.** *The finitely presented group scheme*

$$\mathrm{GL}_{n,B} = \operatorname{Spec} B[x_{ij}, \det(x_{ij})^{-1}]$$

is smooth, and its quotient $B\mathrm{GL}_{n,B} = [\operatorname{Spec} B/\mathrm{GL}_{n,B}]$ *is a displayed Artin stack. Its atlas has self-intersection* $\mathrm{GL}_{n,B}$*. Over any square-zero algebra* $R[\varepsilon]/\varepsilon^2$*, its infinitesimal stabilizer at the identity is*

$$\ker\big(\mathrm{GL}_n(R[\varepsilon]/\varepsilon^2) \to \mathrm{GL}_n(R)\big) = \{1 + \varepsilon M : \ M \in M_n(R)\},$$

with multiplication $(1+\varepsilon M)(1+\varepsilon N) = 1 + \varepsilon(M+N)$*. In particular the displayed atlas is not étale for* $n > 0$*.*

Proof.

(1) **Represent invertible matrices by a smooth principal open.** Localization at the determinant represents invertible matrices. This is a principal open of affine n^2-space over B, hence smooth.

(2) **Compute the infinitesimal group law.** A matrix congruent to the identity modulo a square-zero ideal is uniquely $1 + \varepsilon M$, with inverse $1 - \varepsilon M$. Multiplying and using $\varepsilon^2 = 0$ gives the stated group law.

(3) **Identify the nonzero tangent and quotient atlas.** The group scheme has a nonzero relative tangent module, so its atlas cannot be étale. Its torsor quotient and atlas are Construction 11.18.

□

Theorem 11.21 (The infinitesimal relation on presented Artin groupoids)**.** *For the displayed quotient stack* $\mathcal{X} = [U/R]$ *of Construction 11.18, define on every infinitesimal arrow*

$j = (\operatorname{Spec} R_0 \hookrightarrow \operatorname{Spec} S_0)$ *the groupoid*

$$\mathcal{D}_{\mathcal{X}}(j) = \mathcal{X}(S_0) \overset{\mathrm{h}}{\underset{\mathcal{X}(R_0)}{\times}} \mathcal{X}(S_0).$$

An object is a pair of S_0-families with a specified isomorphism between their reductions to R_0; arrows are pairs of isomorphisms compatible with that boundary isomorphism. The assignment is an étale stack on I_B. Its iterated homotopy fibre products give a simplicial infinitesimal groupoid, with unit and composition induced by identity and composition of the specified boundary isomorphisms. On the ordinary affine sheaf F_A this groupoid reduces to the fibre-product relation of Theorem 10.6, and consequently has its explicit finite principal-parts filtration.

Proof.

(1) **Construct the functorial infinitesimal groupoid relation.** The restriction $\mathcal{X}(S_0) \to \mathcal{X}(R_0)$ is functorial in commutative squares of thickenings. Homotopy pullbacks of groupoids may be constructed as the groupoids of triples (x, y, α) with $\alpha : x|_{R_0} \cong y|_{R_0}$; their morphisms are compatible pairs. This construction proves functoriality.

(2) **Verify descent and the simplicial operators.** On an étale covering diagram of j, each endpoint gives a covering diagram by Lemma 9.4. Torsors and their isomorphisms descend along those covers, so the groupoids at both endpoints satisfy descent; finite homotopy pullbacks preserve the required limit diagrams. Composition and inversion of the boundary isomorphisms obey the groupoid laws, giving all the simplicial operators.

(3) **Recover the affine principal-parts relation.** For a represented ordinary affine scheme F_A, its point groupoids are discrete. An isomorphism between their reductions is precisely equality, so the displayed homotopy pullback is the ordinary kernel pair $F_A(S_0) \times_{F_A(R_0)} F_A(S_0)$. Apply the finite nilpotence proof of Theorem 10.6 for its principal-parts filtration. For a general Artin quotient the boundary isomorphism remains part of the groupoid data.

□

11.8 Étale descent and the smooth Artin groupoid de Rham comparison

The smooth-groupoid construction extends the coefficient calculation to presentations with stabilizers. To retain that information we use the full simplicial atlas of the Artin presentations from Construction 11.18. The following construction provides its internal coefficient diagram over B. The broader de Rham formalism for ordinary Artin stacks is the cited theory of (Behrend 2005; Stacks Project Authors 2026c).

Definition 11.22 (Groupoid de Rham cochains)**.** Let $R \rightrightarrows U$ be a smooth affine groupoid over B whose nerve has finitely presented smooth affine terms U_q, and put $A_q = \mathcal{O}(U_q)$. Its **groupoid de Rham cochains** are

$$\mathrm{DR}_{\mathrm{grp}}([U/R]/B) = \operatorname{Tot}_{q \geq 0} N^q \Omega^{\bullet}_{A_\bullet/B}, \qquad D|_{(q,p)} = \partial + (-1)^q d.$$

Here N^q is normalization, the intersection of degeneracy kernels; d is the de Rham differential and ∂ is the alternating coface differential.

Construction 11.23 (De Rham cochains of a displayed smooth affine groupoid)**.**

(1) **Construct the cosimplicial diagram of forms.** Use the groupoid and notation of Definition 11.22. The face maps of its full simplicial scheme induce a cosimplicial diagram of relative differential graded algebras

$$[q] \longmapsto \Omega^{\bullet}_{A_q/B}.$$

(2) **Form the internal total differential.** In each fixed total degree only finitely many terms occur, so D and all its signs are maps of internal B-modules.

(3) **Compare with completed formal cochains.** For polynomial and principal-localization terms, the natural comparison of Theorems 11.6 and 11.13 replaces each relative-form complex with its completed formal coordinate-cochain complex. Its simplicial naturality gives a morphism of the corresponding total complexes.

Proof.

(1) **Verify the total differential.** All simplicial maps are maps of B-algebras and commute strictly with universal derivations. The simplicial identities imply $\partial^2 = 0$; naturality gives $d\partial = \partial d$, so the displayed signed total differential squares to zero.

(2) **Normalize the simplicial complex.** Degenerate summands split off by the standard finite normalization identities.

(3) **Compare completed infinitesimal cochains with the total de Rham complex.** The comparison ρ from completed normalized infinitesimal cochains to forms is natural under polynomial substitution and localization, hence is compatible with all cofaces wherever those local charts apply. Levelwise chain homotopy equivalence then gives a total-complex quasi-isomorphism by the first-quadrant finite-total-degree filtration.

□

Proposition 11.24 (Étale refinement and infinitesimal stabilizer retention)**.** *The above cosimplicial coefficient construction is functorial under maps of displayed smooth groupoids and under étale refinement of their affine nerve terms. The augmentation of an étale Čech refinement is a quasi-isomorphism of the corresponding total complexes of forms. For the classifying stack $B\mathrm{GL}_{n,B}$, the nerve is*

$$U_q = (\mathrm{GL}_{n,B})^q,$$

and its first nonconstant term has differential module $\Omega^1_{\mathrm{GL}_{n,B}/B}$; its cotangent at the unit contains $\mathfrak{gl}_n(B)^\vee$. Hence this construction retains the nonzero infinitesimal stabilizers which would be lost by using only the atlas $U_0 = \operatorname{Spec} B$.

Proof.

(1) **Construct the map induced by a groupoid morphism.** A morphism of groupoids induces a map of whole nerves, so it induces every coface and degeneracy map and hence a map of the total complexes.

(2) **Apply differential base change to an étale refinement.** Under an étale affine refinement $A \to A'$, the vanishing of $\Omega^1_{A'/A}$ and the flat base-change formula give $\Omega^p_{A'/B} \cong A' \otimes_A \Omega^p_{A/B}$.

(3) **Prove the refinement quasi-isomorphism.** Faithfully flat Čech descent is exact on each displayed coefficient module; the spectral sequence of the resulting first-quadrant double complex proves the claimed quasi-isomorphism.

(4) **Identify the stabilizer tangent retained by the nerve.** For the last assertion the groupoid nerve has $U_0 = \operatorname{Spec} B$, $U_1 = \mathrm{GL}_{n,B}$ and the indicated higher fibre products. Differentiating the matrices $1 + \varepsilon M$ of Proposition 11.20 identifies the identity tangent with $M_n(B)$ and its dual with the identity cotangent module. The full simplicial diagram therefore records this information although $\Omega^1_{B/B} = 0$ at the zero-simplicial atlas.

□

For ordinary smooth Artin stacks, presentation independence follows from Behrend (2005). The internal nerve construction above gives functoriality under displayed groupoid maps, invariance under the stated étale refinements, and the completed-polynomial comparison.

11.9 Effective observations of differential coefficients

The observation structures of Theorem 10.18 and Construction 10.19 apply to the differential coefficient objects. Their finite formulas give the following tracking result.

Proposition 11.25 (Positive tests and completed differential data)**.** *The finite coefficient operations defining P^r, normalized cochains, d, truncation, and formal Taylor transport commute with substitution of all computational parameters. Their realizations on the numerical coefficient covers of Proposition D.3 have uniform trackers obtained by composing the constructed arithmetic operations. Their semantic equalities, closed-form predicates, and cohomology classes remain predicates and quotients in $\mathcal{E}$; their classifiers take values in $\Omega_{\mathcal{E}}$.*

Proof.

(1) **Verify uniform finite coefficient dependence.** The formulas for principal parts, normalization, differentials, and truncated Taylor transport use the specified arithmetic, finite-list, and matrix operations. For completed cochains, δ and ρ have finite coefficient dependence. The same holds for J, H in (11.10): the perturbation raises difference degree, so only finitely many terms contribute to each output coefficient. Displacement differentiation requires one additional input degree. These bounds and finite rational operations are uniform in the indices.

(2) **Apply finite-output tracking.** With the canonical coefficient observations of Construction 10.19, Proposition 10.20 gives the effective-point maps, their implemented finite outputs, and parameter coherence.

(3) **Keep coefficient predicates and quotients typed.** Equality and closedness are kernel and equalizer conditions, and cohomology is formed by internal quotients. These remain constructions in $\mathcal{E}$ with $\Omega_{\mathcal{E}}$-valued predicates. In the cofinite numerical fibre, Proposition 2.30 makes semantic positive tests on the coefficient modules constant; the finite tests supplied here are tests on their numerical names.

□

We now have formal functions, their comparison with differential forms, and transport of coefficient modules along the completed relation. The next section uses that same relation to describe jets: sections over infinitesimal neighbourhoods and the operators determined by their finite coefficients.

12 Crystalline jets and formal differential operators

A resource jet collects the finite response coefficients of a valuation or a section of a resource bundle. Theorem 7.1 identifies these coefficients with Taylor-valued execution measurements.

Principal-parts correspondences now specify their geometric re-centring and composition. Their full compatible system constructs the jet comonad, and flat resource transport gives its crystalline coalgebras. The family of Example 10.14 will realize all these comparisons explicitly in Example 12.12.

The construction of jets from infinitesimal shape follows Khavkine and Schreiber (2027). Here the primary endpoint model supplies represented function objects and resource coefficients. Finite principal parts identify fibrewise-linear jet operators with algebraic differential operators. Effective descent identifies the full coalgebra category on the stated cells, while the connection–stratification comparison supplies the coalgebras arising from finite-projective modules.

12.1 Finite principal-parts correspondences

For a finite-projective coordinate cell E, write $A_E = \mathrm{Sym}_B(E^\vee)$ and $X_E = X_{A_E} = \iota_! F_{A_E}$ in the primary infinitesimal-arrow category. Write $\mathcal{D}_E^{\leq r}$ for the étale sheaf (10.5); its restriction to based Weil probes is the previously constructed finite diagonal of Section 10.2. The two projections

$$s_r, t_r : \mathcal{D}_E^{\leq r} \rightrightarrows X_E$$

form the order-r formal correspondence. At $r = 0$ both projections are the identity; for $r > 0$ they retain nilpotent differences through order r.

Definition 12.1 (Finite formal-disk and jet functors)**.** For the displayed correspondence, the **finite formal-disk functor** and **finite jet functor** are the adjoint pair

$$\mathbb{T}^r_{X_E} = (t_r)_! s_r^* \quad \dashv \quad \mathbb{J}^r_{X_E} = (s_r)_* t_r^*. \tag{12.1}$$

Here $(t_r)_!$ is dependent sum and $(s_r)_* = \Pi_{s_r}$ is dependent product. The adjunction follows by composing $(t_r)_! \dashv t_r^*$ and $s_r^* \dashv (s_r)_*$. The common underlying correspondence is

$$X_E \xleftarrow{\ s_r\ } \mathcal{D}_E^{\leq r} \xrightarrow{\ t_r\ } X_E.$$

Composing two correspondences pulls back over their common endpoint. The next calculation maps that composite to the order-$(r+s)$ neighbourhood; taking mates will give the jet comparison in the reverse direction.

Theorem 12.2 (Addition of formal order)**.** *Composition of adjacent formal displacements defines natural maps*

$$\mathcal{D}_E^{\leq r} \times_{X_E} \mathcal{D}_E^{\leq s} \longrightarrow \mathcal{D}_E^{\leq r+s},$$

compatible with order truncation, unit, associativity, and interchange of the two vertices. Applying dependent sums gives $\mathbb{T}^s_{X_E} \mathbb{T}^r_{X_E} \to \mathbb{T}^{r+s}_{X_E}$. Their right-adjoint mates give the filtered jet comparison maps

$$\mathbb{J}^{r+s}_{X_E} Y \longrightarrow \mathbb{J}^r_{X_E} \mathbb{J}^s_{X_E} Y, \qquad \mathbb{J}^r_{X_E} Y = \Pi_{s_r}(t_r^* Y). \tag{12.2}$$

At order zero $\mathbb{J}^0_{X_E} \cong \mathrm{Id}$. The displayed comparisons obey the filtered comonad associativity and counit identities.

Proof.

(1) **Bound the order of a composite displacement.** At any nilpotent arrow, two successive differences u, v have nilpotence bounds $u^{r+1} = 0$ and $v^{s+1} = 0$ in the sense that all their monomials of the indicated total degrees vanish. Every monomial of total degree $r + s + 1$ in $u + v$ contains at least $r + 1$ factors from u or at least $s + 1$ factors from v. Thus $u + v$ has order at most $r + s$.

(2) **Verify units, associativity, and vertex interchange.** This defines the displayed map, naturally in the stage. The unit is displacement zero, associativity is $(u+v)+w = u+(v+w)$, and vertex interchange sends u to $-u$. These polynomial equations commute with all truncation maps.

(3) **Compose the finite correspondences.** Put $Q_{r,s} = \mathcal{D}_E^{\leq r} \times_{t_r, X_E, s_s} \mathcal{D}_E^{\leq s}$. The cartesian square

$$\begin{array}{ccc} Q_{r,s} & \xrightarrow{\pi_2} & \mathcal{D}_E^{\leq s} \\ {\scriptstyle \pi_1}\downarrow & \lrcorner & \downarrow{\scriptstyle s_s} \\ \mathcal{D}_E^{\leq r} & \xrightarrow[t_r]{} & X_E \end{array}$$

gives dependent-sum Beck–Chevalley. Consequently the composite $(t_s)_! s_s^* (t_r)_! s_r^*$ is the action of the composite correspondence. The map $Q_{r,s} \to \mathcal{D}_E^{\leq r+s}$ therefore induces $\mathbb{T}^s_{X_E} \mathbb{T}^r_{X_E} \to \mathbb{T}^{r+s}_{X_E}$.

(4) **Construct the jet comparison and its coherence.** Taking right-adjoint mates under (12.1) gives (12.2). Mates preserve the equations between these composites, proving the filtered associativity and unit identities.

□

Write $\delta_{r,s} : \mathbb{J}^{r+s} \to \mathbb{J}^r \mathbb{J}^s$. The mate of associativity is the commutative square

$$\begin{array}{ccc} \mathbb{J}^{r+s+t} & \xrightarrow{\delta_{r+s,t}} & \mathbb{J}^{r+s}\mathbb{J}^t \\ {\scriptstyle \delta_{r,s+t}}\downarrow & & \downarrow{\scriptstyle \delta_{r,s}\mathbb{J}^t} \\ \mathbb{J}^r \mathbb{J}^{s+t} & \xrightarrow[\mathbb{J}^r \delta_{s,t}]{} & \mathbb{J}^r \mathbb{J}^s \mathbb{J}^t. \end{array} \tag{12.3}$$

Order-zero factors give the counit identities. Thus the filtered comonad equations are the mates of the associativity and unit equations for formal correspondences. The addition of orders records the bound on a composite displacement.

The corresponding algebra maps of principal parts are the duals of the finite correspondence compositions. They retain the two A_E-actions; replacing their target by an unaugmented symmetric power would discard the source–target information needed for substitution.

12.2 Formal disks and jets in the primary geometry

Definition 12.3 (The formal-disk monad and jet comonad)**.** For $X \in \mathscr{H}_{B,\mathrm{et}}$ let $\eta_X : X \to \mathfrak{I}_{\mathrm{inf}} X$ be the unit of the middle endpoint adjunction, and put

$$\mathcal{D}_X = X \times_{\mathfrak{I}_{\mathrm{inf}} X} X, \qquad s, t : \mathcal{D}_X \rightrightarrows X.$$

Because $\mathscr{H}_{B,\mathrm{et}}$ is an internal sheaf topos, its slices admit dependent sums and products $(\eta_X)_! \dashv \eta_X^* \dashv (\eta_X)_*$. The **formal-disk monad** and **jet comonad** are

$$\mathbb{T}^\infty_X = \eta_X^* (\eta_X)_!, \qquad \mathbb{J}^\infty_X = \eta_X^* (\eta_X)_*.$$

Theorem 12.4 (The jet comonad and coalgebra descent)**.** *The displayed functors are a monad and a comonad on $\mathscr{H}_{B,\mathrm{et}}/X$, with*

$$\mathbb{T}^\infty_X \dashv \mathbb{J}^\infty_X, \qquad \mathbb{J}^\infty_X Y \cong \Pi_s (t^* Y).$$

For every X there is a canonical comparison functor

$$\mathscr{H}_{B,\mathrm{et}}/\Im_{\mathrm{inf}}X \longrightarrow (\mathscr{H}_{B,\mathrm{et}}/X)^{\mathbb{J}_X^\infty}.$$

It is an equivalence if η_X is an effective epimorphism, in particular for the polynomial finite-projective cells $X = X_A$ of Proposition 10.11.

Proof.

(1) **Construct the monad, comonad, and their adjunction.** The adjunctions construct the monad and comonad, with multiplication and comultiplication given by the unit and counit mates. The triangle identities prove both unit and counit equations and the associativity/coassociativity equations. Successive slice adjunctions give

$$\mathrm{Hom}_X(\eta_X^*(\eta_X)_!E, F) \cong \mathrm{Hom}_{\Im_{\mathrm{inf}}X}((\eta_X)_!E, (\eta_X)_*F) \cong \mathrm{Hom}_X(E, \eta_X^*(\eta_X)_*F).$$

(2) **Identify the formal-pair correspondence and comultiplication.** Dependent-product Beck–Chevalley for the defining pullback of $\mathcal{D}_X$ gives the correspondence formula $\Pi_s(t^*Y)$. Comparing the two groupoid action diagrams determines the jet comultiplication by re-centering along composable formal pairs.

(3) **Construct the descent comparison.** The comparison functor sends $H \to \Im_{\mathrm{inf}}X$ to its pullback η_X^*H with the canonical unit-induced coaction.

(4) **Prove comonadicity under effective descent.** If η_X is an effective epimorphism, it is an effective descent morphism in the sheaf topos: its entire Čech kernel-pair diagram realizes the target, and objects with compatible descent action are precisely its pullbacks. The same Čech action is the coalgebra structure of $\eta_X^*(\eta_X)_*$, proving comonadicity in this case. The effective-epimorphism calculation on polynomial cells is Proposition 10.11.

□

The jet counit evaluates a formal-disk section at its centre. Comultiplication re-centres it along composable nilpotent pairs. A section $\sigma : X \to Y$ prolongs by $X \cong \mathbb{J}_X^\infty X \to \mathbb{J}_X^\infty Y$, and coKleisli composition uses that comultiplication. This is the crystalline differential-operator calculus of the primary endpoint geometry. Restricting its relations to $\operatorname{Spec} B \hookrightarrow \operatorname{Spec} C$ recovers the based-Weil calculation of Theorem 7.14.

12.3 Jet coefficients and principal parts

To calculate the jets of a bundle associated to a finite-projective module, we realize that module as an affine bundle sheaf. For a displayed finite-projective A-module N, the fibre at $x : A \to S$ is $S \otimes_{A,x} N$. Equivalently, for a displayed idempotent $e_N \in M_k(A)$, put

$$Y_N^0(\operatorname{Spec} S) = \{(x, z) : x : A \to S,\ z \in S^k,\ e_N(x)z = z\}, \qquad Y_N = \iota_! Y_N^0 \longrightarrow X_A.$$

This bundle is represented by $\operatorname{Spec} \mathrm{Sym}_A(N^\vee)$ over $\operatorname{Spec} A$; in particular it satisfies étale descent. The following calculation takes place in the primary slice throughout.

Theorem 12.5 (Finite jets and modules of principal parts)**.** *Let $j = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} S)$ and let $x : A \to S$ be a generalized point of X_A. Define*

$$Q_{j,x,r} = S \otimes_{A,x,s} P^r_{A/B}, \qquad k_{j,x,r} = (\operatorname{Spec} R \hookrightarrow \operatorname{Spec} Q_{j,x,r}),$$

where $Q_{j,x,r} \to R$ is reduction after the augmentation to S. Then $k_{j,x,r}$ is a primary test and represents the pullback $yj \times_{X_A,s_r} \mathcal{D}_A^{\le r}$. For the bundle Y_N,

$$(\mathbb{J}^r_{X_A} Y_N)(j,x) \cong Q_{j,x,r} \otimes_{A,t} N. \tag{12.4}$$

Here the left side denotes the fibre over the generalized point x. Consequently, with source A-action,

$$P^r_{A/B}(N) := P^r_{A/B} \otimes_{A,t} N, \qquad \mathbb{J}^r_{X_A} Y_N \cong Y_{P^r_{A/B}(N)}. \tag{12.5}$$

For $A = B[x_1, \dots, x_n]$ and $N = A \otimes_B M$ with M displayed finite-projective over B, the fibre formula is

$$(\mathbb{J}^r_{X_A} Y_N)(j,x) \cong \prod_{|\alpha| \le r} (S \otimes_B M).$$

All comparisons preserve truncation and the full Taylor coordinate-substitution law of Theorem 10.3.

Proof.

(1) **Construct the representing nilpotent test.** The algebra $Q_{j,x,r}$ is finitely presented over B by its finite tensor-product presentation. Let L be its displacement ideal and $J = \ker(S \to R)$. Then $L^{r+1} = 0$ and $Q/L = S$. If $J^a = 0$, the kernel $K = \ker(Q \to R)$ satisfies $K^a \subseteq L$, so $K^{a(r+1)} = 0$. The quotient to R is surjective. Thus $k_{j,x,r}$ is an object of I_B.

(2) **Identify the pullback of finite principal parts.** The endpoint pullback of j and $k_{1,r} = (\operatorname{Spec} A \hookrightarrow \operatorname{Spec} P^r)$ over $\iota(\operatorname{Spec} A)$ has larger algebra Q and smaller algebra $R \otimes_A A = R$. It is exactly $k_{j,x,r}$. The representing property of Lemma 10.7 therefore proves the asserted sheaf pullback.

(3) **Compute the primary jet fibre.** Dependent-product adjunction and Yoneda give

$$\underline{\mathrm{Hom}}_{X_A}(yj, \Pi_{s_r} t_r^* Y_N) \cong \underline{\mathrm{Hom}}_{X_A}(yk_{j,x,r}, Y_N) \cong Q \otimes_{A,t} N,$$

where the map from $yk_{j,x,r}$ to X_A on the right is its target vertex. This proves (12.4).

(4) **Identify the represented principal-parts bundle.** Tensor associativity identifies this fibre with $S \otimes_{A,x} P^r_{A/B}(N)$. The module $P^r_{A/B}(N)$ is a retract of a finite power of P^r, which is finite-projective as a source A-module; hence its bundle is defined above and the fibrewise comparison proves (12.5).

(5) **Recover monomial coefficients and coherence.** For free coordinates, the truncated monomials in $u = y - x$ are a finite basis of Q over S, giving the product formula. Every map is induced by a representing pullback or algebra substitution; these universal properties prove all coherence assertions.

□

The coefficient algebra of a jet at $x : A \to S$ is the pushout

$$\begin{array}{ccc} A & \xrightarrow{s} & P^r_{A/B} \\ {\scriptstyle x}\downarrow & & \downarrow \\ S & \longrightarrow & Q_{j,x,r}. \end{array}$$

The source action fixes the base point; the target action $A \xrightarrow{t} P^r_{A/B} \to Q_{j,x,r}$ pulls back the bundle. Its fibre is therefore $Q_{j,x,r} \otimes_{A,t} N$. Keeping the two actions visible explains the tensor convention and the Taylor evaluation at the target.

At the universal target point of $k_{j,x,r}$, a polynomial section f is evaluated by $f(x+u)$. The finite binomial identity therefore gives its primary Taylor coefficients directly:

$$f(x+u) = \sum_{|\alpha|\le r} \frac{u^\alpha}{\alpha!}\partial^\alpha f(x) \pmod{(u)^{r+1}}. \tag{12.6}$$

The symmetric-power expression descends along displayed projective retracts. Thus primary polynomial Taylor transport is proved on the representing test, independently of any preservation of dependent products by restriction to based-Weil probes.

Definition 12.6 (Algebraic differential operators)**.** For A-modules N, M, an **algebraic differential operator of order at most** r is a B-linear map whose $(r+1)$-fold iterated commutators with multiplication by elements of A vanish. Their object is $\mathrm{Diff}^{\le r}_{A/B}(N, M)$.

Theorem 12.7 (Algebraic differential operators and primary jets)**.** *For A-modules N, M, there is a natural isomorphism*

$$\mathrm{Diff}^{\le r}_{A/B}(N, M) \cong \mathrm{Hom}_A(P^r_{A/B} \otimes_{A,t} N, M), \tag{12.7}$$

where the source A-action is used on the tensor product. For displayed finite-projective N, M, these are exactly the fibrewise-linear primary geometric maps $\mathbb{J}^r_{X_A} Y_N \to Y_M$. Their action on sections is composition with jet prolongation, and their composition agrees with the order-additive principal-parts comparisons.

Proof.

(1) **Convert commutators to diagonal powers.** For a B-linear $D : N \to M$, set $\widetilde{D}(a \otimes n) = aD(n)$ on $A \otimes_B N$. This is source-A-linear, and the diagonal generator $\Delta b = 1 \otimes b - b \otimes 1$ satisfies

$$\widetilde{D}\big(\Delta b\,(a \otimes n)\big) = a\big(D(bn) - bD(n)\big).$$

Induction identifies products of $r+1$ generators with iterated commutators. Since these generate I^{r+1}, the order condition is precisely annihilation of $I^{r+1}(A \otimes_B N)$.

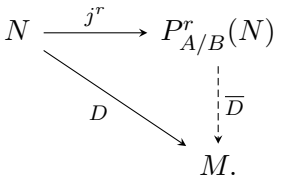
(2) **Apply the quotient universal property.** The quotient is $P^r_{A/B} \otimes_{A,t} N$ by right exactness of tensor. A map from it recovers D by $D(n) = \widetilde{D}(1 \otimes n)$. These inverse constructions prove (12.7), with universal prolongation $n \mapsto 1 \otimes n$ (Stacks Project Authors 2026a, Tag 09CH).

(3) **Identify primary bundle maps.** For finite-projective modules, affine-bundle Yoneda identifies fibrewise-linear maps with degree-one coordinate maps, which finite duality identifies with module maps. Full faithfulness of $\iota_!$ and Theorem 12.5 give the asserted comparison. Prolonging a section evaluates it at the universal target vertex, so its operator action is composition with $n \mapsto 1 \otimes n$.

(4) **Compare composition.** On successive universal vertices, both constructions apply the same two operators. The three-vertex substitution of Theorem 12.2 therefore identifies their composite, its bound $r + s$, and its associativity equations.

□

The operator classification is the following factorization: every order-r operator is obtained uniquely by applying an A-linear map after the universal B-linear jet map.



Here $j^r(n) = 1 \otimes n$, and $\overline{D}$ is the unique A-linear arrow with this composite, using the source A-action on principal parts. The triangle is interpreted in B-modules, with the stronger linearity of its dashed leg specified.

12.4 Finite formal coefficients and Taylor formulas

Put

$$W_{n,r} = \mathbb{Q}[d_1, \dots, d_n]/(d_1, \dots, d_n)^{r+1}, \qquad D_{n,r} = D_{B \otimes_{\mathbb{Q}} W_{n,r}}, \qquad D_{n,\infty} = \bigcup_r D_{n,r}.$$

This is the based-Weil specialization of the primary construction. Its polynomial coefficient formulas agree with Theorem 12.5; the proposition below also treats arbitrary B-modules and arbitrary natural maps in the based-Weil category, on that category's own domain. At a based stage C, the last union consists of the nilpotent n-tuples. A tuple with zero augmentation lies in the nilpotent augmentation ideal and hence in some $D_{n,r}(C)$. Conversely a nilpotent tuple has zero augmentation because B is reduced. Therefore

$$\mathcal{D}_{\mathcal{R}^n_B} \cong \mathcal{R}^n_B \times D_{n,\infty}, \qquad (x, y) \longmapsto (x, y - x). \tag{12.8}$$

For the finite relation let $s_r(x, d) = x$ and $t_r(x, d) = x + d$, and define $\mathbb{J}^r_{\mathcal{R}^n_B} E = \Pi_{s_r}(t_r^* E)$.

Proposition 12.8 (Finite jet coefficients)**.** *For every B-module M, the original based-Weil coefficient functor, considered in its own slice category, has the explicit finite coefficients*

$$\mathbb{J}^r_{\mathcal{R}^n_B}(\mathcal{R}^n_B \times j_B M) \cong \mathcal{R}^n_B \times \prod_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| \le r}} j_B M.$$

For a geometric map $F : j_B M \to j_B N$ and directions $v_1, \dots, v_n$, its finite Taylor formula is

$$F\left(x + \sum_i d_i v_i\right) = \sum_{|\alpha| \le r} \frac{d^\alpha}{\alpha!} D^{|\alpha|} F(x)(\underbrace{v_1, \dots, v_1}_{\alpha_1}, \dots, \underbrace{v_n, \dots, v_n}_{\alpha_n}) \quad (d \in D_{n,r}).$$

The term of degree zero is $F(x)$.

Proof.

(1) **Identify finite jets with monomial coefficients.** The formal-affine trivialization reduces the first assertion to $(j_B M)^{D_{n,r}}$ and its monomial coefficient basis in (7.9).

(2) **Compute coefficients using square-zero substitutions.** For one direction, write its uniquely represented polynomial as $\sum_{j=0}^r a_j d^j$. Substitute $d = \varepsilon_1 + \cdots + \varepsilon_r$ in the Weil algebra with $\varepsilon_i^2 = 0$. Iterating the first-order expansion identifies the coefficient of $\varepsilon_1 \cdots \varepsilon_j$ with $D^j F(x)(v, \dots, v)$. In the polynomial expression this coefficient is $j! a_j$.

(3) **Compare truncations and divide by factorials.** The substitution into j square-zero variables is defined on the order-j truncation, and truncation identifies its coefficients with the first j coefficients of the order-r expression. Rational factorials are invertible.

(4) **Recover symmetric mixed derivatives.** For several directions use disjoint groups of square-zero variables. The same coefficient comparison gives $\alpha!$ and the displayed mixed derivative. Symmetry proved above makes the formula independent of the order within each group.

□

The directed union in (12.8) gives

$$\mathbb{J}^{\infty}_{\mathcal{R}^n_B} E \cong \varprojlim_r \mathbb{J}^r_{\mathcal{R}^n_B} E.$$

Indeed maps out of its union are compatible families of maps out of the finite disks; the dependent-product adjunction gives the assertion in the slice. All countable coefficient objects here are internal dependent products indexed by the NNO.

12.5 Finite-order coefficient change

For a commutative B-algebra A, the truncated polynomial algebra $A[d]/d^{r+1}$ has underlying module A^{r+1} and finite convolution multiplication. For a coefficient map $B \to B'$, there is a canonical isomorphism

$$B' \otimes_B (A[d]/d^{r+1}) \cong (B' \otimes_B A)[d]/d^{r+1}.$$

Indeed tensor preserves the finite direct sum of coefficient modules, and the coefficientwise comparison respects convolution. Several variables use their finite monomial lists. These comparisons commute with truncation and polynomial substitutions; the derivative formula (7.6) commutes with them term by term. Finite Weil coefficient modules commute with gauge and filtered coefficient change by tensor associativity.

At every finite order these comparisons preserve the measured coefficients of Theorem 7.1. At infinite order the distinction is (10.13): the measured family of Example 10.14 has a polynomial budget at each finite order but no single polynomial budget for all its coefficients.

12.6 Formal-order completion and its de Rham comparison

Proposition 12.9 (Crystalline bundles give jet coalgebras)**.** *Let N be a displayed finite-projective module over $A = \mathrm{Sym}_B(E^\vee)$, represented by $e_N \in M_k(A)$. Use the affine bundle $Y^0_N \to F_A$ and its primary realization $Y_N = \iota_! Y^0_N \to X_A$ constructed in Section 12.3. Every formal stratification on N gives naturally a $\mathbb{J}^{\infty}_{X_A}$-coalgebra structure*

$$\gamma_N : Y_N \longrightarrow \mathbb{J}^{\infty}_{X_A} Y_N$$

in the primary étale infinitesimal-arrow geometry. Morphisms of stratifications induce coalgebra morphisms. Its restriction to based B-Weil arrows is exactly the earlier coefficient-functor bundle and Taylor transport.

Proof.

(1) **Transport along finite formal pairs.** Use the affine coefficient bundle constructed in Section 12.3, with fibre $S \otimes_{A,\xi} N$. For two S-points x, y agreeing in R, Theorem 10.6 supplies a finite principal-parts map $P^r_{A/B} \to S$. Pulling back the stratification gives

$$\varepsilon_{x,y} : (Y_N)_x \xrightarrow{\cong} (Y_N)_y.$$

(2) **Construct the coaction.** Truncation compatibility makes this map independent of the nilpotence bound, defining $s^*Y_N \to t^*Y_N$ on the full relation. Transposition through $s^* \dashv \Pi_s$ gives γ_N. The identity pair gives its counit equation. The triple-diagonal cocycle gives the commutative square, with $\mathbb{J} = \mathbb{J}^\infty_{X_A}$,

$$\begin{array}{ccc} Y_N & \xrightarrow{\gamma_N} & \mathbb{J}Y_N \\ {\scriptstyle \gamma_N}\downarrow & & \downarrow{\scriptstyle \delta_{Y_N}} \\ \mathbb{J}Y_N & \xrightarrow[\mathbb{J}(\gamma_N)]{} & \mathbb{J}\mathbb{J}Y_N. \end{array}$$

(3) **Verify morphisms and restriction.** For a stratification morphism, $f_y \varepsilon_{x,y} = \varepsilon'_{x,y} f_x$ transposes to the coalgebra square. At a based arrow $S = C$, the same matrix fibres and transport formulas recover the based-Weil bundle and Taylor transport.

□

Lemma 12.10 (Crystalline restriction to a principal open)**.** *Let $A = \mathrm{Sym}_B(E^\vee)$ be a displayed finite-projective polynomial coordinate algebra and $h \in A$. For its principal open A_h, the finite principal-parts coefficients, formal stratifications, and crystalline jet coactions have the same comparisons after restriction to X_{A_h}. In particular the shape unit of X_{A_h} is an effective epimorphism, and a finite-projective A_h-module with integrable connection gives a primary jet coalgebra. Finite and full jet functors restrict along $X_{A_h} \hookrightarrow X_A$ by the corresponding slice Beck–Chevalley maps.*

Proof.

(1) **Lift units through the nilpotent quotient.** For $S \twoheadrightarrow R_0$, a unit modulo the kernel lifts locally to a unit: lift its inverse and correct the nilpotent product error by a finite geometric series. Reduction preserves units, proving the converse.

(2) **Restrict the infinitesimal correspondences.** At infinitesimally related A-points, h is therefore invertible at one endpoint precisely when it is invertible at the other. Thus the relation over X_{A_h} is both the source and target pullback of the relation over X_A. The finite-order version is Theorem 11.13. These cartesian squares give the finite and full jet comparisons by dependent-product Beck–Chevalley.

(3) **Restrict coefficient bundles and transport.** The finite algebras are $A_h \otimes_A P^r_{A/B}$. A displayed idempotent over A_h defines its affine module bundle, and the representing-test calculation of Theorem 12.5 applies on this open. The localized connection–stratification comparison of Theorem 11.13 and the transposition of Proposition 12.9 give its jet coaction.

(4) **Prove effective lifting.** An A_h-point of R_0 lifts locally as an A-point by Proposition 10.11. Its lifted value of h is a unit by the first step, so the point factors through A_h. The shape unit is consequently effective epic, giving descent by Theorem 12.4. The finite inverse and transport formulas commute with truncation and parameters.

□

Theorem 12.11 (Formal-order completion of the primary crystalline jets)**.** *For $X_E = X_{A_E}$ a finite-projective polynomial resource-coordinate sheaf in the primary étale infinitesimal-arrow geometry,*

$$\mathbb{J}^\infty_{X_E} Y \cong \varprojlim_r \mathbb{J}^r_{X_E} Y \tag{12.9}$$

naturally in every $Y \to X_E$. The order-additive comparison maps of Theorem 12.2 construct the comultiplication of the jet comonad of Theorem 12.4. Its jet-coalgebra descent equivalence holds here by Proposition 10.11. For coefficients in finite-projective modules it is the completed primary principal-parts bundle

$$\mathbb{J}^{\infty}_{X_A} Y_N \cong \varprojlim_r Y_{P^r_{A/B}(N)},$$

with finite coefficients calculated by Theorem 12.5. Stratifications on these modules construct their crystalline jet coactions by Proposition 12.9.

Proof.

(1) **Express full jets as a limit of finite jets.** By Theorem 10.6, the full relation is the internally increasing union $\mathcal{D}_{X_E} = \varinjlim_r \mathcal{D}_E^{\leq r}$. For every object $Z \to X_E$, the universal property of the union gives compatible maps on its finite stages:

$$\begin{aligned} \mathrm{Hom}_{X_E}(Z, \mathbb{J}^{\infty}_{X_E} Y) &\cong \mathrm{Hom}_{\mathcal{D}_{X_E}}(s^*Z, t^*Y) \\ &\cong \varprojlim_r \mathrm{Hom}_{\mathcal{D}_E^{\leq r}}(s_r^*Z, t_r^*Y) \\ &\cong \mathrm{Hom}_{X_E}\left(Z, \varprojlim_r \mathbb{J}^r_{X_E} Y\right). \end{aligned}$$

(2) **Identify the natural isomorphism and comultiplication.** Internal Yoneda proves the natural isomorphism. Addition of displacements and ordinary composition of formal pairs are the same maps in the union, so their dependent-product mates identify the limit comultiplication with the comultiplication of Theorem 12.4.

(3) **Recover crystalline descent and based coefficients.** The effective descent and crystalline coaction clauses are the explicit constructions cited in the statement. Restriction to based B-Weil probes recovers the original finite principal-parts maps by the displayed stagewise coefficient formulas.

□

Example 12.12 (The measured family as a crystalline resource line)**.** Continue with $R = B[t^{-1}]$, $q = t^{-1}$, and $A = R[z, (1-qz)^{-1}] = B[z, (t(t-z))^{-1}]$ from Proposition 11.15. The principal-open comparison of Lemma 12.10 realizes the line $N = A$ and its integrable connection $d - \omega$ in the primary resource geometry. Its stratification $\tau(z, z') = (1-qz)/(1-qz')$ induces the coaction $Y_N \to \mathbb{J}^{\infty}_{X_A} Y_N$.

The horizontal section $f = (1-qz)^{-1}$ has finite jet

$$f(z+u) = \sum_{j=0}^{r} \frac{q^j}{(1-qz)^{j+1}} u^j \pmod{u^{r+1}}. \tag{12.10}$$

This follows by multiplying the finite geometric-series inverse of $1 - qu/(1-qz)$ by $f(z)$. At $z = 0$ it is precisely $F_r(u)$, so Theorem 7.1 identifies this geometric jet with the measured coefficients of the program e_r. The complete jet is the compatible family of these finite expressions. Its fibre at zero has coefficients in $R[[u]]$ and is outside $B[[u]][t^{-1}]$ by Proposition 10.15.

Re-centring from z to z' and then to z'' multiplies the transport factors to $\tau(z, z'')$. Transposition is exactly the coalgebra coassociativity equation. The logarithm of the same factor is the completed cocycle (11.15), whose de Rham image is ω. Thus measured coefficients, completed functions, forms, and crystalline transport are calculated on one resource family.

Definition 12.13 (Geometric differential operators)**.** On a finite-projective polynomial resource cell, an **order-r geometric differential operator** from $Y \to X_E$ to $Z \to X_E$ is a map $\mathbb{J}^r_{X_E} Y \to Z$ over X_E. A **complete geometric differential operator** is a map $\mathbb{J}^\infty_{X_E} Y \to Z$ over X_E.

Corollary 12.14 (Finite principal-parts differential operators)**.** *For a finite-projective polynomial resource cell, the maps* (12.2) *compose an order-s operator after an order-r operator into an operator of order at most $r + s$. The corresponding complete operators are maps $\mathbb{J}^\infty_{X_E} Y \to Z$ and compose by the jet comonad. Polynomial resource operators have their previously constructed finite Taylor coefficients under this correspondence.*

Proof.

(1) **Construct composition using order addition.** The first assertion is the definition via the finite formal disk adjunction. For $\alpha : \mathbb{J}^r Y \to Z$ and $\beta : \mathbb{J}^s Z \to W$, compose

$$\mathbb{J}^{r+s} Y \longrightarrow \mathbb{J}^s \mathbb{J}^r Y \xrightarrow{\mathbb{J}^s \alpha} \mathbb{J}^s Z \xrightarrow{\beta} W.$$

The comparison is (12.2) with its indices interchanged if necessary.

(2) **Verify finite and completed composition laws.** The finite associativity theorem proves the operator composition laws; the completed comonad gives the same argument without an order bound.

(3) **Compare polynomial and fibrewise-linear algebraic operators.** Primary polynomial agreement is the representing-test Taylor calculation (12.6) and Theorems 7.24 and 12.5. For fibrewise-linear algebraic operators, the universal-property identification and its composition coherence are Theorem 12.7. Their based-Weil specializations agree with Proposition 12.8.

□

The completed relation now has both a modal and an algebraic description. Where the shape unit is an effective epimorphism, including polynomial finite-projective cells, its jet coalgebras are identified with $\mathscr{H}_{B,\mathrm{et}}/\mathfrak{I}_{\mathrm{inf}} X$. For bundles associated to finite-projective modules, Theorem 11.11 identifies finite-projective stratifications with integrable B-connections. These comparisons use the same infinitesimal relation but have different domains; an arbitrary semantic jet coalgebra need not arise from a connection on a finite-projective module.

The measured family now has an operational coefficient calculation, a completed function, a de Rham cocycle, and crystalline transport. The remaining instances relate this geometry to numerical clocks, input sizes, and the logarithmic connections of asymptotic scales.

13 Effective instances and geometric calculations

This section connects the constructions to two kinds of examples. The numerical execution instance supplies measured costs, input-size comparisons, and measured overheads. The later geometric calculations exhibit normal resource directions, asymptotic scales, and logarithmic connections. Each example keeps its specified implementation family and coefficient context, so its conclusion can be read directly against the general classification.

We begin with an operational realization of the measured span. The machine configuration and update map determine its histories; the first-hit clock then identifies the numerical cost of a returning execution.

13.1 The numerical execution instance

Definition 13.1 (Numerical configurations)**.** A **numerical configuration** consists of a finite register-program code, an instruction pointer, finite register data, and a Boolean halted flag. The configuration carrier is Conf. The instruction set consists of increment, zero test with decrement, jumps, and halt.

Construction 13.2 (Numerical configurations and measured return)**.**

(1) **Construct configurations and transition maps.** In the effective numerical fibre, use the finite register-program syntax of Proposition D.9: increment, zero test with decrement, jumps, and halt. Finite instruction lists have primitive-recursive codes; invalid codes denote a permanently looping program. Use the configurations of Definition 13.1. Reading an instruction and updating these finite data constructs the total primitive-recursive maps

$$h : \mathsf{Conf} \to 2, \qquad s : \mathsf{Conf} \to \mathsf{Conf}, \qquad \mathrm{init} : \mathbb{N} \times \mathbb{N} \to \mathsf{Conf},$$

and an output map $o : h^{-1}(1) \to \mathbb{N}$. Extend s by the identity on halted configurations. These are the numerical code and configuration objects specified by these finite instruction clauses.

(2) **Construct measured return and time.** Define an equation morphism with additive natural cost by

$$b : \mathsf{Conf} \to L_{\Sigma,\mathbb{N}}(\mathbb{N} + \mathsf{Conf}), \qquad b(c) = \begin{cases} \eta(0, \mathsf{inl}\, o(c)), & h(c) = 1, \\ \eta(1, \mathsf{inr}\, s(c)), & h(c) = 0. \end{cases}$$

Finite case analysis constructs this map; the discreteness lemma makes the indicated total data maps computational algebra morphisms. Then

$$\mathrm{ev}_{\mathrm{time}} = b^{\ddagger_{\mathbb{N}}}\, \mathrm{init} : \mathbb{N} \times \mathbb{N} \to L_\Sigma(\mathbb{N} \times \mathbb{N})$$

returns the number of updates before the halted state and the output. The cost is zero for an initially halted configuration. This measurement counts the specified register updates. Its relation to binary-input time is governed by the explicit input and output convention; Proposition 6.13 proves the numeric-counter obstruction, while Construction 6.4 and Proposition 6.5 construct the bit-input time instance.

(3) **Record the configuration history.** Tagging each branch by its current configuration instead constructs

$$\mathrm{ev}_H : \mathbb{N} \times \mathbb{N} \to L_\Sigma(\mathsf{Conf}^* \times \mathbb{N})$$

by Theorems 4.6 and 4.9. Its word includes the initial, intermediate, and final configurations. The valuation that assigns 1 to a nonhalted configuration and 0 to a halted one recovers $\mathrm{ev}_{\mathrm{time}}$.

(4) **Measure peak occupancy and other finite folds.** A fixed total primitive-recursive occupancy function $\mathrm{occ} : \mathsf{Conf} \to \mathbb{N}$, defined from the chosen finite register representation, folds these same recorded configurations by maximum. The product monoid records time and that peak occupancy on one execution. Other finite measurement expressions built from the recorded fields, constants, addition, maximum, and finite folds have total interpretations $\nu : \mathsf{Conf}^* \to \mathbb{N}$ by finite recursion. Here the expression is selected before forming its implementation image.

(5) **Compare measured and unmeasured execution.** Algorithm 13.1 gives the configuration procedure for these measurements. Its counter counts simulated register updates; its own pseudocode lines are not additional updates of that program. The first halting stage defines the partial evaluator of Proposition D.9. Its finite run is the corresponding return component of Lemma 1.11, so erasure has the same value and domain.

Algorithm 13.1 Measured register execution

Input: Program code e, input x, configuration maps h, s, o, and occupancy occ

$c \leftarrow \mathrm{init}(e, x)$; $k \leftarrow 0$; $H \leftarrow [c]$
$p \leftarrow \mathrm{occ}(c)$
while $h(c) = 0$ **do**
    $c \leftarrow s(c)$; $k \leftarrow k + 1$
    $H \leftarrow H{+}{+}[c]$
    $p \leftarrow \max\{p, \mathrm{occ}(c)\}$
end while
return $(k, o(c), H, p)$

The procedure returns precisely when the simulated program halts. The regular epimorphism $\mathbb{N} \twoheadrightarrow \mathsf{PC} = J_{\Sigma}\mathsf{FG}$ of that proposition supplies the numerical code presentation. The added time and occupancy values are the folds just defined.

Proposition 13.3 (The bounded numerical clock)**.** *The NNO recursion* $\mathrm{run}(e, x, k) = s^k(\mathrm{init}(e, x))$ *constructs the monotone Boolean clock*

$$\alpha(e, x, k) = h(\mathrm{run}(e, x, k)).$$

Its first-hit domain is the return domain of the preceding execution, and its unique first-hit index is the time value of $\mathrm{ev}_{\mathrm{time}}$*. Write its domain and index as*

$$D_\alpha = \{(e, x) : \exists k : \mathbb{N},\ \alpha(e, x, k) = 1\} \hookrightarrow \mathbb{N} \times \mathbb{N}, \qquad \tau_\alpha : D_\alpha \to \mathbb{N}.$$

Proof.

(1) **Establish monotonicity and define the first-hit test.** The halted state is absorbing, so the clock is monotone. Define the Boolean predicate

$$\mathrm{First}_\alpha(e, x, k) = \alpha(e, x, k) \wedge \bigwedge_{j<k} \neg\alpha(e, x, j).$$

(2) **Construct the unique first hit.** Finite induction through a successful stage produces a least one, and linearity of the NNO order proves uniqueness.

(3) **Identify the classified return domain and its cost.** The corresponding subobject therefore projects isomorphically onto $\{(e, x) : \exists k\, \alpha(e, x, k) = 1\}$, whose classifier lies in Σ. On its kth summand there are exactly k update branches followed by the zero-cost return branch. This is R_k of Theorem 4.6, with the same output and cost k.

□

13.2 Numerical cost adequacy

Theorem 13.4 (Numerical cost adequacy)**.** *In the numerical instance, the measured evaluator erases to the partial evaluator* $\varphi(e, x)$ *of Proposition D.9, with exactly the same domain. Its partial time map* $T(e, x)$ *has that domain and satisfies*

$$T(e, x) = k \quad \Longleftrightarrow \quad \mathrm{First}_\alpha(e, x, k) = 1. \tag{13.1}$$

The graph predicate on the right is primitive recursive, uniformly in e, x, k. The bounded-return predicate

$$\mathsf{Within}(e, x, b) = \alpha(e, x, b)$$

is primitive recursive and is true exactly when the execution returns with time at most b.

Proof.

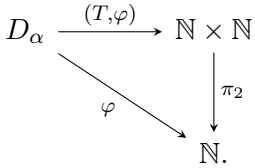
(1) **Identify erasure and the cost graph.** The first-hit evaluator of Proposition D.9 and the weighted evaluator have the same return components. Proposition 13.3 identifies their domains and the time on R_k with its first-hit index k, proving (13.1).

(2) **Verify primitive recursiveness.** Bounded iteration of the primitive-recursive configuration step gives run and α. The first-hit predicate uses only the bounded conjunction over earlier stages, so is primitive recursive uniformly in e, x, k.

(3) **Identify bounded return.** If $\alpha(e, x, b)$ is true, finite search gives a first hit $k \leq b$. Conversely, a first hit at $k \leq b$ makes the clock true at b by monotonicity. This proves the bounded-time statement.

□

On the returning domain, the measured and unmeasured evaluators are related by the commuting triangle



Both partial maps use the same domain mono $D_\alpha \hookrightarrow \mathbb{N} \times \mathbb{N}$ of program–input pairs. Thus the measured span is obtained from an operational comparison preserving both domains and returned values.

Definition 13.5 (Blum complexity measures)**.** For the displayed program enumeration $(\varphi_e)_e$, a **Blum complexity measure** is a partial-recursive numerical measurement $T_e(x)$ whose domain equals that of φ_e and for which the relation $T_e(x) = k$ is decidable uniformly in e, x, k.

Corollary 13.6 (The numerical time is a Blum measure)**.** *For the register-program enumeration of Proposition D.9, the numerical time $T_e(x)$ is a Blum complexity measure: its domain is the domain of φ_e, and the relation $T_e(x) = k$ is decidable uniformly in e, x, k.*

Proof.

(1) **Verify the two defining measure properties.** Domain equality and the primitive-recursive first-hit test are Theorem 13.4. They are the two defining properties of an abstract complexity measure in (Blum 1967).

□

This corollary concerns the numerical time defined above; an arbitrary valuation of histories need not have decidable cost-equality predicates. For example, assigning zero to every event gives cost zero on precisely the halting domain.

The classical external interpretation of the numerical resource space also gives a geometric reading of Blum speedup. Blum's theorem provides computable functions for which every implementation admits an asymptotic improvement, with a prescribed recursive speedup.

Choosing an unbounded multiplicative improvement rules out an O-optimal implementation. In the ordinary external growth locale, Theorem 5.3 then identifies its feasibility open as non-supercompact: an optimal implementation would generate that open and contradict speedup. The numerical measurement throughout this instance counts the updates of its displayed register machine. Other valuations use their own history folds, as in Theorem 4.9.

Measured execution gives a function on inputs. Complexity bounds expressed in an input size must be pulled back along the chosen size map. Keeping that map explicit distinguishes, for example, polynomial bounds in numerical value from polynomial bounds in binary length.

13.3 Input sizes and effective bound functions

Proposition 13.7 (Size-indexed bounds)**.** *An implemented total size map $\ell : X \to \mathbb{N}$ and a displayed total numerical bound expression $b_\beta : \mathbb{N} \to \mathbb{N}$ construct the bound function $g_\beta = b_\beta \ell$ and its germ on the supplied proper-filter context of inputs. They therefore construct the principal admitted budget $\downarrow [g_\beta]$. For numerical inputs,*

$$\forall x : \mathbb{N}\ \alpha(e, x, g_\beta(x)) = 1 \quad \Longleftrightarrow \quad e \in P \ \text{ and } \ \forall x : \mathbb{N}\ T_e(x) \le g_\beta(x). \tag{13.2}$$

Every such pointwise bound gives $\Phi(e) \in B g_\beta$ and hence the corresponding upper-feasibility judgment for $p(e)$.

Proof.

(1) **Construct the size-indexed bound function and admitted budget.** Composition of the specified size and bound descriptions computes the bound function; rational inclusion and the germ quotient give its germ. The unit of J_Σ from Theorem 1.8 gives $\eta_L([g_\beta]) = \downarrow [g_\beta]$. Its pairing is

$$\chi_{\downarrow [g_\beta]}(u) = (\exists a \le [g_\beta]\ u(a)) = u([g_\beta]),$$

so admission is explicit and uniform in the supplied parameters. For the name presentation, let C be a partitioned carrier of the supplied description names, let $f : C \to J_\Sigma L$ be the resulting budget map, and let $q : C_{J_\Sigma L} \twoheadrightarrow J_\Sigma L$ be a target name cover. Applying Proposition D.2, take the canonical partitioned cover of $C \times_{J_\Sigma L} C_{J_\Sigma L}$. Its tracked projections construct the covered-span presentation

$$C \xleftarrow[\text{epi}]{p} \widehat{P} \xrightarrow{v} C_{J_\Sigma L}, \qquad qv = fp.$$

(2) **Identify the bounded clock with the time inequality.** For each x, Theorem 13.4 identifies the bounded clock with return and the stated time inequality.

(3) **Deduce incidence and the represented-input version.** Quantifying over x proves (13.2); a global bound is eventual with constant 1, giving the incidence. For represented inputs the same function construction is on $X = C_X$, with no quotienting of their costs.

□

Finite-word length and binary bit length are constructed by finite recursion on the chosen representations. Thus a polynomial in numerical value uses $\ell(n) = n$, whereas a polynomial in binary input length uses $\ell(n) = \mathrm{bitlength}(n)$, with the encoding's declared convention at zero. These are different functions. On higher-type or redundantly named inputs the pointwise formulation does not require finite size fibres or an enumeration of all names of a

given size. An extensional size map is pulled back along q_X when available; an encoding size may instead depend on the name. Computing a bound for purposes of checking it also has its own recorded execution cost.

For finite binary words, Construction 6.1 and Proposition 6.6 construct the length fibres and their worst-case maxima explicitly. Theorem 6.2 gives the exact equivalence between an upper bound pulled back along length and an upper bound on those maxima. The lower input quantifier is characterized separately in Proposition 6.8.

13.4 A numerical program and two size conventions

The following program makes the distinction concrete. Work in the register-machine environment of Construction 13.2 and Theorem 13.4. Place the natural input n in an input register and 0 in an output register. The instruction scheme in Algorithm 13.2 uses the existing zero-test/decrement, increment, jump, and halted-state conventions.

Algorithm 13.2 The doubling register program

Input: Input register $R_0 = n$; output register $R_1 = 0$

1: loop : **if** $R_0 = 0$ **then goto** halt;
   otherwise $R_0 \leftarrow R_0 - 1$
2: $R_1 \leftarrow R_1 + 1$
3: $R_1 \leftarrow R_1 + 1$
4: **goto** loop
5: halt : **return** R_1

Its finite code $e_{2\cdot}$ has denotation $v(n) = 2n$. For each input n there are exactly n nonzero loop passes followed by the final zero branch. Counting the fixed finite instruction block gives fixed integers $a > 0$ and $b \geq 0$, independent of n, with

$$T_{e_{2\cdot}}(0) = b, \qquad T_{e_{2\cdot}}(n+1) = T_{e_{2\cdot}}(n) + a, \qquad T_{e_{2\cdot}}(n) = an + b. \tag{13.3}$$

For the four-update loop consisting of a test/decrement, two increments, and a jump, with one final zero test and zero cost for the halted state, $a = 4$ and $b = 1$. If the fixed encoding expands one of these commands into several primitive updates, its finite expansion merely changes a, b and not the growth class. The clock adequacy theorem identifies each first-hit index with this update count, so (13.3) is a statement about *measured execution*. As $a > 0$,

$$\kappa(e_{2\cdot}) = [n+1].$$

Write

$$I_{\mathrm{lin}} = {\downarrow}\,[n+1], \qquad I_{\mathrm{poly,val}} = {\downarrow}\,\{[(n+1)^k] : k \in \mathbb{N}\}.$$

Both are ideal points of $\mathfrak{C}_{\mathrm{Ev}}$, and the latter is admitted in the overt-NNO instance: its positive pairing with a monotone $u : L \to \Sigma$ is $\exists k\, u([(n+1)^k])$, which factors through Σ by Proposition 1.4. The displayed code gives concrete witnesses

$$I_{\mathrm{lin}} \models \mathcal{U}_v, \qquad I_{\mathrm{poly,val}} \models \mathcal{U}_v. \tag{13.4}$$

Indeed $\Phi(e_{2\cdot}) \in B(n+1)$, and this same cost belongs to the larger polynomial module. The first assertion tests the principal linear budget; the second tests the nonprincipal ideal of all polynomial bounds in *numerical input value*.

Now put $\lambda(n) = 1 + \mathrm{bitlength}(n+1)$, as in Section 2.7, and define the different polynomial budget

$$I_{\mathrm{poly,bit}} = {\downarrow}\,\{[\lambda^k] : k \in \mathbb{N}\}.$$

For every fixed k, the inequality $n+1 \le c\lambda(n)^k$ fails on arbitrarily large inputs, as can be seen along $n = 2^m$ from exponential growth against fixed polynomials in m. Hence

$$\kappa(e_{2\cdot}) \notin I_{\mathrm{poly,bit}}.$$

The displayed code therefore meets the polynomial budget for numerical value and fails it for binary length. The problem's feasible open is the union over its implementation fibre, as in Theorem 4.25.

The numerical implementation over a budget family. We can also evaluate this measured code on the nonconstant budget family of (3.7). There $b : \mathbb{S} \to \mathfrak{C}_{\mathrm{Ev}}$ uses $a_0 = [1]$, $a_1 = [n+1]$, and distinguished open u. The one-description presentation containing $e_{2\cdot}$ has denotation image $\{v\}$ and feasible open U_{a_1}. By (3.8), its pullback is u: the code is feasible exactly where the budget becomes linear. In the full implementation fibre this same witness proves $u \le b^*\mathcal{U}_v$; equality additionally depends on the other implementations of v.

13.5 Measured overhead constructors

An overhead bound must account for the events introduced by its implementation. The basic operation is concatenation of recorded words. In the fixed finite-description signature, include the idle primitive idle $= \eta_1 : 1 \to L_\Sigma 1$ with event $a = \iota_{\mathrm{idle}}(*,*)$. For the supplied count $m = b(x)$, Algorithm 13.3 prefixes $\mathrm{rep}(a,m)$ to the original execution.

Algorithm 13.3 Prefixing a supplied idle count

Input: Original description e; input x; supplied count $m = b(x) \in \mathbb{N}$

1: **for** $j = 1, \dots, m$ **do**
2: IDLE
3: **end for**
4: **return** RUN(e, x)

The resulting history transformation is

$$D_b(H) = \mathrm{rep}(a, b(x)) {+\!\!+} H, \qquad D_b D_c = D_{b+c}$$

on a fixed input. For the selected valuation its formula is

$$T_{\nu, D_b e}(x) = \nu(\mathrm{rep}(a, b(x)) {+\!\!+} H_e(x)). \tag{13.5}$$

Here $D_b e$ denotes the execution description that prefixes these events. For a monoid fold this is $\overline{w}(\mathrm{rep}(a, b(x)))\, T_{\overline{w},e}(x)$. For additive time with $w(a) = 1$ it is $b(x) + T_e(x)$; for a maximum fold it is the maximum of the recorded prefix and original occupancies. The equality follows from the fold clauses, so no additive formula is assigned to an arbitrary measurement ν.

If $b(x)$ must itself be computed, the execution of that computation is prefixed too. In a numerical code translation, the dispatch and loop instructions of its finite wrapper also contribute their recorded events. Thus (13.5) describes the specified execution word; a concrete wrapper's bound is obtained by evaluating the valuation on all of its events. The cost of executing a code transformation and the execution cost of its output program have different inputs and are kept separate.

For the bounded clock in Proposition 13.3, a numerical reindexing is also available. Define

$$\beta(e, x, k) = \begin{cases} 0, & k < b(x), \\ \alpha(e, x, k - b(x)), & k \ge b(x). \end{cases}$$

Its return extent is unchanged, and its first-hit index satisfies

$$\tau_\beta(e,x) = b(x) + \tau_\alpha(e,x). \tag{13.6}$$

This follows by the two Boolean cases and uniqueness of the first hit. For a nonnegative integral polynomial $P(x,c)$, the inflationary map $r_x(c) = c + P(x,c)$ gives the monotone Boolean clock

$$\beta_P(e,x,k) = \bigvee_{j \le k} (\mathrm{First}_\alpha(e,x,j) \wedge r_x(j) \le k).$$

A first hit at j becomes a first hit at $r_x(j)$, since $j \le r_x(j)$ and the original first hit is unique. Conversely a hit contains the original returning witness. This proves equality of extents and the reindexed clock value. Algorithm 13.4 computes this Boolean predicate by bounded first-hit search. The procedure has its own execution history; its runtime is not inferred from the reindexed clock value.

Algorithm 13.4 Bounded first-hit search and clock regrading

Input: $e, x, k \in \mathbb{N}$; bounded clock α; supplied inflationary map r_x

```
for j = 0, ..., k do
    if α(e, x, j) = 1 then
        return the Boolean value of r_x(j) ≤ k
    end if
end for
return false
```

13.6 Polynomial valuations and compiler comparisons

For $F(x) = \sum_{j=0}^{d} b_j x^j$ and a marked resource M,

$$F(M) \subseteq Bb_0 + \sum_{j=1}^{d} (Bb_j) M^j, \qquad DF(x,v) = \sum_{j=1}^{d} j b_j x^{j-1} v.$$

The first follows from resource multiplication, and the second from Theorem 7.5. Substituting polynomials commutes with their coefficient and geometric interpretations by Theorem 7.24.

Let $\gamma : P \to P'$ be a finite implemented translation with the measured comparison obtained from its compiled histories, $T_{\gamma(e)} \le a_e T_e + b_e$ eventually, where $a_e \in \mathbb{Q}_{\ge 0}$ is a supplied numerical constant (more generally a supplied nonnegative bounded coefficient) and b_e is a nonnegative remainder function. The bound $b_e \in O(T_e)$ gives $T_{\gamma(e)} \in O(T_e)$. A reverse measured comparison gives Θ. The remainder condition $T_{\gamma(e)} - T_e \in o(T_e)$ gives $\sim$. These implications follow from Proposition 2.8 with the implementation parameter retained.

Description measurements use the same calculus with their own measurement functions. The prefix-free word family $1^n 0$ has length $n+1$. Prefixing a fixed word of length c gives length $n+1+c \sim n+1$, with error constant c. Execution cost is measured separately from description length by the recorded histories and their selected valuations.

13.7 Numerical and oscillating examples

Example 13.8 (Leading coefficients and errors)**.** Let $g = n+1$. Then $n+2 \sim g$, whereas $2g \in \Theta(g)$ and $2g \not\sim g$. The first difference is $1 \in o(g)$; the second is g, which fails its smallness inequality at tolerance $1/2$. In the orbit description the first multiplier is $1+t$, and the second is the unit 2 outside $1+\mathfrak{m}$.

Example 13.9 (Polynomial, logarithmic, and exponential orders)**.** For integers $0 \leq r < s$, $(n+1)^r \in o((n+1)^s)$, by the direct cutoff for $(n+1)^{s-r}$. The function $(n+1)\lambda$ is in $\widetilde{\Theta}(n+1)$ and in $\omega(n+1)$. Every fixed polynomial belongs to $o(2^n)$ by the finite binomial estimate of Section 13.9. The parity examples of Section 2.4 give comparisons which have no single limiting ratio.

Example 13.10 (Tight bounds and asymptotic equivalence)**.** At a reference resource g, first-order variation within the tight-bound orbit has direction module Bg. Variation within its asymptotic-equivalence class has direction module $\mathfrak{m}g$ by (7.11). For $g = t^{-1}$ these are $t^{-1}B$ and $t^{-1}\mathfrak{m}$. The direction g is allowed by the first and excluded by the second, whereas the constant direction $1 = tg$ belongs to both. This is the differential form of the leading-coefficient distinction.

The next calculation returns to the geometric interpretation of resource constraints. It exhibits a direction excluded by a bounded budget and shows how allowing a chosen overhead removes that normal obstruction.

13.8 An explicit normal resource direction

For cofinite input put $H = t^{-1}B$ and take a fixed implementation of bounded cost $c = \Phi(e) \in B$. The valuation family $F_e(b) = c + b$ has budget-B tangent $B \subset H$ at $b = 0$. The direction t^{-1} gives a nonzero class in H/B, since $n+1$ is not bounded. After localization at t, this quotient becomes zero by Theorem 2.37. This computes an explicit normal resource obstruction and its change under overhead inversion.

A numerical asymptotic scale organizes resource coefficients by order of decay. Absolute differentials describe logarithmic connections along this scale.

13.9 Asymptotic expansions, scale filtrations, and connections

13.9.1 The numerical scale and its remainders

Work in the cofinite numerical fibre and put $t(n) = 1/(n+1)$. It is a regular element of B, belongs to $\mathfrak{m}$, and is not a unit: its reciprocal $n+1$ is unbounded. Thus its powers form a strict asymptotic scale, $t^{r+1} \in o(t^r)$.

Definition 13.11 (Asymptotic expansion)**.** For a rational sequence $(a_j)_{j\in\mathbb{N}}$, the **asymptotic expansion relation** (distinct from relative-error equivalence $\sim$) is written

$$f \sim_t \sum_{j\geq 0} a_j t^j$$

and means the family of finite remainder statements

$$f - \sum_{j=0}^{r} a_j t^j \in o(t^r) \qquad (r \in \mathbb{N}).$$

No value for an infinite sum is part of this definition. More generally an **asymptotic scale** (φ_j) with $\varphi_{j+1} \in o(\varphi_j)$ defines its expansion by the corresponding remainders. This is the finite-remainder use of asymptotic expansion (NIST 2026).

Proposition 13.12 (Coefficient uniqueness and arithmetic)**.** *The rational coefficients of a t-expansion are unique. Expansions add and multiply with the usual finite coefficient formulas. On the subobject of expansions with $a_0 \in \mathbb{Q}^\times$, their germs are units of B and their inverses have the recursively determined reciprocal expansion.*

Proof.

(1) **Exclude nonzero rational constants from the vanishing ideal.** A nonzero rational constant cannot belong to $o(1)$: choose k with $k|a| > 1$; its required eventual extent would be empty, contrary to properness.

(2) **Prove coefficient uniqueness by induction.** Suppose coefficients agree below degree r in two expansions of f. Subtract their order-r remainder equations and divide by the everywhere nonzero t^r. Then $a_r - b_r \in o(1)$, hence $a_r = b_r$. Induction proves uniqueness.

(3) **Verify addition of expansions.** Addition follows from the small-error sum law.

(4) **Compute the product expansion.** For multiplication fix order r. The order-zero estimates make both germs bounded. Multiplying the two order-r remainders by bounded germs gives errors in $o(t^r)$. Terms of the finite product with degree greater than r belong to $t^{r+1}B \subseteq o(t^r)$. The remaining coefficients are the finite convolution.

(5) **Construct the reciprocal expansion.** For $a_0 \neq 0$, the order-zero expansion gives $f = a_0(1 + a)$ with $a \in \mathfrak{m}$. Hence f is a unit by Theorem 2.19. Set $b_0 = a_0^{-1}$ and $b_r = -a_0^{-1} \sum_{j=1}^{r} a_j b_{r-j}$. At each finite order, convolution gives $f \sum_{j=0}^{r} b_j t^j = 1 + o(t^r)$. Multiply the remainder by the bounded inverse of f. This is the reciprocal expansion.

□

An expansion need not determine the germ. The rational-valued function 2^{-n} is nonzero but belongs to $o(t^r)$ for every r. Indeed for $k = r + 1$ and $n \geq 2k$,

$$2^n \geq \binom{n}{k} \geq (n/2)^k/k!,$$

so $(n + 1)^r 2^{-n}$ is bounded by a rational constant times $1/n$ eventually. Each tolerance has an explicit integer cutoff. Thus its expansion has all coefficients zero, while its nonzero germ remains distinct.

13.9.2 Filtered coefficients and differentiation

The filtration $F^r B = t^r B$ records a specified scale of error. It is finer than declaring an error merely vanishing. For example $b(n) = 1/(1 + \lfloor\sqrt{n + 1}\rfloor)$ belongs to $\mathfrak{m}$ but not to tB: square inputs make b/t arbitrarily large. The floor square root is a bounded finite search, so this is a rational-valued function.

Proposition 13.13 (Scale boundary)**.** *There are natural isomorphisms and exact sequences*

$$t^r B/t^{r+1}B \cong B/tB, \qquad \operatorname{gr}_{tB} B \cong (B/tB)[z],$$

$$0 \longrightarrow M/tM \xrightarrow{t} M/t^2M \longrightarrow M/tM \longrightarrow 0$$

for every embedded resource $M \subseteq K$.

Proof.

(1) **Identify the associated graded algebra by cancellation.** Multiplication by t^r is injective on K and hence on B and M. It identifies the quotient in degree r with B/tB. Products of these identifications add degrees, giving the polynomial graded algebra.

(2) **Verify exactness of the scale sequence.** In the displayed sequence, the first map sends $[m]$ to $[tm]$. If $tm \in t^2M$, cancellation gives $m \in tM$. Its image is exactly the kernel of reduction modulo tM. Surjectivity of the last map is quotient surjectivity.

□

For a finite polynomial parameter algebra $A = \mathrm{Sym}_B(E)$ and an ideal $J \subseteq B$, relative differentiation gives

$$d_{/B}(JA) \subseteq J(A \otimes_B E).$$

Indeed $d_{/B}(bp) = b\, d_{/B}p$ for $b \in J$. This proves coefficientwise stability of remainders in $t^r A$ and $\mathfrak{m}A$ under polynomial parameter differentiation. It supplies an explicit differential calculus for resource-valued asymptotic expansions.

The nilpotent Taylor parameter d of Section 12 and the regular asymptotic scale t have different roles. Relative Weil differentiation fixes t as a coefficient. Absolute coefficient differentiation is constructed next.

13.9.3 Logarithmic differentials and pole resources

Definition 13.14 (Logarithmic differentials)**.** The **module of logarithmic differentials** for the scale t is

$$L_t = (\Omega^1_{B/\mathbb{Q}} \oplus B\ell)/(dt - t\ell),$$

Write $d : B \to L_t$ for the universal differential followed by this quotient.

A B-linear map $L_t \to N$ is exactly a pair (δ, ω) with $\delta : B \to N$ a $\mathbb{Q}$-derivation and $\delta(t) = t\omega$. This follows directly from the generators and the one displayed relation.

For $k \geq 0$ let $M_k = t^{-k}B \subseteq K$.

Proposition 13.15 (Logarithmic connections)**.** *The formula*

$$\nabla_k(bt^{-k}) = t^{-k} \otimes (db - kb\ell) \tag{13.7}$$

defines an L_t-connection on M_k. It is compatible with $M_k \hookrightarrow M_{k+1}$ and with $M_k \otimes_B M_l \cong M_{k+l}$. Its residue in the generator t^{-k} is $-k$.

Proof.

(1) **Verify the connection and Leibniz identity.** The coefficient b is unique because t is regular. The derivation identity gives $\nabla_k(am) = a\nabla_k(m) + m \otimes da$.

(2) **Verify compatibility with inclusions.** Under the inclusion, $bt^{-k} = tb\, t^{-(k+1)}$. Substitute $d(tb) = t\, db + b\, dt$ and $dt = t\ell$ into the right-hand formula at $k+1$; the result is exactly the formula at k.

(3) **Verify tensor compatibility.** For a product $bt^{-k} \cdot ct^{-l}$, the Leibniz rule for $d(bc)$ adds the two logarithmic terms, giving the tensor compatibility.

(4) **Construct the residue map.** Define $\mathrm{res} : L_t \to B/tB$ by $db \mapsto 0$ and $\ell \mapsto 1$. The relation maps to zero, so this is well-defined and factors through L_t/tL_t.

(5) **Compute the residue of the resource line.** Applying it to the coefficient in (13.7) gives $-k\bar{b}$. Under the generator identification of M_k/tM_k with B/tB, this is multiplication by $-k$.

□

Corollary 13.16 (Residues distinguish logarithmic resource lines)**.** *In the cofinite numerical fibre, the residue of ∇_k is $-k$ in every bounded-unit frame of $M_k = t^{-k}B$. The logarithmic resource lines (M_k, ∇_k) and (M_l, ∇_l) are isomorphic as L_t-connections exactly when $k = l$. Tensor product adds their residues.*

Proof.

(1) **Show that the residue is independent of the generator.** Every generator of the free rank-one module M_k has the form ut^{-k} with $u \in B^\times$. Its connection form is $u^{-1}du - k\ell$. Since $\operatorname{res}(du) = 0$, its residue is $-k$.

(2) **Use residues to distinguish the indices.** An isomorphism $M_k \to M_l$ has the form $t^{-k} \mapsto ut^{-l}$ with $u \in B^\times$. Compatibility with the connections gives $du + (k-l)u\ell = 0$; applying residue yields $(k-l)\bar{u} = 0$ in B/tB. This quotient is nonzero because t is not a unit, and every nonzero rational remains a unit in it. As $\bar{u}$ is also a unit, $k = l$.

(3) **Verify the converse and tensor rule.** Conversely the identity is an isomorphism when the indices agree. Tensor compatibility follows from Proposition 13.15, and its residues are $-(k+l)$.

□

13.9.4 Principal parts and geometric realization

Definition 13.17 (Logarithmic first principal parts)**.** Put $A_t = B \ltimes L_t$ and use the two algebra maps

$$\lambda(b) = (b, 0), \qquad \rho(b) = (b, db).$$

The **logarithmic first principal-parts module** of M is $P_t^1(M) = A_t \otimes_{B,\rho} M$, with its left B-module structure induced by λ.

The right-B sequence $0 \to L_t \to A_t \to B \to 0$ splits through ρ. Tensoring therefore gives a short exact sequence of left B-modules

$$0 \to L_t \otimes_B M \to P_t^1(M) \to M \to 0. \tag{13.8}$$

Its canonical $\mathbb{Q}$-linear map $j(m) = 1 \otimes m$ obeys $j(bm) = b\,j(m) + db \otimes m$. Consequently a connection $\nabla : M \to L_t \otimes_B M$ gives the B-linear splitting $j - \nabla$, and a splitting s gives the connection $j - s$. These formulas are mutually inverse. Exactness and full faithfulness of j_B realizes the sequence and this correspondence inside $\mathcal{G}_B$. This constructs the differential resource bundle with its extension class and splittings.

14 Conclusion

The construction gives a differential geometry of computational resources with a common coefficient ring. Measured execution determines cost functions and implementations; embedded resource modules and their locale classify budgets and feasibility. Taylor-valued history instrumentation identifies finite geometric jets with measured response coefficients. Resource deformation complexes then describe constrained variations, including compatibility between moving budgets and the effect of permitted overheads on obstructions.

The finite-word instance also identifies conventional worst-case complexity inside this geometry. Fibrewise maximum is left adjoint to length pullback; the constructed machine instance yields the standard deterministic time and workspace upper classes and the polynomial class **P**. Classical universal lower predicates are recovered by implementationwise double-negation closure, without identifying them with effective witness selection (Theorems 6.2, 6.7, 6.10, 6.12 and 6.18).

The primary infinitesimal geometry represents the function objects needed for completion, de Rham cochains, and crystalline jets. The comparison with Kähler forms, the equivalence

between connections and stratifications, and the principal-parts description of differential operators retain the same resource parameters and uniformly tracked finite coefficient names.

The examples exercise these structures together. A measured family has polynomial bounds at every finite order without a common polynomial degree for its complete coefficient family. Its generating function carries a logarithmic de Rham class and a crystalline resource line. Event instrumentations exhibit a mixed-jet curvature defect and an explicit coherent ordered transport. Coupled budgets can obstruct a joint lift despite separate feasibility, and logarithmic residues distinguish asymptotic pole orders. These calculations connect the geometric foundations to the measured resources from which they were constructed.

Table 1 summarizes the objects and their roles.

Table 1: Objects and their roles in the construction.

Notation	Setting	Role
$\mathcal{E},\ \Sigma \hookrightarrow \Omega_{\mathcal{E}}$	Hyland's effective topos; finite-hit truth values	Computational names, partial returns and positive observations
$J_\Sigma,\ \mathcal{A}_\Sigma$	Selected ideal completion and its algebras	Admitted directed suprema and recursive execution
$K,\ B,\ \mathfrak{m}$	Rational germs, bounded germs, vanishing germs	Resource coefficients; $O(g) = Bg$, $o(g) = \mathfrak{m}g$
$L,\ \mathfrak{C}_{\mathrm{Ev}}$	Growth lattice and budget locale	Ordinary families of marked resources
$\mathscr{S} = \mathrm{Sh}_{\mathcal{E}}(\mathfrak{C}_{\mathrm{Ev}})$	Budget sheaves, equivalently $[L, \mathcal{E}]$	Universal budget and implementation witnesses
$\mathscr{H}_{B,\mathrm{et}},\ \mathcal{O}_{\mathrm{inf}}$	Primary étale infinitesimal-arrow sheaves	Represented functions, formal nerve, de Rham comparison and jets
$\mathcal{G}_B,\ j_B M$	Based projective Weil diagrams	Finite resource coefficients and marked deformations

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A Support, representatives, and finite denominator witnesses

The resource calculations use formulas on representatives and then descend them to germs. This appendix records the support operations, common principal stages, and finite denominators that justify those steps. The pointwise operations are rational formulas with decidable cases:

$$e_g(x) = \begin{cases} 1, & g(x) \neq 0, \\ 0, & g(x) = 0, \end{cases} \qquad q(f,g)(x) = \begin{cases} f(x)/g(x), & g(x) \neq 0, \\ 0, & g(x) = 0. \end{cases}$$

Changing either representative on a noneventual complement changes these formulas only there. Intersecting the representative-equality extents proves that they descend. Under a domination witness $f \preceq g$, the equation $q(f,g)g = f$ holds on the same eventual comparison extent.

For a finite list $f_1, \dots, f_r$ in a resource module, the nonnegative germ $g = \sum_i |f_i|$ belongs to that resource and dominates the list. Each f_i is a bounded multiple of g. This is the common principal stage used to verify finite tensor relations and exactness. The stage is constructed from the finite list; no simultaneous selection of principal generators for arbitrary submodules is involved.

A quotient argument with local representatives is an internal argument over the effective epimorphism from functions to germs. To define a function on the quotient we prove its formula invariant under the equivalence relation. To prove existence of a multiplier, we construct it over a representative and then use the image predicate of its defining equation. This proves internal existence.

Similarly, a finite list of localization equations has a common denominator given by the product of its denominators. For equations involving a fixed scale t, take a common maximum exponent. These finite operations justify the flatness and finite-cover arguments. The common-denominator condition for an infinite coefficient family is retained as an existential statement about one exponent, as in (10.13); it is not obtained by taking a maximum of infinitely many witnesses.

B Finite Weil duality and coefficient coherence

The linear adjoints and coefficient-change comparisons in the main text use finite-projective duality. We record the transpose formulas and check their independence of the chosen dual basis, then verify compatibility with stage and coefficient maps. Let C be a finite projective B-algebra, F a C-module, and M a B-module. For a dual basis (c_i, ϕ_i), the transpose of a balanced map $h : C^\vee \otimes_C F \to M$ is

$$\widetilde{h}(x) = \sum_i c_i \otimes h(\phi_i \otimes x).$$

For $\psi \in C^\vee$,

$$(\psi \otimes 1)\widetilde{h}(x) = h\left(\sum_i \psi(c_i)\phi_i \otimes x\right) = h(\psi \otimes x).$$

The same identity with the right C-action on $C^\vee$ shows $\widetilde{h}(cx) = c\widetilde{h}(x)$. It also proves independence of the basis: every contraction with a functional has the specified value, and finite duality detects equality in $C \otimes_B M$. This gives the inverse equations used in (7.10).

For a stage map $u : C \to C'$, contracting the naturality equation of a family $v_C : F(C) \to C \otimes_B M$ with $\phi \in (C')^\vee$ gives

$$h_C((\phi u) \otimes x) = h_{C'}(\phi \otimes F(u)x).$$

These are exactly the coequalizer relations defining Λ_{lin}. Thus its adjunction is natural in both the coefficient module and the geometric module.

For a coefficient map $\beta : B \to B'$ and a rational Weil presentation W, tensor associativity gives

$$(B' \otimes_{\mathbb{Q}} W) \otimes_{B'} (B' \otimes_B M) \cong B' \otimes_B ((B \otimes_{\mathbb{Q}} W) \otimes_B M).$$

On a pure tensor both sides have the same coefficient, Weil basis element, and resource element. The diagrams for two coefficient maps, two Weil substitutions, truncation, and

interchange therefore commute on these generators. This proves the finite coefficient coherence used for gauge and input transport.

C History substitution and numerical clock regrading

Two distinct comparisons are used in the execution examples: reassociating a recorded history preserves its fold, while regrading a numerical clock changes its index. The following calculations establish their composition laws and keep their respective measurements explicit.

C.1 Coherence of the resource valuation

For a monoid (C, e, μ), an event map $w : A \to C$, and returning words $s, t, u \in A^*$, the recursive fold satisfies

$$\overline{w}(s{+\!+}t) = \overline{w}(s)\overline{w}(t), \qquad \overline{w}((s{+\!+}t){+\!+}u) = \overline{w}(s{+\!+}(t{+\!+}u)).$$

The first equation is induction on s; the second follows from word associativity and monoid associativity. Empty words give the two unit equations. Each interface-cut transformation in Theorem 4.7 preserves the flattened word, so these formulas prove its resource coherence. Input substitution precomposes the state, event, and valuation maps in these same formulas; induction on the finite descriptions proves its compatibility with composition and recursive execution.

For a finite-history measurement ν that is not a monoid fold, the assembled cost remains $\nu(H_{\text{assembled}})$. No monoidal law is deduced for ν. A richer finite accumulator may instead construct its stated law. For the sequential time convention, an overhead execution of cost $H(e, x)$ returning a waiting amount $b(e, x)$ contributes $H(e, x) + b(e, x) + T_e(x)$ together with every recorded dispatch transition. This is the sum of the component words' time folds.

For two verified measured comparisons $T_2 \le aT_1 + b$ and $T_3 \le cT_2 + d$, on the intersection of their eventual extents,

$$T_3 \le caT_1 + (cb + d).$$

The comparison retains the same implementation parameter and the nonnegative coefficients. For general monotone growth transformations, their composition is the one in Construction 4.36.

C.2 The bounded numerical calculation

For the numerical clock $\alpha(e, x, k)$ and a supplied inflationary regrading $r_x : \mathbb{N} \to \mathbb{N}$ with its finite Boolean tests, set

$$R_r\alpha(e, x, k) = \bigvee_{j \le k} (\mathrm{First}_\alpha(e, x, j) \wedge r_x(j) \le k).$$

This is monotone and Boolean by its finite formula. A first hit of α at j becomes the first hit of $R_r\alpha$ at $r_x(j)$; inflationarity puts j inside the bounded search and uniqueness excludes another contributor. A hit of the new clock contains a first-hit witness of the original, so their classified extents agree.

For inflationary r_x, s_x, the two clocks

$$R_s(R_r\alpha) = R_{s \circ r}\alpha$$

are true at exactly the indices at least $s_x(r_x(j))$ when the original first hit is j. Existence and uniqueness of that j prove the equality in every parameter context. The identity regrading

gives the original monotone clock. Input substitution acts on the same bounded formula, so commutes with regrading. This is a clock-index calculation for the constructed numerical instance; measured compiler costs are supplied by the recorded histories and their valuations above.

D Name presentations, finite descriptions, and numerical codes

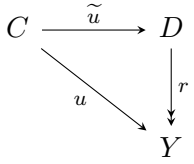
The implementation construction requires names whose partial implementations preserve domains, finite typed descriptions, and a numerical program presentation. This appendix supplies those three constructions from the realizability and exact-completion constructions of (Hyland 1982; van Oosten 2008). Costs are subsequently assigned by the execution histories and valuations of the main text.

Lemma D.1 (Lifting to a modest name object)**.** *Let C be a partitioned assembly, let D be a modest assembly, and let $r : D \twoheadrightarrow Y$ be a regular epimorphism in* **Eff***. Every arrow $u : C \to Y$ has a tracked lift $\widetilde{u} : C \to D$ with $r\widetilde{u} = u$. This construction uses no external choice.*

Proof.

(1) **Extract a numerical output witness.** Regular epimorphy realizes the total relation $r(d) = u(c)$. Apply its totality realizer to the distinguished name $\varepsilon_C(c)$. The output contains a name n_c of some $d \in D$ and evidence of the displayed equation.

(2) **Recover the unique denotation.** Modesty says that a numerical name realizes at most one element of D. Hence the relation $n_c \Vdash_D d$ is total and functional in c. Unique-value comprehension defines $\widetilde{u}(c) = d$. The program extracting n_c tracks this map, and the retained equality evidence proves $r\widetilde{u} = u$.



(3) **Retain parameters.** Apply the same construction on the canonical partitioned cover of the parameter object times the input cover. One realizer handles all valid parameter names. The resulting equality descends; no element of a multiply inhabited fibre has been selected.

□

Proposition D.2 (Canonical names, covered arrows, and partial lifts)**.** *Every object of* **Eff** *has a canonical regular-epimorphic partitioned-assembly presentation relative to its realizing relation. Every semantic arrow has a tracked covered-span presentation with exact denotation; every classified partial arrow has such a presentation with exactly its pulled-back domain. For a modest target name cover, first-hit search constructs an actual partial name map with that domain and value. Composition, finite products, and parameter substitution are implemented on partitioned refinements, and their semantic equations descend through effective quotients.*

Proof.

(1) **Construct the canonical cover.** For an object presented by realizability equality $E(x, y)$, put

$$\widehat{X} = \{(x, n) : n \in E(x, x)\}, \qquad \varepsilon_{\widehat{X}}(x, n) = n.$$

Projection to x is a regular epimorphism. The carrier retains all realizing witnesses, so it chooses no name for any semantic element. For an assembly this is the same construction with $n \Vdash_X x$.

(2) **Present total arrows by covered spans.** For $q : C \twoheadrightarrow X$, $q' : C' \twoheadrightarrow Y$, and $f : X \to Y$, form $P = C \times_Y C'$ using fq and q', and take its canonical partitioned cover $\widehat{P} \twoheadrightarrow P$. Its projections give $p : \widehat{P} \twoheadrightarrow C$ and $v : \widehat{P} \to C'$ with $q'v = fqp$.

$$\begin{array}{ccc} \widehat{P} & \xrightarrow{v} & C' \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q'} \\ C & \xrightarrow[fq]{} & Y \end{array}$$

Both name maps are tracked. The map $q'v$ coequalizes the kernel pair of p and descends to fq; its further descent through q is exactly f. No splitting of p is used.

(3) **Recover the exact partial domain.** For $f : X \rightharpoonup Y$ with domain mono $D_f \hookrightarrow X$, pull that domain back to C. Cover $q^{-1}D_f$ partitionedly, then pull back the target cover and cover that pullback. The resulting left leg into C has image exactly $q^{-1}D_f$, and the right leg denotes f there. Thus descent preserves both domain and value.

(4) **Construct first-hit lifts for modest target names.** For a Σ-classified domain, Lemma D.1 lifts its classifier on C through the modest object $2^{\mathbb{N}} \twoheadrightarrow \Sigma$. Write the resulting stream as α_c. The hit assembly $H = \{(c, n) : \alpha_c(n) = 1\}$ is partitioned. When C' is modest, that lemma lifts the semantic value on H to a tracked map $g : H \to C'$. The first-hit operation is Algorithm D.1.

Algorithm D.1 Tracked partial evaluation on a name

Input: Name c; total tracked stream α_c; tracked value map g on hit names

```
n ← 0
while α_c(n) = 0 do
    n ← n + 1
end while
return g(c, n)
```

The search terminates exactly on $q^{-1}D_f$ and returns a name of the original value. Its delayed recursion evaluates g only on valid hit names, giving the claimed classified partial map.

(5) **Compose spans and construct refinements.** Compose covered spans by pulling back the second left leg along the first right leg and taking a canonical partitioned cover. Pullback stability of regular epimorphisms makes the resulting domain image exactly the composite semantic domain. Finite products and parameter pullbacks use the same construction. Two covers have a common refinement obtained by covering their fibre product over the denotation object.

(6) **Descend equations and finite returns.** Identity and composition have their specified denotations on common refinements, so their equations descend through the effective

quotients. For iteration, cover the witnesses returning after each finite number of transitions and take the image of their $\mathbb{N}$-indexed sum. Induction on return length identifies each component with the semantic return component; their union is exactly the semantic finite-return domain. Actual implemented primitives compose by the first-hit and finite-return algorithms; their trackers and histories are retained as implementation data.

□

The distinction from unrestricted partitioned-assembly projectivity is essential. Give an external set-surjection constant name 0 on both carriers. It is a regular epimorphism of assemblies; its splitting would split the original set-surjection. Thus arbitrary fixed-carrier projective lifting entails external choice (Hofstra 2003, Section 2.4). Canonical covers, the modest-target lemma, and the effective descent above require no such splitting.

For a modest object its PER support is a numerical name cover; the kernel pair is the PER. Extensional trackers descend through that pair, and Lemma D.1 supplies lifts to the modest support. For a represented relation, pull back the name cover along its membership mono and retain the resulting proof names. General relations use the covered-span construction.

Proposition D.3 (Numerical covers of coefficient objects and towers)**.** *The coefficient objects built from $B \hookrightarrow K$ and resource submodules by finite products, finite lists, effective quotients, subobjects, and displayed finite-projective retracts have modest numerical name covers. The compatible coefficient towers used in Section 10 have numerical name covers with uniform coordinate evaluation. Arithmetic and the stated finite coefficient formulas have tracked lifts on these covers, uniformly in their numerical orders and coefficient presentation codes.*

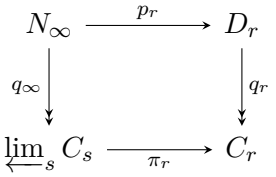
Proof.

(1) **Present germs and their subobjects.** The modest sequence object $\mathbb{Q}^X$ has its numerical PER support cover, and its quotient onto K is regular epic. Modest objects are closed under exponentials and subobjects in **Eff** (van Oosten 2008). Pull the sequence object back along $B \hookrightarrow K$ or $M \hookrightarrow K$ and take the numerical support of the resulting modest subobject. Membership witnesses are retained; their validity is not asserted to be Σ-classified.

(2) **Construct finite coefficient presentations.** Products of the covers cover finite products. Tagged finite lists of names cover finite lists; composing with a quotient covers its quotient. Pullback along a mono covers a subobject. A displayed retract is a subobject of its finite free presentation, so the same construction applies. Polynomial coefficients use finite lists and their effective equation quotients. Finite-source Hom objects are retracts of finite powers by the displayed dual-basis formulas. All constructions are uniform in the finite presentation codes.

(3) **Cover compatible towers without selecting coordinates.** Let $q_r : D_r \twoheadrightarrow C_r$ be the resulting uniform numerical covers. The tagged object $D = \sum_{r:\mathbb{N}} D_r$ is modest. Take a canonical partitioned cover $P \twoheadrightarrow \varprojlim_r C_r$. The coordinate evaluation $P \times \mathbb{N} \to \sum_r C_r$ lifts through D by Lemma D.1. Transpose this lift and restrict to the subobject H of $D^{\mathbb{N}}$ whose rth name has tag r and whose denotations satisfy the tower equations. The resulting map $P \to H$ proves that $H \to \varprojlim_r C_r$ is epic. As a subobject of a modest

exponential, H is modest and has a numerical support cover $N_\infty \twoheadrightarrow H$.



Evaluation of the coded name sequence defines p_r uniformly in r, and $q_r p_r = \pi_r q_\infty$.

(4) **Implement formulas and retain parameter contexts.** Apply Lemma D.1 to each coefficient operation with its modest numerical target cover. Compose the resulting arithmetic, finite-list, and matrix trackers. An output of order r using input order $b(r)$ first applies $p_{b(r)}$ and then its finite tracker. For an arbitrary parameter context, retain that context's partitioned cover and pull back these coefficient presentations. The numerical coordinates present coefficients; they do not enumerate an arbitrary parameter carrier. All domain and value equations descend on the retained covers.

□

Definition D.4 (Typed finite descriptions)**.** For a fixed finite typed computational signature, a **parameterized syntax tree** is a finite prefix list of constructor nodes with finite binding and child-position data, together with the tuple of supplied parameter labels required at its leaves. Its type annotations record the types at every node, including operation slots. A **well-typed finite description** is such a tree satisfying the constructor typing equations. Semantic equalities of arbitrary parameter labels are not Boolean grammar tests.

Construction D.5 (Typed finite descriptions)**.**

(1) **Present the constructor data and parse finite trees.** For a fixed finite typed computational signature, let $\mathcal{S}$ contain its constructor tags, arities, source and target sorts, and supplied parameter slots. Use the syntax-tree data of Definition D.4. The valid-tree condition is the finite stack-parsing condition: each node consumes one pending place and opens the number specified by its tag; the stack is empty exactly at the end.

(2) **Construct the internal carrier and its presentation.** Thus the carrier is a subobject of a sum of finite products of label objects and numerical data. It exists using the NNO and finite lists and has a partitioned presentation in **Eff**.

(3) **Retain typing and environment parameters.** Take the well-typed subobject using the constructor typing equations. For each primitive slot, form the object of its semantic operation, actual partial name map, and numerical tracker, subject to the domain-and-value equations. These are subobjects defined by the partial-map objects, numerical application, and equality predicates of $\mathcal{E}$. Their finite product with the label objects constructs the implemented environment object Env; no section onto all semantic operations is used. Structural and arithmetic primitives are supplied by their explicit algorithms, and numerical primitives by Proposition D.9. Its well-typed descriptions over $\theta : \mathsf{Env}$ form $E_{X,Y,\theta}$.

Lemma D.6 (Interpretation and substitution)**.** *Finite recursion constructs the interpretation of the displayed descriptions in the computational function objects, and tracked interpretations on their name presentations. Interpretation commutes with finite typed substitution. Pure structural maps, composition, branching, sequential pairing, and recursive execution have constructors.*

Proof.

(1) **Interpret the finite description tree.** A leaf is its specified parameter or primitive; a node applies the designated constructor morphism to the interpretations of its children. Products and exponentials of Theorem 1.8, lifted strength, and Proposition 1.10 give those morphisms. A recursion node denotes the least solution of its body, rather than a recursive definition of the tree interpreter itself. Algorithm D.2 carries out this finite traversal with the supplied constructor maps. It constructs the interpreted operation; a recursion constructor forms its evaluator rather than running it.

Algorithm D.2 Interpretation of a finite typed description

Input: Well-typed finite tree c; environment θ; supplied constructor maps

procedure INTERPRET(c, θ)
  if c is a parameter or primitive leaf **then**
    return its supplied interpretation in θ
  else
    for each child c_i of c **do**
      $v_i \leftarrow$ INTERPRET(c_i, θ)
    end for
    return $\mathrm{op}_{\mathrm{tag}(c)}((v_i); \theta)$
  end if
end procedure

(2) **Implement the constructors on names.** On names, use the actual primitive maps, trackers, and classified-domain squares in the constructed environment of Construction D.5. The modest-target and first-hit constructions of Propositions D.2 and D.3 supply the coefficient primitives. Composition runs the first implementation and then the second on the returned name; branching inspects its finite tag; a recursive execution runs its body until its exit tag is returned. The finite-return construction supplies the latter partial map.

(3) **Prove domain and value adequacy.** Induction on the number of transitions identifies its domain and returned denotation with the semantic iteration. Structural induction then proves the same for the whole finite description.

(4) **Verify identity and composite substitution.** Substitution replaces slot leaves by their designated trees and shifts bound indices by the depth of the enclosing context. At a leaf the interpretation equation is its definition. At a compound node it follows from the equation for the children and the same constructor morphism. This proves identity and composite substitution, including the iteration node.

(5) **Retain parameters and construct code-output descriptions.** The algorithms traverse only finite trees; arbitrary supplied names remain labels. A code-output extension of the signature uses these same trees as data sorts, and its identity and composition constructors provide the compiler descriptions used in Construction 4.36.

□

For pure syntax, all numerals are constructor strings over a finite alphabet. Prefix serialization and stack parsing therefore give total primitive-recursive maps $\mathrm{ser} : \mathsf{Tree} \to \mathbb{N}$ and $\mathrm{parse} : \mathbb{N} \to \mathsf{Tree}$, with $\mathrm{parse}\,\mathrm{ser} = \mathrm{id}$ (invalid words parse to a fixed dummy tree). Enumerating finite words up to a given length gives finite lists $E(s)$ containing every tree at some stage. Consequently pure trees are overt. This assertion concerns pure descriptions, not a numerical enumeration of arbitrary supplied parameter carriers.

D.1 The numerical program presentation

Definition D.7 (Finite functional graphs)**.** Let FG be the decidably ordered poset of **finite functional graphs** in $\mathbb{N} \times \mathbb{N}$, ordered by inclusion. Its empty graph is least.

An ideal of FG determines a partial function by taking its union, and every partial function determines its ideal of finite subgraphs. Directedness proves functionality and reconstructs every finite subgraph; these operations are inverse.

Lemma D.8 (Recognizing admitted finite-information ideals)**.** *For an overt poset P with Σ-classified order, an ideal belongs to $J_\Sigma P$ exactly when its membership predicate is uniformly Σ-valued.*

Proof.

(1) **Recover uniform membership from admitted pairings.** The positive upper test $u_p(q) = (p \leq q)$ has pairing $\chi_I(u_p) \leftrightarrow I(p)$, by lower closure and reflexivity. Admission therefore gives membership uniformly in p.

(2) **Construct admitted pairings from membership.** Conversely, uniform membership and an upper test have Σ-valued conjunction; overtness of P gives their existential pairing uniformly in the test.

□

Proposition D.9 (Numerical codes and extensional denotation)**.** *In* **Eff***, finite register programs construct a partial evaluator $\varphi(e, x)$ and a regular epimorphism*

$$q : \mathbb{N} \twoheadrightarrow \mathsf{PC} = J_\Sigma \mathsf{FG}, \qquad q(e) = \{g : g \subseteq \mathrm{graph}(\varphi_e)\}.$$

Its kernel pair is extensional equality of partial functions. Specialization and sequential composition have total recursive code constructors.

Proof.

(1) **Construct the register-program enumeration.** Encode finite instruction lists with instructions for increment, zero-test with decrement, jumps, and halt. A configuration stores the program, instruction pointer, and finite register array. The total primitive-recursive transition reads one instruction, updates that finite data, and leaves a halted configuration fixed. Invalid codes are assigned a fixed looping configuration. Primitive recursion constructs bounded runs. Minimizing the first halting stage and reading the output defines φ.

(2) **Implement the partial-recursive constructors.** Conversely, zero, successor, projections, composition, primitive recursion, and minimization have register-program implementations: use finite register blocks for intermediate values, counted loops for recursion, and an increasing search loop for minimization. Thus this is the ordinary partial-recursive enumeration underlying K_1 (van Oosten 2008).

(3) **Recognize finite subgraphs and prove admission.** For fixed e, g, the total stream whose stage-s value tests whether every pair in g is verified by an s-step run of e has hit support $g \subseteq \mathrm{graph}(\varphi_e)$. The empty graph is accepted immediately. The preceding lemma proves that $q(e)$ is admitted, uniformly in e.

(4) **Construct a uniform numerical code from a name.** To prove epicity, cover $J_\Sigma \mathsf{FG}$ by a partitioned assembly. Its uniform membership classifier on that cover times FG lifts to a total stream $\theta(a, g, k)$. Algorithm D.3 performs the finite diagonal search.

Directedness makes all discovered outputs agree, and each pair in the represented partial function is eventually found. The supplied name a is a parameter of this one program, so specialization compiles it uniformly into a numerical code.

Algorithm D.3 Evaluation from a named finite-graph ideal

Input: Name a; input x; total membership stream $\theta(a, g, k)$

$s \leftarrow 0$
while true **do**
  for each valid finite functional graph g with code at most s **do**
    for $k = 0, \ldots, s$ **do**
      if $\theta(a, g, k) = 1$ and $(x, y) \in g$ for some y **then**
        return the unique such y
      end if
    end for
  end for
  $s \leftarrow s + 1$
end while

Its composite with q is the original cover, proving that q is regular epic.

(5) **Identify extensional equality and its name comparison.** Equality of the finite-subgraph ideals is exactly equality of their unions. Uniformly, a finite convergence witness for one code can be transported to an extensionally equal code by running the latter on the finitely many named inputs until the same outputs appear. This gives the equality comparison on names as well as the semantic kernel-pair equation.

(6) **Implement parameter specialization.** Specialization prefixes instructions loading the fixed numerical parameter and then runs the original code on its paired input.

(7) **Implement composition.** Composition renames the scratch registers of two finite programs, runs the first, copies its output to the second input block, and runs the second. These are total recursive operations on instruction lists; their return domains and values follow from the two executions.

□

Lemma D.10 (Diagonal nontermination is not positive)**.** *In the numerical effective model, the complement of* $H = \{e : \varphi_e(e) \downarrow\}$ *is not* Σ*-classified.*

Proof.

(1) **Recognize diagonal termination positively.** H is classified by its bounded-run hit stream.

(2) **Derive a total decision from positive nontermination.** By Lemma D.1, a Σ-classifier of its complement lifts on the numerical presentation through $2^{\mathbb{N}}$ to a total recursive hit stream. Let α, β be the total recursive hit streams for H and its assumed complement. In Algorithm D.4, DECIDE must terminate because precisely one stream has a hit.

(3) **Obtain the diagonal contradiction.** Under this assumption the procedure DIAGONAL has a register index d by Proposition D.9. Evaluating at d contradicts either Boolean answer. This proves the claim.

Algorithm D.4 The diagonal program under the assumed complement classifier

Input: Hypothetical total recursive hit streams α for H and β for $\mathbb{N} \setminus H$

```
procedure DECIDE(e)
    k ← 0
    while α(e, k) = 0 and β(e, k) = 0 do
        k ← k + 1
    end while
    return α(e, k)
end procedure
procedure DIAGONAL(e)
    if DECIDE(e)= 1 then
        while true do
            skip
        end while
    end if
    return 0
end procedure
```

□

The name comparisons above concern denotations and partial domains. Costs are supplied by the recorded instruction events and valuations in Section 4.5 and Construction 13.2. An extensional quotient identifies denotations; equality of the costs of its representatives holds exactly under the kernel-pair criterion of Theorem 4.16.

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